

What Optimizing Labs Reveal: Scaling Laws as Production Functions

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Language-model developers choose model size and training data to minimize training and serving costs. This optimization cuts both ways for measurement. It removes the variation that identifies the technology—information about curvature is fourth order in allocation errors—but makes choices informative about objectives. Designed experiments give, without functional-form restrictions, an elasticity of substitution between parameters and data of about 0.7. Inverting first-order conditions with this technology shows that open-weight models released since 2024 were trained as if compactness—for developers that serve them, lower serving cost—accounted for most of lifetime cost, up from near zero before 2023.

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Training a frontier language model has become one of the largest single investments in the technology sector. The compute used to train frontier models has grown about fivefold a year since 2018 (Section V), in line with [Sevilla and Roldán \(2024\)](#), and the amortized cost of the largest training runs about 2.4-fold a year since 2016 ([Cottier et al., 2024](#)). Developers plan these runs with neural scaling laws, empirical relations between a model’s loss L on held-out text, its number of parameters N and the number of tokens D it is trained on ([Kaplan et al., 2020](#); [Hoffmann et al., 2022](#)). Training costs about $6ND$ floating-point operations (FLOP), and a deployed model costs about $2N$ FLOP for every token it processes. A scaling law is therefore a production function with a known cost side, and the choice of N and D is a cost-minimization problem whose answer depends on expected use: a developer that expects heavy use trains a smaller model on more data ([Sardana et al., 2024](#)). Two objects govern this choice: how

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easily parameters and data substitute, and how much developers value a compact model. Both enter forecasts of data demand, of the energy and capital that serving will absorb, and of the returns to compute in models of AI and growth.

This paper asks what developers' data reveal about these two objects. The data are released models, whose sizes and token counts developers chose, and the experiments run to choose them. Our organizing idea is that optimization cuts both ways for measurement. A developer that minimizes cost places its models on the expansion path, where they carry almost no information about the curvature of the technology. The same optimization makes its choices informative about its objectives: the further a model lies from the path that minimizes training cost, the more the developer must value something that training cost leaves out. Designed experiments break the circle: they supply the technology, which then turns choices into revealed objectives.

Theory.—Because training compute is multiplicative in parameters and data, isocost lines are straight lines in logs, and cost minimization equates the output elasticities of the two inputs, ε_N and ε_D . An interior optimum therefore requires the elasticity of substitution between parameters and data, σ , to lie between zero and one (Lemma 1). In logs a Cobb–Douglas isoquant is itself a straight line, so cost has an interior minimum on an isoquant only where the isoquant is more curved than that. The U-shaped IsoFLOP profiles that practitioners fit, loss against model size at fixed compute, are the empirical content of this restriction. The ratio $w \equiv \varepsilon_N/\varepsilon_D$ measures what else the developer values. A developer stops adding parameters when their marginal product equals their full marginal cost, which includes whatever it loses when a model of given quality is larger; tokens cost only training. Under any objective that rewards lower loss, and ignoring data costs and size-dependent FLOP prices, $w - 1$ is therefore the marginal value of compactness at given quality divided by training expenditure (Proposition 2). We report it as a share of lifetime cost, $s \equiv (w - 1)/w$. For a developer that serves its own models, s is the planned serving share of lifetime cost. For a developer whose users bear the serving cost, it is the value the developer places on compactness, through adoption, memory tiers or latency.

The same geometry limits what optimizing developers' data can reveal. On the expansion path, where parameters and data are functions of compute alone, the data identify the path and the loss–compute frontier but the curvature of the technology only through functional form (Proposition 3). Off the path, the information about the compute-optimal ratio of tokens to parameters, M^* , is second order in developers' allocation errors, and the information about curvature is fourth order: curvature affects loss only through the loss from misallocation, which is itself second order. The better developers optimize, the less their data reveal. This binds in compute-optimal ladders run by a single laboratory; released models lie far off the path, and their heterogeneous values of compactness supply variation that is chosen rather than designed. Designed experiments restore identification without a functional form. With σ^* the elasticity of substitution on the

expansion path, $1/\sigma^* - 1$ equals the curvature of the IsoFLOP profile in log model size at its minimum, divided by twice the slope of the loss–compute frontier in log compute (Proposition 4). Along an IsoFLOP line, loss changes only through the input mix, and the frontier slope converts that curvature into a proportion of the marginal product, so neither the irreducible loss nor the functional form enters.

The technology.—We apply this estimator to the IsoFLOP experiments of three laboratories, Chinchilla (Hoffmann et al., 2022), Llama 3 (Grattafiori et al., 2024) and Marin’s ladders on three corpora (Marin Community, 2026), and to the near-factorial Farseeer sweep (Li et al., 2025b). It gives a common σ^* of about 0.70, with a 95 percent confidence interval of $[0.67, 0.72]$ and no detectable heterogeneity across designs. Parametric fits that free the outer curvature of the loss function come close to this value, while the standard Chinchilla form overstates σ^* on Llama 3’s profiles. The level of the technology does not travel: the compute-optimal ratio M^* differs by more than an order of magnitude across designs, and the allocation exponent, which governs how compute-optimal model size grows with compute, also differs across designs and published laws. Our own pre-registered experiment trains the same architectures on two corpora of different quality with a shared tokenizer. [TBD-m9: one-sentence headline—model-free and flexible-form σ^* by corpus; whether the quality filter is Hicks-neutral or factor-biased on both validation sets and on WikiText; the sign of the Chinchilla form’s extrapolation error in w at high tokens per parameter.]

What over-training reveals.—We invert developers’ first-order conditions on a clean inference-demand sample of 77 open-weight base models released in 2023–2025, which excludes research suites, mixture-of-experts models, and distilled or pruned models. The inversion needs two features of the technology, the zero point $M^*(C)$ at the model’s own compute C and the curvature $1/\sigma^* - 1$, because in the fitted families $\ln w = (1/\sigma^* - 1) \ln[M/M^*(C)]$ with $M \equiv D/N$. Under our reference technology, the Chinchilla data fitted with the outer curvature free, the median model is trained as if compactness were worth three-quarters of lifetime cost ($s = 0.75$), and almost every model is over-trained relative to the training-optimal path. For developers that publish their own scaling laws, the median under their own technology is 0.61, slightly below the reference for the same models. The trend is more robust than the level. Among production-scale open-weight releases, the compute-weighted share under the reference rose from 0.04 in 2019–2022 to 0.82 in 2025, and it rises under every technology we consider.

These objects matter for economic questions (Section V). The allocation exponent governs forecasts of data demand: across its estimates, compute-optimal data demand grows by a factor of 2.0 to 2.8 a year at the current growth rate of compute. The curvature σ^* , by contrast, barely affects the cost of a binding cap on unique training data; how quickly repeated data lose value matters far more (Muennighoff et al., 2023; Villalobos et al., 2024).

What is robust.—The model-free σ^* is stable across the annealed designs of

three laboratories. It is lower, about 0.51, on the small-scale ladder of [Porian et al. \(2024\)](#), whose runs were not annealed, and parametric estimates from small sweeps without IsoFLOP profiles are heterogeneous. The sign of over-training is also robust. For any technology with U-shaped IsoFLOP profiles, a model is over-trained exactly when its M exceeds $M^*(C)$, so the sign needs only bounds on M^* beyond the largest designed budget (Proposition 5). With M^* anchored at the largest IsoFLOP budgets and its drift bounded by published expansion paths, $w > 1$ is identified for 86 percent of the clean sample and $w < 1$ for none. The level of s is not robust. It depends on M^* at frontier scale, which no public design observes, and the median share ranges from 0.17 to 0.85 across the technologies we specify ex ante. Inside Farseer’s design, where extrapolation in tokens per parameter can be checked, parametric forms understate w at high M ; beyond the designs in compute, nothing is observed. Open-weight models are more over-trained than closed ones at given compute, and with few developers in the sample, serving one’s own models does not significantly predict over-training, so for developers that do not serve their models we read s only as a value of compactness. We treat s as ordinal and conditional on a technology.

Related literature.—[Hao and Merrill \(2026\)](#) model a profit-maximizing developer that pays for training and serving, and derive the elasticity of substitution implied by the Chinchilla form. They solve the developer’s problem under a Leontief approximation, which rules out over-training. Proposition 2 inverts the first-order conditions of such a problem when $\sigma > 0$ and extends it to developers that do not bear serving costs. [Sardana et al. \(2024\)](#), [Bian, Yan and Venkataraman \(2025\)](#) and [Roberts et al. \(2026\)](#) solve the forward problem: optimal size and data given inference demand, latency constraints or repeated sampling at test time. We solve the inverse problem; their mechanisms are sources of the value of compactness it recovers. [Kricheli et al. \(2026\)](#) show that designs with a fixed ratio of tokens to parameters leave scaling-law parameters ill-conditioned when the two exponents are close. Proposition 3 shows that on a compute-optimal path the same degeneracy arises whatever the exponents, because equal decay rates of the two loss terms are what compute optimality means, and it gives the rate at which allocation errors restore information. [Mertens, Fischl-Lanzoni and Thompson \(2026\)](#) estimate developer-specific compute efficiency from cross-developer regressions, and [Whitfill \(2025\)](#) shows that observational estimates of algorithmic progress are subject to selection bias. Online Appendix E takes our framework to such observational data.

In industrial organization, the observation that optimizing firms facing common prices generate data that do not identify the technology goes back to [Marschak and Andrews \(1944\)](#). [Bond and Söderbom \(2005\)](#) show that identification then requires variation in relative prices, adjustment costs or optimization errors, and [Akerberg, Caves and Frazer \(2015\)](#) and [Gandhi, Navarro and Rivers \(2020\)](#) study the consequences for proxy and first-order-condition estimators. Language-model training is the cleanest case, because FLOP accounting gives every developer

the same log-cost weights. We add the rates at which information returns with allocation errors, the design-theoretic orders of [Box and Lucas \(1959\)](#) generated here by economic wedges, and a designed-experiment estimator that needs no functional form. Our inversion follows the production approach of [De Loecker and Warzynski \(2012\)](#), but w is not a markup: it prices model size beyond its cost in training. Its elasticities come from experiments rather than from estimation on the same choices, which avoids the circularity stressed by [Doraszelski and Jaumandreu \(2021\)](#) at the cost of external validity. The critiques of [Bond et al. \(2021\)](#) and [Raval \(2023\)](#), on the output concept and factor-biased technology, carry over, and Section [IV](#) takes them up.

Section [I](#) sets out the training problem, Section [II](#) what optimizing developers’ data identify, Section [III](#) the technology, Section [IV](#) what over-training reveals, and Section [V](#) the economic implications; Section [VI](#) concludes. Online Appendices [A–F](#) contain proofs, data construction, Monte Carlo designs and robustness results.

I. The Training Problem

This section writes a scaling law as a production technology with a known, multiplicative cost and states the three results used throughout: duality, the requirement that interior compute optima have an elasticity of substitution below one, and the inversion of a developer’s first-order conditions. Proofs are in Online Appendix [A](#).

A. Technology and Cost

A developer trains a language model with N parameters on D tokens (tokens processed, counting repetitions) and measures its cross-entropy loss L , in nats per token, on a held-out corpus. Following [Hoffmann et al. \(2022\)](#), with productivity terms added, we write

$$(1) \quad L = E + e^{-\omega} \left[A(e^{\psi_N} N)^{-\alpha} + B(e^{\psi_D} D)^{-\beta} \right]^{\kappa} e^{\epsilon}, \quad A, B, \alpha, \beta, \kappa > 0.$$

Here E is the irreducible loss, and the bracket collects the losses from finite parameters and finite data. Productivity is Hicks-neutral (ω) or factor-augmenting (ψ_N and ψ_D ; a cleaner corpus raises ψ_D), and ϵ is seed and evaluation noise. The developer knows its productivity up to the error of its own small-scale experiments; Proposition [2](#) shows what a gap between belief and technology does. The Chinchilla law has $\kappa = 1$ and no productivity terms. The outer exponent κ scales both output elasticities and leaves the isoquants unchanged, so it separates returns to scale from substitution; $\kappa = 1$ is testable (Section [III](#)). Where the noise enters matters only for estimation, and the information rates of Section [II](#) are the same with noise on $\ln L$.

Lower-case letters denote logs, $n \equiv \ln N$, $d \equiv \ln D$ and $c \equiv \ln C$, and $y \equiv$

$-\ln(L - E)$ is the log output index. With $u \equiv A(e^{\psi_N} N)^{-\alpha}$ and $v \equiv B(e^{\psi_D} D)^{-\beta}$, the output elasticities are

$$(2) \quad \varepsilon_N \equiv \frac{\partial y}{\partial n} = \frac{\kappa \alpha u}{u + v}, \quad \varepsilon_D \equiv \frac{\partial y}{\partial d} = \frac{\kappa \beta v}{u + v},$$

and neither depends on ω , ϵ or E . When $\alpha = \beta$, reducible loss is a power of a constant-returns CES aggregate of N and D . When $\alpha \neq \beta$ the technology is not homothetic, but it is quasi-homothetic with respect to the non-uniform scaling $(N, D) \mapsto (t^\beta N, t^\alpha D)$, which multiplies reducible loss by $t^{-\kappa\alpha\beta}$ for every $t > 0$ (Online Appendix A, Lemma A4). Every isoquant is therefore a rescaled copy of every other, the compute-optimal mix drifts with compute as a power law, and all curvature lies across the expansion path. This is why data on the path carry no information about curvature (Section II).

Training a dense transformer takes about six floating-point operations (FLOPs) per parameter per token, so training compute is $C \approx 6ND$ (Kaplan et al., 2020). This is a FLOP-accounting approximation, not an identity: it omits attention and embedding FLOPs, and FLOPs are not dollars. Proposition 2 carries size-dependent FLOP prices and data costs. In FLOP units the cost is multiplicative: both log-cost weights, $\partial c / \partial n$ and $\partial c / \partial d$, equal one, and the implicit price of a parameter, $6D$, rises with the quantity of the other input. Minimizing compute for a given loss therefore equates the output elasticities, $\varepsilon_N = \varepsilon_D$, without the price data that the first-order-condition approaches of De Loecker and Warzynski (2012) and Gandhi, Navarro and Rivers (2020) require: FLOP accounting supplies the cost shares. Isocosts are straight lines in logs, $n + d = c - \ln 6$.

In this language, iso-loss curves are isoquants, an IsoFLOP profile traces loss along an isocost, the compute-optimal allocation $(N^*(C), D^*(C))$ is a conditional factor demand, and the loss–compute frontier $L^*(C)$ is the inverse of a cost function (Figure 1, panel a). We write $M \equiv D/N$ for tokens per parameter and $M^*(C)$ for its compute-optimal value. Isoquants, the elasticity of substitution, the compute-optimal allocation and the wedge defined below are ordinal: they are unchanged by any strictly increasing transformation of the same output (loss, bits per byte, or accuracy on a benchmark that is monotone in loss) and do not depend on E , ω or κ (Online Appendix A, Lemma A1). The frontier elasticity and returns to scale are cardinal, defined in units of reducible loss. Input units matter too: M^* depends on the tokenizer and on whether embedding parameters are counted, so it is comparable across studies only after harmonization (Section III).

Online Appendix Table D1 gives a full dictionary between scaling-law practice and production economics. Three analogies break down. There are no market prices for parameters and tokens: FLOP accounting supplies the log-cost weights, and differences in the price of a FLOP appear as wedges. Returns to scale measured in loss depend on the cardinalization of output, so “diminishing returns to compute” has no unit-free content. And the inputs of a training sweep are assigned by an experimenter, not chosen: an IsoFLOP sweep is designed variation

in the input mix, not a shock to relative prices, so the endogeneity problems of IO arise for released models, not for experiments.

B. Duality

Proposition 1 (Duality). *Let $\kappa = 1$ in (1), set productivity and noise to zero except in part (iii), and let $a \equiv \beta/(\alpha + \beta)$, $\gamma \equiv \alpha\beta/(\alpha + \beta)$ and $G \equiv (\alpha A/\beta B)^{1/(\alpha + \beta)}$.*

- (i) *The allocation that minimizes $6ND$ subject to $L \leq \bar{L}$, or maximizes y at given C , satisfies $\varepsilon_N = \varepsilon_D$. The conditional factor demands are $N^*(C) = G(C/6)^a$ and $D^*(C) = G^{-1}(C/6)^{1-a}$, so $M^*(C) = G^{-2}(C/6)^{1-2a}$.*
- (ii) *The loss-compute frontier and the cost function are $L^*(C) = E + K(C/6)^{-\gamma}$ and $C^*(L) = 6[K/(L - E)]^{1/\gamma}$, with $K \equiv \frac{\alpha + \beta}{\beta}AG^{-\alpha}$. Along the expansion path, $\varepsilon_N = \varepsilon_D = \gamma$.*
- (iii) *At given compute, Hicks-neutral productivity ω leaves the allocation unchanged, and factor-augmenting productivity moves it only through the bias index $\chi \equiv \beta\psi_D - \alpha\psi_N$: $\ln M^*(C)$ falls by $2\chi/(\alpha + \beta)$.*

The proof is in Online Appendix A (Lemma A3). The proposition is Shephard–Uzawa duality for the scaling law (Shephard, 1953; Uzawa, 1964): the technology, the conditional factor demands and the cost function carry the same information, linked by a and γ . With $\kappa \neq 1$, part (i) is unchanged and γ is multiplied by κ . By part (ii), the elasticity of compute with respect to output, measured as $1/(L - E)$, is $1/\gamma$ along the frontier; its inverse, γ , is the returns-to-scale measure that Nerlove (1963) estimated from a cost function for electricity supply and found to vary with scale. Under the parameters of Besiroglu et al. (2024), $\gamma = 0.178$: a one percent reduction in reducible loss requires about 5.6 percent more compute.

Scaling studies estimate each of these objects. A fit of (1) to all runs (Approach 3 of Hoffmann et al., 2022) is a primal estimator. A regression of the loss-minimizing $\ln N$ of each IsoFLOP profile on $\ln C$ (their Approach 2) estimates the factor demand, and a frontier fitted through the same minima, which we call the IsoFLOP-minima frontier, estimates the cost function; their Approach 1 reports the minimizing N and D as well as the minimal loss, so it estimates both. Gaps among these estimates test duality (Klump, McAdam and Willman, 2007); Online Appendix D reports such tests. The minima of IsoFLOP profiles are outcomes of inputs that an experimenter assigned, not choices.

C. Substitution

The Hicks elasticity of substitution between parameters and data depends on the input mix only through the ratio of the output elasticities, $w \equiv \varepsilon_N/\varepsilon_D = \alpha u/(\beta v)$, which we call the wedge. For every member of the family (1),

$$(3) \quad \sigma(w) = \frac{1 + w}{1 + \alpha + w(1 + \beta)}, \quad \sigma^* \equiv \sigma(1) = \frac{2}{2 + \alpha + \beta},$$

which follows from the formula of Hao and Merrill (2026) (Online Appendix A, Lemma A1). The local elasticity lies between $1/(1 + \alpha)$ and $1/(1 + \beta)$ and equals σ^* on the compute-optimal path, where $w = 1$. With two inputs, the Hicks, Allen–Uzawa and Morishima elasticities coincide (Blackorby and Russell, 1989). Under the Besiroglu et al. (2024) parameters ($\alpha = 0.348$, $\beta = 0.366$), $\sigma^* = 0.737$, and σ varies only between 0.732 and 0.742 over all input mixes.

Lemma 1 (Interior compute optima require $0 < \sigma < 1$). *Let $y = F(n, d)$ be any twice continuously differentiable technology with $F_n, F_d > 0$, and let $C = 6ND$. At an allocation with $\varepsilon_N = \varepsilon_D$:*

- (i) *the second-order condition for minimizing compute at given output holds strictly if and only if $0 < \sigma < 1$ there;*
- (ii) *if $\sigma > 1$ or $\sigma < 0$ there, the allocation minimizes rather than maximizes output along its isocost; if $\sigma > 1$ everywhere, compute-optimal training is a corner solution.*

Every member of the family (1) has $\sigma < 1$ at every input mix, so the first-order condition in Proposition 1 is sufficient.

The proof is in Online Appendix A (Lemma A2 and Corollary A1). With a linear cost $q_N N + q_D D$, isocosts are straight lines in levels, and any convex isoquant ($\sigma > 0$) gives an interior tangency. With $C = 6ND$ they are straight lines in logs, so the isoquant must be convex in logs, which requires $\sigma < 1$. Because the implicit price of each input rises with the quantity of the other, shifting toward one input raises its own price, and that pays only if the inputs are complements in this sense.

The lemma’s observable implication is that IsoFLOP profiles, loss against $\ln N$ at fixed compute, are U-shaped with an interior minimum, as the designs of Section III show wherever their grids bracket the minimum. That $\sigma^* < 1$ is implied by this fact and imposed by every functional form we fit; it is not a finding. Its level is: Proposition 4 turns the curvature of the U into σ^* without a functional form. And $\sigma > 0$ is what makes over-training a sensible response to a value of compactness. In the Leontief limit of Hao and Merrill (2026) the optimum sits at the kink whatever that value, whereas for $0 < \sigma < 1$ parameters can be traded for tokens at a cost governed by σ^* (Figure 1, panel b).

D. What Over-Training Reveals

Most language models released since 2023 lie far from the expansion path, with M well above $M^*(C)$ (Section IV). A developer that minimized training cost would have $w = 1$; the next result says what a wedge above one reveals about the developer’s objective.

Proposition 2 (What over-training reveals). *A developer chooses (N, D) to maximize $V(L(N, D), N) - X_T - c_D D$. Here V is any objective that falls with loss*

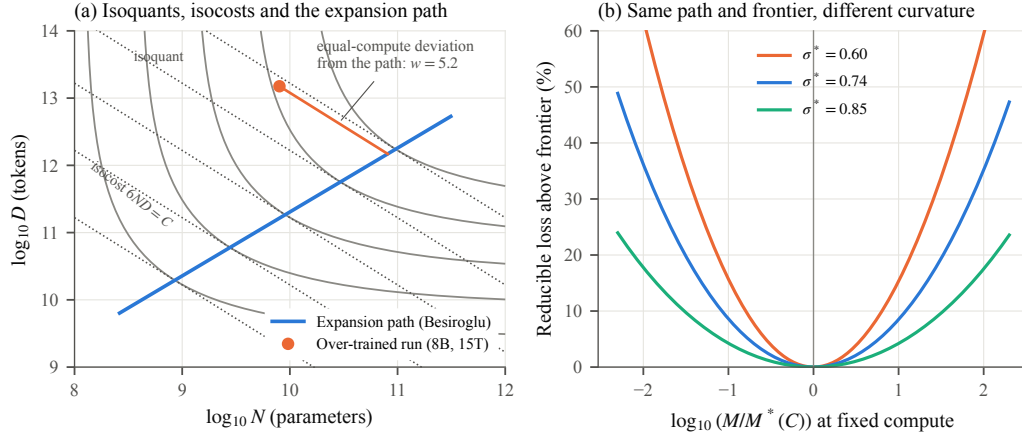


FIGURE 1. THE GEOMETRY OF THE TRAINING PROBLEM

Note: Panel (a): the $(\log_{10} N, \log_{10} D)$ plane under the Besiroglu et al. (2024) parameters. Solid gray curves are isoquants at the frontier losses for $C = 10^{20}, \dots, 10^{24}$ FLOP; dotted lines are isocosts $6ND = C$; the blue line is the compute-optimal expansion path. The orange dot is a run with $N = 8 \times 10^9$ (total) parameters and $D = 1.5 \times 10^{13}$ tokens, nominally Llama 3 8B, joined to the path at equal compute; its wedge is $w = 5.2$. Panel (b): three members of the κ family with $\sigma^* = 0.60, 0.74$ and 0.85 , constructed to share the expansion path and the loss–compute frontier (Online Appendix A, Proposition A1(iv)). The vertical axis is reducible loss in excess of the frontier at $C = 10^{22}$ FLOP, as a function of the deviation of tokens per parameter from the compute-optimal ratio. The technologies coincide on the path and differ off it.

and may depend on size directly; $X_T \equiv c_T(N) 6ND$ is training expenditure at the (shadow) price $c_T(N)$ of a training FLOP, whose size elasticity is δ ; and $c_D \geq 0$ is a cost per token. Let $m_N \equiv -(\partial V / \partial \ln N)_L / X_T$ be the value of compactness at fixed quality per unit of training expenditure, and $m_D \equiv c_D D / X_T$.

(i) At an interior optimum with no binding constraint,

$$(4) \quad w = \frac{\varepsilon_N}{\varepsilon_D} = \frac{1 + \delta + m_N}{1 + m_D}.$$

With $\delta = m_D = 0$, $w - 1 = m_N$, and the expenditure share $s \equiv (w - 1)/w$ equals $m_N/(1 + m_N)$. The wedge is computed from (N, D) and the technology alone; it does not depend on ω , ϵ , E or κ .

- (ii) (Serving.) Let $\delta = m_D = 0$, and let the developer bear a share λ of the expenditure X_S on serving T tokens over the model's life, at $2N$ FLOPs per token and p times the price of a training FLOP, with T independent of size at given loss. Then $w = 1 + \lambda p T / (3D)$, $w - 1 = \lambda X_S / X_T$, and $s = \lambda X_S / (X_T + \lambda X_S)$ is the internalized serving share of lifetime expenditure.
- (iii) (Constraints and misspecification.) A binding memory, tier or latency cap multiplies the right side of (4) by $1 + \nu$ with $\nu \geq 0$; a binding cap on tokens adds $\mu \geq 0$ to its denominator; and logit distillation from a teacher with N_T parameters is a data cost with $m_D = N_T / (3N)$. If the developer's first-order conditions involve a technology L' other than the econometrician's L (its own productivity, a law it believes, or the output it values), the measured wedge is $\tilde{w} = w w_L / w_{L'}$, which equals $w e^{-\chi}$ under factor-augmenting productivity.
- (iv) (Family token budgets.) If a family trains models of sizes $N_1, \dots, N_K$ on one token budget D , member wedges differ only through the technology: $\ln w_i - \ln w_j = -\alpha(\ln N_i - \ln N_j)$. If the sizes are chosen at the margin and there is no data cost, the family's first-order conditions identify one number, $\sum_i \pi_i m_{N,i} = \bar{w}_H - 1$, where $\bar{w}_H$ is the compute-weighted harmonic mean of the member wedges and the weight π_i is proportional to member i 's share of the family's training compute divided by w_i . If the sizes come from a menu, the wedges carry no information about the $m_{N,i}$.

The proof is in Online Appendix A (Propositions A8 and A9), which also treats demand that responds to size and minimum-size and time-to-train constraints. The developer adds parameters until their marginal value in loss equals their marginal cost: training expenditure plus whatever it loses when the model is larger at the same quality, such as the cost of serving it, adoption among users who cannot run it, latency or a memory tier. Tokens cost only training and data. The ratio of the two first-order conditions therefore puts the value of compactness in the numerator, and an econometrician who sees only training cost observes a wedge: the developer acts as if parameters were w times as expensive, relative to tokens, as training accounting says.

What $w - 1$ measures depends on the developer’s objective; Online Appendix Table F1 tabulates eleven cases. Three models of conduct matter most. For a developer that serves its own model ($\lambda = 1$), s is planned serving expenditure as a share of lifetime expenditure, with no conversion of FLOPs into dollars; a count of tokens, $T = 3D(w - 1)/(\lambda p)$, needs the price ratio p , which is likely above one because decoding is often bound by memory bandwidth. Compute proportional to N that is planned at training time, such as reinforcement-learning rollouts, and test-time scaling (Roberts et al., 2026) enter X_S in the same way. For an open-weight developer whose users bear the serving cost, $w - 1$ is the value of compactness through adoption. Under a binding tier, memory or latency cap (Bian, Yan and Venkataraman, 2025), $w - 1$ only bounds m_N from above. Section IV tests the three readings as a problem of conduct (Berry and Haile, 2014; Backus, Conlon and Sinkinson, 2021; Duarte et al., 2024). We therefore call s the share of the value of compactness, and planned serving expenditure only for developers that serve their models.

The other sources of a wedge have known signs (Online Appendix A, Proposition A8(iv)). Caps, a price of a FLOP that rises with size, and parameter-augmenting productivity ($\chi < 0$) raise the measured wedge. Data costs, caps on tokens, distillation (Busbridge et al., 2025), minimum-size floors and data-augmenting productivity ($\chi > 0$) lower it. Serving demand cannot produce a wedge below one as long as serving cost rises with size. Under the Besiroglu et al. (2024) parameters, Gopher has $w = 0.36$, which reveals a binding data constraint, better data, or optimization against another technology, plausibly the law of Kaplan et al. (2020). Heterogeneity in these wedges is what moves released models off the expansion path, so it is also the variation that could identify the technology from observational data, the counterpart of the relative-price variation of Bond and Söderbom (2005) (Section II).

The construction parallels the production approach to markups (De Loecker and Warzynski, 2012). The wedge is the ratio statistic of Raval (2023) with FLOP-accounting weights; a ratio different from one, which Raval reads as misspecification, is here the object of interest. The two classic critiques are the last case of part (iii): factor-augmenting heterogeneity (Demirer, 2020; Raval, 2023) and a mismatch between the measured output and the output the developer values (Bond et al., 2021). As in De Ridder, Grassi and Morzenti (2026), levels are more fragile than ranks.

Part (iv) matters because many families train every size on one token budget (Section IV). Within such a family a small sibling’s large wedge is fixed by the technology, so member-level wedges are not revealed preferences; the family reveals a weighted average that puts less weight on its more over-trained members or, if sizes come from a menu of tiers, nothing about compactness. A common budget under member-by-member choice would be a knife-edge, so its prevalence suggests that D is chosen at the family level.

In the κ family the wedge has a closed form. For a model with compute C and

tokens per parameter M ,

$$(5) \quad \ln w = \left(\frac{1}{\sigma^*} - 1 \right) \ln \frac{M}{M^*(C)},$$

where $M^*(C)$ is the compute-optimal ratio at the model’s own compute (Online Appendix A, Lemma A4 and Corollary A3). The log wedge is the distance from the expansion path at equal compute, scaled by curvature (Figure 1, panel a). The technology supplies the zero point $M^*(C)$ and the scale $1/\sigma^* - 1$; at given compute, it ranks models by M . A model with 8 billion parameters trained on 15 trillion tokens (nominally Llama 3 8B, $M \approx 1,875$) has $w \approx 5.2$ under the Besiroglu et al. (2024) parameters: its developer behaves as if making it one percent smaller at the same loss were worth 4.2 percent of the training bill, and $s \approx 0.81$. (Section IV estimates $w \approx 8.4$ under Meta’s own technology.) Such estimates are only as good as $M^*(C)$, which on-path data identify only by assuming optimality and which no public design observes at frontier scale (Sections II and IV).

II. What Optimizing Labs’ Data Identify

Section I described the technology by four objects: the slope a of the expansion path, the frontier elasticity γ , the elasticity of substitution at the optimum σ^* and the compute-optimal ratio $M^*(C)$. Reading developers’ choices requires the last two: by equation (5), $M^*(C)$ is the zero point of the wedge and σ^* sets its scale. This section asks which data identify them. Optimization removes the variation that identifies curvature (subsection A). Designed IsoFLOP variation identifies σ^* without a functional form (subsection B). Beyond the largest designed budget, $M^*(C)$ is only partially identified, and for released models what survives is chiefly the sign of the wedge (subsection C). Proofs are in Online Appendix A.

The results use technology (1) with Hicks-neutral productivity only ($\psi_N = \psi_D = 0$), and the notation of Section I: $a \equiv \beta/(\alpha + \beta)$ and $b \equiv 1 - a$ are the slopes of the expansion path in $(\ln N, \ln D)$, $\gamma \equiv \kappa\alpha\beta/(\alpha + \beta)$ is the frontier elasticity, $\sigma^* = 2/(2 + \alpha + \beta)$, and G and K are the constants of the path and the frontier (Proposition 1). A sample is *on-path* if its runs share one technology, each run minimizes training compute for its loss, so that every wedge equals one, and compute varies over an interval.

A. Optimization Removes the Variation That Identifies Curvature

Proposition 3 (What optimizing labs’ data identify, and how fast). *Let runs follow (1) with $\psi_N = \psi_D = 0$.*

- (i) *(On the path.) In an on-path sample, $n_i = \ln G + a(c_i - \ln 6)$ and $d_i = -\ln G + b(c_i - \ln 6)$: choices reveal the path, that is, a and $M^*(C)$, and nothing else. Outcomes satisfy $\mathbb{E}[y \mid c] = \gamma(c - \ln 6) - \ln K + \mathbb{E}[\omega \mid c]$, so within the model γ is identified if and only if $\mathbb{E}[\omega \mid c]$ is known up to a*

constant, as in a single-lab sweep. Curvature is not identified: for every $\sigma^* \in (0, 1)$ some member of family (1) generates the same joint distribution of (n, d, y) , and by Lemma 1 no value $\sigma^* \geq 1$ does.

- (ii) (Global versus local identification.) Suppose also that $\mathbb{E}[\omega \mid c]$ is constant and $\kappa = 1$. Globally, $\alpha = \gamma/a$ and $\beta = \gamma/b$ are identified by the functional form, and A/B , hence the technology's own cost-minimizing mix, only by imposing that the observed path is optimal. Locally, identification fails: perturbations that move the decay rates of the two loss terms along the path, αa and βb , in opposite directions leave on-path outcomes unchanged to first order, so the least-squares criterion is of fourth order along them. The errors of estimators of σ^* should then not be expected to shrink faster than the inverse fourth root of the number of runs, whether or not $\alpha = \beta$, and Wald intervals are invalid. Imposing $\beta/(\alpha + \beta) = a$ restores the square-root rate.
- (iii) (Off the path.) Let runs deviate from the path with log wedges $\ln w_i$, with E known and Gaussian noise. To leading order, the information about $\ln M^*$ is proportional to the residual dispersion of the $\ln w_i$. The information about σ^* is bounded by a multiple of $\sum_i \Phi_i^2$, where $\Phi_i \equiv \ln(C_i/C_{\min}(R_i))$ is run i 's Farrell allocative loss: the log of its compute relative to the least compute that attains its reducible loss $R_i = L_i - E$. If the deviations of a fixed design are scaled by $v \rightarrow 0$, the two informations are $O(v^2)$ and $O(v^4)$, so the standard error of $\ln \hat{M}^*$ grows like v^{-1} and that of $\hat{\sigma}^*$ like v^{-2} .

A ray has no curvature. Because the FLOP accounting gives every lab the same log-cost weights, cost-minimizing labs that share a technology place every run on the expansion path, along which the family is quasi-homothetic: reducible loss is log-linear along the path, and all curvature lies across it. Choices trace the path and outcomes trace the frontier, but neither reveals how fast loss rises off the path, which is what σ^* measures. A one-parameter set of technologies shares the path and the frontier and differs only off it. At five times the compute-optimal ratio, reducible loss lies 2.0, 4.1 and 7.6 percent above the common frontier for the members with $\sigma^* = 0.85, 0.74$ and 0.60 drawn in Figure 1, panel b.

Part (ii) explains why fits of the Chinchilla form to compute-optimal data can nonetheless look precise. Compute-optimality means that the two loss terms decay at the same rate along the path, so the data see a single exponential in compute; they cannot tell how its level splits between the terms, and a different split absorbs decay rates that move apart. The on-path model is thus the analogue of an over-fitted mixture, whose parameters can be learned only at the fourth-root rate (Chen, 1995). What precision remains under $\kappa = 1$ comes from the restriction $\beta/(\alpha + \beta) = a$, which ties $\sigma^* = 2/(2 + \gamma/(ab))$ to the path and the frontier: functional form plus optimality. Simulations in Online Appendix A confirm the fourth-root rate, and the square-root rate with the restriction imposed; the rate rests on such simulations and a heuristic argument, not on a limit

theorem (Sargan, 1983; Rotnitzky et al., 2000). The mechanism also reconciles our result with Kricheli et al. (2026), who find fixed tokens-per-parameter designs ill-conditioned as $|\alpha - \beta| \rightarrow 0$: the two terms decay at the same rate along a fixed-ratio ray if and only if $\alpha = \beta$, and along the compute-optimal path always.

Real ladders and labs deviate from the path. Because the frontier is a power law, output decomposes exactly as

$$(6) \quad y_i = \gamma(c_i - \ln 6) - \ln K - \gamma \Phi_i - \epsilon_i$$

with ω normalized to zero. Curvature enters output only through the Farrell losses, which are Harberger triangles, $\Phi_i \approx \sigma^*(\ln w_i)^2/[4(1 - \sigma^*)]$. A run reveals where the path lies to the extent that it is off the path, which is first order in its log wedge, and the path’s curvature only to the extent that it wastes compute, which is second order. Information squares these: halving the dispersion of log wedges quarters the information about M^* and divides the information about σ^* by sixteen. The better labs optimize, the less their data reveal.

Proposition 3 is the point of Marschak and Andrews (1944) in its cleanest form: without wedges, every allocation lies on one ray, and identification requires the variation in relative prices, adjustment frictions or optimization errors of Bond and Söderbom (2005) (see also Gandhi, Navarro and Rivers, 2020). Here these are heterogeneous wedges from serving demand, data costs, size-dependent prices of a FLOP and binding constraints (Proposition 2), and belief errors such as allocation by the Kaplan et al. (2020) law. IsoFLOP sweeps are different: the experimenter assigns the inputs, so their minima estimate the path from outcomes, not from choices. The rates in part (iii) are the design-theoretic property of estimating a curvature parameter (Box and Lucas, 1959; Kiefer and Wolfowitz, 1959); what is specific here is that economic wedges generate the design’s spread. The problem is distinct from the functional dependence of Akerberg, Caves and Frazer (2015), which concerns the first stage of proxy estimators.

The result binds for compute-optimal ladders, fixed tokens-per-parameter designs run by one lab (DataDecide trains every size at about 100 tokens per parameter; Magnusson et al., 2025) and releases of the compute-optimal era. Released models since 2023 lie far off the path. Their heterogeneous values of compactness supply transverse variation, but using it requires the exclusion restriction that these values move the mix without moving the technology, and the variation may be confounded with factor-biased technology. We therefore take the technology from designed sweeps and use released models’ wedges to read objectives (Section IV).

The Chinchilla data show part (i) directly. On the 41 runs within 0.15 log points of the fitted compute-optimal size, the profile likelihood for σ^* with κ free is flat: its 95 percent confidence set runs from 0.20 to 0.99, the edge of the grid searched. The full 240-run design, which adds transverse variation, gives [0.68, 0.72] (Online Appendix D).

Figure 2 shows what the rates mean at realistic noise and design sizes, in sim-

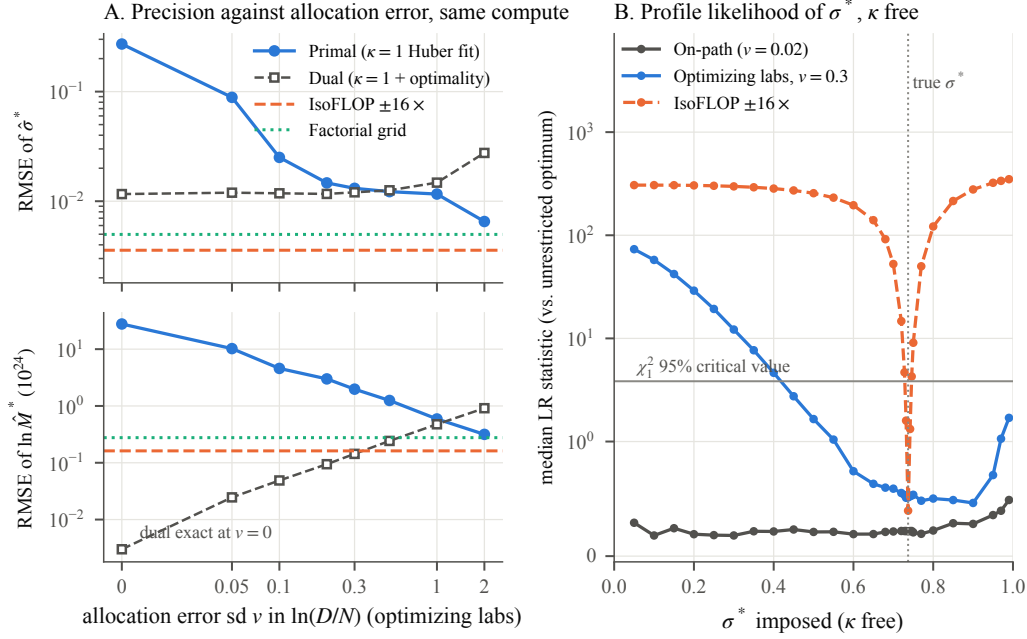


FIGURE 2. IDENTIFICATION BY DESIGN: MONTE CARLO EVIDENCE

Notes: Simulations of the Besiroglu et al. (2024) technology ($\sigma^* = 0.737$; $M^* = 18.1$ at 10^{24} FLOP) with normal noise of standard deviation 0.0075 in $\ln L$. Every design has 90 runs on nine budgets matching Chinchilla’s nominal IsoFLOP budgets (6×10^{18} to 3×10^{21} FLOP) and the same total compute, 5.1×10^{22} FLOP. Optimizing labs set $\ln(D/N)$ at its compute-optimal value plus a normal allocation error with standard deviation v ; the IsoFLOP design places ten sizes log-spaced over $[N^*/16, 16N^*]$ at each budget, and the factorial design crosses nine values of N with ten of D . Panel A: root mean squared error of $\hat{\sigma}^*$ and of $\ln \hat{M}^* (10^{24})$ for optimizing labs, 500 replications per design. Primal: Huber ($\delta = 10^{-3}$) fit of the Hoffmann et al. (2022) form with $\kappa = 1$, started from nine values drawn at random over the region of Hoffmann et al.’s initialization grid, none at the truth. Dual: exponents from the path of the optima and the frontier under $\kappa = 1$ and optimality. Horizontal lines: primal estimator under the IsoFLOP and factorial designs. Panel B: median profile likelihood-ratio statistic for σ^* with κ free, relative to the unrestricted optimum, on a 29-point grid over $[0.05, 0.99]$ (150 replications), for on-path labs ($v = 0.02$), optimizing labs with $v = 0.3$ and the IsoFLOP design; horizontal line, the 95 percent χ^2_1 critical value; dotted line, the truth.

Source: Authors’ simulations; design, estimators and full tables in Online Appendix C.

ulated designs of 90 runs with equal total compute that differ only in where runs sit on their isocost lines. Optimizing labs miss the compute-optimal $\ln(D/N)$ by a normal error with standard deviation v , so their log wedges have standard deviation $(1/\sigma^* - 1)v$. With $v = 0$, a Huber fit of the Chinchilla form ends on a corner, with one power-law term switched off, in 87 percent of replications. As v rises from 0.05 to 0.2, the root mean squared error of $\hat{\sigma}^*$ falls from 0.089 to 0.015 (panel A); an IsoFLOP design with the same compute gives 0.0036. The compute-optimal ratio is harder still: the root mean squared error of $\ln \hat{M}^*$ is 4.6 at $v = 0.1$ and 0.59 at $v = 1$, against 0.16 for the IsoFLOP design.

For moderate allocation errors this precision is functional-form information, as part (ii) predicts. With κ free (panel B), the 95 percent profile confidence set for σ^* covers the entire grid searched, $[0.05, 0.99]$, in 91 percent of on-path replications. At $v = 0.3$ it rules out low elasticities but still contains all of $[0.50, 0.95]$ in 65 percent of replications, against widths of 0.016 and 0.019 for the IsoFLOP and factorial designs. Standard inference also fails on the path: the Wald covariance is singular, or the fit sits on a corner, in 79 percent of on-path replications, and a bootstrap that restarts every draw from all starting values covers the true σ^* only 78 percent of the time. Online Appendix C reports the full design, the sensitivity to starting values, noise levels and clustered noise, and a simulated industry.

B. Designed Variation Identifies Curvature without a Functional Form

Proposition 3 says what a design must supply: movement across the path. The next result reads the technology off that movement without a functional form. Let $x \equiv \ln M$, so that an IsoFLOP profile is loss as a function of x at fixed c ; subscripts denote partial derivatives.

Proposition 4 (Model-free σ^* and w). *Let L be any measure of output that falls as quality rises (loss, bits per byte, or $\ln(L - E)$ for any E), and let the technology be smooth with positive marginal products.*

(i) *At any point, $w = (L_c - L_x)/(L_c + L_x)$.*

(ii) *At a compute-optimal point,*

$$(7) \quad \frac{1}{\sigma^*} - 1 = \frac{L_{nn}|_C}{2|dL^*/dc|},$$

where $L_{nn}|_C$ is the curvature of the IsoFLOP profile in $\ln N$ at its minimum and $L^(c)$ is the loss-compute frontier. The ratio is invariant to strictly monotone transformations of output.*

(iii) *At $M^*(C)$, $\partial \ln w / \partial \ln M|_C = 1/\sigma^* - 1$. In family (1), $\ln w$ is exactly linear in $\ln(M/M^*(C))$ with this slope; in general the slope varies with M .*

- (iv) Under $\kappa = 1$ the curvature of every IsoFLOP profile is pinned by the slopes of the frontier and the path, $-y_{nn}|_C = \gamma^2/(ab)$; with κ free it is a separate parameter.
- (v) Hence σ^* is identified by IsoFLOP profiles together with the frontier slope, and w at a point by local variation in two non-collinear directions around it, but only inside the design's support.

Part (i) is a change of coordinates: the slopes of any output measure along the isocost and along a ray of constant M give the two marginal products, and their ratio is the wedge. Part (ii) is the elasticity of substitution at the optimum written in observables. Along an IsoFLOP line, loss changes only through the mix. At the minimum the marginal products of parameters and tokens are equal; the curvature of the profile measures how fast they diverge as the mix moves, and the frontier slope, which by the envelope theorem is the marginal product of compute, measures their common level. A profile that is sharply curved relative to the frontier slope signals poor substitutes. Because both terms are measured in the same output units, the irreducible loss, the transformation of output and κ cancel. Part (iv) shows what $\kappa = 1$ does: it ties the curvature of every profile to (a, γ) , which is the functional-form information of Proposition 3(ii). Part (iii) gives a test of the family, which Section IV applies where a factorial design reaches high M .

Both terms of (7) are familiar: the quadratic coefficient of the parabola fitted to each IsoFLOP profile, as in Approach 2 of Hoffmann et al. (2022), and the slope of the frontier through the minima. Quadratics fitted over wide grids overstate the curvature slightly (Czech et al., 2026); narrow windows or quartic fits remove the bias at some cost in variance (Online Appendix Remark A10). Section III applies the estimator to every public IsoFLOP design.

C. Beyond the Design: What Remains Identified at Frontier Scale

Revealed demand needs the wedge at released models, and most were trained with more compute than any design. The largest public IsoFLOP budget is 10^{22} FLOP (Grattafiori et al., 2024); 72 of the 77 models in the clean sample of Section IV used more, and only two lie inside the Chinchilla design's range of both compute and tokens per parameter. The zero point must therefore be extrapolated. Let $x^*(c) \equiv \ln M^*(C)$, with path slope $e \equiv d \ln M^*/d \ln C = 1 - 2a$, be identified at the largest design budget C_J , and let $c_J \equiv \ln C_J$.

Proposition 5 (Partial identification beyond the design). *Consider a model with compute $C > C_J$ and ratio M , and write $x = \ln M$ and $\Delta c = c - c_J$.*

- (i) *If both factor demands are nondecreasing in compute (normal inputs), then $|e| \leq 1$ and $x^*(c) \in [x^*(c_J) - \Delta c, x^*(c_J) + \Delta c]$.*
- (ii) *If $e \in [e_L, e_U]$ beyond the design, the sharp identified set for $x^*(c)$ is $\mathcal{X}^*(c) = [x^*(c_J) + e_L \Delta c, x^*(c_J) + e_U \Delta c]$; its width grows linearly in the*

log extrapolation distance. With sampling error, the anchor's confidence interval replaces $x^(c_J)$.*

(iii) *For any technology with single-peaked IsoFLOP profiles, which $0 < \sigma < 1$ everywhere guarantees, $\text{sign}(w - 1) = \text{sign}(x - x^*(c))$. Hence $w > 1$ is identified if and only if $x > \sup \mathcal{X}^*(c)$, $w < 1$ if and only if $x < \inf \mathcal{X}^*(c)$, and otherwise the sign is not identified. Neither σ^* nor a functional form is needed.*

(iv) *Magnitudes require a range $[k_L, k_U]$ for the slope of $\ln w$ along the isocost, which in family (1) is $1/\sigma^* - 1$. If $x > \sup \mathcal{X}^*(c)$, then*

$$\ln w \in [k_L(x - \sup \mathcal{X}^*(c)), k_U(x - \inf \mathcal{X}^*(c))].$$

(v) *Bounds $w \in [w_L, w_U]$ imply $s \in [1 - 1/w_L, 1 - 1/w_U]$ for the expenditure share $s = (w - 1)/w$ and, under part (ii) of Proposition 2, $\lambda pT/(3D) \in [w_L - 1, w_U - 1]$.*

The sign result needs only the location of the minimum. On a single-peaked profile, a model with more tokens per parameter than is optimal at its compute has a higher marginal product of parameters than of tokens, so $w > 1$ whatever the curvature. Magnitudes need the curvature as well as the distance to the minimum, and the distance is uncertain because the minimum is extrapolated. At frontier scale the data therefore identify a sign, not a level. With the path anchored at the largest IsoFLOP budgets (Section IV), Llama 3 8B, trained on 1,868 tokens per parameter with 7×10^{23} FLOP, lies above every admissible compute-optimal ratio, so its over-training is identified whatever σ^* is. Llama 3.1 405B, at 38 tokens per parameter and 4×10^{25} FLOP, lies inside the identified set under every assumption considered there. A confidence interval from one fitted technology, by contrast, reflects only sampling error within one design and one functional form.

In sum, near-optimal choices reveal the path slope a , and a single lab's outcomes the frontier elasticity γ . The curvature σ^* requires transverse variation but no functional form. The zero point $M^*(C)$ requires optimality or transverse variation, and beyond the designs it is only partially identified.

III. The Technology from Designed Variation

Section II showed that IsoFLOP profiles, which hold compute fixed and vary the input mix, identify the curvature of the technology without a functional form (Proposition 4). This section takes that result to the data and establishes three facts. First, on every public IsoFLOP design with annealed runs, the elasticity of substitution on the expansion path, σ^* , lies between 0.66 and 0.71, and the estimates from three developers' independent designs are consistent with a common value of about 0.70. Second, parametric scaling laws match this benchmark,

where the data can tell them apart, only when the outer exponent κ is free. Third, the other objects that the inversion of Section IV needs, the allocation exponent a and the compute-optimal ratio M^* , are not portable across designs, conventions and data. Subsection III.E describes our own experiment and its pre-registered analysis.

A. Designs

Table 1 lists the designs. The most informative are IsoFLOP designs: at each of several compute budgets, the experimenter trains models of different sizes on the isocost line $6ND = C$. Each budget yields a profile of loss against $\ln N$ whose minimum lies on the expansion path; these minima are outcomes of assigned inputs, not choices.

Four studies provide such profiles. Epoch AI’s digitization of the runs in Figure 4 of Hoffmann et al. (2022), the sample of Besiroglu et al. (2024), has 245 runs with training compute from 1.4×10^{18} to 1.3×10^{22} FLOP: 137 on nine IsoFLOP profiles from 6×10^{18} to 3×10^{21} FLOP, and 108 from 35 fixed-size trunks with different training horizons. Model-free estimates use the 137 profile runs; parametric fits use the full design without the five highest-loss runs ($n = 240$). Meta’s Llama 3 profiles (Grattafiori et al., 2024), digitized by Czech (2026), cover ten budgets up to 10^{22} FLOP; their unstated loss units do not matter, because the model-free estimator is invariant to monotone transformations of output. Marin’s released ladders (Marin Community, 2026) repeat the design on three corpora with budgets up to 3×10^{20} FLOP. Porian et al. (2024) release loss curves of 16 architectures trained at a constant learning rate, from which we read IsoFLOP-like profiles at 12 budgets up to 2.6×10^{19} FLOP; because the runs are not annealed, these are losses of unfinished schedules.

Farseer (Li et al., 2025b) is not an IsoFLOP design, but its 404 released runs (of about 1,000 trained) form a near-factorial grid dense enough to locate the expansion path inside it. Three smaller sweeps of 30–35 runs have no IsoFLOP profiles: the three web corpora of Gadre et al. (2025), all evaluated on C4, the OLMo ladder (Bhagia et al., 2025), and the single-epoch runs of Muennighoff et al. (2023). With Chinchilla’s full design and Farseer, they form the seven sweep–corpus technologies from five studies on which we compare functional forms. DataDecide (Magnusson et al., 2025), whose final checkpoints all have $D \approx 100N$, serves only to compare data recipes.

The model-free estimator requires each profile to be an exact isocost, so N is measured in the convention that makes $C = 6ND$ hold along it (Table 1, notes), and only unit-free objects are compared across rows. The off-path spread, the residual standard deviation of $\ln M$ given $\ln C$ with $M = D/N$, is zero for compute-optimal runs and by Proposition 3 governs how much a design reveals about curvature: 1.4–2.5 in the IsoFLOP designs and 0.94 on the OLMo ladder.

TABLE 1—DESIGNS AND THE ELASTICITY OF SUBSTITUTION ON THE EXPANSION PATH

Design	Runs	N	$M = D/N$	Off-path spread	σ^*		
					Model-free	κ free	$\kappa = 1$
<i>Panel A. IsoFLOP designs and Farseer’s factorial grid</i>							
Chinchilla, IsoFLOP runs	137	57M–16B	0.04–294	1.67	0.673 (0.027)	0.667 (0.023)	0.692 (0.013)
Llama 3	133	59M–17B	0.44–643	1.38	0.660 (0.023)	0.693 (0.020)	0.769 (0.006)
Marin, Comma	85	0.12–12.6B	0.09–3,706	2.46	0.700 (0.027)	0.678 (0.015)	0.690 (0.015)
Marin, DCLM	85	0.12–12.6B	0.09–3,706	2.46	0.713 (0.027)	0.663 (0.013)	0.673 (0.013)
Marin, Nemotron-CC	88	0.12–12.6B	0.09–3,706	2.42	0.705 (0.023)	0.677 (0.014)	0.684 (0.014)
Porian, RefinedWeb ^a	121	5.2M–0.90B	0.02–346	2.29	0.518 (0.017)	0.345 (0.141)	0.599 (0.032)
Porian, OpenWebText2 ^a	116	5.2M–0.90B	0.02–194	2.20	0.505 (0.014)	0.347 (0.160)	0.609 (0.039)
Farseer	404	0.10–6.4B	0.31–2,570	1.97	0.708 ^b	0.710 (0.003)	0.772 (0.010)
<i>Panel B. Other sweep–corpus technologies (parametric fits to all runs)</i>							
Chinchilla, all runs	240	57M–16B	0.46–341	1.35	–	0.701 (0.014)	0.737 (0.006)
Gadre, C4	34	11M–6.9B	5–640	1.46	–	0.607 (0.054)	0.795 (0.020)
Gadre, RedPajama	35	11M–6.9B	5–640	1.47	–	0.623 (0.059)	0.795 (0.013)
Gadre, RefinedWeb	35	11M–6.9B	5–640	1.47	–	0.594 (0.062)	0.796 (0.015)
OLMo ladder	30	0.19–3.2B	10–200	0.94	–	0.544 (0.021)	0.825 (0.034)
Muennighoff	33	7.1M–8.7B	0.04–303	1.97	–	0.511 (0.056)	0.735 (0.031)
<i>Panel C. Heterogeneity across designs (random effects)</i>							
Set		k	Mean [95% CI]	Between s.d.	I^2	Cochran Q (p)	
Model-free, Panel A excluding Porian	6	0.695	[0.673, 0.717]	0.000	0.00	4.1 (0.54)	
Model-free, all of Panel A	8	0.646	[0.562, 0.730]	0.098	0.96	188.9 (<0.001)	
κ free, IsoFLOP designs of Panel A	7	0.672	[0.636, 0.707]	0.019	0.47	11.4 (0.08)	
κ free, sweep–corpus technologies	7	0.622	[0.549, 0.695]	0.068	0.93	82.1 (<0.001)	
$\kappa = 1$, sweep–corpus technologies	7	0.777	[0.745, 0.809]	0.030	0.83	36.3 (<0.001)	

Note: N : parameters in the convention that makes each IsoFLOP profile an exact isocost (Chinchilla: total; Llama 3, Marin: $C/(6D)$; Porian et al.: non-embedding plus output head); Farseer non-embedding; OLMo ladder without the input embedding; Gadre et al. total; Muennighoff et al. the authors’ count. Output: validation loss on MassiveText (Chinchilla), in unstated units (Llama 3), on Paloma (Marin), on the training corpus (Porian et al.), in bits per character (Farseer), on C4 (others). Off-path spread: residual s.d. of $\ln M$ given $\ln C$. Model-free: equation (7) per budget, random-effects mean over budgets, design-conditional wild-bootstrap standard errors. Parametric columns: Huber loss on $\ln L$; Panel A on the same runs (Feng–He–Hu wild standard errors), Farseer and Panel B pairs or cell bootstrap. Panel C: DerSimonian–Laird mean with Hartung–Knapp–Sidik–Jonkman interval; sweep–corpus = Farseer and Panel B. ^aLosses read along constant-learning-rate runs without cooldown. ^bFirst-derivative estimate on the local expansion path (0.701–0.711 across bandwidths); Panel C and Figure 3 use the Hessian-based estimate, 0.703 (0.013).

Source: Hoffmann et al. (2022); Besiroglu et al. (2024); Grattafiori et al. (2024); Czech (2026); Marin Community (2026); Porian et al. (2024); Li et al. (2025b); Gadre et al. (2025); Bhagia et al. (2025); Muennighoff et al. (2023); authors’ calculations (Online Appendix D).

B. The Elasticity without a Functional Form

By Proposition 4(ii), $1/\sigma^* - 1$ equals the curvature of the IsoFLOP profile in $n = \ln N$ at its minimum divided by twice the slope of the IsoFLOP-minima frontier, the curve traced by the minima (equation 7). Both are in the same units of output, so the irreducible loss E , the outer exponent and the loss units drop out.

We estimate both terms budget by budget. A quadratic in $\ln N$ fitted to the runs within one log point of the path gives the curvature and the minimum, a cubic in $\ln C$ through the logged minima gives the frontier slope, and a design’s estimate is the random-effects mean over budgets (DerSimonian and Laird, 1986). Standard errors come from a design-conditional wild bootstrap that holds the design fixed and re-estimates every step in each of 999 draws. On Farseeer we fit a kernel-weighted local quadratic of $\ln L$ in $(\ln N, \ln D)$, find the points where the local wedge w equals one on eight isocosts, and read σ^* off the slope of $\ln w$ in $\ln M$ there, which equals $1/\sigma^* - 1$ for any smooth technology (Proposition 4(iii)). Online Appendix D gives details and robustness.

Figure 3 and Table 1 report the results. On Chinchilla’s profiles $\sigma^* = 0.673$ (standard error 0.027), on Meta’s 0.660 (0.023), and on Marin’s three corpora 0.700, 0.713 and 0.705. Farseeer’s local expansion path gives 0.708, and 0.701–0.711 at every bandwidth we try. The six estimates are consistent with a common value. Their random-effects mean is 0.695 (95 percent confidence interval [0.673, 0.717]), the estimated between-design standard deviation is zero, and Cochran’s homogeneity statistic is 4.1 on five degrees of freedom ($p = 0.54$). With one estimate per study the mean is 0.693. The test has little power against a between-design standard deviation below about 0.02, so “homogeneous” means within two or three hundredths of 0.70. The summary enters Farseeer’s Hessian-based estimate, 0.703, which is sensitive to the bandwidth: at the narrower cross-validated bandwidth it is 0.664, and the mean 0.679.

The estimator is accurate on these grids: on each design’s own fitted technology it is biased by at most 0.006, and its interval covers the truth in 92–98 percent of Monte Carlo replications (Online Appendix D3). The larger threat is FLOP accounting: if FLOPs per token of the measured parameter count vary with size with elasticity η , σ^* shifts by about 0.44η . In Marin’s configuration parameter count ($\eta = 0.087$) its estimates fall to 0.64–0.66; Chinchilla’s and Meta’s FLOP accounting is unobserved, which adds uncertainty of about ± 0.04 for $|\eta| \leq 0.1$.

Porian et al.’s profiles are the exception, at 0.518 (0.017) and 0.505 (0.014), and we keep them out of the summary. Their losses come from constant-learning-rate runs without cooldown, and the gap between unannealed and annealed loss varies with D/N at fixed compute, which bends the profile: the estimates move between 0.45 and 0.55 with the window. They also come from the smallest models and budgets in the table. Including them lowers the mean to 0.646 and makes the set strongly heterogeneous (Table 1, panel C). Our experiment, with annealed runs at a similar scale, can separate schedule from scale (subsection III.E).

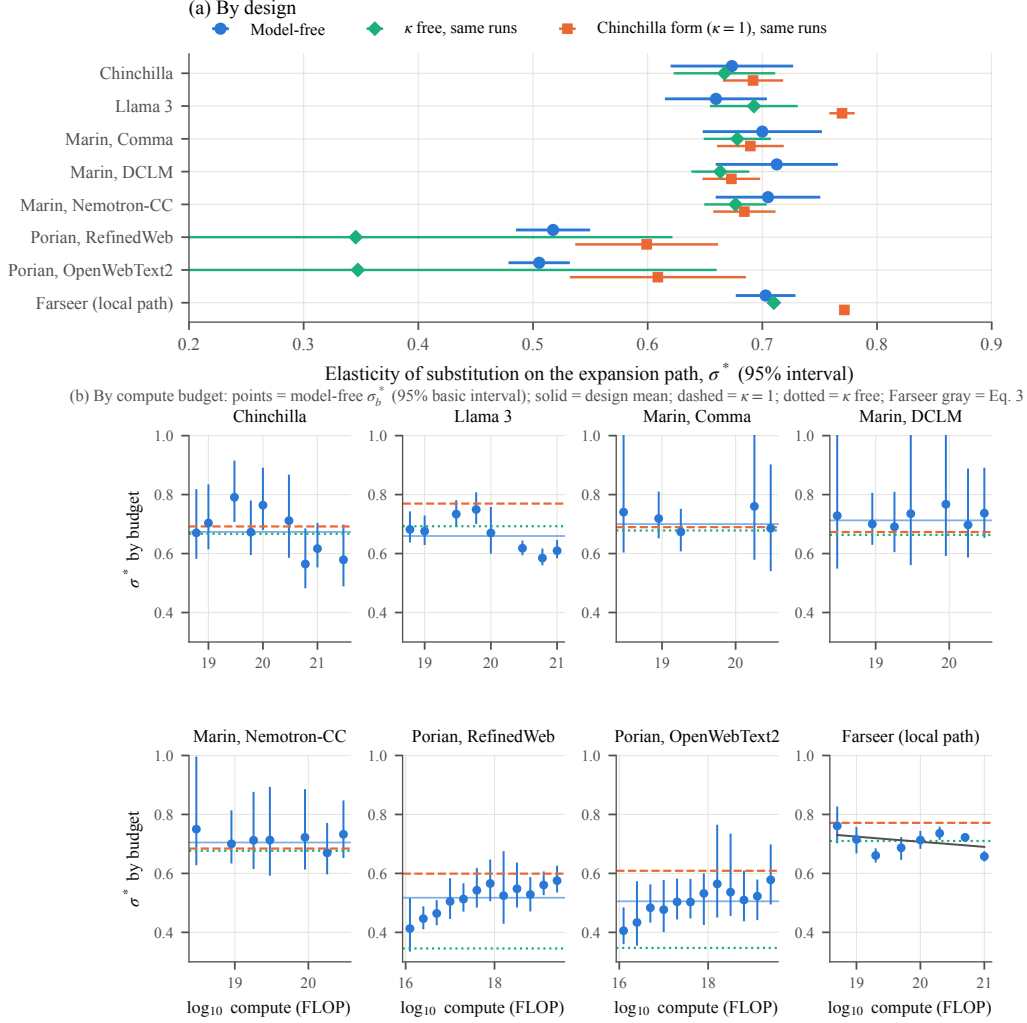


FIGURE 3. THE ELASTICITY OF SUBSTITUTION BY DESIGN, WITH AND WITHOUT A FUNCTIONAL FORM

Note: Panel a: σ^* by design with 95 percent intervals (± 1.96 standard errors). Circles: model-free estimate, equation (7), random-effects mean over budgets; diamonds and squares: κ family and Chinchilla form ($\kappa = 1$) fitted by Huber loss to the same runs. Farseer: Hessian-based estimate on the local expansion path (0.703; the first-derivative estimate is 0.708) and full-grid parametric fits. Panel b: model-free estimate by budget with 95 percent bootstrap intervals; solid line, design mean; dashed, $\kappa = 1$; dotted, κ free. Farseer (last panel): local-path estimates at eight compute levels, with the path value implied by Farseer's own functional form (gray). Porian et al.'s losses come from runs without cooldown.

Source: Designs of Table 1; authors' calculations.

Within designs, σ^* falls with compute: by 0.072 per decade in Chinchilla (standard error 0.026), by 0.053 in Llama 3 (0.010) and by 0.015 in Farseeer (0.008), but not detectably in Marin. Neither parametric family allows such drift, and the estimator shows none when the truth has none, so this is evidence of non-homotheticity, although weak: across windows the Chinchilla drift is 0.034–0.096 per decade. If it is real, σ^* at frontier compute is below 0.70.

That $\sigma^* < 1$ is not a finding: by equation (7) it holds exactly when IsoFLOP profiles are U-shaped under a falling frontier (Lemma 1). The level matters because it prices departures from the compute-optimal mix. With $\sigma^* = 0.70$ (and equal exponents), a model trained on ten times the compute-optimal tokens per parameter needs 73 percent more compute than the compute-optimal model of the same loss, against 57 percent at $\sigma^* = 0.74$ (Online Appendix D). A developer who makes that choice reveals a value of compactness 1.7 times its training expenditure ($w = 2.7$, $s = 0.63$; Proposition 2). Because D counts tokens processed, σ^* is an elasticity between parameters and training tokens, not between compute and a stock of unique text (Section V).

C. Parametric Forms

Scaling laws are usually fitted in the Chinchilla form, the $\kappa = 1$ member of the family $L = E + [AN^{-\alpha} + BD^{-\beta}]^\kappa$ of Section I. In every member $\sigma^* = 2/(2 + \alpha + \beta)$ depends only on the inner exponents, but imposing $\kappa = 1$ ties the curvature of IsoFLOP profiles to the slopes of the frontier and the path (Proposition 4(iv)). The restriction is rejected wherever it can be tested. On Chinchilla’s full design $\hat{\kappa} = 0.774$, and $\kappa = 1$ is rejected at 1 percent under five of the six inference schemes we consider, including a cluster bootstrap over the nine budgets and the 35 off-profile trunks ($p = 0.004$); the exception, the original clustering that assigns off-profile runs to the nearest budget, gives $p = 0.052$. It is rejected in all seven sweep-corpus technologies ($p < 0.001$) and on every IsoFLOP design but Chinchilla’s profile runs, where the quasi-likelihood-ratio statistic of 7.8 is borderline.

Rejection concerns fit; the question here is whether the restriction distorts σ^* . The model-free estimates answer it on the same runs (Figure 3, panel a). Where the comparison discriminates, the κ -free form agrees with the benchmark and the Chinchilla form does not. On Llama 3, $\kappa = 1$ gives 0.769, 4.8 standard errors above the model-free 0.660, while κ free gives 0.693, 1.5 standard errors away. On Farseeer, $\kappa = 1$ gives 0.772, against 0.708 on the local path and 0.710 with κ free. Elsewhere the comparison does not discriminate: on Chinchilla’s profile runs both forms lie within one standard error of the benchmark, on Marin the Chinchilla form is slightly closer, and on Porian et al.’s profiles the benchmark lies between the two. On Chinchilla’s full design the two forms give 0.737 and 0.701, and the bootstrap-calibrated profile-likelihood set for σ^* with κ free, $[0.68, 0.72]$, excludes 0.74 ($p = 0.003$, the floor with 299 draws); the contrast between 0.737 and the model-free 0.673 compares different samples. The form’s other restriction, equal

exponents, survives at 5 percent in every sweep under tests robust to heavy-tailed residuals (Online Appendix D).

With κ free, the three sweeps without IsoFLOP profiles give lower values: 0.59–0.62 on Gadre et al.’s corpora, 0.544 on the OLMo ladder and 0.511 on Muennighoff et al.’s runs. Across the seven sweep–corpus technologies the estimates are strongly heterogeneous: Cochran’s statistic is 82.1 on six degrees of freedom ($p < 0.001$), the between-technology standard deviation is 0.068, and the random-effects mean is 0.622 (95 percent confidence interval [0.549, 0.695]). The heterogeneity survives dropping the OLMo ladder or keeping one estimate per study. Chinchilla and Farseeer agree ($p = 0.51$), and each differs from the OLMo ladder and from Muennighoff et al. by 3.3 to 7.8 standard errors. On the seven IsoFLOP designs, by contrast, the κ -free fits are close to homogeneous ($p = 0.08$).

We cannot tell whether the low values reflect the technology or the κ family’s fit to small designs dominated by off-path runs. With 30–35 runs, E and κ are hard to separate: over the confidence set for E , σ^* moves by up to 0.08 on these sweeps, against at most 0.006 on Chinchilla, Farseeer and the OLMo ladder (Online Appendix Table D10). The OLMo ladder is precise and still low, also with task bits per byte as output (0.51), so a difference in recipe cannot be ruled out. The Chinchilla form raises every estimate, to 0.735–0.825. One more restriction of the family fails where it can be checked: the log wedge is linear in $\ln(M/M^*(C))$ in every member, but convex inside Farseeer’s grid (Section IV).

In sum, where IsoFLOP profiles from annealed runs exist, σ^* is about 0.70 with or without a functional form; small sweeps fitted parametrically give 0.51–0.62; and the Chinchilla restriction overstates σ^* wherever the data can detect it. Section IV therefore uses the κ -free Chinchilla technology as its reference.

D. What Is Not Portable

The inversion of Section IV needs a scale, $1/\sigma^* - 1$, and a zero point, $M^*(C)$. The scale is estimated above. The zero point, and the allocation exponent a that carries it across compute, are where designs disagree. Under the Chinchilla form, a ranges from 0.37 (OLMo ladder) to 0.57 (Gadre et al., RefinedWeb) across the seven sweep–corpus technologies, and from 0.36 to 0.57 once lab laws and Farseeer’s own functional form are added (Section V). The compute-optimal ratio $M^*(10^{21})$ ranges from 3.4 to 60 tokens per parameter, an 18-fold spread. Even the model-free minima paths differ, with slopes of 0.50 on Chinchilla and Llama 3 and 0.35–0.43 on Marin’s three corpora.

M^* is a level: it depends on the ratio A/B , which absorbs every difference in tokenizers, parameter counts and recipes, and a is its slope in compute. Counting Farseeer’s embedding parameters moves a from 0.53 to 0.41 and $M^*(10^{23})$ from 19.5 to 45; counting non-embedding parameters on Chinchilla raises a from 0.514 to 0.556. This is not evidence that σ^* is intrinsically better determined than a : the same change of convention moves Farseeer’s κ -free σ^* from 0.710 to 0.668. The Kaplan–Chinchilla disagreement is the canonical case. On the 975 released runs

of [Porian et al. \(2024\)](#), the allocation exponent falls from 0.84 in Kaplan et al.’s setup to 0.50 (RefinedWeb). Counting the output head’s parameters, a correction of input measurement, accounts for 39–47 percent of the fall; re-tuning warmup, learning rate, schedule and batch size, flexible inputs that were not re-optimized at each scale, accounts for the remaining 53–61 percent (Online Appendix D). Both problems are familiar from production-function estimation ([Collard-Wexler and De Loecker, 2016](#); [Akerberg, Caves and Frazer, 2015](#)).

Data quality moves allocation through one channel. Better data can lower the irreducible loss, scale reducible loss neutrally, or tilt the ratio A/B ; only the tilt changes the isoquant map, M^* and the wedge. Across Gadre et al.’s three web corpora, all evaluated on C4, corpus-specific asymptotes with a neutral scale are not rejected ($p = 0.90$), an asymptote shift alone is ($p = 0.004$), and the implied tilts move M^* by only 8–18 percent, although the tests have little power. DataDecide’s 25 recipes have the power, and every restriction is rejected, yet the tilt adds only 0.05 points to the 99.7 percent of cross-recipe variation in fit that asymptotes and a neutral scale explain. The tilt is small in loss but not in allocation: it spans 0.22–0.26 across recipes, which moves M^* 2.9–3.4-fold and the wedge at given inputs 1.25–1.29-fold, and it is carried by the heavily filtered DCLM recipes. DataDecide identifies the tilt from checkpoints taken before the schedule ends, and on a common validation set part of any corpus difference is distribution match (Online Appendix D4). DeepSeek’s IsoFLOP experiments point the same way: their allocation exponent rose from 0.45 to 0.58 as data quality improved ([DeepSeek-AI et al., 2024](#)).

The allocation exponent and M^* are thus properties of a recipe, a tokenizer and a counting convention as much as of the technology. Section IV uses developers’ own technologies where they exist and reports the range across all technologies; Section V shows that a , not σ^* , drives forecasts of data demand.

E. A Controlled Experiment

The public designs confound the technology with recipes, units and schedules, and none offers annealed runs at the tokens-per-parameter ratios of released small models. We therefore ran our own (Online Appendix B4). Two corpora, FineWeb-Edu and FineWeb (sample-10BT subsets; [Penedo et al., 2024](#)), draw on the same English web crawl and differ in filtering: FineWeb-Edu keeps the documents that an educational-quality classifier rates highly. Both share one 8,192-token byte-level BPE tokenizer, so losses are in common units. The main grid crosses eight decoder-only transformers of width 128 to 640, with 0.39M–49M non-embedding parameters (1.4M–54M with the tied embedding), with token budgets from 25M to 800M: 44 endpoints per corpus. M runs from 0.51 to 2,031 with non-embedding parameters and from 0.46 to 555 with total parameters, and the off-path spread, 1.96, matches Farseer’s. Each width has one constant-learning-rate trunk, from which a copy branches off for each budget and cools down to zero ([Hägele et al., 2024](#)), so every endpoint is an annealed run. Training uses contexts of 256 tokens,

batches of 16,384 tokens and bf16 matrix products on one Apple M5 Max (Hannun et al., 2023). Four-layer models trained to 3.2B tokens raise the largest M to 4,063 (1,743 with total parameters), so that high M is not reached only by a two-layer network; learning-rate checks at the corners of the grid, seed replicates in both corpora and evaluation on WikiText-103 (Merity et al., 2016), drawn from neither corpus, address the other threats specific to small experiments. The largest run uses about 10^{17} FLOP, five orders of magnitude below Chinchilla’s largest, so the experiment tests functional form, neutrality and the direction of extrapolation errors, not production-scale levels.

We fixed the analysis before looking at the results. The pre-analysis plan was committed to the project’s version-control repository on 24 September 2026 (on such plans, see Olken, 2015), when part of the FineWeb-Edu grid had finished but no model had been fitted and no statistic computed. The primary output is loss in bits per byte, and a difference between corpora counts as quality only if it has the same sign on the other corpus’s validation set and on WikiText-103. Every result is reported with both parameter counts. The estimators are the model-free σ^* and local wedge of subsection III.B and Huber fits of the Chinchilla form and the κ family; a conclusion is called robust only if it holds both under a wild cluster bootstrap by width, the unit that shares a trunk (Webb, 2023), and under a parametric bootstrap with the within-trunk covariance estimated from seed replicates. The plan poses six questions, each with a decision rule. Data quality is “factor-biased” if the 95 percent interval of the tilt excludes zero with the same sign on both validation sets, “neutral” if it contains zero on both with a half-width below 0.10, and inconclusive otherwise. The extrapolation test fits each parametric form on $M \leq 100$, predicts the wedge at every endpoint with $M > 100$, and reports the slope of $\ln(w_{\text{restricted}}/w_{\text{local}})$ in $\ln M$, whose sign is the object. One deviation, made for time after the plan was committed, reduces the reach of the extrapolation test on FineWeb and changes no estimator or decision rule (Online Appendix B4). Figure 4 summarizes the results.

[TBD-m9: completed endpoints; the power calculation, dated before the results. Q1: model-free and κ -free σ^* by corpus, both parameter counts, against 0.70 and Porian et al.’s 0.51. Q2: tilt between corpora on the three validation sets, with the decision and the implied factors for M^* and $\hat{w}$. Q3: sign of the extrapolation slope, against Farseeer (Section IV). Q4–Q6: seed noise, learning-rate excess loss and the bound on a , on-path versus full-grid profile likelihood.]

[TBD-m9: two panels. (a) Loss in bits per byte against D by width for both corpora on a common validation set, with fitted curves and the four-layer high- M runs. (b) Model-free σ^* by budget and the local wedge against $\ln M$ for both corpora, with the $M \leq 100$ extrapolations. File: output/figures/m9_*.]

FIGURE 4. A CONTROLLED EXPERIMENT: TWO CORPORA THAT DIFFER IN DATA QUALITY

Note: [TBD-m9: design, output, parameter convention, estimator and bootstrap, as pre-registered.]

IV. What Over-Training Reveals

This section uses the technology estimated in Section III to read the choices of the developers who release models. A developer that trains a model on more tokens per parameter than the training-optimal ratio gives up training efficiency to obtain a smaller model of the same quality. By Proposition 2 it does so only if compactness has value to it, and the technology converts the size of the sacrifice into that value. Under our reference technology, the median open-weight base model released in 2023–2025 was built as if the value of compactness were three quarters of lifetime cost. The level rests on an extrapolation that no public design observes; for most models, the sign does not.

A. From Tokens per Parameter to the Value of Compactness

Proposition 2 gives the first-order conditions of a developer that chooses parameters N and training tokens D under any objective $V(L, N)$ that falls with loss L and may depend directly on size. When the price of a training FLOP does not vary with model size, data cost nothing beyond compute and no constraint binds, the wedge $w \equiv \varepsilon_N/\varepsilon_D$ satisfies $w - 1 = m_N$, the marginal value of compactness at fixed quality per unit of training expenditure. In the κ family the wedge depends on a model's inputs only through its tokens per parameter, $M = D/N$, and its compute, $C = 6ND$ (Online Appendix Corollary A3):

$$(8) \quad \ln w = \left(\frac{1}{\sigma^*} - 1 \right) \ln \frac{M}{M^*(C)}, \quad s \equiv \frac{w - 1}{w} = 1 - \left(\frac{M}{M^*(C)} \right)^{-(1/\sigma^* - 1)}.$$

The data supply M and C . The technology supplies two numbers: the zero point $M^*(C)$, the training-optimal ratio at the model's own compute, and the scale $1/\sigma^* - 1$, which turns log distance from the expansion path into a log wedge. The irreducible loss E , the outer exponent κ and Hicks-neutral productivity do not enter (Proposition A8(ii)).

Everything ordinal in equation (8) is in $M/M^*(C)$. At given compute every technology in the family ranks models by M , and in the sample below a regression of $\ln \hat{w}$ under the reference technology on $\ln M$ alone has an R^2 of 0.9999. That small models are trained on far more tokens per parameter than large ones can be read from model cards. What the technology adds is the translation of M into an economic quantity, and this section is organized around defending the two numbers that make it.

We report the expenditure share $s = m_N/(1 + m_N)$, the value of compactness as a share of training expenditure plus that value, which needs no conversion of FLOPs into dollars. For a developer that serves its own model and bears the cost, s is the planned serving share of lifetime cost; for an open-weight developer whose users bear the serving cost, it is an internalized value of compactness, through adoption, memory tiers or latency, in the same units (Section I.D). A count of lifetime serving tokens T also requires the price ratio p of a serving to a training FLOP: at the median wedge reported below, $T/D = 3(w - 1)/p$ is 9.0 at $p = 1$ and 0.9 at $p = 10$.

Proposition 4 identifies w without a functional form, but only where a design varies inputs locally in two directions. Released models lie outside every design, in M and in C , so equation (8) is an extrapolation. Section IV.D checks its direction in M and bounds it in C .

B. Sample and Technologies

Models.—The clean inference-demand sample is defined ex ante, before any wedge was computed. It starts from the 173 open-weight base models of the verified sample, whose parameter and token counts are documented by their developers (Online Appendix B). Nine sequential steps then drop models whose (N, D) is not a choice of the kind Proposition 2 describes: continued-pretraining re-releases (one observation per pretraining run); research suites and replications whose D was fixed by design, such as Pythia, OPT and BLOOM; mixture-of-experts models; models that were distilled, pruned or initialized from another model; multimodal pretraining; releases without a base checkpoint; non-transformer architectures; and token counts without a primary source. The result is 77 models from 36 families and 18 developers, released 2023–2025. Online Appendix F documents every step and its source; along the way the median reference wedge stays between 3.77 and 4.46.

Technologies.—An ex-ante rule admits 32 technologies: every technology estimated on a designed sweep in Section III (each corpus and parameter convention, with $\kappa = 1$ and with κ free), the path of each IsoFLOP design combined with its model-free curvature, the literature’s estimates on the Chinchilla data, and the published allocation laws of DeepSeek LLM (DeepSeek-AI et al., 2024) and MiniCPM (Hu et al., 2024). The reference is the Chinchilla technology with κ free. Its curvature, $\sigma_\kappa^* = 0.701$, matches the model-free estimate of Section III, and its zero point is $M^*(10^{21}) = 22.7$. Its intervals come from a design-conditional wild

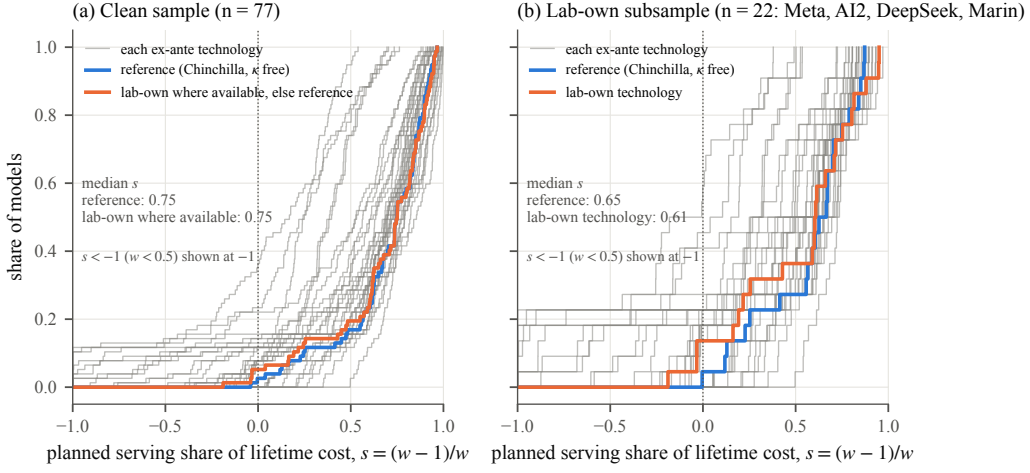


FIGURE 5. THE DISTRIBUTION OF REVEALED EXPENDITURE SHARES ACROSS TECHNOLOGIES

Note: Empirical distribution functions of $s = (w - 1)/w$ across models. Panel a: the 77 models of the clean inference-demand sample. Panel b: the 22 models covered by a lab-own technology (Meta, AI2, DeepSeek, Marin). Gray: each of the 32 ex-ante technologies. Blue: the reference technology (Chinchilla, κ free). Orange: lab-own technology where available, otherwise the reference. Values of s below -1 ($w < 0.5$) are drawn at -1 . For developers that serve their models, s is the planned serving share of lifetime cost; for others, the internalized value of compactness in the same units.

bootstrap of the technology (Feng, He and Hu, 2011), and statements about the sample, such as medians and shares, use the joint distribution across bootstrap draws. Imposing $\kappa = 1$, which Section III rejects, would lower the median wedge from 3.99 to 3.34.

Where a developer has published its own scaling evidence, its technology is primary for its models. For Meta this is its IsoFLOP path, which reproduces its published law (Grattafiori et al., 2024), with the model-free curvature of Meta’s own IsoFLOP profiles ($\sigma^* = 0.660$). For AI2 it is the OLMo ladder with κ free (Bhagia et al., 2025); for DeepSeek, its published law with the reference curvature; and for Marin, its ladder path with its model-free curvature ($\sigma^* = 0.705$; Marin Community 2026). These lab-own technologies cover 22 models. For 11 of them the lab’s law predates the model; for the other 11 (Llama 1 and 2, OLMo 1, Marin 8B) it is ex post. Each technology is evaluated in its own parameter convention, with the output head counted in FLOPs where parameters exclude embeddings. The convention matters for sub-billion-parameter models, but embeddings are at most 28 percent of parameters in the clean sample, and the sample median moves little (Online Appendix F).

C. Revealed Shares

Figure 5 plots the distribution of s across the clean sample, and Table 2 reports selected models. Under the reference technology, the median model has $\hat{w} = 3.99$

and $s = 0.75$ (95 percent joint bootstrap interval $[0.69, 0.79]$), and 97 percent of models are over-trained ($\hat{w} > 1$). The median model was built as if a model one percent smaller at the same quality were worth three percent of its training cost. For a developer that serves the model, this is a planned serving share of three quarters of lifetime cost.

The interval measures sampling error within one technology; the choice of technology matters more. Across the 32 ex-ante technologies the median s ranges from 0.17 to 0.85 (the gray lines in Figure 5). The spread has a pattern. Technologies that let the data choose the curvature give medians of 0.69 to 0.79: the κ -free Chinchilla and Farseer fits, Meta’s and Marin’s paths with model-free curvature, and DeepSeek’s law with the reference curvature. The lowest medians come from $\kappa = 1$ curvatures far above the model-free estimates, such as those of Marin’s Approach-2 laws (σ^* of 0.87 to 0.90), or from a zero point far above the rest, such as MiniCPM’s $M^*(10^{21}) = 192$, which lies outside MiniCPM’s own design. The highest, up to 0.85, come from the κ -free fits of the small sweeps, whose curvatures (σ_κ^* of 0.51 to 0.62) are heterogeneous. Every technology puts the point estimate above one for 64 percent of models, but the union of the 32 technologies’ 95 percent intervals lies above one for only 18 percent.

Curvature governs levels, not signs. Holding the reference zero point fixed, lowering σ^* from 0.80 to 0.60 raises the median wedge from 2.25 to 8.64 and the median s from 0.55 to 0.88. The share of models with $\hat{w} > 1$ does not move, because it depends only on $M^*(C)$. This is why the model-free σ^* of about 0.70 from Section III, and not a $\kappa = 1$ form, should set the scale.

Lab-own technologies give similar answers. On the 22 models they cover, the median s is 0.61 under the lab’s own technology and 0.65 under the reference (Figure 5, panel b). Meta is the sharpest case. Under Meta’s own law and curvature, Llama 3 8B has $\hat{w} = 8.4$ (95 percent interval $[5.7, 12.8]$) and $s = 0.88$. Meta served Llama 3 through its own assistant, so this is a planned serving share. The 405-billion-parameter flagship of the same herd has $\hat{w} = 0.97$; that it lies on Meta’s training-optimal path is by construction, since Meta sized it with this law. Within one developer and one data pipeline, then, the flagship was trained as if for training efficiency and the smallest model as if for serving. The level for the small model depends on the technology Meta is taken to have used: across Meta’s own alternatives, the 8B’s wedge runs from 3.1 under the $\kappa = 1$ law to 12.2 with the path refitted on the budgets whose IsoFLOP minima are bracketed.

D. Extrapolation and Partial Identification

Released models lie beyond the designs in two directions.

Tokens per parameter.—Forty-seven of the 77 models have M above 341, the largest ratio in the Chinchilla design. Inside that range the median s is 0.56, and no model has all 32 technologies’ intervals above one; beyond it the median is 0.85. Only two models lie inside the design in both M and compute. The agreement among technologies at high M therefore owes much to a shared functional form,

TABLE 2—REVEALED WEDGES AND EXPENDITURE SHARES FOR SELECTED MODELS

Model	Serves	N (B)	D (T)	M	M/M^*	Reference $\hat{w}$ [95%]	s	Lab-own $\hat{w}$ (s)	Band of $\hat{w}$
<i>Panel A. Clean inference-demand sample</i>									
Llama 3 8B	Y	8.03	15.0	1,868	86.4	6.73 [5.16, 8.84]	0.85	8.36 (0.88)	[0.47, 168]
Llama 3 70B	Y	70.6	15.0	213	10.0	2.67 [2.05, 3.48]	0.63	2.51 (0.60)	[0.17, 11.7]
Llama 3.1 405B	Y	406	15.6	38	1.8	1.29 [0.99, 1.69]	0.23	0.97 (−0.03)	[0.04, 8.2]
Llama 2 70B	N	69.0	2.0	29	1.3	1.13 [0.93, 1.37]	0.12	0.97 ^e (−0.03)	[0.07, 4.1]
Qwen2.5 7B	Y	7.62	18.0	2,364	109	7.44 [5.66, 9.87]	0.87	—	[0.49, 231]
Qwen2.5 72B	Y	72.7	18.0	248	11.7	2.86 [2.18, 3.74]	0.65	—	[0.17, 13.4]
Qwen3 0.6B	Y	0.60	36.0	60,398	2,759	29.6 [21.1, 43.0]	0.97	—	[1.8, 15,362]
Qwen3 14B	Y	14.8	36.0	2,438	114	7.57 [5.60, 10.2]	0.87	—	[0.38, 218]
DeepSeek LLM 7B	Y	7.00	2.0	286	13.0	2.99 [2.49, 3.60]	0.67	3.42 (0.71)	[0.43, 15.8]
DeepSeek LLM 67B	Y	67.0	2.0	30	1.4	1.15 [0.94, 1.39]	0.13	1.34 (0.26)	[0.07, 4.1]
Gemma 2 27B	Y	27.2	13.0	477	22.3	3.77 [2.92, 4.84]	0.73	—	[0.27, 29.5]
OLMo 2 7B	N	7.30	4.05	555	25.4	3.99 [3.21, 4.94]	0.75	4.05 (0.75)	[0.44, 35.5]
OLMo 2 32B	N	32.2	6.15	191	8.9	2.54 [2.01, 3.20]	0.61	1.28 (0.22)	[0.23, 9.7]
Marin 8B	N	8.03	12.7	1,582	73.1	6.26 [4.83, 8.16]	0.84	5.08 ^e (0.80)	[0.46, 133]
SmolLM2 1.7B	N	1.71	11.0	6,428	293	11.3 [8.69, 15.1]	0.91	—	[0.96, 864]
SmolLM2 135M	N	0.13	2.0	14,868	658	16.0 [12.4, 21.2]	0.94	—	[1.9, 2,701]
<i>Panel B. Mixture-of-experts models, excluded from the sample (bounds at total and active N)</i>									
DeepSeek-V3	Y	685/37 B, 14.8 T				[1.01, 3.50]		[0.01, 0.71]	
Kimi K2	Y	1,026/32 B, 15.5 T				[0.87, 3.79]		[−0.15, 0.74]	
Qwen3 235B-A22B	Y	235/22 B, 36.0 T				[2.34, 6.39]		[0.57, 0.84]	
Qwen3 30B-A3B	Y	30.5/3.3 B, 36.0 T				[5.56, 14.3]		[0.82, 0.93]	

Note: Serves: Y if the developer served its own language models through a first-party API or product when the model was released. N : total parameters; D : documented training tokens; $M = D/N$; M^* : the reference technology’s training-optimal ratio at the model’s own compute. Reference: Chinchilla with κ free ($\sigma_\kappa^* = 0.701$); $\hat{w}$ from equation (8); brackets: 95 percent design-conditional wild-bootstrap interval of one technology, not the uncertainty of the wedge. $s = (w - 1)/w$, negative when $w < 1$. Lab-own: Meta, Meta’s IsoFLOP path with its model-free curvature; DeepSeek, its published law; AI2, the OLMo ladder with κ free; Marin, its ladder path with its model-free curvature. ^eLab law postdates the model. Band: union of the 95 percent intervals (points for published laws) over the 32 ex-ante technologies.

Source: Per-model values under every technology and convention are in the replication package; see Online Appendix F.

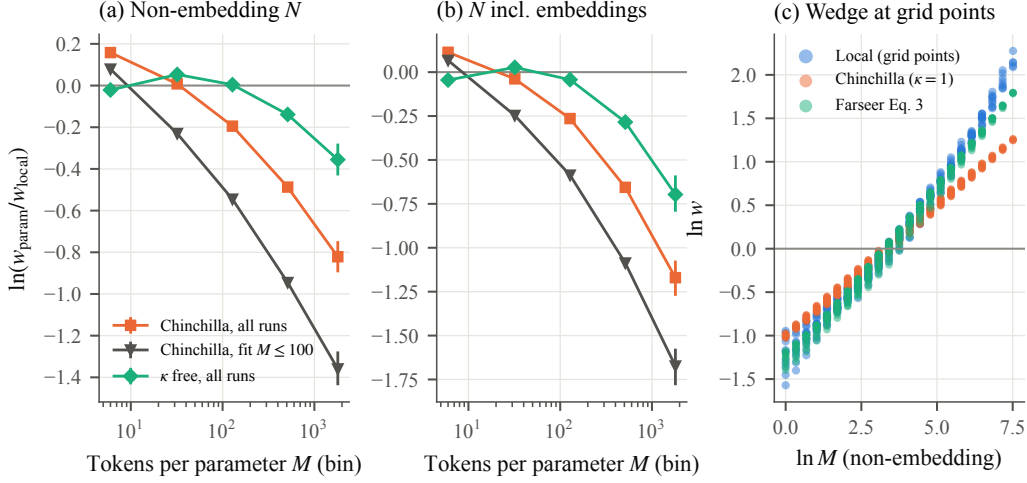


FIGURE 6. PARAMETRIC EXTRAPOLATION AGAINST THE MODEL-FREE WEDGE

Note: Farseeq’s 404 released runs (Li et al., 2025b). The local wedge is $w = (L_c - L_x)/(L_c + L_x)$ (Proposition 4(i)), from kernel-weighted local quadratic fits of loss on $(\ln N, \ln D)$, evaluated at grid points inside each model size’s token range. Panels a and b: mean of $\ln(w_{\text{param}}/w_{\text{local}})$ by bin of M for parametric forms fitted to the same runs (Chinchilla form on all runs and on runs with $M \leq 100$; κ family on all runs), with N excluding (a) or including (b) embeddings; bars are 95 percent intervals from a wild cluster bootstrap by model size (Webb weights, 999 draws) and exclude smoothing bias. Panel c: $\ln w$ at grid points against $\ln M$, locally and under the Chinchilla form and Farseeq’s own non-homothetic form.

under which $\hat{w} \rightarrow \infty$ as $M \rightarrow \infty$ at given compute.

Farseeq’s near-factorial design (Li et al., 2025b), which reaches $M = 2,570$ in non-embedding parameters, allows a direct check. Proposition 4(i) identifies the wedge at each grid point from local slopes of the loss surface, without a functional form. Figure 6 compares this local wedge with those of parametric forms fitted to the same runs. Every form understates the wedge at high M , and the gap grows with M . At $M \geq 1,024$, where the local wedge is 7.5, the κ -free form’s wedge is 0.70 to 0.78 times the local one (the range over kernel bandwidths) and the Chinchilla form’s is 0.44 to 0.50 times. The Chinchilla form fitted only on runs with $M \leq 100$, the extrapolation that equation (8) performs for small over-trained models, gives 0.26 to 0.30 times. Panel c shows the source: $\ln w$ is convex in $\ln(M/M^*(C))$, whereas the κ family makes it linear (Proposition 4(iii)), and linearity is rejected. Applying Farseeq’s gaps to the 57 clean models inside Farseeq’s range of M raises their median s under the κ -free Farseeq technology from 0.68 to 0.71 or 0.72, depending on the bandwidth.

Where it can be checked, then, parametric extrapolation in M errs on the conservative side: it understates the wedge relative to the true technology. Three limits apply. The evidence comes from one recipe, and at the highest ratios from the smallest models in the design, of 0.10 to 0.34 billion parameters. Nothing is observed beyond Farseeq’s range, where 20 clean models lie. And revealed prefer-

ence recovers the technology a developer believed when it chose (N, D) (Proposition A8(ii)): if developers planned with fitted parametric laws, the parametric wedge is the one their first-order conditions equate, and a correction toward the true technology would move $\hat{w}$ away from it (Remark A12). [TBD-m9: Our controlled experiment repeats the check at ratios up to $M \approx X$ in non-embedding parameters (pre-registered question Q3): fitted on $M \leq 100$, the Chinchilla form over/understates the local wedge at $M > 1,000$ by a factor of Y (Section III).]

Compute.—The zero point is the harder problem. No design observes $M^*(C)$ above 10^{22} FLOP, while 88 percent of the clean sample used more compute than the largest Chinchilla run (1.3×10^{22} FLOP). Proposition 5 replaces a point extrapolation with an identified set. We anchor M^* at the largest budget of each IsoFLOP design: 22.7 at 3×10^{21} FLOP for Chinchilla, 22.3 at 10^{22} for Llama 3, 9.4 to 12.4 at 3×10^{20} for Marin’s three corpora, and DeepSeek’s published law. We then let the elasticity of M^* with respect to compute, $e = 1 - 2a$, lie anywhere in the range spanned by these paths and the published laws, $[-0.16, 0.19]$. The identified set for $M^*(10^{24})$ is then $[2.3, 89]$. With Meta’s own anchor alone it narrows to $[10.1, 57]$.

What the data identify at frontier scale is a sign, not a level. For any technology with single-peaked IsoFLOP profiles, which hold whenever $0 < \sigma < 1$ (Lemma 1), the wedge exceeds one exactly when M exceeds $M^*(C)$ (Proposition 5(iii)). For 86 percent of the clean sample, M exceeds every compute-optimal ratio in the identified set, so over-training is identified with no assumption on curvature or functional form. With each lab’s own anchor the share is 77 percent. Admitting every ex-ante technology as an anchor, with the widest range of path slopes, it falls to 23 percent. No model is identified as under-trained under any of these assumptions. The 11 models whose sign is not identified under the IsoFLOP anchors all have $M \leq 96$, and they include most large flagships: Llama 2 70B, Llama 3.1 405B, Qwen2 72B, Falcon 180B and DeepSeek LLM 67B. Magnitudes need a curvature. With $1/\sigma^* - 1$ between 0.40 and 0.52, the range of the κ -free and model-free estimates, the medians of the models’ lower and upper bounds on s are 0.59 and 0.93.

E. Conduct: Who Values Compactness

Proposition 2 admits several objectives, and they predict different patterns (Remark A8). If developers value compactness because they bear serving costs, the wedge should rise with a developer’s own serving footprint, and providers of closed models, which bear their serving cost in full, should over-train at least as much as open-weight developers. If open developers value compactness through adoption by users who bear the cost, the wedge should rise with the sensitivity of adoption to users’ cost. If sizes are set by a menu of memory tiers, sizes should bunch at tier boundaries and the wedge should be unrelated to demand given the tier. Inference below uses a restricted wild cluster bootstrap by developer (Cameron, Gelbach and Miller, 2008; Webb, 2023).

Open-weight models are more over-trained than closed ones. Among 186 models of production scale from 67 developers, the open-weight premium in $\ln \hat{w}$ is 0.52 given release year and compute ($p = 0.011$). A developer that bears all of its serving cost should not over-train less than one that bears part of it, so serving costs alone do not explain open developers’ choices. The comparison is limited to disclosed allocations, and closed developers have disclosed few since 2023.

Within the clean sample the tests have little power. The coefficient on the developer serving its own models is 0.45 (standard error 0.26), the sign the serving story predicts, but with 18 developer clusters, 7 of them serving, the bootstrap p -value is 0.39. Stated on-device targets are carried by two developers and cannot be tested. Sizes do bunch: 51 percent of the 220 open-weight production-scale models released in 2023–2026 sit in five windows around quantized memory tiers, against 13 percent under a smooth counterfactual (Chetty et al., 2011; Kleven, 2016). Given compute, however, a model’s position relative to the tiers predicts its wedge no better than placebo tier menus do (placebo p -values between 0.37 and 1.00). Tiers shape the choice of N ; the tier model’s prediction for the wedge is not visible.

Because the data do not discriminate between these objectives, we report the estimand separately by who serves. The 33 clean models whose developers served their own language models at release, from 7 developers including Meta, Alibaba, Google and DeepSeek, have a median s of 0.80, and every one is over-trained. For them s reads as planned serving expenditure. The other 44 models, from 12 developers including AI2, Hugging Face and TII, have a median s of 0.74. For them s is an internalized value of compactness, not serving expenditure.

Common token budgets need separate treatment. Thirty-two clean models belong to 11 families that train every size on one token budget, so their member wedges are not revealed preferences (Proposition 2(iv)). The family-level shares that these families reveal run from 0.18 for DeepSeek LLM to 0.89 for Qwen3 (Online Appendix Proposition A9). Excluding the 11 families leaves the median s at 0.75, and families with common and size-specific budgets have similar median wedges (3.90 and 4.01).

The cost side matters little for s . If the price of a FLOP, in training and serving alike, varies with model size with an elasticity between -0.1 and 0.1 , the median s is 0.72 to 0.77. If serving cost scales as N^ϕ with ϕ between 0.5 and 1.25, it is 0.70 to 0.86, and it never depends on p . The rival wedges of Section I.D have known signs. The two that concern markup estimation, factor-biased recipes and a mismatch between the loss we measure and the output a developer values (Demirer, 2020; Raval, 2023; Bond et al., 2021), are reduced by lab-own technologies for the 22 models they cover, and the spread across the 32 technologies indicates their size for the rest. Dropping the nine models trained mainly on synthetic data, whose generation is compute planned at training time, leaves the median s at 0.74.

F. Validation and Trends

Levels cannot be validated with public data. Per-model token volumes on OpenRouter, the natural test, are not available, and one year of developer-level traffic (Aubakirova et al., 2026) equals 0.04 to 4.6 percent of the lifetime tokens that the same developers’ releases imply at $p = 1$, which says only that the platform carries a small part of a model’s lifetime use (Online Appendix F10). Hugging Face derivative counts give a weaker check. At given compute, models with higher M have more quantizations, fine-tunes and other derivatives (elasticity 0.80, $p = 0.031$; 74 models from 16 developers). Smaller models of equal compute are adapted more, as planned demand predicts but also as accessibility predicts, so this does not validate levels.

The trend is clearer than the level. Among open-weight releases of production scale (141 models with at least 10^{21} FLOP; Table 3, panel C), the compute-weighted share $\sum_i (w_i^+ - 1)C_i / \sum_i w_i^+ C_i$, with $w_i^+ = \max(w_i, 1)$, rose from 0.04 for releases in 2019–2022 to 0.35 in 2023, 0.54 in 2024 and 0.82 in 2025. It is planned serving expenditure for developers that serve and an internalized value of compactness for the rest. The early value reflects two under-trained models, OPT-175B and GLM-130B, with 92 percent of that period’s compute. On the clean sample the aggregate share rises from 2023 to 2025 under every ex-ante technology; the level does not travel, and for 2025 it ranges from 0.24 to 0.96 across technologies. Llama 3.1 405B carries 39 percent of 2024 compute, and without it the 2024 share is 0.63. Part of the rise may reflect more inference tokens per query, from reasoning and repeated sampling (Roberts et al., 2026), rather than more queries; both are serving demand. Section V compares these shares with firms’ disclosures.

Two findings of this section are robust: over-training, identified at frontier scale under weak assumptions for 86 percent of the clean sample, and the steep rise between 2022 and 2025 in the share of lifetime cost that developers attached to compactness. The level, about three quarters for the median model, is conditional on a zero point that no public design observes at the relevant compute.

V. Economic Implications

Three economic questions turn on the objects estimated in Sections III and IV: how fast the demand for training data grows, what a binding limit on unique text costs, and how much of AI compute goes to serving models rather than training them. Each loads on a different object. The allocation exponent a , the least portable object, governs the first. The curvature σ^* , the best-measured object, barely matters for the second. The level of the third rests on the zero point $M^*(C)$. Table 3 and Figure 7 collect the numbers.

A. The Growth of Data Demand

Along the expansion path, compute-optimal data grow as $D^*(C) \propto C^{1-a}$, where a is the compute elasticity of the optimal model size (Proposition 1). Frontier training compute has grown 5.1-fold a year since 2018 (95 percent confidence interval 4.3 to 5.9), close to the four- to fivefold rate reported by Sevilla and Roldán (2024).¹ Across ten technologies, a runs from 0.36 (Farseeer’s own form, locally at 10^{22} to 10^{23} FLOP) to 0.57 (the RefinedWeb sweep of Gadre et al. 2025). Compute-optimal data demand therefore grows 2.0-fold to 2.8-fold a year, or 34-fold to 183-fold over five years; σ^* does not enter. The level is not portable either: at the fitted frontier compute of September 2026, 9.4×10^{26} FLOP, compute-optimal data range from 9 to 519 trillion tokens. The designs of Section III agree on curvature but not on the expansion path, and that disagreement becomes a factor of more than five in five-year data needs.

These forecasts bear on whether frontier training will exhaust public text, whose quality-adjusted stock Villalobos et al. (2024) put at about 100 trillion tokens in 2024 (95 percent interval 22 to 490 trillion). Because the technologies disagree about levels, we anchor each at the data use of the four disclosed dense training runs above 10^{25} FLOP. Frontier runs then reach 100 trillion unique tokens between mid-2026 and late 2027 under every technology, and the repetition-adjusted stock of 320 trillion tokens between late 2027 and mid-2029, close to Villalobos et al.’s median of 2028 (Figure 7, panel a). The size of the stock dominates the uncertainty: under the Chinchilla technology, the 5–95 percent range of the first date runs from 2025.7 to 2029.4, whereas a moves the dates by only about a year.

B. The Data Wall

A data wall concerns unique tokens, whereas D counts tokens processed, including repetitions. We therefore combine the technology with the repetition model of Muennighoff et al. (2023), in which repeated tokens lose value exponentially: effective data are $D' = U + UR_D^*(1 - e^{-R/R_D^*})$, where U is unique tokens, $R = D/U - 1$ the number of repetitions, and $R_D^* = 15.4$ their estimated decay scale. Scarcity is $r \equiv D^*(C)/U$, the ratio of compute-optimal to unique tokens; the cost of a cap on U and the shadow value of a unique token, $-\partial C/\partial U$ at fixed loss, depend on r alone (Table 3, panel B).

The curvature barely matters. At $r = 4$, reaching the unconstrained frontier loss costs 7.2 percent more compute under the Chinchilla refit with $\kappa = 1$ ($\sigma^* = 0.74$), 7.4 percent with κ free ($\sigma^* = 0.70$), and 7.7 percent under a member of the $\kappa = 1$ refit’s observationally equivalent family that shares its expansion path and frontier but has $\sigma^* = 0.60$ (Proposition 3). Curvature matters only near the hard wall,

¹We regress $\log_{10} C$ on the release date of the running top-ten training runs in Epoch AI (2026) (snapshot of September 23, 2026; 92 runs, 2018–2026); the interval is a wild cluster bootstrap over 27 developers (Webb, 2023). Excluding runs whose compute Epoch rates as speculative gives 4.8-fold. For January 2018 to May 2024 we obtain 5.2-fold, against Epoch’s published 4.2-fold, probably a vintage effect of the database.

the scarcity beyond which the frontier loss is unattainable: $r = 117, 83$, and 47 for the three technologies. The repetition technology matters far more. Our primary specification, which uses Muennighoff et al.’s data-decay scale without their decay of excess parameters, is the most favorable case. Under their full model, in which excess parameters also lose value, the cost at $r = 4$ is 43 percent and the hard wall falls to $r \approx 9$; the gap between $\sigma^* = 0.74$ and 0.70 stays small (43 versus 40 percent).

The elasticity that governs the wall is thus not σ^* , which describes substitution between parameters and tokens processed, but the elasticity of substitution between compute and unique tokens, which combines σ^* with the returns to repetition. Under Muennighoff et al.’s estimates it rises from 0.28 to 0.58 as r increases from 1.5 to 16 , almost independently of σ^* ; it is implied by the two models, not estimated. Behind the wall the return to compute falls, but slowly: the compute elasticity of reducible loss is 0.95γ at $r = 4$ and 0.82γ at $r = 16$, where γ is the frontier elasticity. Whether the wall binds today depends on M^* rather than σ^* : at the 2026 frontier, compute-optimal r against a 100-trillion-token stock ranges from 0.09 to 5.2 across technologies, while frontier runs at observed data use process 0.6 to 1.2 times the stock.

At 10^{26} FLOP and $r = 4$, a unique token is worth 0.43 of the compute cost of processing one token: about $\$3$ per million tokens at $\$10^{-18}$ per FLOP, the median amortized hardware and energy cost of 2024–2025 frontier runs (Epoch AI, 2026). At $r = 16$ it is worth $\$88$ per million; at $r \leq 1$, nothing (Figure 7, panel c). These are private values to one run; because data are nonrival (Jones and Tonetti, 2020), a token that several developers use is worth more to society.

Where σ^* does matter is in translating serving demand into data demand. At given compute, a model with wedge w uses $D/D^*(C) = w^{\sigma^*/[2(1-\sigma^*)]}$ times the compute-optimal data, because $\ln w = (1/\sigma^* - 1) \ln[M/M^*(C)]$ in the κ family (Proposition 4). At $w = 3$ the multiple is 4.7 under $\sigma^* = 0.74$ but 2.3 under $\sigma^* = 0.60$. Forecasts that start from observed tokens per parameter do not depend on σ^* ; forecasts that start from projected serving demand do.

C. The Serving Share of AI Compute

Aggregated over models, the share $s = (w - 1)/w$ of Section IV measures the value developers placed on compact models relative to lifetime cost; for developers that serve their models, it is the planned serving share of lifetime cost (Proposition 2). Weighted by training compute, it rose from 0.04 for open-weight models released in 2019–2022 to 0.54 in 2024 and 0.82 in 2025 (Table 3, panel C). Fleet disclosures are of the same order: inference took about three-fifths of Google’s machine-learning energy in 2019–2021 (Patterson et al., 2022), 70 percent of Meta’s AI power capacity (Wu et al., 2022), and about 40 percent of Nvidia’s data-center revenue in fiscal 2024 (NVIDIA Corporation, 2024).

The objects differ, however, and the differences cut against agreement. The share s compares planned serving with a model’s own final training run, whereas

TABLE 3—ECONOMIC IMPLICATIONS UNDER ALTERNATIVE TECHNOLOGIES

Panel A. Data demand and the allocation exponent a					Year frontier runs reach		
Technology	a	Growth of D^* ($\times$)		D^* in 2026 (T)	100T unique		320T
		per year	5 years		$D^*(C)$	obs. M	obs. M
Chinchilla refit, $\kappa = 1$	0.51	2.20	52	48	2027.9	2027.4	2029.0
Chinchilla refit, κ free	0.50	2.24	56	57	2027.7	2027.4	2028.9
Chinchilla IsoFLOP minima	0.50	2.26	59	58	2027.6	2027.3	2028.8
Llama 3 (Meta’s law)	0.46	2.39	78	90	2027.0	2027.1	2028.5
DeepSeek LLM (published)	0.52	2.16	47	40	2028.2	2027.5	2029.1
Farseer, $\kappa = 1$	0.53	2.16	47	43	2028.1	2027.5	2029.1
Farseer, own form (local)	0.36	2.84	183	419	2025.4	2026.6	2027.8
Gadre et al., RefinedWeb	0.57	2.02	34	9	2030.6	2027.8	2029.6
OLMo ladder (AI2)	0.37	2.78	165	519	2025.2	2026.7	2027.9
Marin, DCLM	0.42	2.55	107	120	2026.7	2026.9	2028.2
Panel B. The data wall and the curvature σ^*							
Technology	σ^*	Extra compute (%)		Hard wall $r_{\max}$	Shadow value $r = 4$	σ_{CU} $r = 4$	$\gamma_{\text{eff}}/\gamma$ $r = 16$
		$r = 4$	$r = 16$				
<i>Repetition of data only ($R_D^* = 15.4$; primary)</i>							
Chinchilla refit, $\kappa = 1$	0.74	7.2	58	117	0.43	0.49	0.82
Equivalent, $\sigma^* = 0.70$	0.70	7.4	63	84	0.45	0.48	0.81
Equivalent, $\sigma^* = 0.60$	0.60	7.7	77	47	0.51	0.47	0.78
Chinchilla refit, κ free	0.70	7.4	63	83	0.45	0.48	0.80
<i>Muennighoff et al.’s full model (excess parameters also decay, $R_N^* = 5.3$)</i>							
Chinchilla refit, $\kappa = 1$	0.74	43.4	∞	9.1	4.28	0.49	–
Chinchilla refit, κ free	0.70	40.3	∞	9.5	3.76	0.49	–
<i>Muennighoff et al.’s data-only fit ($R_D^* = 2.9$)</i>							
Chinchilla refit, $\kappa = 1$	0.74	35.6	1,244	28	1.50	0.54	0.56
Panel C. Aggregate expenditure share $s = (w - 1)/w$ of open-weight releases							
Release period	Open-weight universe				Clean sample		
	Models	s	95% CI	$s, \kappa = 1$	Models	s	Range
2019–2022	8	0.04	[0.03, 0.05]	0.03	–	–	–
2023	53	0.35	[0.25, 0.46]	0.33	24	0.28	[0.01, 0.58]
2024	56	0.54	[0.41, 0.64]	0.50	40	0.58	[0.05, 0.74]
2025	24	0.82	[0.76, 0.86]	0.77	13	0.83	[0.24, 0.96]
All, 2019–2025	141	0.64	[0.54, 0.72]	0.59	77	0.66	[0.10, 0.86]

Notes: Panel A: a , allocation exponent ($N^* \propto C^a$; for Farseeer’s own form, the local slope at 10^{22} – 10^{23} FLOP). Growth of D^* at frontier compute growth of 5.06-fold per year. D^* in 2026: compute-optimal data at the fitted frontier compute of September 2026 (9.4×10^{26} FLOP), in trillions (T) of each technology’s own tokens. Years: first date at which frontier runs reach the quality-adjusted (100T) or repetition-adjusted (320T) stock of public text, growing 5 percent per year, under compute-optimal allocation ($D^*(C)$) or at the median $D/D^*(C)$ of the four disclosed dense runs above 10^{25} FLOP (obs. M). Panel B: $r = D^*(C)/U$; extra compute to reach the unconstrained frontier loss; $r_{\max}$, scarcity beyond which that loss is unattainable; shadow value, $-\partial C/\partial U$ at fixed loss in units of the cost of processing one token; σ_{CU} , elasticity of substitution between compute and unique tokens; $\gamma_{\text{eff}}/\gamma$, compute elasticity of reducible loss behind the wall relative to the frontier elasticity. Equivalent: members of the $\kappa = 1$ refit’s observationally equivalent family. Panel C: $s = \sum_i (w_i^+ - 1)C_i / \sum_i w_i^+ C_i$, $w^+ = \max(w, 1)$, under the reference technology unless stated; open-weight universe: models with $6ND \geq 10^{21}$ FLOP and confident N and D , excluding mixture-of-experts and non-transformer models; 95 percent CI from a design-conditional wild bootstrap ($B = 399$); Range: minimum and maximum over the 32 ex-ante technologies.

Source: Epoch AI (2026); Villalobos et al. (2024); Muennighoff et al. (2023); authors’ calculations.

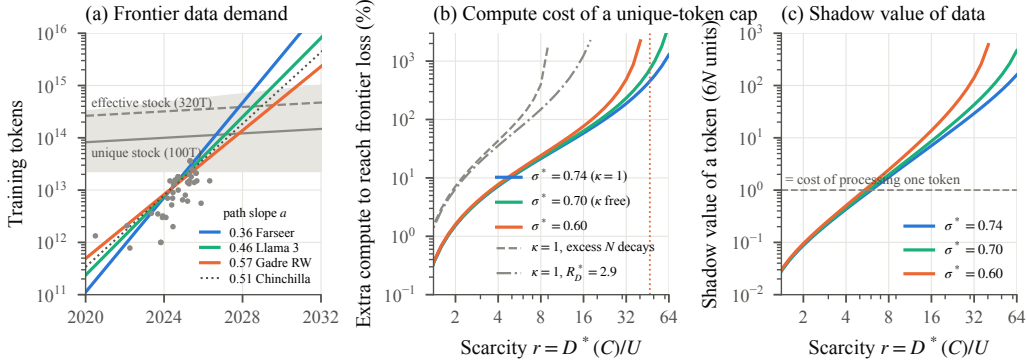


FIGURE 7. DATA DEMAND, THE DATA WALL, AND THE SHADOW VALUE OF DATA

Note: Panel a: training tokens of frontier runs, 2020–2032, anchored at the data use of disclosed dense frontier runs (“obs. M ” in Table 3) and growing at 5.06^{1-a} per year, for four allocation exponents a . Gray dots: dense models in the verified sample above 10^{24} FLOP. Solid and dashed gray lines: the quality-adjusted (100T) and repetition-adjusted (320T) stocks of public text of Villalobos et al. (2024), growing 5 percent per year; shading: the 95 percent interval of the unique stock with stock growth between 0 and 10 percent. Panel b: extra compute needed to reach the unconstrained frontier loss as a function of scarcity $r = D^*(C)/U$, under data-only repetition ($R_D^* = 15.4$) for the Chinchilla refit with $\kappa = 1$ ($\sigma^* = 0.74$), the refit with κ free ($\sigma^* = 0.70$), and the member of the $\kappa = 1$ refit’s equivalent family with $\sigma^* = 0.60$; and, for $\kappa = 1$, under the full repetition model of Muennighoff et al. (2023) (dashed) and their data-only fit (dash-dotted). Dotted vertical line: the hard wall $r_{\max} = 47$ for $\sigma^* = 0.60$ (the others lie beyond the axis). Panel c: shadow value of a unique token, $-\partial C/\partial U$ at fixed loss, in units of the compute cost of processing one token.

fleet shares mix vintages and include research compute. For OpenAI in 2024, press-reported inference spending was about a quarter of all compute spending but four-fifths of inference plus final training runs (You, 2025). Two adjustments convert the 2025 planned multiple, $w - 1 = 4.5$, into a fleet share. Fivefold annual growth of training compute roughly halves the serving flow relative to current training when models are served for a year, and final training runs account for only 10 to 23 percent of developers’ research compute (Denain and Wu, 2026). Together they imply fleet shares of 0.18 to 0.34 at equal cost per FLOP; the disclosed range is reached only if research compute is small. The comparison is therefore a loose check, not a validation. Because the level of s depends on the technology but its rising trend does not (Section IV.F), s is a revealed-preference indicator of expected deployment—for energy and capital-expenditure planning and for the market for inference (Demirer et al., 2025)—in its trend if not in its level.

D. Growth Models

For models of AI and growth that map compute into capability, the relevant elasticity is γ , not σ^* : near the expansion path, compute is a sufficient statistic for the input bundle (Proposition 1), and σ^* enters only when the mix is constrained, as behind a data wall. Halving reducible loss requires $2^{1/\gamma}$ times the compute: 49 times under the Chinchilla refit with $\kappa = 1$ ($\gamma = 0.178$) and 161 times under Meta’s Llama 3 IsoFLOP frontier ($\gamma = 0.136$), or 2.4 to 3.1 years of frontier compute growth. Integrated-assessment models such as GATE (Erdil et al., 2025) map compute into automation through capability thresholds rather than loss, so these numbers discipline only their input side.

VI. Conclusion

Optimization cuts both ways for measurement. Developers who choose model size and training data to minimize cost generate data that say little about the technology they optimize over—along the expansion path, information about curvature is fourth order in their allocation errors (Proposition 3)—and choices that say a good deal about what they optimize for. Designed training experiments break the circle. IsoFLOP designs identify the elasticity of substitution between parameters and data without a functional form (Proposition 4), and the designs of three laboratories agree on about 0.7. Given the technology, over-training reveals the value developers place on compact models (Proposition 2). Under our reference technology, the median model in the clean sample of open-weight releases is built as if that value, which for developers that serve their models is planned serving expenditure, were three-quarters of lifetime cost; across open-weight releases, weighted by training compute, the share rose from 0.04 in 2019–2022 to 0.82 in 2025. [TBD-m9: one sentence on what the controlled two-corpus experiment adds— σ^* by corpus (model-free and κ free), the neutrality verdict for

FineWeb-Edu versus FineWeb, and the direction of the high- M extrapolation error.]

What economists should use.—For substitution between parameters and data in frontier-scale language models, an elasticity of about 0.7: the model-free random-effects estimate across the large designs is 0.70, with a 95 percent confidence interval of 0.67 to 0.72, and parametric estimates that free the outer exponent agree with it there. That the elasticity is below one is not a separate finding; it is what U-shaped IsoFLOP profiles imply (Lemma 1). Values of 0.5 to 0.6 from small-scale sweeps are heterogeneous and are best read as design-specific. For mapping compute into loss, take the frontier elasticity γ from the design whose compute range and data are closest to the application; for frontier-scale language models the large designs give 0.14 to 0.18. The allocation exponent and the compute-optimal ratio M^* are not portable, and forecasts that depend on them, above all of data demand, should carry their range across designs. The expenditure share s is best used as an ordinal and conditional measure. Its ranks and its trend are robust, and its sign is identified for 86 percent of the clean sample under weak assumptions (Proposition 5); its level depends on $M^*(C)$ at frontier scale, which no public design observes, and it measures planned serving expenditure only for developers that serve their models.

Implications for scaling studies.—Studies that aim to measure the technology rather than to tune a recipe should place runs off the expansion path, estimate curvature from IsoFLOP profiles with the model-free estimator of Proposition 4, and test rather than impose $\kappa = 1$. They should also report design-conditional intervals and the uncertainty in $M^*(C)$, the object on which allocation decisions and economic forecasts depend most.

Limitations.—Four caveats bound these conclusions. First, units: losses and token counts come from different corpora, tokenizers, and counting conventions, so only unit-free objects compare across designs. Second, extrapolation: 88 percent of the clean sample use more compute than any design, and a quarter lie beyond the tokens per parameter of the Farseer design. Where it can be checked, parametric extrapolation in tokens per parameter understates w , but nothing is observed at frontier compute. [TBD-m9: what the high- M extrapolation test adds.] Third, disclosure: developers that do not report training tokens, including most closed developers, are absent, so the revealed-demand results describe open-weight releases. Fourth, the economic applications inherit outside inputs: the data-wall costs rest on repetition estimates from models of at most nine billion parameters, and loss is not capability.

Extensions.—The duality and identification arguments carry over to the isoquant between training and test-time compute (Snell et al., 2025; Brown et al., 2024), whose elasticity growth models calibrate only loosely (Erdil et al., 2025), and to market structure, where a cost of quality that rises 50- to 160-fold per halving of reducible loss is an endogenous sunk cost in the sense of Sutton (1991). More broadly, designed training runs give production-function econometrics a

rare setting in which what optimizing agents reveal, and what they conceal, can be checked against experimental variation.

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ONLINE APPENDIX

FORMAL RESULTS AND PROOFS

This appendix states and proves the formal results used in the paper. All statements are exact unless an order of approximation is given. Every algebraic identity below was also verified symbolically and every comparative-static, bias and information statement numerically (by brute-force optimization, simulation, or direct computation of Fisher information); the replication package reports each check.

Results here are numbered A1, A2, ...; the header of each result names the main-text statement it proves. In brief: Proposition 1 (duality) is Lemma A3; Lemma 1 ($\sigma < 1$ at interior optima) combines Lemmas A1 and A2 with Corollary A1; Proposition 2 (the generalized wedge) is Proposition A8, with Lemmas A4 and A5, Corollary A3 and Proposition A9; Proposition 3 (identification and information rates) combines Propositions A1 and A2 with Lemma A4(iv); Proposition 4 (model-free σ^* and w) is Proposition A10; and Proposition 5 (partial identification of $M^*(C)$ and of the wedge) is Proposition A11. Propositions A3–A7 and Corollaries A2, A4 and A5 support Online Appendix E (observational production functions and algorithmic progress). Throughout, E denotes irreducible loss and $\mathbb{E}$ the expectation operator.

A1. Setup and notation

Definition A1 (Technology). A training run uses N parameters and D training tokens (tokens processed, including repetitions) and attains held-out cross-entropy

$$(A1) \quad L = E + e^{-\omega} \left[A(e^{\psi_N} N)^{-\alpha} + B(e^{\psi_D} D)^{-\beta} \right] e^{\epsilon}, \quad A, B, \alpha, \beta > 0.$$

Lower-case letters are logs: $n = \ln N$, $d = \ln D$, $c = \ln C$ with training compute $C = 6ND$, so $c = \ln 6 + n + d$; $M \equiv D/N$. Hicks-neutral productivity ω and factor-augmenting productivity (ψ_N, ψ_D) are known to the lab when it chooses inputs; ϵ is not. Write $u \equiv Ae^{-\alpha(n+\psi_N)}$, $v \equiv Be^{-\beta(d+\psi_D)}$ and $R \equiv u + v$, so that $L - E = e^{-\omega+\epsilon}R$. (The main text writes R for $L - E$; the two coincide when $\omega = \epsilon = 0$.) The log output index is $y \equiv -\ln(L - E)$, and $F(n, d) \equiv -\ln(Ae^{-\alpha n} + Be^{-\beta d})$. The output elasticities are $\varepsilon_N \equiv -\partial \ln(L - E)/\partial n$ and $\varepsilon_D \equiv -\partial \ln(L - E)/\partial d$; the wedge is $w \equiv \varepsilon_N/\varepsilon_D$ and the elasticity share is $\varsigma \equiv \varepsilon_N/(\varepsilon_N + \varepsilon_D)$ (not to be confused with the expenditure share $s = (w - 1)/w$). Expansion-path objects are $a \equiv \beta/(\alpha + \beta)$, $b \equiv \alpha/(\alpha + \beta)$, $\gamma \equiv \alpha\beta/(\alpha + \beta)$, $G \equiv (\alpha A/\beta B)^{1/(\alpha+\beta)}$, $K \equiv \frac{\alpha+\beta}{\beta}AG^{-\alpha}$ and $\sigma^* \equiv 2/(2 + \alpha + \beta)$. Finally $\chi \equiv \beta\psi_D - \alpha\psi_N$ (the bias index of productivity) and $\Omega \equiv \omega + \gamma(\psi_N + \psi_D)$ (productivity in compute-equivalent units).

Remark A1 (Exact log form). Taking logs of (A1) gives, exactly and for any known E , $y = \omega + F(n + \psi_N, d + \psi_D) - \epsilon$, $\varepsilon_N = \alpha u/(u + v)$ and $\varepsilon_D = \beta v/(u + v)$.

None of the three depends on ω , ϵ or E . (Two precision notes on the model specification: the identity is not an approximation; and “ $\varepsilon_N = \alpha u/R$ ” requires R to denote the ω -free sum $u + v$, not the observed $L - E = e^{-\omega+\epsilon}(u + v)$.)

Definition A2 (Generalized family). The generalized family replaces the bracket in (A1) by $[Ae^{-a_1(n+\psi_N)} + Be^{-b_1(d+\psi_D)}]^\kappa$ with $a_1, b_1, \kappa > 0$. Chinchilla is $\kappa = 1$ ($a_1 = \alpha$, $b_1 = \beta$). The joint law of Kaplan et al. (2020, eq. 1.5) is the member with $E = 0$, $a_1 = \alpha_N/\alpha_D \approx 0.738$, $b_1 = 1$ and $\kappa = \alpha_D \approx 0.103$. For this family write $Q \equiv u + v$ with $u = Ae^{-a_1 n}$, $v = Be^{-b_1 d}$, $S \equiv a_1 + b_1$, $a \equiv b_1/S$, $b \equiv a_1/S$, $\gamma \equiv \kappa a_1 b_1/S$ and $\sigma^* \equiv 2/(2 + S)$; with $\kappa = 1$ these coincide with Definition A1.

A2. The elasticity of substitution and the second-order condition

Lemma A1 (Elasticity of substitution; used in Lemma 1). *For the technology (A1) (and for the generalized family with (a_1, b_1) in place of (α, β)):*

(i) *The Hicks elasticity of substitution between N and D is*

$$\sigma = \frac{\alpha u + \beta v}{\alpha u(1 + \beta) + \beta v(1 + \alpha)}, \quad \text{equivalently} \quad \frac{1}{\sigma} = 1 + \varsigma\beta + (1 - \varsigma)\alpha.$$

(ii) *σ depends on (N, D) only through the wedge,*

$$\sigma(w) = \frac{1 + w}{1 + \alpha + w(1 + \beta)},$$

which is monotone in w between $1/(1 + \alpha)$ ($w \rightarrow 0$) and $1/(1 + \beta)$ ($w \rightarrow \infty$).

(iii) *On the compute-optimal expansion path ($w = 1$), $\sigma = \sigma^* = 2/(2 + \alpha + \beta)$.*

(iv) *σ is invariant to any strictly increasing transformation of output; in particular it does not depend on E , ω , ϵ , κ , or on whether output is measured by loss, bits per byte, or a monotone benchmark link.*

(v) *For the Kaplan et al. (2020) law, $\sigma \in [1/2, 1/1.738] = [0.500, 0.575]$ and $\sigma^* = 2/3.738 = 0.535$.*

(vi) *With two inputs the Hicks, Allen–Uzawa and Morishima elasticities coincide (Blackorby and Russell, 1989), so (i) is all three.*

Proof. (i) With output $f(N, D) = -(AN^{-\alpha} + BD^{-\beta})$ in levels, $f_N = \alpha u/N$, $f_D = \beta v/D$, $f_{NN} = -\alpha(1 + \alpha)u/N^2$, $f_{DD} = -\beta(1 + \beta)v/D^2$ and $f_{ND} = 0$. The Hicks elasticity for two inputs, $\sigma = -f_N f_D (N f_N + D f_D) / \{N D (f_{NN} f_D^2 - 2 f_{ND} f_N f_D + f_{DD} f_N^2)\}$, then equals $(\alpha u + \beta v) \alpha u \beta v / \{\alpha u \beta v [\alpha u(1 + \beta) + \beta v(1 + \alpha)]\}$, which is the stated expression; dividing numerator and denominator by $\alpha u + \beta v$ gives $1/\sigma$. (ii) Substitute $\alpha u = w \beta v$. The derivative of $\sigma(w)$ has the sign of $\alpha - \beta$, so σ is monotone between the two limits. (iii) is (ii) at $w = 1$. (iv) For g strictly

increasing, the first derivatives of $g \circ f$ are g' times those of f , and the second derivatives add terms proportional to $g'' f_i f_j$; in the denominator these terms sum to $g'' g'^2 (f_N^2 f_D^2 - 2 f_N^2 f_D^2 + f_D^2 f_N^2) = 0$, so numerator and denominator both scale by g'^3 . Since L is a strictly increasing transformation of $-R$ (and of $-Q$ in the generalized family) for given $(E, \omega, \epsilon, \kappa)$, the claim follows. (v) Apply (ii) with $(a_1, b_1) = (0.738, 1)$. $\square$

Lemma A2 (Log coordinates and the second-order condition; Lemma 1 in the main text). *Let $y = F(n, d)$ be twice continuously differentiable with $F_n, F_d > 0$, and define $P \equiv F_n F_d (F_n + F_d) > 0$ and $Q \equiv -(F_{nn} F_d^2 - 2 F_{nd} F_n F_d + F_{dd} F_n^2)$. Then*

- (i) $\sigma = P/(P + Q)$; hence $\sigma \in (0, 1)$ if and only if $Q > 0$, that is, if and only if the isoquant is strictly convex in $(\ln N, \ln D)$ coordinates.
- (ii) Consider $\max y$ subject to $n + d = c - \ln 6$ (equivalently $\min 6ND$ subject to $y \geq \bar{y}$). At a point with $F_n = F_d$, the second derivative of y along the isocost is $F_{nn} - 2F_{nd} + F_{dd} = -Q/F_n^2$. The point is a strict local maximum of output (minimum of compute) if $\sigma \in (0, 1)$ and a strict local minimum of output if $Q < 0$, which requires $\sigma > 1$ or $\sigma < 0$.

Proof. (i) With $f(N, D) = F(\ln N, \ln D)$: $f_N = F_n/N$, $f_D = F_d/D$, $f_{NN} = (F_{nn} - F_n)/N^2$, $f_{DD} = (F_{dd} - F_d)/D^2$ and $f_{ND} = F_{nd}/(ND)$. Substituting into the Hicks formula, the powers of N and D cancel and $\sigma = -F_n F_d (F_n + F_d) / [(F_{nn} - F_n) F_d^2 - 2 F_{nd} F_n F_d + (F_{dd} - F_d) F_n^2] = P/(P + Q)$. If $Q > 0$ then $P + Q > P > 0$ and $\sigma \in (0, 1)$; if $\sigma \in (0, 1)$ then $P + Q > P$, so $Q > 0$. The curvature of the log-isoquant $d = \phi(n)$ is $\phi'' = Q/F_d^3$, so $Q > 0$ is strict convexity. (ii) Along $(n, d) = (n_0 - t, d_0 + t)$, $d^2 y / dt^2 = F_{nn} - 2F_{nd} + F_{dd}$; at $F_n = F_d$ the bracket in Q equals $F_n^2 (F_{nn} - 2F_{nd} + F_{dd})$. If $Q < 0$, then $\sigma = P/(P + Q)$ lies outside $[0, 1]$. $\square$

Corollary A1 (Gross complementarity; Lemma 1 in the main text). *Under the multiplicative cost $C = 6ND$, every interior compute-optimal allocation that satisfies the second-order condition strictly has $\sigma < 1$ at the optimum. If $\sigma > 1$ everywhere, compute-optimal training puts all compute on one input, subject to bounds: for the CES technology $Y = (N^r + D^r)^{1/r}$ with $r \in (0, 1)$ ($\sigma = 1/(1 - r) > 1$), output on the isocost $ND = X$ is $\sqrt{X} (2 \cosh(rt))^{1/r}$ with $t = \frac{1}{2} \ln(D/N)$, which is minimized, not maximized, at the symmetric point. Every member of the generalized family with $a_1, b_1 > 0$ has $\sigma < 1$ everywhere.*

Proof. The first two statements are Lemma A2(ii). The CES computation is direct. For the generalized family, Lemma A1(ii) gives $\sigma \leq 1/(1 + \min\{a_1, b_1\}) < 1$. $\square$

Remark A2. With a linear cost $p_N N + p_D D$ the second-order condition only requires convex isoquants in levels ($\sigma > 0$). With the multiplicative cost the isocost is a straight line in log coordinates, so the isoquant must be convex in logs, which

is the stronger requirement $\sigma < 1$. The implicit level price of parameters, $6D$, rises with the quantity of the other input; this is why interior optima require complementarity. The empirical content of Corollary A1 is therefore that IsoFLOP profiles are U-shaped in $\ln N$ with an interior minimum; $\sigma^* < 1$ is implied by that fact, not estimated, and every functional form fitted in the paper imposes it. Proposition A10 turns the curvature of the U into the level of σ^* .

A3. Duality, allocation and productivity

Lemma A3 (Cost function and conditional factor demands; Proposition 1 in the main text). *Consider the problem of minimizing $6ND$ subject to $L \leq \bar{L}$, or equivalently of maximizing y at given c , for the technology (A1).*

- (i) *The first-order condition is $\alpha u = \beta v$ ($\varepsilon_N = \varepsilon_D$), and it is sufficient because $\sigma < 1$ everywhere.*
- (ii) *With $\psi_N = \psi_D = 0$: $N^* = G(C/6)^a$, $D^* = G^{-1}(C/6)^b$ and $R^*(C) = K(C/6)^{-\gamma}$ with $K = \frac{\alpha+\beta}{\beta} AG^{-\alpha} = (\alpha+\beta)(A/\beta)^a(B/\alpha)^b$. The cost function is $C^*(R) = 6(K/R)^{1/\gamma}$, with $\partial \ln C^* / \partial \ln R = -1/\gamma$; along the path $\varepsilon_N = \varepsilon_D = \gamma$.*
- (iii) *In general,*

$$\begin{aligned}
 n^* &= \ln G + a(c - \ln 6) + a\psi_D - b\psi_N \\
 &= \ln G + a(c - \ln 6) + \chi/(\alpha + \beta), \\
 d^* &= -\ln G + b(c - \ln 6) + b\psi_N - a\psi_D, \\
 y^* &= \omega + \gamma(c - \ln 6 + \psi_N + \psi_D) - \ln K \\
 &= \Omega + \gamma(c - \ln 6) - \ln K, \\
 \ln M^*(C) &= -2 \ln G + (b - a)(c - \ln 6) - 2\chi/(\alpha + \beta).
 \end{aligned}$$

- (iv) *At given compute, ω does not enter (n^*, d^*) ; $\partial n^* / \partial \psi_D = a > 0$, $\partial d^* / \partial \psi_D = -a < 0$ and $\partial \ln M^* / \partial \psi_D = -2a < 0$: better data raise parameters and lower both tokens and tokens per parameter. Only the bias index χ moves the mix; productivity with $\chi = 0$ ($\alpha\psi_N = \beta\psi_D$) is equivalent to a Hicks-neutral shift $\Delta\omega = \alpha\psi_N$.*
- (v) *At a given loss target instead of given compute, $c^* - \ln 6 = (\bar{y} - \omega + \ln K)/\gamma - \psi_N - \psi_D$, so $\partial n^* / \partial \psi_D = 0$ and $\partial d^* / \partial \psi_D = -1$ (effective data $e^{\psi_D} D$ is held fixed), and $\partial(d^* - n^*) / \partial \omega = (a - b)/\gamma$, which is nonzero unless $\alpha = \beta$: Hicks-neutral productivity leaves the expansion path (the mix as a function of compute) unchanged but moves the mix chosen for a given loss target because the technology is non-homothetic. This co-movement runs entirely through compute: conditional on c the mix is still flat in ω (part (iv) and Proposition A7(ii)); it is the conditioning on compute that rules out a mix-based proxy for neutral productivity.*

- (vi) The sign in (iv) survives inference demand: under the lifetime-compute objective (case (iii)(a) of Proposition A8 with $\lambda = p = 1$), $\partial \ln M / \partial \chi = -2/(\alpha + \beta + \theta_T) < 0$ at given compute, where $\theta_T \equiv (T/3D)/(1 + T/3D) \in [0, 1)$.

Proof. (i) The Lagrangian of $\max\{-\ln(u + v)\}$ subject to $n + d = c - \ln 6$ gives $\alpha u/R = \beta v/R$. Sufficiency: the upper contour sets of y are convex in (n, d) (log-sum-exp is convex), and the constraint is linear. (ii) Substituting $n = \ln G + a \ln(C/6)$ and $d = -\ln G + b \ln(C/6)$ gives

$$\frac{\alpha u}{\beta v} = \frac{\alpha A}{\beta B} G^{-(\alpha+\beta)} (C/6)^{-\alpha a + \beta b} = 1,$$

because $\alpha a = \beta b = \gamma$. At the optimum $v = \alpha u/\beta$, so $R^* = u(\alpha + \beta)/\beta = \frac{\alpha+\beta}{\beta} A G^{-\alpha} (C/6)^{-\gamma}$. The second expression for K follows by substituting G . Inverting $R^*(C)$ gives $C^*(R)$. On the path $\varepsilon_N = \alpha u/R^* = \alpha\beta/(\alpha + \beta)$. (iii) Effective inputs $\tilde{n} = n + \psi_N$ and $\tilde{d} = d + \psi_D$ satisfy $\tilde{n} + \tilde{d} = c - \ln 6 + \psi_N + \psi_D$ and enter R exactly as (n, d) do when $\psi = 0$. Apply (ii) with effective compute and subtract ψ_N and ψ_D . Output is $y^* = \omega - \ln R^*$ evaluated at effective compute. The form with χ uses $a\psi_D - b\psi_N = (\beta\psi_D - \alpha\psi_N)/(\alpha + \beta)$. (iv) Differentiate (iii). If $\alpha\psi_N = \beta\psi_D$ then $u + v = e^{-\alpha\psi_N}(Ae^{-\alpha n} + Be^{-\beta d})$. (v) Solve $y^* = \bar{y}$ for c and substitute into (iii). (vi) By Proposition A8 and Corollary A3, the lifetime optimum at compute c satisfies $\ln M = m_0(c) - 2\chi/S + (2/S)\ln(1 + \frac{T}{3}e^{-d})$ with $S = \alpha + \beta$, $d = \frac{1}{2}(c - \ln 6 + \ln M)$ and $m_0(c) = -2\ln G + (b - a)(c - \ln 6)$. The implicit function theorem gives $\partial \ln M / \partial \chi = -(2/S)/(1 + \theta_T/S)$. $\square$

Remark A3 (Chinchilla's approaches). Lemma A3(ii) links the three objects that scaling studies estimate: the frontier $R^*(C)$ (a cost function), the demands $N^*(C), D^*(C)$, and the primal $L(N, D)$. The mapping to Hoffmann et al. (2022) is not one to one. Their Approach 1 takes the envelope of training curves over FLOPs and reports the argmin N and D as well as the minimal loss, so it estimates the conditional factor demands as well as the frontier; the frontier fitted through IsoFLOP minima (their Approach 2 data) is what this paper calls the IsoFLOP-minima frontier. Approach 3 is the primal.

A4. Geometry and the wedge as a sufficient statistic

Lemma A4 (Geometry of the technology; used in Propositions 2, 3 and 4). *For the generalized family (in effective inputs):*

- (i) $\nabla_{(n,d)}^2 \ln R = \kappa \frac{uv}{Q^2} \begin{pmatrix} a_1^2 & -a_1 b_1 \\ -a_1 b_1 & b_1^2 \end{pmatrix}$ has rank one, and its null vector (b_1, a_1) is the direction of the expansion path $(\partial n^*/\partial c, \partial d^*/\partial c) = (a, b)$.
- (ii) With the transverse coordinate $\tau \equiv a_1 n - b_1 d$, $\ln R = \kappa[-\frac{1}{2}(a_1 n + b_1 d) + g(\tau)]$ with $g(\tau) = \ln(Ae^{-\tau/2} + Be^{\tau/2})$ and $g''(\tau) = s_u(1 - s_u) \in (0, \frac{1}{4}]$, $s_u = u/Q$.

Thus $\ln R$ is affine along every line with direction (b_1, a_1) , and w , s and σ are constant along such lines.

(iii) $\ln w = \tau^* - \tau$ with $\tau^* = \ln(a_1 A / b_1 B) = S \ln G$; the expansion path is the level set $\{\tau = \tau^*\}$; and for any run with compute C and ratio M ,

$$(A2) \quad \ln w = \frac{S}{2} \ln(M/M^*(C)) = \left(\frac{1}{\sigma^*} - 1\right) \ln(M/M^*(C)),$$

where $M^*(C) = G^{-2}(C/6)^{b-a}$ is the training-optimal ratio at the run's own compute.

(iv) (Quasi-homotheticity.) $R(\lambda^{b_1} N, \lambda^{a_1} D) = \lambda^{-\kappa a_1 b_1} R(N, D)$ for every $\lambda > 0$: the family is quasi-homogeneous of degree $-\kappa a_1 b_1$ with respect to the non-uniform dilation $(N, D) \mapsto (\lambda^{b_1} N, \lambda^{a_1} D)$, whose orbits are the lines of direction (b_1, a_1) in (n, d) . More generally, let F be C^2 with $F_n, F_d > 0$ and let $v = (v_n, v_d)$ with $v_n, v_d > 0$. Then (a) translation by tv maps the isoquants of F onto isoquants for every t (the technology is quasi-homothetic with respect to the dilation along v) if and only if (b) $\ln w$ is constant along v ; and (c) if some strictly increasing transformation $g \circ F$ has v in the null space of its Hessian everywhere, then (a) and (b) hold. Under (a)–(b) and the multiplicative cost, the set of compute-optimal points $\{w = 1\}$ is a union of orbits; if the optimum at each compute is unique, the expansion path is a single orbit, a straight line in logs with slope $a = v_n/(v_n + v_d)$, so factor demands are power laws in compute. The rank-one Hessian in (i) is (c) with g the identity and $v = (b_1, a_1)$: all curvature, and hence all information about σ , is transverse to the dilation orbits.

Proof. (i) $\partial \ln Q / \partial n = -a_1 u / Q$, so $\partial^2 \ln Q / \partial n^2 = a_1^2 u / Q - a_1^2 u^2 / Q^2 = a_1^2 uv / Q^2$; the other entries follow in the same way, and $\ln R = \kappa \ln Q$. The determinant is zero and $(a_1^2 b_1 - a_1 b_1 a_1, -a_1 b_1^2 + b_1^2 a_1) = 0$. (ii) Since $-\frac{1}{2}(a_1 n + b_1 d) \mp \frac{1}{2}\tau$ equals $-a_1 n$ and $-b_1 d$ respectively, $Q = e^{-(a_1 n + b_1 d)/2} (Ae^{-\tau/2} + Be^{\tau/2})$. Along $(n_0 + b_1 t, d_0 + a_1 t)$, τ is constant and $a_1 n + b_1 d$ is affine in t . Differentiating g twice gives $pq/(p+q)^2$ with $p = Ae^{-\tau/2}$ and $q = Be^{\tau/2}$, which is $s_u(1 - s_u)$ because $p/(p+q) = u/Q$. (iii) $w = a_1 u / (b_1 v) = (a_1 A / b_1 B) e^{-\tau}$. At given compute write $n = n^* - x/2$ and $d = d^* + x/2$ with $x = \ln(M/M^*(C))$; then $\tau - \tau^* = -(a_1 + b_1)x/2$. (iv) $A(\lambda^{b_1} N)^{-a_1} + B(\lambda^{a_1} D)^{-b_1} = \lambda^{-a_1 b_1} (AN^{-a_1} + BD^{-b_1})$; raise to the power κ . (a) $\Leftrightarrow$ (b): the log-isoquant through a point has slope $dd/dn = -F_n/F_d = -w$. If w is constant along v , the translate of an isoquant solves the same differential equation as the isoquant through the translated point, so the two coincide; conversely, if translates of isoquants are isoquants, their slopes agree at corresponding points. (c) $\Rightarrow$ (b): if $H_{g \circ F} v = 0$ everywhere, then $\frac{d}{dt} \nabla(g \circ F)(x + tv) = H_{g \circ F}(x + tv) v = 0$, so the gradient, and with it the ratio F_n/F_d , is constant along v . Under (b), $w = 1$ at a point implies $w = 1$ on its orbit; with a unique optimum at each compute, the orbit through one optimum contains the

optimum at every compute, since each orbit crosses every isocost $n + d = \text{const}$ exactly once ($v_n + v_d > 0$). $\square$

Lemma A5 (Allocative loss is a function of the wedge; used in Propositions 2 and 3). *Let $C_{\min}(R)$ denote the least training compute $6ND$ that attains reducible loss R . For any run with wedge w ,*

$$(A3) \quad \frac{C}{C_{\min}(R)} = \left(\frac{a_1 + b_1 w}{a_1 + b_1} \right)^{S/(a_1 b_1)} w^{-1/a_1} \stackrel{\kappa=1}{=} \left(\frac{\alpha + \beta w}{\alpha + \beta} \right)^{1/\gamma} w^{-1/\alpha},$$

which depends neither on (A, B, E, κ) nor on productivity. If $\alpha = \beta = \rho$, $C/C_{\min} = [\cosh(\frac{\rho}{2} \ln(M/M^*))]^{2/\rho}$. To second order, $\ln(C/C_{\min}) = \frac{(\ln w)^2}{2S} + O(|\ln w|^3) = \frac{\sigma^*}{4(1 - \sigma^*)} (\ln w)^2 + O(|\ln w|^3)$: the allocative loss is a Harberger triangle whose area is governed by the elasticity of substitution.

Proof. Let $Q_0 = u + v$ at the run. The cheapest bundle producing Q_0 has $a_1 u^* = b_1 v^*$ and $u^* + v^* = Q_0$, so $u^* = aQ_0$ and $v^* = bQ_0$. Because $u \propto N^{-a_1}$ and $v \propto D^{-b_1}$, $C/C_{\min} = (u^*/u)^{1/a_1} (v^*/v)^{1/b_1} = (a/s_u)^{1/a_1} (b/(1 - s_u))^{1/b_1}$. Substituting $s_u = b_1 w / (a_1 + b_1 w)$ gives $a/s_u = (a_1 + b_1 w)/(Sw)$ and $b/(1 - s_u) = (a_1 + b_1 w)/S$, and $1/a_1 + 1/b_1 = S/(a_1 b_1)$. With $a_1 = b_1 = \rho$, (A3) equals $[(w^{1/2} + w^{-1/2})/2]^{2/\rho} = \cosh(\frac{1}{2} \ln w)^{2/\rho}$ and $\ln w = \rho \ln(M/M^*)$ by (A2). For the expansion, let $x = \ln w$: $\ln((a_1 + b_1 e^x)/S) = ax + \frac{1}{2} abx^2 + O(x^3)$; multiplying by $S/(a_1 b_1) = 1/(abS)$ and subtracting $x/a_1 = x/(bS)$ leaves $x^2/(2S)$. Finally $S = 2(1 - \sigma^*)/\sigma^*$. $\square$

A5. What on-path data identify

Definition A3 (On-path sample). A sample of runs is on-path if (a) all runs share the technology (generalized family, common parameters); (b) productivity is Hicks-neutral ($\psi \equiv 0$); (c) every run minimizes training compute for its loss ($T = 0$, no binding constraints); and (d) compute varies over an interval (or at least five distinct values).

Proposition A1 (What on-path data identify; Proposition 3(i) in the main text). *In an on-path sample:*

- (i) $n_i = \ln G + a(c_i - \ln 6)$ and $d_i = -\ln G + b(c_i - \ln 6)$. The inputs are deterministic affine functions of compute, and the choice data reveal a and G (the observed compute-optimal ratio $M^*(C)$).
- (ii) $\mathbb{E}[y \mid c] = \gamma(c - \ln 6) - \ln K + \mathbb{E}[\omega \mid c]$. Within this model (without instruments for compute or panel restrictions on ω), the frontier elasticity γ is identified if and only if $\mathbb{E}[\omega \mid c]$ is known up to an additive constant (for example, ω mean-independent of c , as in a single-lab sweep).

(iii) Suppose in addition $\mathbb{E}[\omega \mid c]$ is constant and $\kappa = 1$. Write $c' \equiv c - \ln 6$, $r_N \equiv \alpha a$ and $r_D \equiv \beta b$ for the rates at which the two terms decay along the observed path, and $P \equiv Ae^{-\alpha n_0}$, $Q \equiv Be^{-\beta d_0}$ (with K normalized by $\mathbb{E}[e^{-\omega+\epsilon}] = 1$), so that on the path $\mathbb{E}[L \mid c] = E + Pe^{-r_N c'} + Qe^{-r_D c'}$: a two-rate exponential mixture whose rates coincide at the truth ($r_N = r_D = \gamma$).

- (a) (Global identification, by functional form.) E , γ and K are identified, and so are $\alpha = \gamma/a$ and $\beta = \gamma/b$: every parameter with $A, B > 0$ that generates the same on-path outcomes has $r_N = r_D = \gamma$. The ratio A/B , and hence the technology's own cost-minimizing ratio, is identified only by imposing that the observed path is optimal ($G^{\alpha+\beta} = \alpha A/\beta B$); outcomes identify only $P + Q$.
- (b) (No local identification.) The Jacobian of the on-path mean of $\ln L$ with respect to $(E, \ln A, \ln B, \alpha, \beta)$ has rank three; its null space contains the direction that changes A/B at fixed K and a direction with $\beta^2 \delta \alpha + \alpha^2 \delta \beta = 0$. Because A/B is not identified, more is true: every perturbation that moves r_N and r_D in opposite directions leaves the on-path mean unchanged to first order (the change is absorbed by re-splitting $P + Q$), so along each such direction the least-squares or likelihood criterion is of fourth order in the step. In particular $\sigma^* = 2/(2 + \alpha + \beta)$ is identified only at second order whether or not $\alpha = \beta$; the gradient of σ^* lies in the row space of the Jacobian if and only if $\alpha = \beta$, but that Jacobian is evaluated at the true A/B , which on-path data do not reveal. (If A/B is known, or pinned by imposing the intercept restriction $G^{\alpha+\beta} = \alpha A/\beta B$, the only first-order-flat direction is the one that keeps the split P/Q at its true value, and σ^* is then first-order identified if and only if $\alpha = \beta$, which is the first draft's statement; it does not survive estimating A/B from outcomes.) Moreover, the twin map $(\alpha, \beta, A, B) \mapsto (\beta b/a, \alpha a/b, Be^{-\beta d_0 + \beta b n_0/a}, Ae^{-\alpha n_0 + \alpha a d_0/b})$ leaves the on-path outcome distribution exactly unchanged (a, b are the observed path slopes), so every parameter value off the ray $\alpha a = \beta b$ has an observationally equivalent twin, whose σ^* differs unless $a = b$ (the truth is its own twin). On the boundary of the parameter space identification fails altogether: with $A \rightarrow 0$ ($B \rightarrow 0$) the on-path distribution converges to the true one for any α (β).
- (c) (Inference.) The on-path model is an over-fitted mixture: the spread $|r_N - r_D|$ enters the mean only through a second-order term, so, by the analogue of the optimal rate for over-fitted mixtures (Chen, 1995), no estimator of (α, β) or σ^* should be expected to converge faster than $n^{-1/4}$ uniformly near the truth. This is a heuristic argument, not a theorem for this model (a nonlinear regression, not a mixture density); see Sargan (1983) on identification without first-order identification and Rotnitzky et al. (2000) on likelihood inference with a singular information matrix, whose results do not apply directly because P/Q is

not identified at the truth. In a compact parameter space bounded away from $A = 0$ and $B = 0$, least squares attains this rate (by simulation, below); its distribution is non-normal (a mass of estimates with $\hat{r}_N = \hat{r}_D$, which converge at $n^{-1/2}$, and a mass with split rates); Wald and delta-method intervals based on the (near-)singular Hessian are invalid; and the likelihood-ratio statistic does not have the standard χ^2 limit. Imposing the cross-equation restriction $\beta/(\alpha + \beta) = a$ (the fitted technology's allocation exponent equals the observed path slope) forces $r_N = r_D$ and restores $n^{-1/2}$ rates. The precision of on-path estimates of σ^* under $\kappa = 1$ is therefore the precision of that restriction: functional form plus optimality.

(iv) Without the restriction $\kappa = 1$: for every $S > 0$, the member

$$(A4) \quad \begin{aligned} a_1 &= (1 - a)S, & b_1 &= aS, & \kappa_S &= \frac{\gamma}{a(1 - a)S}, \\ A_S &= \frac{b_1}{S} K^{1/\kappa_S} G^{a_1}, & B_S &= \frac{a_1 A_S}{b_1 G^S} \end{aligned}$$

generates exactly the same joint distribution of (n, d, y) as the true technology. Hence $\sigma^* = 2/(2 + S)$ is not identified: every value in $(0, 1)$ is consistent with the data, whereas by Lemma A2 no value $\sigma^* \geq 1$ is, because such technologies make the observed allocations cost maxima.

Proof. (i) is Lemma A3(ii). (ii) On the path $y = \omega + \gamma(c - \ln 6) - \ln K - \epsilon$ with $\mathbb{E}[\epsilon | c] = 0$; if $\mathbb{E}[\omega | c] = h(c)$ with h unknown, then $\gamma(c - \ln 6) + h(c)$ and $\tilde{\gamma}(c - \ln 6) + h(c) + (\gamma - \tilde{\gamma})(c - \ln 6)$ are indistinguishable. (iii)(a) With ω independent of c , $\mathbb{E}[L | c] = E + \tilde{K}e^{-\gamma c'}$ with $\tilde{K} = K \mathbb{E}[e^{-\omega + \epsilon}]$; a three-parameter exponential is identified from an interval of c . Suppose $(\tilde{\alpha}, \tilde{\beta}, \tilde{A}, \tilde{B})$ with $\tilde{A}, \tilde{B} > 0$ fits the same curve on the observed path $n = n_0 + ac'$, $d = d_0 + bc'$. Then $\tilde{P}e^{-\tilde{\alpha}ac'} + \tilde{Q}e^{-\tilde{\beta}bc'} = \tilde{K}e^{-\gamma c'}$ on an interval with $\tilde{P}, \tilde{Q} > 0$. Exponentials with distinct rates are linearly independent, so $\tilde{\alpha}a = \tilde{\beta}b = \gamma$, which gives $\tilde{\alpha} = \gamma/a$ and $\tilde{\beta} = \gamma/b$. The remaining condition, $\tilde{P} + \tilde{Q} = \tilde{K}$, is one equation in two unknowns; imposing optimality adds $G^{\alpha+\beta} = \alpha A/\beta B$. (b) For the Jacobian, on the path $u = u_0 e^{-\gamma c'}$ and $v = v_0 e^{-\gamma c'}$ with $u_0/v_0 = \beta/\alpha$. The scores of $\ln L$ are $1/L$ times: 1 for E ; u and v for $\ln A$ and $\ln B$; $-(n_0 + ac')u$ for α ; and $-(d_0 + bc')v$ for β . They span $\{1, e^{-\gamma c'}, c'e^{-\gamma c'}\}$, so the rank is three. A null vector must satisfy $\delta E = 0$, $au_0\delta\alpha + bv_0\delta\beta = 0$ (the coefficient on $c'e^{-\gamma c'}$), and $u_0\delta\ln A + v_0\delta\ln B - u_0n_0\delta\alpha - v_0d_0\delta\beta = 0$; with $u_0/v_0 = \beta/\alpha$ the second condition is $\beta^2\delta\alpha + \alpha^2\delta\beta = 0$, along which $\delta(\alpha+\beta) = \delta\alpha(1-\beta^2/\alpha^2)$. This local analysis holds P/Q at its true value. Every (P, Q) with $P+Q = P_0+Q_0$ and equal rates generates the true distribution, and at such a point the first-order change of the mean from $(\delta r_N, \delta r_D)$ is $-c'e^{-\gamma c'}(P\delta r_N + Q\delta r_D)$, which vanishes for $P/Q = -\delta r_D/\delta r_N > 0$ whenever $\delta r_N \delta r_D < 0$. The second-order term, $\frac{1}{2}c'^2 e^{-\gamma c'}(P\delta r_N^2 + Q\delta r_D^2)$, is not in the span of $\{1, e^{-\gamma c'}, c'e^{-\gamma c'}\}$, so the criterion is exactly of fourth order

along such directions. Since $\delta(\alpha + \beta) = \delta r_N/a + \delta r_D/b$, generic opposite-sign directions move σ^* at first order, including when $a = b$. The twin claim follows by substitution: the twin's two terms are $Qe^{-r_D c'}$ and $Pe^{-r_N c'}$ with the roles of the inputs exchanged. On the boundary, $A = 0$ removes the N -term whatever α is, and the remaining term fits the true curve with $\beta = \gamma/b$. (c) Writing the on-path mean as $E + (P + Q) \int e^{-r c'} dG(r)$ with G the two-point distribution on $\{r_N, r_D\}$ with weights proportional to (P, Q) , the data identify the moments of G around γ through the coefficients on $c'^k e^{-\gamma c'}$: the mean from the first-order term and the variance from the second-order term, each at best at rate $n^{-1/2}$, so the spread $|r_N - r_D|$, and with it (α, β) and σ^* , at best at rate $n^{-1/4}$, as in [Chen's](#) bound for over-fitted mixtures (a heuristic argument; the simulation below confirms the rate for least squares). The restriction $\beta/(\alpha + \beta) = a$ is $r_N = r_D$, which leaves the regular model $E + (P + Q)e^{-r c'}$. A Monte Carlo with 40 on-path compute levels confirms the rates: the 75th and 90th percentiles of $|\hat{\sigma}^* - \sigma^*|$ scale as $n^{-0.24}$ and $n^{-0.27}$ to $n^{-0.28}$ both for $\alpha \neq \beta$ and for $\alpha = \beta$, and as $n^{-0.50}$ with the restriction imposed (replication package). (iv) Apply Lemma [A3\(ii\)](#) to the inner aggregator of member S . Its path slope is $b_1/S = a$. Its path intercept is $(a_1 A_S / b_1 B_S)^{1/S} = G$ by construction of B_S . Its inner frontier is $K_1 (C/6)^{-a_1 b_1 / S}$ with $K_1 = (S/b_1) A_S G^{-a_1} = K^{1/\kappa_S}$, by construction of A_S . Hence $R^* = K (C/6)^{-\kappa_S a_1 b_1 / S} = K (C/6)^{-\gamma}$. Inputs and outputs therefore coincide run by run for every (ω, ϵ) . Values $\sigma^* \geq 1$ correspond to $S \leq 0$ (with $\kappa < 0$) and violate Lemma [A2\(ii\)](#). $\square$

Remark A4 (Identification in practice). Proposition [A1\(iii\)](#) concerns on-path designs: compute-optimal ladders and fixed tokens-per-parameter designs. The Chinchilla data are not such a design; they have transverse IsoFLOP variation, and the large standard errors that [Besiroglu et al. \(2024\)](#) report for A and B (125 and 1,293) mostly reflect the parameterization at $N = D = 1$ (normalizing at the geometric means of the inputs, as in [Klump, McAdam and Willman, 2007](#), removes most of the ill-conditioning). The distinction that matters is between extrapolation in compute and on-path non-identification. IsoFLOP-minima estimators (Approach 2, and the frontier through the minima) estimate (a, G) from the argmins of designed profiles, that is, from outcomes of assigned inputs; with behavioral on-path data the same objects come from choices. The primal (Approach 3) estimates (E, A, B, α, β) from outcomes; the two are linked by $G^{\alpha+\beta} = \alpha A / \beta B$, a restriction that Hoffmann et al.'s Approach 3 violates ([Hoffmann et al., 2022](#)).

Proposition A2 (Information rates; Proposition [3\(ii\)](#) in the main text). *Consider a sweep with $y_i = F(n_i, d_i; \phi) - \epsilon_i$, ϵ_i i.i.d. $N(0, V_\epsilon)$, E known, and the generalized family parameterized by $\phi = (S, a, \gamma, \ln G, \ln K)$. For run i let w_i be its wedge, $\Delta_i \equiv b n_i - a d_i - \ln G = -(\ln w_i)/S$ its transverse deviation (Lemma [A4](#)), and $\Phi_i \equiv \ln(C_i / C_{\min}(R_i))$ its allocative loss (Lemma [A5](#)).*

(i) (Exact decomposition.) $y_i = \gamma(c_i - \ln 6) - \ln K - \gamma \Phi_S(\Delta_i) - \epsilon_i$, where

$\Phi_S(\Delta) = [\ln(b + ae^{-S\Delta}) + aS\Delta]/(abS) = \Phi_i \geq 0$ and $\Phi_S(0) = \Phi'_S(0) = 0$. The curvature parameter S , and hence σ^* , enters output only through the runs' allocative losses.

(ii) (Scores.) The scores for the path level and for curvature are

$$\begin{aligned} \frac{\partial y_i}{\partial \ln G} &= \frac{\gamma(1 - w_i)}{b + aw_i} = -\gamma \ln w_i + O((\ln w_i)^2), \\ \frac{\partial y_i}{\partial S} &= -\gamma \frac{\partial \Phi_S(\Delta_i)}{\partial S} = -\frac{\gamma}{S} \Phi_i [1 + O(|\ln w_i|)] = -\frac{\gamma(\ln w_i)^2}{2S^2} + O(|\ln w_i|^3). \end{aligned}$$

(iii) (Efficient information.) Let ζ_k be the vector of scores for parameter k and M_{-k} the residual-maker for the other four score vectors. Then

$$\begin{aligned} \mathcal{I}^*(\ln G) &= V_\epsilon^{-1} \|M_{-G} \zeta_G\|^2 \\ &= (\gamma^2/V_\epsilon) \|M_{-G} \ln w\|^2 [1 + O(\max_i |\ln w_i|)], \\ \mathcal{I}^*(S) &= V_\epsilon^{-1} \|M_{-S} \zeta_S\|^2 \\ &\leq (\gamma^2/(S^2 V_\epsilon)) \sum_i \Phi_i^2 [1 + O(\max_i |\ln w_i|)] \\ &= O(\sum_i (\ln w_i)^4). \end{aligned}$$

If every run lies on the expansion path ($w_i \equiv 1$), both informations are zero. If the transverse deviations of a fixed design are scaled by $v \rightarrow 0$, then $\mathcal{I}^*(\ln G) = O(v^2)$ and $\mathcal{I}^*(\sigma^*) = O(v^4)$, so the standard error of $\hat{\sigma}^*$ grows like v^{-2} and that of $\ln \hat{M}^*$ like v^{-1} . Both rates are attained by factorial designs.

(iv) (Estimated E , noise on $\ln L$.) If E is estimated and the noise is Gaussian on $\ln L$ instead of y , the score for S is multiplied by $R_i/L_i \in (0, 1)$, so the bound $\mathcal{I}^*(S) = O(\sum_i (\ln w_i)^4)$ still holds; in the nine-budget, five-offset factorial design the orders v^4 (for S) and v^2 (for $\ln M^*$ at the design's central compute) are attained (numerical log-log slopes 4.00 and 2.00), and estimating E costs no information about S to two decimals.

Proof. (i) Because the frontier is a power law, $R = K(C_{\min}(R)/6)^{-\gamma}$ for every R , so $\ln R = \ln K - \gamma(c - \ln 6) + \gamma \ln(C/C_{\min}(R))$. By Lemma A5 for the generalized family and Lemma A4(iii), $\ln(C/C_{\min}) = \ln((a_1 + b_1 w)/S)/(abS) - \ln w/a_1$ with $w = e^{-S\Delta}$, $a_1 = bS$ and $b_1 = aS$; this is $\Phi_S(\Delta)$. It is nonnegative because $C \geq C_{\min}$, and $\Phi_S(0) = 0$. Its derivative is $\Phi'_S(\Delta) = (1 - e^{-S\Delta})/(b + ae^{-S\Delta})$, which vanishes at zero. (ii) Holding (n, d) fixed, Δ depends on $\ln G$ (with derivative -1) but not on S . Hence $\partial y/\partial \ln G = \gamma \Phi'_S(\Delta) = \gamma(1 - w)/(b + aw)$. Write $\Phi_S = h(S\Delta)/(abS)$ with $h(x) = \ln(b + ae^{-x}) + ax$; then $\partial \Phi_S/\partial S = [xh'(x) - h(x)]/(abS^2)$, and $h(x) = \frac{1}{2}abx^2 + O(x^3)$, $xh'(x) - h(x) = \frac{1}{2}abx^2 + O(x^3)$. So $\partial \Phi_S/\partial S = \frac{1}{2}\Delta^2 + O(|\Delta|^3) = \Phi_S/S + O(|\Delta|^3)$. (iii) Under Gaussian errors the information matrix is $V_\epsilon^{-1} Z'Z$ with Z the score matrix; the efficient information

for one parameter is the Schur complement $V_\epsilon^{-1} \zeta'_k M_{-k} \zeta_k$. The inequality uses $\|M_{-S} \zeta_S\| \leq \|\zeta_S\|$ and (ii). The orders follow from (ii): $\zeta_G = O(v)$ and $\zeta_S = O(v^2)$. Attainment: for a design that crosses several compute levels with symmetric transverse offsets, Δ_i^2 is not in the span of the leading-order nuisance scores $\{1, c_i - \ln 6 - \Phi_i, \Delta_i, \Delta_i(c_i - \ln 6)\}$ and Δ_i is not in the span of the others. The numerical exponents are 4.00 and 2.00 for a nine-budget, five-offset factorial design. (iv) $\ln L = \ln(E + R)$, so $\partial \ln L / \partial S = (R/L) \partial \ln R / \partial S = -(R/L) \partial y / \partial S$ and the bound in (iii) applies with the factor $(R_i/L_i)^2 \leq 1$; the attainment claims are numerical (finite-difference scores with E as a sixth parameter). $\square$

Remark A5 (Relation to the literature, and where the results bind). Propositions A1 and A2 are instances of the classical point that optimizing firms facing common relative prices generate inputs that lie on a lower-dimensional set (Marschak and Andrews, 1944). Bond and Söderbom (2005) show that in a Cobb–Douglas model with common input prices and no adjustment costs the input elasticities are not identified, and that identification requires variation in relative prices, adjustment frictions or optimization errors; Gandhi, Navarro and Rivers (2020) show that flexible-input elasticities are not identified from the proxy structure alone. Scaling laws are the cleanest instance: because the FLOP accounting $C \approx 6ND$ is common, the log-cost weights are identical across labs, and without wedges every allocation lies on one ray (Proposition A1(i)). This is distinct from the functional dependence of Akerberg, Caves and Frazer (2015), which concerns the first stage of proxy estimators (Proposition A7(iii)). The escape routes map one to one: relative-price variation corresponds to heterogeneous wedges (serving demand, data costs, size-dependent training prices; Proposition A8); adjustment frictions to binding memory, latency or data constraints; optimization errors to belief errors, such as allocations planned with the Kaplan et al. (2020) law. Designed IsoFLOP sweeps are a different escape: the experimenter assigns (N, D) directly, so there is neither a price nor a behavioral response.

The rates in Proposition A2 are the design-theoretic property of estimating a location parameter versus a curvature (second-order) parameter in nonlinear regression: information about a curvature parameter scales with the fourth moment of the design’s spread (Box and Lucas, 1959; Kiefer and Wolfowitz, 1959). What is specific here is that the spread is generated by economic wedges, and that the curvature enters output only through the runs’ Farrell allocative losses (Proposition A2(i)). Proposition A1(iii) separates identification by functional form, which holds globally, from local (first-order) identification, which fails, with the non-standard inference that follows. It also reconciles two statements that look contradictory. Kricheli et al. (2026) show that fixed tokens-per-parameter designs become ill-conditioned as $|\alpha - \beta| \rightarrow 0$; the first draft of this paper said that on the compute-optimal path σ^* is first-order identified if and only if $\alpha = \beta$. The mechanism is one: the degeneracy arises whenever the two terms decay at the same rate along the design’s ray, $\alpha a' = \beta b'$ for a ray of slope $a' = 1 - b'$. On a fixed-ratio ray ($a' = b' = \frac{1}{2}$) that happens if and only if $\alpha = \beta$; on the

compute-optimal path it happens always, because equal decay rates are what compute-optimality means ($\varepsilon_N = \varepsilon_D$ along the path), and then, with A/B estimated, σ^* is not first-order identified whether or not $\alpha = \beta$ (the first draft's statement holds only with A/B known).

Where the results bind: on compute-optimal ladders and fixed tokens-per-parameter designs run by a single lab, and on releases of the compute-optimal era. Released models since 2023 lie far off the training-optimal path. Their heterogeneous wedges supply transverse variation, but that variation is generated by the objectives of Proposition A8 and is confounded with factor-biased technology (Proposition A7(iv)), so it identifies the technology only jointly with a model of the wedges. (For σ^* itself, multiply the information bound for S by $(dS/d\sigma^*)^2 = 4/\sigma^{*4}$; the order is unchanged.)

A6. Technical change over time

Proposition A3 (What time series of compute-optimal models identify; used in Online Appendix E). *Suppose compute-optimal models are observed at several dates.*

- (i) *(Nonparametric sign result.) Let $y = F(n, d, t)$ be smooth with $F_n, F_d > 0$, and let the compute-optimal allocation be interior at each date. Define the Hicks bias $\mathcal{B}_t \equiv \partial \ln(F_n/F_d)/\partial t$ at fixed (n, d) (the change in the marginal rate of technical substitution at fixed inputs; $\mathcal{B}_t > 0$ is parameter-using). Then*

$$(A5) \quad \frac{\partial \ln M^*(C, t)}{\partial t} = -\frac{\sigma_t^*}{1 - \sigma_t^*} \mathcal{B}_t, \quad \sigma_t^* \in (0, 1).$$

Hence the sign of the Hicks bias is identified from compute-optimal models alone: $\mathcal{B}_t > 0$ if and only if the compute-optimal tokens-per-parameter ratio falls at given compute, and neutrality ($\mathcal{B}_t = 0$) is testable. The magnitude of $\mathcal{B}_t$ is not identified without σ_t^ .*

- (ii) *(Generalized family with factor augmentation $\psi_{N,t} = g_N t$, $\psi_{D,t} = g_D t$.) The frontier shifts as $\ln K_t = \ln K_0 - \gamma(g_N + g_D)t$; the path intercept moves as $\ln G_t = \ln G_0 + (ag_D - bg_N)t$; the mix drifts as $\partial \ln M^*/\partial t = 2(bg_N - ag_D)$; and $\mathcal{B} = b_1 g_D - a_1 g_N = S(ag_D - bg_N)$, which equals $(1 - 1/\sigma)(g_N - g_D)$ only when $a_1 = b_1$. The drift of M^* therefore signs $b_1 g_D - a_1 g_N$ ($\beta g_D - \alpha g_N$ when $\kappa = 1$), which is the sign of $g_D - g_N$ only when $a_1 = b_1$. Because a and γ are identified at each date (Proposition A1), g_N and g_D are identified separately, by the family's functional form. S , and therefore σ^* and $|\mathcal{B}|$, are not: the family (A4) with $(A_{S,t}, B_{S,t}) = (A_S e^{-a_1 g_N t}, B_S e^{-b_1 g_D t})$ is observationally equivalent at every date for every $S > 0$.*
- (iii) *Hicks neutrality $\Leftrightarrow a_1 g_N = b_1 g_D \Leftrightarrow bg_N = ag_D \Leftrightarrow M^*(C)$ is constant over time. Neutral change is a proportional shift of both A and B ; a shift in E*

is not technical change of this kind (it is not a multiplicative shift of any homogeneous output index).

- (iv) If $\kappa = 1$ is imposed, then $S = \gamma/(ab)$ and every object in (ii), including $\mathcal{B}$, is identified, by functional form.

Proof. (i) At given compute the optimum solves $r(n, d, t) \equiv \ln(F_n/F_d) = 0$ with $n + d = c - \ln 6$. Differentiating, $dn = -dd$ and $(r_d - r_n)dd + r_t dt = 0$. At a point with $F_n = F_d$ the isoquant is tangent to the isocost, so the derivative of r in the direction $(-1, 1)$ is the same along both curves. Along the isoquant, $\sigma = d \ln M / d \ln \text{MRTS}$ and $\ln \text{MRTS} = \ln w + \ln M$ imply $d \ln w = (1/\sigma - 1) d \ln M$; since $d \ln M = 2 dd$, $r_d - r_n = 2(1/\sigma^* - 1)$. Hence $d \ln M^*/dt = 2 dd/dt = -r_t \sigma^*/(1 - \sigma^*)$, and $r_t = \mathcal{B}_t$ because $\text{MRTS} = (F_n/F_d)(D/N)$. Lemma A2 gives $\sigma^* \in (0, 1)$. (ii) Apply Lemma A3 with $\psi_t = gt$ to the inner aggregator: $\ln K_{1,t}$ falls at $(a_1 b_1 / S)(g_N + g_D)$, so $\ln K_t = \kappa \ln K_{1,t}$ falls at $\gamma(g_N + g_D)$; $\ln G_t = \ln G_0 + (b_1 g_D - a_1 g_N)t/S$. The Hicks bias is $\partial_t \ln(a_1 u / b_1 v)$ at fixed (n, d) . Neither drift depends on S once (a, γ) are fixed, so the two drifts give two linear equations in (g_N, g_D) (for $g_N + g_D$ and $ag_D - bg_N$) with determinant $a + b = 1$. A separate Hicks-neutral trend $\omega_t = g_\omega t$ does not break this: in member S it equals augmentation at $(g_N, g_D) = (a, b) g_\omega / \gamma$, since $\kappa_S a_1 = \gamma/a$ and $\kappa_S b_1 = \gamma/b$. This is the same for every S , so (g_N, g_D) are the total augmentation rates, with the neutral part included. Observational equivalence follows from Proposition A1(iv) date by date. (iii) $M^*(C) = G_t^{-2}(C/6)^{b-a}$; $\mathcal{B} = 0$ iff $a_1 u / b_1 v$ is constant at fixed inputs, iff A_t/B_t is constant. (iv) Immediate from $\gamma = abS$ when $\kappa = 1$. $\square$

Remark A6 (Relation to Diamond, McFadden and Rodriguez). [Diamond, McFadden and Rodriguez \(1978\)](#) show that time series of output, inputs and marginal products cannot separate σ from the bias of technical change. With constant relative input prices, the drift of the input ratio still reveals the sign of the Hicks bias, because $0 < \sigma^* < 1$ (part (i)); in scaling laws the FLOP accounting fixes relative log-cost weights at one, so the premise holds. With a CES technology the bias is $(1 - 1/\sigma)(g_N - g_D)$, and $\sigma^* < 1$ (Corollary A1) would then also reveal which input is augmented faster. That second step does not carry over to the Chinchilla family, where $\mathcal{B} = \beta g_D - \alpha g_N$ (with $\kappa = 1$): the drift signs $\beta g_D - \alpha g_N$, which coincides with the sign of $g_D - g_N$ only when $\alpha = \beta$. For example, with $(\alpha, \beta) = (0.30, 0.45)$ and $(g_N, g_D) = (0.5, 0.4)$, $\mathcal{B} = 0.03 > 0$ and the compute-optimal D/N falls, although parameters are augmented faster. Within the family g_N and g_D are nonetheless identified separately (part (ii)), but through the family's functional form, not through the drift alone. Proposition A3 is a methodological result; it is not applied to vintage data here, because published compute-optimal laws from different dates differ in tokenizers, parameter conventions and outputs. The argument holds relative prices fixed. If the effective cost of tokens relative to parameters moves over time, the drift of M^* mixes price responses with bias, and the DMR problem returns. Such moves come from data-acquisition costs, a tightening data constraint (Proposition A8(iii)(d)), or

inference demand. The sign result therefore needs training-optimal allocations (IsoFLOP minima) under FLOP-only costs at each date. A designed IsoFLOP sweep at each vintage identifies σ_t^* and hence $|\mathcal{B}_t|$ (compare the insignificant, negatively correlated individual augmentation terms with a well-identified sum in [Ho et al., 2024](#)).

Corollary A2 (Compute-equivalent gains; used in Online Appendix E). *Compare an old and a new technology, each evaluated on its own frontier $L_j^*(C) = E_j + K_j(C/6)^{-\gamma_j}$. The compute-equivalent gain $f(C) \equiv C_{\text{old}}^*(L_{\text{new}}^*(C))/C$ is constant in C if and only if $E_{\text{old}} = E_{\text{new}}$ and $\gamma_{\text{old}} = \gamma_{\text{new}}$, in which case $f = (K_{\text{old}}/K_{\text{new}})^{1/\gamma}$. The individual exponents may differ: for example $(\alpha, \beta) = (0.3, 0.4)$ and $(0.4, 0.3)$ have the same γ . Factor-augmenting change gives $f = e^{\psi_N + \psi_D}$, and Hicks-neutral change gives $f = e^{\Delta\omega/\gamma}$. If instead both technologies are evaluated at a common allocation rule $D = mN$, f is constant only if E and the exponent sets $\{\alpha, \beta\}$ coincide, and, given that, if and only if the new law is the old one with N rescaled, $L_{\text{new}}(N) = L_{\text{old}}(\lambda N)$ for a fixed λ , in which case $f = \lambda^2$. Equal exponents are therefore necessary but not sufficient at a common rule. Factor augmentation gives a constant f on a common ray only if $\psi_N = \psi_D$, and Hicks-neutral change does so only if $\alpha = \beta$.*

Proof. With $x = C/6$, $\ln f = \gamma_o^{-1}[\ln K_o - \ln(E_n - E_o + K_n x^{-\gamma_n})] - \ln x$ and $d \ln f / d \ln x = (\gamma_n / \gamma_o) K_n x^{-\gamma_n} / (E_n - E_o + K_n x^{-\gamma_n}) - 1$. This vanishes for all x if and only if $E_n = E_o$ and $\gamma_n = \gamma_o$. Factor augmentation multiplies K by $e^{-\gamma(\psi_N + \psi_D)}$ (Lemma A3(iii)), and ω multiplies it by $e^{-\Delta\omega}$. On a common ray, $L_j(N) = E_j + A_j N^{-\alpha_j} + B_j m^{-\beta_j} N^{-\beta_j}$. Because $C = 6mN^2$ on the ray, a constant gain f requires $L_n(N) = L_o(\lambda N)$ for all N with $\lambda = f^{1/2}$; linear independence of power functions then forces equal E and equal exponent sets, and matching the coefficient on each power N^{-e} requires the same λ for both terms. With equal exponents and $A_n = A_o e^{-\alpha\psi_N}$, $B_n = B_o e^{-\beta\psi_D}$ this gives $\lambda = e^{\psi_N} = e^{\psi_D}$. With $A_n = A_o e^{-\Delta\omega}$ and $B_n = B_o e^{-\Delta\omega}$ it gives $\lambda^\alpha = \lambda^\beta = e^{\Delta\omega}$, which requires $\alpha = \beta$ when $\Delta\omega \neq 0$. $\square$

Proposition A4 (Farrell decomposition). *Let run i have compute-equivalent productivity Ω_i and true wedge w_i , and let $C_{\min, F}(L)$ be the compute that best practice (Ω_F) needs to reach loss L . Then*

$$\ln(6N_i D_i) - \ln C_{\min, F}(L_i) = \underbrace{\Phi(w_i)}_{\text{allocative}} + \underbrace{(\Omega_F - \Omega_i)/\gamma + \epsilon_i/\gamma}_{\text{technical}},$$

with $\Phi(w) = \ln(C/C_{\min})$ given by Lemma A5. For two models at the same compute, the log compute-equivalent gain is $\Delta\Omega/\gamma - \Delta\Phi$. In particular, a reallocation that keeps Ω fixed, such as the Kaplan-to-Chinchilla move from Gopher-like to Chinchilla-like allocations, is purely allocative. Under the [Besiroglu et al. \(2024\)](#) parameters, $\Phi(w_{\text{Gopher}}) = \ln 2.01$.

Proof. For a lab with (ω, ψ) , the minimal compute that reaches L is $6[Ke^{-\Omega}/(L-E)]^{1/\gamma}$. This follows from Lemma A3(iii): effective compute is $Ce^{\psi_N + \psi_D}$ and R is scaled by $e^{-\omega}$. Hence $\ln C_{\min,i}(L) - \ln C_{\min,F}(L) = (\Omega_F - \Omega_i)/\gamma$. Evaluating at the observed L_i , whose reducible part is $e^{-\omega_i + \epsilon_i} R_i$, and using $C_i/C_{\min,i}(e^{-\omega_i} R_i) = e^{\Phi(w_i)}$ (Lemma A5 in effective units) gives the decomposition. The Gopher number applies (A3) at Gopher's wedge under those parameters, $w = 0.362$ ($N = 280B$, $D = 300B$). $\square$

A7. Observational data: transmission, selection and proxies

Throughout this subsection runs are on the expansion path of a common technology with Hicks-neutral productivity, so $y_i = \omega_i + \gamma c_i - \epsilon_i$ up to a constant (Lemma A3(iii)). Compute follows $c_i = \pi_0 + \pi_1 \omega_i + \eta_i$ with (ω, η, ϵ) mutually uncorrelated, variances $V_\omega, V_\eta, V_\epsilon$. Special cases are the *target rule* (every lab trains to a common loss: $\pi_1 = -1/\gamma$), *exogenous budgets* ($\pi_1 = 0$) and *funding* (more productive labs raise more compute: $\pi_1 > 0$).

Proposition A5 (Transmission bias; used in Online Appendix E). *Under the assumptions of this subsection:*

- (i) *The OLS slope of y on c converges to $b_f = \gamma + \pi_1 V_\omega / (\pi_1^2 V_\omega + V_\eta)$, and its bias has the sign of π_1 . Under the target rule, $b_f = \gamma V_\eta / (V_\omega / \gamma^2 + V_\eta) \in [0, \gamma]$. If all labs hit the target exactly ($V_\eta = 0$), y is constant up to noise and cross-lab data carry no information about γ .*
- (ii) *The ω -component of the residual variance is $V_\omega V_\eta / \text{Var}(c) < V_\omega$ whenever $\pi_1 \neq 0$: TFP dispersion is understated.*
- (iii) *The reverse-regression estimate $\gamma_r \equiv \text{Var}(y) / \text{Cov}(c, y)$ satisfies*

$$(\gamma_r - \gamma) \text{Cov}(c, y) = (1 + \gamma \pi_1) V_\omega + V_\epsilon.$$

Hence, when $\text{Cov}(c, y) > 0$, the interval $[b_f, \gamma_r]$ contains γ if and only if $-(1 + V_\epsilon/V_\omega)/\gamma \leq \pi_1 \leq 0$. The bounds hold under target and exogenous-budget behavior and fail under the funding rule, where both estimates exceed γ . The forward–reverse bracket familiar from errors-in-variables problems (Klepper and Leamer, 1984) therefore does not survive transmission that runs from productivity to compute.

- (iv) *The OLS slope of n on c equals a for any π_1 : Hicks-neutral transmission biases returns to compute, not the allocation exponent. With factor-biased productivity, the slope converges to $a + \text{Cov}(\chi, c) / [(\alpha + \beta) \text{Var}(c)]$.*
- (v) *(Panel remedies.) With lab fixed effects and time-invariant ω_f , the within estimator is consistent (Mundlak, 1961). With $\omega_{ft} = \rho_\omega \omega_{f,t-1} + \xi_{ft}$ and compute chosen before ξ_{ft} is realized, $y_{ft} = \lambda_0(1 - \rho_\omega) + \gamma c_{ft} + \rho_\omega(y_{f,t-1} - \gamma c_{f,t-1}) + \xi_{ft} - \epsilon_{ft} + \rho_\omega \epsilon_{f,t-1}$. The moments $\mathbb{E}[(\xi_{ft} - \epsilon_{ft} + \rho_\omega \epsilon_{f,t-1})z] = 0$ with*

$z \in \{1, c_{ft}, c_{f,t-1}, y_{f,t-2}\}$ over-identify $(\gamma, \rho_\omega, \lambda_0)$, where λ_0 is the intercept of y . The instrument c_{ft} is valid only if compute does not respond to the previous run's noise $\epsilon_{f,t-1}$ (as when the lab knows $\omega_{f,t-1}$ and treats ϵ as evaluation noise). If the budget responds to the realized outcome $y_{f,t-1}$, drop c_{ft} and use $\{1, c_{f,t-1}, c_{f,t-2}, y_{f,t-2}\}$. The first-stage proxy inversion of [Olley and Pakes \(1996\)](#) and [Levinsohn and Petrin \(2003\)](#) is not available here ([Proposition A7](#)), so the lagged-outcome form replaces it.

Proof. (i) $\text{Cov}(y, c) = \gamma \text{Var}(c) + \pi_1 V_\omega$ and $\text{Var}(c) = \pi_1^2 V_\omega + V_\eta$; substitute $\pi_1 = -1/\gamma$ for the target rule. With $V_\eta = 0$ under the target rule, $y = \text{const} - \epsilon$. (ii) The residual variance is $\text{Var}(y) - \text{Cov}(y, c)^2 / \text{Var}(c) = V_\omega + V_\epsilon - \pi_1^2 V_\omega^2 / \text{Var}(c)$. (iii) $\text{Var}(y) - \gamma \text{Cov}(c, y) = \gamma \pi_1 V_\omega + V_\omega + V_\epsilon$. $b_f \leq \gamma$ iff $\pi_1 \leq 0$; $\gamma_r \geq \gamma$ iff $(1 + \gamma \pi_1) V_\omega + V_\epsilon \geq 0$. (iv) By [Lemma A3\(iii\)](#), $n = \ln G + a(c - \ln 6) + \chi/(\alpha + \beta)$ exactly, and ω does not enter. (v) Substitute $\omega_{f,t-1} = y_{f,t-1} - \gamma c_{f,t-1} + \epsilon_{f,t-1}$ (up to the constant). Here c_{ft} and $c_{f,t-1}$ are chosen before ξ_{ft} and ϵ_{ft} are realized. $c_{f,t-1}$ is also chosen before $\epsilon_{f,t-1}$ is realized, while c_{ft} is uncorrelated with $\epsilon_{f,t-1}$ under the stated behavioral condition. $y_{f,t-2}$ is uncorrelated with $(\xi_{ft}, \epsilon_{ft}, \epsilon_{f,t-1})$ when ϵ is serially independent. $\square$

Proposition A6 (Selection on the outcome; used in Online Appendix E). *Suppose a run is released (reported) only if $y_i \geq \bar{y}$. Let $S_i \equiv \omega_i - \epsilon_i$ and $m(k) \equiv \mathbb{E}[S \mid S \geq k]$.*

(i) (Exogenous compute, $\pi_1 = 0$.) *In the released sample $\mathbb{E}[y \mid c] = \gamma c + m(\bar{y} - \gamma c)$, whose slope is $\gamma[1 - m'(\bar{y} - \gamma c)]$. Since $m' \geq 0$ for every distribution, the released-sample OLS slope is at most γ . If S has a log-concave density, then $0 \leq m' \leq 1$ ([Bagnoli and Bergstrom, 2005](#)) and the slope lies in $[0, \gamma]$; without log-concavity it can be negative. If moreover $\mathbb{E}[\omega \mid S]$ is nondecreasing (for example ω and ϵ independent with log-concave densities, [Efron, 1965](#)), then $\mathbb{E}[\omega \mid c, \text{released}]$ is nonincreasing in c .*

(ii) (Gaussian case, any π_1 .) *Let (ω, η, ϵ) be jointly normal, $\delta \equiv \text{Cov}(c, y) / \text{Var}(y)$ and $V_{c|y} \equiv \text{Var}(c) - \delta^2 \text{Var}(y)$. Selection on y leaves the reverse regression of c on y unchanged ($\delta_{\text{sel}} = \delta$). The forward slope becomes*

$$b_{\text{sel}} = \frac{\delta r \text{Var}(y)}{\delta^2 r \text{Var}(y) + V_{c|y}}, \quad r \equiv \frac{\text{Var}(y \mid y \geq \bar{y})}{\text{Var}(y)},$$

where $r = 1 + z\lambda(z) - \lambda(z)^2 \in (0, 1)$ with $z = (\bar{y} - \mathbb{E}y)/\text{sd}(y)$ and $\lambda(z) = \phi(z)/[1 - \Phi(z)]$. The slope b_{sel} is increasing in r .

(iii) (Net sign.) *If $\pi_1 \leq 0$, then $b_{\text{sel}} \leq \gamma$, and $b_{\text{sel}} \leq b_f \leq \gamma$ when $\text{Cov}(c, y) > 0$: the net bias is negative. If $\pi_1 > 0$, then $\gamma\delta < 1$ and $b_{\text{sel}} > \gamma$ if and only if*

$$r > r^* \equiv \frac{\gamma V_{c|y}}{\delta \text{Var}(y)(1 - \gamma\delta)}.$$

Mild selection leaves the upward transmission bias dominant; severe selection reverses it.

- (iv) *Since selection on y lowers the forward slope and leaves $1/\delta$ unchanged, the bracket of Proposition A5(iii) remains valid in released samples whenever $-(1 + V_\epsilon/V_\omega)/\gamma \leq \pi_1 \leq 0$.*

Proof. (i) Given c (independent of S), released iff $S \geq \bar{y} - \gamma c$. m is nondecreasing because $\mathbb{E}[S \mid S \geq k_1]$ is a mixture of $\mathbb{E}[S \mid k_1 \leq S < k_2] \leq k_2$ and $\mathbb{E}[S \mid S \geq k_2] \geq k_2$ for $k_1 < k_2$. The CEF is therefore γc plus a nonincreasing function of c , and the linear projection of a nonincreasing function on c has a nonpositive slope. Log-concavity implies a nonincreasing mean residual life $m(k) - k$, hence $m' \leq 1$. For the last claim, $\mathbb{E}[\omega \mid S \geq k] = \mathbb{E}[h(S) \mid S \geq k]$ with h nondecreasing, and $S \mid S \geq k$ is stochastically increasing in $k = \bar{y} - \gamma c$. (ii) Under joint normality, $c = \mathbb{E}c + \delta(y - \mathbb{E}y) + \nu$ with ν independent of y , so selection on y does not change the distribution of ν or its independence from y . Then $\text{Cov}_{\text{sel}}(c, y) = \delta \text{Var}_{\text{sel}}(y)$ and $\text{Var}_{\text{sel}}(c) = \delta^2 \text{Var}_{\text{sel}}(y) + V_{c|y}$. The truncated-normal variance gives r . Differentiating in r gives a derivative with the sign of $\delta V_{c|y} > 0$. (iii) For $\pi_1 \leq 0$ and $\delta > 0$ combine (ii) with Proposition A5(i); if $\delta \leq 0$, then $b_{\text{sel}} \leq 0 < \gamma$ directly. For $\pi_1 > 0$, $\text{Var}(y) - \gamma \text{Cov}(c, y) = (1 + \gamma\pi_1)V_\omega + V_\epsilon > 0$ gives $\gamma\delta < 1$; rearranging $b_{\text{sel}} > \gamma$ gives the threshold. (iv) follows from (ii) and Proposition A5(iii). $\square$

Remark A7. Part (i) is the classical attenuation of least squares under truncation on the dependent variable (Goldberger, 1981; Hausman and Wise, 1977); the new elements are its interaction with transmission (parts (ii)–(iii)) and the survival of the forward–reverse bracket (part (iv)). The model specification’s statement that $\mathbb{E}[\omega \mid c, \text{released}]$ is decreasing in c holds for exogenous compute (part (i)). Under the funding rule, $\mathbb{E}[\omega \mid c, \text{released}] = \mathbb{E}[\omega \mid c] + \Delta(c)$, where only the selection component Δ is decreasing; with mild selection the total is increasing (verified by simulation). Dropping the highest-loss runs from a designed sweep (Besiroglu et al., 2024) is truncation on the dependent variable of the same kind, with $\pi_1 = 0$ and selection operating through ϵ and the design.

Proposition A7 (Allocation-based proxies; used in Online Appendix E). *Suppose runs minimize cost with a common and known wedge $\bar{w}$ (for example $T = 0$), share (A, B, α, β) , and differ in unobserved $(\omega_i, \psi_{N,i}, \psi_{D,i})$.*

- (i) $\ln M_i = m_0(c_i) + \frac{2}{\alpha+\beta}(\ln \bar{w} - \chi_i)$ with $m_0(c) = -2 \ln G + (b-a)(c - \ln 6)$. The pair (c, M) reveals the bias index exactly, $\chi_i = \ln \bar{w} - \frac{\alpha+\beta}{2}[\ln M_i - m_0(c_i)]$: a Levinsohn–Petrin-type invertibility holds with a known linear inverse.
- (ii) *Given compute, M does not depend on ω or on any productivity with $\chi = 0$. No allocation-based proxy exists for Hicks-neutral productivity: the policy function is flat in ω .*
- (iii) *Even for χ , the proxy step does not identify the technology. The map $(n, d) \mapsto (c, \ln M)$ is a bijection, so any first-stage control function $\varphi(c, \ln M)$*

absorbs $F(n, d)$ entirely; this is the functional-dependence problem of [Akerberg, Caves and Frazer \(2015\)](#). Identification must come from timing restrictions (for example, χ Markov and compute chosen before the innovation) or from a known mapping between χ and level productivity Ω . The same holds for compute as a proxy for ω under a strictly monotone budget rule $c = h(\omega, z)$, because on the path the inputs are functions of c alone.

- (iv) If wedges are heterogeneous, the allocation reveals only $\ln w_i - \chi_i$. Inference demand and factor bias cannot be separated from (N, D) alone; this is the contamination in Proposition [A8\(ii\)](#).

Proof. (i)–(ii) Lemma [A4\(iii\)](#) and Proposition [A8\(ii\)](#): the econometrician's $\hat{w} = (M/M_0^*(C))^{(\alpha+\beta)/2}$, computed with $\chi = 0$, equals $we^{-\chi}$; solve for $\ln M$. The derivative in ω is zero by Lemma [A3\(iv\)](#). (iii) $n = \frac{1}{2}(c - \ln 6 - \ln M)$ and $d = \frac{1}{2}(c - \ln 6 + \ln M)$. (iv) is (i) with $\bar{w}$ replaced by w_i . $\square$

A8. What over-training reveals: the generalized wedge

Proposition A8 (Generalized wedge; Proposition [2](#) in the main text). A developer chooses (N, D) to maximize

$$(A6) \quad \begin{aligned} \Pi(N, D) &= V(L(N, D), N) - X_T(N, D) - c_D D \\ \text{subject to } g_k(n, d) &\leq 0, \quad k = 1, \dots, K, \end{aligned}$$

where V is differentiable with $\partial V / \partial L < 0$; $X_T \equiv c_T(N) 6ND$ is training expenditure at the (shadow) price $c_T(N)$ of a training FLOP, whose size elasticity is $\delta \equiv d \ln c_T / d \ln N$; $c_D \geq 0$ is a cost per training token; L is given by [\(A1\)](#) or by any technology with $\varepsilon_N, \varepsilon_D > 0$; and the g_k are differentiable constraints. Let $m_N \equiv -(\partial V / \partial \ln N)_L / X_T$ be the value of compactness at fixed quality per unit of training expenditure, $m_D \equiv c_D D / X_T$, and $\xi_k \geq 0$ the multipliers (value per log point).

- (i) At any interior Karush–Kuhn–Tucker point,

$$(A7) \quad w = \frac{1 + \delta + m_N + \sum_k \xi_k g_{k,n} / X_T}{1 + m_D + \sum_k \xi_k g_{k,d} / X_T}.$$

With $\delta = m_D = 0$ and no binding constraint, $w - 1 = m_N$, where $m_N = -(\partial V / \partial \ln N)_L / (c_T 6ND)$ is the marginal value of compactness at fixed quality divided by training expenditure. The expenditure share $s \equiv (w - 1) / w$ is then $m_N / (1 + m_N)$.

- (ii) (Invariance and contamination.) w is computed from (N, D) and the technology alone; it is invariant to ω , ϵ , E and κ . If the developer's first-order conditions involve a technology L' (its own productivity, a law it believes, or its own output concept) while the econometrician uses L , then $\hat{w} = w \cdot$

$w_L(N, D)/w_{L'}(N, D)$. With factor-augmenting productivity $w_L/w_{L'} = e^{-\chi}$, so $\hat{w} = we^{-\chi}$.

(iii) (Special cases.) Let T be lifetime inference tokens in present value ($T = q/(r + h)$ for a token flow q , discount rate r and obsolescence hazard h), $c_S(N)$ the price of a serving FLOP, $X_S \equiv c_S 2NT$ serving expenditure and $p \equiv c_S/c_T$ the price ratio.

- (a) (Serving cost borne at rate λ .) If $V = R(L) - \lambda X_S$ with $T = T(L, N)$, then $w = 1 + \delta + \lambda p(\phi + \eta) T/(3D)$, where $\phi \equiv 1 + d \ln c_S / d \ln N$ and $\eta \equiv (\partial \ln T / \partial \ln N)_L$; a direct effect of size on revenue at fixed quality adds $-(\partial R / \partial \ln N)_L / X_T$. With $\delta = \eta = 0$ and $\phi = 1$: $w = 1 + \lambda p T / (3D)$, $w - 1 = \lambda X_S / X_T$ and $s = \lambda X_S / (X_T + \lambda X_S)$, the internalized serving share of lifetime expenditure; no conversion of FLOPs into dollars is needed. $\lambda = 1$ for a developer that serves its own model, $\lambda < 1$ when users bear part of the cost. Compute proportional to N that is planned at training time (reinforcement-learning rollouts, synthetic-data generation, evaluation) enters X_S in the same way.
- (a') (Adoption.) If the developer bears no serving cost and $V = v(A)$, where adoption $A(L, P_u)$ falls with users' serving cost per token $P_u \propto N$, then $m_N = v'(A)A|\varepsilon_{A,P}|/X_T$: $w - 1$ is the adoption value of compactness, not a token count. It is case (a) with $\lambda \equiv v'(A)A|\varepsilon_{A,P}|/X_u$, where X_u is users' serving expenditure.
- (b) (Tier or memory cap.) If $N \leq \bar{N}$ binds, $w = (1 + \nu)(1 + \delta + m_N)/(1 + m_D)$ with $\nu \equiv \xi/[X_T(1 + \delta + m_N)] \geq 0$. Any $w \geq (1 + \delta + m_N)/(1 + m_D)$ is rationalizable, including with $m_N = 0$, so $w - 1$ bounds m_N from above (when $\delta = m_D = 0$) but does not identify it.
- (c) (Latency.) A binding service-level objective $\tau(N) \leq \bar{\tau}$, with elasticity $\ell \equiv d \ln \tau / d \ln N > 0$, is case (b) with $\nu = \ell \xi_\tau / [X_T(1 + \delta + m_N)]$. A latency cost inside V instead enters m_N .
- (d) (Data cost and data cap.) $w = (1 + \delta + m_N)/(1 + m_D)$, so a data cost lowers w for given m_N . With $\delta = 0$ and $\lambda = p = 1$, $\hat{T}/D \equiv 3(w - 1) = (T/D - 3m_D)/(1 + m_D) < T/D$, and $s = (X_S - X_D)/(X_T + X_S)$ with $X_D \equiv c_D D$. A binding cap $D \leq \bar{D}$ adds $\mu \equiv \xi_D / X_T \geq 0$ to the denominator, $w = (1 + \delta + m_N)/(1 + m_D + \mu)$, which can fall below one with $T = 0$.
- (e) (Logit distillation.) Teacher forward passes cost $2N_T D$ FLOPs at the training price, a per-token cost with $m_D = N_T/(3N)$: with $\delta = 0$ and $\phi = 1$, $w = (1 + \lambda p T / (3D)) / (1 + N_T / (3N))$.
- (f) (Utilization varying with size.) If the prices of training and serving FLOPs have size elasticities δ and δ_S ($\phi = 1 + \delta_S$), then $w = 1 + \delta + \lambda(1 + \delta_S)pT/(3D)$, which equals $(1 + \delta)(1 + \lambda p T / (3D))$ if and only if $\delta_S = \delta$.

(iv) (*Signs.*) Let $w_0 \equiv (1 + \delta + m_N)/(1 + m_D)$. A single binding constraint raises w above w_0 if and only if $g_n > w_0 g_d$. Memory or tier caps and latency objectives raise w ; data caps and deadlines that bind on tokens lower it; deadlines proportional to compute ($g_n = g_d$) pull it toward one; minimum-size floors ($g_n < 0 = g_d$) lower it. Hence $\hat{T} \equiv 3D(\hat{w}-1)$ overstates λpT under caps, latency objectives, $\delta > 0$ (with $\delta_S \geq 0$) and $\chi < 0$, and understates it under data costs, data caps, distillation, minimum-size floors, $\delta < 0$ (with $\delta_S \leq 0$) and $\chi > 0$. A wedge below one cannot be generated by serving demand as long as serving expenditure rises with size at given quality ($\phi + \eta \geq 0$).

Proof. In (n, d) the Lagrangian is $\Pi - \sum_k \xi_k g_k$. Since $\partial L / \partial n = -(L - E)\varepsilon_N$ and $\partial L / \partial d = -(L - E)\varepsilon_D$, with $\Lambda \equiv -(\partial V / \partial L)(L - E) > 0$ and $\partial X_T / \partial n = (1 + \delta)X_T$, $\partial X_T / \partial d = X_T$, the first-order conditions are $\Lambda \varepsilon_N + (\partial V / \partial \ln N)_L - (1 + \delta)X_T - \sum_k \xi_k g_{k,n} = 0$ and $\Lambda \varepsilon_D - X_T - c_D D - \sum_k \xi_k g_{k,d} = 0$. Dividing the first by the second gives (A7). (ii) Remark A1; in the generalized family $w = a_1 u / (b_1 v)$ does not involve κ . If the first-order conditions involve L' , (A7) determines $w_{L'}(N, D)$, and the econometrician computes $w_L(N, D) = w_{L'}(N, D) \cdot w_L / w_{L'}$. With factor-augmenting productivity the true elasticities use $u = A e^{-\alpha(n+\psi_N)}$ and $v = B e^{-\beta(d+\psi_D)}$, so $w_{L'} / w_L = e^{\beta\psi_D - \alpha\psi_N} = e^\chi$. (iii)(a) $-(\partial V / \partial \ln N)_L = \lambda \partial X_S / \partial \ln N|_L = \lambda X_S(\phi + \eta)$ because $\ln X_S = \ln c_S + \ln 2 + \ln N + \ln T$; divide by X_T and use $X_S / X_T = pT / (3D)$. The share follows from $w - 1 = \lambda X_S / X_T$. (a') $(\partial V / \partial \ln N)_L = v'(A) \partial A / \partial \ln P_u$. (b) $g = n - \bar{n}$, so $g_n = 1$ and $g_d = 0$. (c) $g = \ln \tau(n) - \ln \bar{\tau}$, so $g_n = \ell$. (d) Definition of m_D ; the cap is $g = d - \bar{d}$. (e) The teacher's cost $2N_T D c_T$ has elasticity zero in N and one in D , so it acts as $c_D = 2N_T c_T$ and $m_D = 2N_T D / (6ND)$. (f) The elasticity of X_T in N is $1 + \delta$ and that of X_S is $1 + \delta_S$, so $w = [(1 + \delta)X_T + \lambda(1 + \delta_S)X_S] / X_T$, and $(1 + \delta)(1 + \lambda pT / (3D)) - w = (\delta - \delta_S)\lambda pT / (3D)$, which vanishes iff $\delta_S = \delta$. (iv) With $A \equiv 1 + \delta + m_N$ and $B \equiv 1 + m_D$, $\partial_\xi [(A + \xi g_n / X_T) / (B + \xi g_d / X_T)]$ has the sign of $g_n B - g_d A$, that is, of $g_n - w_0 g_d$, for every ξ . The first-order conditions are necessary, and the inversion uses nothing else; they are also sufficient when Π is concave in (n, d) , for example in the lifetime-compute problem, where $\ln(6ND + 2NT)$ is convex in logs and $\{y \geq \bar{y}\}$ is convex in the generalized family. $\square$

Remark A8 (Conduct, expenditure shares, and the parallel with markup estimation). Proposition A8 inverts the first-order conditions of a profit-maximizing developer in the spirit of Hao and Merrill (2026), with an imperfectly substitutable technology; in their Leontief limit over-training does not arise. Cases (a), (a') and (b) are three models of conduct: a developer that serves its own model and bears the serving cost ($\lambda = 1$); an open-weight developer that internalizes users' costs only through adoption; and size-class competition with N set by a menu of deployment tiers. They make different predictions. Under (a), $w - 1$ rises with the developer's own serving footprint; under (a'), with the sensitivity

of adoption to users' serving cost; under (b), sizes bunch at tier boundaries and $w - 1$ is unrelated to demand given the tier. The tools of [Berry and Haile \(2014\)](#), [Backus, Conlon and Sinkinson \(2021\)](#) and [Duarte et al. \(2024\)](#) for testing conduct apply. Without knowing p , the object that (a) identifies is the ratio of internalized serving expenditure to training expenditure, $w - 1 = \lambda\phi X_S/X_T$, or the share s ; a count of tokens needs p : $T = 3D(w - 1)/(\lambda\phi p)$.

The wedge is the ratio of the two inputs' output elasticities to their cost elasticities, which is the statistic [Raval \(2023\)](#) compares across inputs, here with FLOP-accounting weights. Part (ii) contains the two classic critiques of the production approach to markups: factor-augmenting heterogeneity ([Demirer, 2020](#); [Raval, 2023](#)) and a mismatch between the output the econometrician measures and the output the firm values ([Bond et al., 2021](#)). [Doraszelski and Jaumandreu \(2021\)](#) show that consistent estimation of the output elasticity by proxy methods generally requires controlling for the markup itself. Here the elasticities come from designed experiments, not from proxy estimation on the same choices, which removes that circularity at the cost of the experimental technology's external validity for each developer. As in [De Ridder, Grassi and Morzenti \(2026\)](#), levels are fragile and ranks are more robust: at given compute, the rank of $\hat{w}$ across models is the rank of M for every member of the generalized family.

Corollary A3 (Sufficient statistics; used in Proposition 2). *In the generalized family, for a model with compute C and ratio M at an interior optimum of Proposition A8 with $\delta = m_D = 0$ and no binding constraint,*

$$m_N = w - 1 = \left(\frac{M}{M^*(C)} \right)^{1/\sigma^*-1} - 1, \quad s = 1 - \left(\frac{M}{M^*(C)} \right)^{-(1/\sigma^*-1)},$$

$$\frac{C}{C_{\min}} = \left(\frac{\alpha + \beta w}{\alpha + \beta} \right)^{1/\gamma} w^{-1/\alpha},$$

where $M^*(C)$ is the training-optimal ratio at the model's own compute (the last formula with $\kappa = 1$; in general Lemma A5). Under Proposition A8(iii)(a) with $\phi = 1$ and $\eta = 0$, $\lambda p T/D = 3[(M/M^*(C))^{1/\sigma^*-1} - 1]$. In the homothetic case $\alpha = \beta = \rho$, $M^* = (B/A)^{1/\rho}$ is constant, $1/\sigma^* - 1 = \rho$ and $C/C_{\min} = \cosh(\frac{\rho}{2} \ln(M/M^*))^{2/\rho}$. With (α, β) unequal, ρ is replaced by the mean exponent $(\alpha + \beta)/2 = 1/\sigma^* - 1$ and M^* by $M^*(C)$. This justifies evaluating a technology's own compute-optimal law at each model's own compute.

Proof. Combine Proposition A8(i), equation (A2) and Lemma A5. $\square$

Proposition A9 (Family token budgets; Proposition 2 in the main text). *A developer trains K models of sizes $N_1, \dots, N_K$ on a common token budget D and chooses D (and, where stated, the sizes) to maximize $\sum_i V_i(L(N_i, D), N_i) - \sum_i X_{T,i} - c_D D$ subject to $n_i \leq \bar{n}_i$, where $X_{T,i} = c_T 6N_i D$ and the data cost is incurred once. Let w_i be member i 's wedge computed from the technology, $\omega_i \equiv N_i / \sum_j N_j$ its compute share, $s_i \equiv (w_i - 1)/w_i$, $m_{N,i} \equiv -(\partial V_i / \partial \ln N_i)_L / X_{T,i}$,*

$m_D \equiv c_D D / \sum_j X_{T,j}$, $\Lambda_i \equiv -(\partial V_i / \partial L)(L_i - E)$, and $\rho_i \equiv \Lambda_i \varepsilon_{D,i} / X_{T,i}$, the marginal value of data to member i per unit of its marginal training cost. Let $\bar{w}_H \equiv (\sum_i \omega_i / w_i)^{-1}$ be the compute-weighted harmonic mean of the member wedges and $\pi_i \equiv (\omega_i / w_i) / \sum_j (\omega_j / w_j)$.

- (i) (Member wedges are technological.) Given D , $\ln w_i - \ln w_j = \ln w(n_i, d) - \ln w(n_j, d)$, which equals $-a_1(n_i - n_j)$ in the generalized family: the profile of $\hat{T}_i \equiv 3D(w_i - 1)$ across sizes is fixed by the technology.
- (ii) (Family first-order condition.) At an interior optimum in D , $\sum_i \omega_i \rho_i = 1 + m_D$. If the sizes are set by a menu, this is the only first-order condition, and the w_i carry no information about the $m_{N,i}$.
- (iii) (Sizes chosen, caps slack.) If each n_i is also chosen at an interior optimum, $\sum_i \pi_i (1 + m_{N,i}) = (1 + m_D) \bar{w}_H$, where $\bar{w}_H$ is computed from the technology alone and satisfies the identity $1 - 1/\bar{w}_H = \sum_i \omega_i s_i$ (so the family's first-order condition pins the π -weighted average of the members' values of compactness, not $\bar{w}_H$, which is observed). Under Proposition A8(iii)(a) with common (λ, p) , $\delta = \eta = 0$, $\phi = 1$ and $m_D = 0$: $\sum_i \pi_i \hat{T}_i = \lambda p \sum_i \pi_i T_i$. Only this π -weighted average of the members' inversions is a revealed-preference object; the member-level T_i are not identified (one equation for K unknowns).
- (iv) (Binding caps and floors.) If every size is at a binding cap, the equality in (iii) becomes $\sum_i \pi_i (1 + m_{N,i}) \leq (1 + m_D) \bar{w}_H$, so that $\sum_i \pi_i \hat{T}_i \geq \lambda p \sum_i \pi_i T_i$: $\bar{w}_H - 1$ bounds the family's average value of compactness from above. Binding minimum-size floors reverse the inequality.
- (v) (A common D under member-level choice is a knife-edge.) If instead each member chooses its own D_i (Proposition A8 with $\delta = m_D = 0$ and no binding constraint), equal token counts $D_i = \bar{D}$ occur only if $1 + m_{N,i} = w(n_i, \bar{d})$ for every i ; in the generalized family, only if $(1 + m_{N,i}) / (1 + m_{N,j}) = (N_i / N_j)^{-a_1}$ exactly.

Proof. (i) In the generalized family $\ln w = \ln(a_1 A / b_1 B) - a_1 n + b_1 d$ (Lemma A4(iii)). (ii) The first-order condition in d is $\sum_i \Lambda_i \varepsilon_{D,i} = \sum_i X_{T,i} + c_D D$; divide by $\sum_j X_{T,j}$ and note $X_{T,i} / \sum_j X_{T,j} = \omega_i$. (iii) The condition in n_i is $\Lambda_i \varepsilon_{N,i} = X_{T,i} (1 + m_{N,i})$, so $\rho_i = (1 + m_{N,i}) / w_i$; substitute into (ii) and multiply by $\bar{w}_H$. For the shares, $1 - \sum_i \omega_i / w_i = \sum_i \omega_i (1 - 1/w_i)$. With $m_D = 0$, $\sum_i \pi_i m_{N,i} = \bar{w}_H - 1 = \sum_i \pi_i (w_i - 1)$ because $\sum_i \pi_i w_i = \bar{w}_H$; multiply by $3D$ and use $m_{N,i} = \lambda p T_i / (3D)$. (iv) With a binding cap the condition in n_i is $\Lambda_i \varepsilon_{N,i} = X_{T,i} (1 + m_{N,i}) + \xi_i$ with $\xi_i \geq 0$, so $\rho_i \geq (1 + m_{N,i}) / w_i$; a floor has $\xi_i \leq 0$. (v) Proposition A8(i) member by member, and (i). $\square$

Remark A9. In a brute-force example with three members whose sizes are chosen on a common budget, the member inversions $\hat{T}_i / T_i$ are 0.86, 0.91 and 1.06 while the π -weighted average is exact (replication package). Families that train every

size on the same budget are therefore informative about serving demand only at the family level, and only if the sizes are chosen at the margin; if sizes come from a menu of tiers they are informative about nothing beyond the bound in (iv).

A9. *Model-free identification of the elasticity of substitution and the wedge*

Proposition A10 (Model-free σ^* and w ; Proposition 4 in the main text). *Let $y = F(n, d)$ be C^2 with $F_n, F_d > 0$, where y is any strictly increasing transformation of $-L$. Write $x \equiv \ln M = d - n$ and $c \equiv \ln C = \ln 6 + n + d$, let $Y(x, c) \equiv F((c - \ln 6 - x)/2, (c - \ln 6 + x)/2)$, let L also denote loss as a function of (x, c) , and let $L^*(c) \equiv \min_x L(x, c)$ be the loss-compute frontier with $\operatorname{argmin} x^*(c) = \ln M^*(C)$.*

(i) (Wedge.) *At any point, $F_n = Y_c - Y_x$, $F_d = Y_c + Y_x$ and*

$$w = \frac{Y_c - Y_x}{Y_c + Y_x} = \frac{L_c - L_x}{L_c + L_x}.$$

w is invariant to strictly monotone transformations of output, so it requires neither E nor κ nor a functional form: only the slopes of any output measure along the isocost and along the ray of constant M .

(ii) (Elasticity of substitution at the compute optimum.) *At $x = x^*(c)$, where $L_x = 0$,*

$$(A8) \quad \frac{1}{\sigma^*} - 1 = \frac{2(-Y_{xx})}{Y_c} = \frac{2L_{xx}}{|dL^*/dc|} = \frac{L_{nn}|_C}{2|dL^*/dc|},$$

where $L_{nn}|_C$ is the second derivative of the IsoFLOP profile with respect to $\ln N$ at its minimum and $dL^/dc = L_c$ by the envelope theorem. The ratio is invariant to strictly monotone transformations of output at the argmin , but not elsewhere: the curvature of the IsoFLOP profile divided by twice the slope of the loss-compute frontier, in any common output units, identifies σ^* .*

(iii) (Slope of the log wedge.) *At $x^*(c)$, $\partial \ln w / \partial \ln M|_C = 1/\sigma^* - 1$. In the generalized family $\ln w$ is exactly linear in $\ln(M/M^*(C))$ with this slope at every compute (Lemma A4(iii)); in general the slope varies with M , which gives a test of the family.*

(iv) (Local σ anywhere.) *$F_{nn} = Y_{cc} - 2Y_{cx} + Y_{xx}$, $F_{dd} = Y_{cc} + 2Y_{cx} + Y_{xx}$ and $F_{nd} = Y_{cc} - Y_{xx}$, so $\sigma = P/(P + Q)$ (Lemma A2) is identified at any point from the second-order expansion of any output in (x, c) .*

(v) (What $\kappa = 1$ imposes.) *In the generalized family, with $y = -\ln(L - E)$, $dy^*/dc = \gamma$ and $-y_{nn}|_C = S\gamma$ at the argmin , so (A8) returns $S/2$ for every κ and E . The restriction $\kappa = 1$ imposes $S = \gamma/(ab)$, that is, $-y_{nn}|_C =$*

$\gamma^2/(ab)$: the curvature of every IsoFLOP profile in log reducible loss is pinned by the slopes of the frontier and of the path. With κ free the curvature is a separate parameter, which (A8) measures without the restriction.

- (vi) (Designs.) σ^* is identified by IsoFLOP profiles: the curvature of a profile at its minimum, and the frontier slope from the minima at two or more budgets (or locally). On-path designs identify the frontier slope but not the curvature, hence not σ^* without a functional form (Proposition A1). The wedge at a point is identified by local variation in two non-collinear directions around it (for example, a local factorial in N and D), and only inside the design's support; its precision falls as $\varepsilon_D \rightarrow 0$, since $w \rightarrow \infty$ when $L_c + L_x \rightarrow 0$.

Proof. (i) $\partial/\partial n = \partial_c - \partial_x$ and $\partial/\partial d = \partial_c + \partial_x$. With $L = h(Y)$ and $h' < 0$, $L_x = h'Y_x$ and $L_c = h'Y_c$, and h' cancels; any strictly monotone g multiplies both first derivatives by g' . (ii) At $F_n = F_d \equiv f$, $P = 2f^3$ and $Q = -f^2(F_{nn} - 2F_{nd} + F_{dd}) = -4f^2Y_{xx}$, since $Y_{xx} = (F_{nn} - 2F_{nd} + F_{dd})/4$. By Lemma A2, $1/\sigma - 1 = Q/P = -2Y_{xx}/f$ with $f = Y_c$. For $L = h(Y)$, $L_{xx} = h''Y_x^2 + h'Y_{xx} = h'Y_{xx}$ at $Y_x = 0$ and $dL^*/dc = L_c = h'Y_c$, with $h' < 0$. At fixed c , $dn = -dx/2$, so $L_{nn}|_C = 4L_{xx}$. For a monotone g , $(g \circ L)_{xx} = g''L_x^2 + g'L_{xx} = g'L_{xx}$ at $L_x = 0$ and $d(g \circ L^*)/dc = g'dL^*/dc$; away from the argmin the term $g''L_x^2$ does not vanish. (iii) $\partial_x \ln w = (Y_{cx} - Y_{xx})/(Y_c - Y_x) - (Y_{cx} + Y_{xx})/(Y_c + Y_x)$, which is $-2Y_{xx}/Y_c$ at $Y_x = 0$. (iv) Second-order chain rule. (v) Along the path $y^* = \gamma(c - \ln 6) - \ln K$ (Lemma A3 applied to the inner aggregator), so $dy^*/dc = \gamma$; by (ii) and $1/\sigma^* - 1 = S/2$, $-Y_{xx} = \gamma S/4$ and $-y_{nn}|_C = -4Y_{xx} = S\gamma$. With $\kappa = 1$, $\gamma = abS$. (vi) Part (ii) uses only derivatives of a profile at its minimum and of the frontier; part (i) uses two directional derivatives at the point. $\square$

Remark A10 (Finite grids). IsoFLOP profiles are sampled on finite grids and fitted by low-order polynomials (Czech et al., 2026). If the profile $f(t)$, with t the distance from its minimum in $\ln N$ (or $\ln M$), is sampled at points symmetric about the minimum and fitted by least squares on $(1, t, t^2)$, the quadratic coefficient is $f''/2 + (f'''/24)\kappa_4 + O(h^4)$, with $\kappa_4 \equiv (m_6 - m_2m_4)/(m_4 - m_2^2)$, m_k the k -th moment of the grid and h its half-width, and the fitted minimum is displaced by $-(f'''/6)(m_4/m_2)/f'' + O(h^3)$. For the Besiroglu et al. (2024) technology at 10^{21} FLOP and nine equally spaced sizes, a grid spanning a factor of 9 (100) in N overstates the curvature by 1.3 (5.9) percent, so the model-free estimate is $\hat{\sigma}^* = 0.734$ (0.726) against 0.737; a quartic fit removes the leading term (0.737 in both cases). Quadratic fits on grids narrower than a factor of about 3 in N , or local polynomials of order four, keep the bias below 0.001. These are noise-free calculations of bias only: with seed and evaluation noise, a narrower grid or a quartic fit raises the variance of the estimated curvature, so the choice is a bias-variance trade-off to be made with the design's noise level.

A10. Partial identification beyond the design

Proposition A11 (Partial identification of $M^*(C)$ and of the wedge; Proposition 5 in the main text). *Let $x^*(c) \equiv \ln M^*(C)$ be the unique argmin of the IsoFLOP profile at compute C , absolutely continuous in c , with path slope $e(c) \equiv dx^*/dc = 1 - 2a(c)$, where $a(c) \equiv d \ln N^*/d \ln C$. Suppose $x^*(c_J)$ is identified at the largest design budget c_J (from IsoFLOP minima, or from a path fitted to the minima at the largest budgets), and consider a model with compute $c > c_J$ and log ratio $x = \ln M$.*

(i) (Normal inputs.) *If $N^*(C)$ and $D^*(C)$ are nondecreasing, then $e \in [-1, 1]$ and $x^*(c) \in [x^*(c_J) - (c - c_J), x^*(c_J) + (c - c_J)]$.*

(ii) (Bounded path slope.) *If $e(c') \in [e_L, e_U]$ for $c' \in [c_J, c]$, the sharp identified set is*

$$\mathcal{X}^*(c) = [x^*(c_J) + e_L(c - c_J), x^*(c_J) + e_U(c - c_J)],$$

whose width grows linearly in the log extrapolation distance; monotonicity of M^ is the case $e_L \geq 0$. When the true technology is in the generalized family, every point of $\mathcal{X}^*(c)$ is generated by a technology with positive marginal products and single-peaked IsoFLOP profiles that coincides with the true one for $c \leq c_J$. If confidence intervals CI_j for $x^*(c_j)$ at budgets $c_1 < \dots < c_J$ hold jointly with probability $1 - p$ and the slope bounds hold from c_1 on, then $\bigcap_j [\inf \text{CI}_j + e_L(c - c_j), \sup \text{CI}_j + e_U(c - c_j)]$ covers $x^*(c)$ with probability at least $1 - p$.*

(iii) (Sign of the wedge.) *If each IsoFLOP profile is single-peaked in x , which holds whenever $0 < \sigma < 1$ throughout (Lemma A2), then $\text{sign}(w - 1) = \text{sign}(x - x^*(c))$ at every (c, x) , for any technology. Hence $w > 1$ is identified if and only if $x > \sup \mathcal{X}^*(c)$, $w < 1$ if and only if $x < \inf \mathcal{X}^*(c)$, and the sign is not identified otherwise. Neither σ^* nor the functional form is needed.*

(iv) (Magnitude.) *Suppose $\partial \ln w / \partial x|_c \in [k_L, k_U]$ with $k_L > 0$ between $x^*(c)$ and x ; in the generalized family $k_L = k_U = 1/\sigma^* - 1$, so a range for σ^* gives $[k_L, k_U]$. With $\mathcal{X}^*(c) = [\ell, u]$: $\ln w \in [k_L(x - u), k_U(x - \ell)]$ if $x > u$; $\ln w \in [k_U(x - u), k_L(x - \ell)]$ if $x < \ell$; and $\ln w \in [k_U(x - u), k_U(x - \ell)]$ if $\ell \leq x \leq u$.*

(v) (Expenditure share and T .) *Since $s = 1 - 1/w$ is increasing in w , bounds $[w_L, w_U]$ give $s \in [1 - 1/w_L, 1 - 1/w_U]$. Under Proposition A8(iii)(a) with $\phi = 1$ and $\eta = \delta = 0$, $\lambda p T / (3D) \in [w_L - 1, w_U - 1]$; under the other objectives of Proposition A8 the same bounds apply to w and map into T as described there.*

(vi) (A bound inside the support.) *Let $F_{nn}, F_{dd} \leq 0$ and $F_{nd} \geq 0$, which hold throughout the generalized family. Then $\ln w$ is nonincreasing in n and nondecreasing in d , so for any support point with $n_b \geq n_0$ and*

$d_b \leq d_0$, $w(n_0, d_0) \geq w(n_b, d_b)$, and under (iii)(a) with $\lambda = p = 1$, $T_0 \geq 3D_0[w(n_b, d_b) - 1]$, where $w(n_b, d_b)$ is identified without extrapolation. No finite upper bound follows from these shape restrictions, and without $F_{nd} \geq 0$ the bound can fail.

Proof. (i) $d \ln N^* + d \ln D^* = d \ln C$ with both terms nonnegative gives $a \in [0, 1]$; integrate $e = 1 - 2a$. (ii) Integrate e over $[c_J, c]$; the linear continuations $x^*(c_J) + e(c - c_J)$, $e \in [e_L, e_U]$, attain every point. For the technology: in the generalized family let $\tilde{L}(n, d) = E + [Ae^{-a_1(n+\psi_N(c))} + Be^{-b_1(d+\psi_D(c))}]^\kappa$ with $c = \ln 6 + n + d$ and nondecreasing ψ_N, ψ_D that vanish for $c \leq c_J$. At fixed c the profile is a translate of a profile of the original technology (at effective compute $c + \psi_N + \psi_D$), hence single-peaked, with argmin $x^*(c) + 2b\psi_N(c) - 2a\psi_D(c)$ (Lemma A3(iii)); $\tilde{F}_n = F_n(1 + \psi'_N) + F_d\psi'_D > 0$ and $\tilde{F}_d = F_n\psi'_N + F_d(1 + \psi'_D) > 0$. Choosing $\psi'_N = \Delta'_+/(2b)$ and $\psi'_D = \Delta'_-/(2a)$ generates any continuation $x^*(c) + \Delta(c)$ with $\Delta(c_J) = 0$. On the event that every CI_j covers, every cone contains $x^*(c)$. (iii) By Proposition A10(i), $w - 1 = -2Y_x/(Y_c + Y_x) = -2Y_x/F_d$ with $F_d > 0$, and single-peakedness means $Y_x > 0$ for $x < x^*$ and $Y_x < 0$ for $x > x^*$. If $0 < \sigma < 1$ throughout, every critical point of a profile is a strict local maximum of y ($Y_{xx} = -Q/(4F_n^2) < 0$ at $Y_x = 0$, Lemma A2), so there is at most one. (iv) $\ln w(x) = \int_{x^*}^x \partial_x \ln w$, with $\ln w(x^*) = 0$ and $x^* \in [\ell, u]$. (v) Monotonicity. (vi) $\partial \ln w / \partial n = F_{nn}/F_n - F_{nd}/F_d \leq 0$ and $\partial \ln w / \partial d = F_{nd}/F_n - F_{dd}/F_d \geq 0$; move from (n_b, d_b) to (n_b, d_0) and then to (n_0, d_0) . In the generalized family $F = -\kappa \ln Q$, whose Hessian has entries $a_1^2 uv/Q^2$, $-a_1 b_1 uv/Q^2$ and $b_1^2 uv/Q^2$ times $-\kappa$. A saturating continuation of the data term ($\varepsilon_D \rightarrow 0$ beyond the support) satisfies the shape restrictions and sends $w \rightarrow \infty$; the concave quadratic $F = 0.05n + 0.3d - 0.025n^2 - 0.005d^2 - 0.02nd$ has $F_{nd} < 0$ and $w(0, 0.5) < w(0, 0)$. $\square$

Remark A11 (An illustration at frontier scale). This remark illustrates the logic of Proposition A11 only; the application in Section IV harmonizes units and uses lab-own anchors. Anchor the path at the largest budgets of the two lab designs with the most compute: the Chinchilla IsoFLOP minima (nine budgets up to 2.9×10^{21} FLOP) and the Llama 3 minima (ten budgets up to 10^{22} FLOP). A line fitted to each set of minima gives $M^* = 20.5$ (95 percent interval $[11.4, 36.8]$) and 22.3 ($[12.2, 40.2]$) at the largest budget, with in-sample path slopes $e = 0.005$ ($[-0.16, 0.17]$) and 0.074 ($[-0.09, 0.24]$). The intervals are equal-tailed wild bootstrap- t intervals conditional on the designed budgets (HC2-rescaled residuals and HC2 standard errors, Webb weights, 9,999 draws); with nine or ten points and the anchor at the high-leverage end of the design, even this construction covers only about 90 percent in simulations at these designs, and a straight path within the design is maintained. With the path slope confined to the range of allocation exponents across the paper's sweeps, $a \in [0.37, 0.57]$ or $e \in [-0.14, 0.26]$, the identified set for $M^*(10^{24})$ is $[5.0, 168]$ (Chinchilla anchor) or $[6.4, 133]$ (Llama 3 anchor). Among the verified-sample models beyond each anchor's largest budget

(149 and 137 models), the sign of $w - 1$ is identified as positive for 115 and 111, negative for 2 and 2, and not identified for 32 and 24. With only normal inputs, positive signs are identified for 46 and 47 models, and none is identified as negative. Point extrapolation classifies every model, and the two anchors then disagree about the largest one (Llama 3.1 405B: $w > 1$ with the Chinchilla anchor, $w < 1$ with the Llama 3 anchor), whose sign no set-valued assumption identifies. Magnitudes are wide: with $\sigma^* \in [0.6, 0.75]$ and the cross-sweep slope range, the medians of the models' lower and upper bounds on w are 1.58 and 16.1 (Chinchilla anchor). The models are measured in their own parameter and token conventions, which are not harmonized with the anchors'.

Remark A12 (Parametric projection, calibration and the direction of extrapolation errors). A confidence set for $(\ln(A/B), \alpha, \beta)$ projects into one for T . At fixed compute, $\text{Var}(\ln \hat{w}) = \frac{1}{4} \text{Var}(\hat{S})(\ln M)^2 - \frac{1}{2} \text{Cov}(\hat{S}, \hat{S} \ln \hat{M}^*) \ln M + \frac{1}{4} \text{Var}(\hat{S} \ln \hat{M}^*)$ is exactly quadratic in $\ln M$, so the band widens linearly in the log extrapolation distance with slope $\frac{1}{2} z_{1-p/2} \text{sd}(\hat{S})$. For the Chinchilla extraction, a pairs bootstrap ($B = 300$) gives $\text{sd}(\ln \hat{w}) = 0.161$ at $M = 1,875$ and $C = 7.2 \times 10^{23}$, of which 0.143 comes from $\ln M^*$ and 0.047 from $\alpha + \beta$. That is sampling error within one sweep and one functional form ($\kappa = 1$): the implied 95 percent interval for $\ln M^*$ at that compute is about 1.6 log points wide, while the identified set of Remark A11 at 10^{24} FLOP is about 3.5 log points wide and the estimates of $M^*(10^{24})$ across the fifteen technology estimates used for the wedge range from 1.4 to 291 (5.4 log points); the range of σ^* adds more. Normalizing a family's flagship to $w_f = 1$ is a calibration assumption, not the approach of Gandhi, Navarro and Rivers (2020), which uses an observed expenditure share; by Proposition A8 it is valid only if the flagship's m_N , m_D , δ and binding constraints all vanish. Finally, evidence that fitted laws misstate losses at extreme ratios (Sardana et al., 2024) does not sign the error in $\hat{w}$: revealed preference recovers the technology the developer believed when it chose (N, D) (Proposition A8(ii)), so if developers planned with fitted laws, correcting toward the true technology moves $\hat{w}$ away from their first-order condition.

A11. Growth-accounting corollaries

Corollary A4 (Frontier-release regressions). *Let frontier releases satisfy $\ln R_t = \ln K - \gamma(c_t + g_A t)$, where g_A is algorithmic progress in compute-equivalent units, and $c_t = c_0 + gt + u_t$ with u_t independent of t . The OLS slope of $-\ln R_t$ on c_t converges to $\gamma[1 + (g_A/g)R_{c,t}^2]$, with $R_{c,t}^2 = g^2 \text{Var}(t)/(g^2 \text{Var}(t) + \text{Var}(u))$. With deterministic trends this is $\gamma/(1 - s_A)$, $s_A \equiv g_A/(g + g_A)$: for $s_A \in [0.05, 0.40]$ the naive scaling exponent is inflated by a factor of 1.05–1.67 (Nagy et al., 2013). Adding a time trend recovers γ from deviations of compute from trend, and does so only if $\text{Var}(u) > 0$.*

Proof. $\text{Cov}(-\ln R_t, c_t) = \gamma \text{Var}(c) + \gamma g_A \text{Cov}(t, c)$ and $\text{Cov}(t, c) = g \text{Var}(t)$. The last statement is Frisch–Waugh. $\square$

Corollary A5 (Hall-type bias in cost-share growth accounting). *To first order, $\Delta y = \varepsilon_N \Delta n + \varepsilon_D \Delta d + \Delta \Omega$. A residual computed with equal weights $\bar{\varepsilon} = (\varepsilon_N + \varepsilon_D)/2$, as the compute accounting $C = 6ND$ suggests, equals*

$$\Delta \Omega + \frac{1}{2}(\varepsilon_N - \varepsilon_D)(\Delta n - \Delta d) = \Delta \Omega + \bar{\varepsilon} \frac{w-1}{w+1} (\Delta n - \Delta d).$$

Over-trained models ($w > 1$) that shift further toward tokens ($\Delta d > \Delta n$) make the residual understate progress (the analog of [Hall, 1988](#)). With weights γ on each input, the bias is $(\varepsilon_N - \gamma)\Delta n + (\varepsilon_D - \gamma)\Delta d$.

Proof. Subtract $\bar{\varepsilon}(\Delta n + \Delta d)$ and use $\varepsilon_N = 2\bar{\varepsilon}w/(1+w)$ and $\varepsilon_D = 2\bar{\varepsilon}/(1+w)$. For discrete changes the error is second order; Törnqvist (average-of-endpoint) weights reduce it to third order. $\square$

DATA CONSTRUCTION

This appendix documents the samples summarized in [Table B1](#): their sources, the conventions we impose, the corrections we make to source data, and the sample restrictions. [Table 1](#) in [Section III](#) reports the designs that identify the technology. A single script per module regenerates every sample from the raw files with fixed seeds, and the replication package lists the download commands. The snapshots of living databases are from September 23 and 24, 2026.

B1. Public Sweeps

COMMON CONVENTIONS.

For every sweep we build a run-level panel with the fields (N, D, L) , a cluster identifier and the derived quantities $M = D/N$ and $C = 6ND$. We use $C = 6ND$ even when a sweep reports a more detailed FLOP count, so that compute means the same thing across sweeps; the model-free estimator, which needs exact isocosts, is the exception ([Section III](#)). The sweeps do not share a parameter-count convention, a tokenizer or an output measure ([Table B2](#)). We keep each sweep’s own conventions and compare only unit-free objects across sweeps. The exponents α and β and the derived a , γ and σ^* are invariant to the unit of loss. They are not invariant to the parameter-count convention, which [Section III.D](#) treats as a measurement problem. The asymptote E , the levels A and B , and the compute-optimal ratio M^* are not comparable across rows.

DESIGN STATISTIC.

The off-path spread in [Table B1](#) is the residual standard deviation of an OLS regression of $\ln M$ on $\ln(6ND)$ over the estimation sample. [Figure B1](#) plots six public designs with run-level losses against their isocost lines. Farseer is a near-factorial grid in $(\ln N, \ln D)$ with $\text{corr}(\ln N, \ln D) = -0.04$. The Chinchilla runs

TABLE B1—SAMPLES

Sample	Units	N	$M = D/N$	Off-path spread	Use
<i>Panel A. Public designed sweeps (unit: training run)</i>					
Chinchilla (digitized)	245 ^a	57M–16B	0.04–341	1.35	III; IV; D
Llama 3 (digitized)	133	59M–17B	0.44–643	1.38	III; IV; D
Marin, 3 corpora	258	0.12–12.6B	0.09–3,706	2.42–2.46	III; IV; D
Porian et al.	975 ^b	5M–0.90B	0.02–346	2.20–2.29	III; D
Farseer	404	0.10–6.4B ^c	0.31–2,570	1.97	III; IV; D
Gadre et al., 3 corpora	104	11M–6.9B	5–640	1.44	III; D; E
OLMo ladder	30	0.19–3.2B ^d	10–200	0.94	III; IV; E
Muennighoff et al.	33	7M–8.7B	0.04–303	1.97	III; D
DataDecide, 25 recipes	1,100 ^e	4M–1.2B ^d	5–110	0.73	III; D
Step Law	1,911 ^f	0.21–1.07B ^c	19–466	–	D
<i>Panel B. Our controlled experiment, FineWeb-Edu and FineWeb (unit: annealed endpoint)</i>					
Main grid, 2 corpora	2 × 44	0.39–49M ^c	0.51–2,031	1.96	III.E
High- M runs	12	0.79–3.2M ^c	64–4,063	–	III.E
<i>Panel C. Released models (unit: model)</i>					
Verified sample	173	70M–406B ^g	2.1–60,398	–	IV; F
Clean sample	77	0.13–406B	19–60,398	–	IV; F
Production-scale universe	335	0.13–600B ^g	0.2–60,398	–	IV; V; F
Benchmark panel	128	70M–180B	2.1–24,290	1.64	E
Algorithmic-progress sample	231	2M–280B	0.001–204 ^h	–	E
Epoch language models	247	65M–540B	0.3–24,000	–	E

Note: N is parameters (total, including embeddings, unless marked), D is tokens processed, $M = D/N$ and $C = 6ND$; ranges are over the estimation sample. Off-path spread: residual standard deviation of an OLS regression of $\ln M$ on $\ln C$, zero for compute-optimal runs; “–”: not computed. Roman numerals refer to sections of the main text, letters to Online Appendices. Llama 3 and Marin: $N = C/(6D)$ at the nominal budget; Porian et al.: non-embedding plus output head. ^a137 runs on nine IsoFLOP profiles and 108 off-profile runs; parametric fits drop the five highest-loss runs ($n = 240$), model-free estimates use the 137 profile runs. ^b16 architectures; losses read along each run’s curve at 12 IsoFLOP budgets (121 and 116 profile points). ^cNon-embedding parameters. In our experiment N including the tied embedding is 1.4–54M and M is 0.46–555 in the main grid (up to 1,743 in the high- M runs). ^dExcludes the input embedding only. ^e25 recipes × 14 sizes × 3 seeds (5 at 1B); 21,888 intermediate checkpoints enter estimation. ^f17 (N, D) cells, each with a grid of learning rates and batch sizes. ^gActive parameters for mixture-of-experts models. ^h D is dataset size (unique tokens), not tokens processed. Algorithmic-progress sample: [Ho et al. \(2024\)](#); clean sample: the clean inference-demand sample.

Source: Authors’ calculations from the sources cited in the text of this appendix; Panel B: this paper.

TABLE B2—CONVENTIONS OF THE PUBLIC SWEEPS

Dataset	N	D	Output (tokenizer)	Cluster unit
Chinchilla	Total	$C/(6N)$, C digitized	MassiveText validation, nats/token (SentencePiece, 32k)	Budget (9); off-profile trunk (35)
Parseer	Non-embedding (two untied 65,536-row embeddings excluded)	Tokens, single epoch	English bits per character, IntelliValSet	Run; model size (robustness)
Gadre et al.	Total	$20 \times \text{multiplier} \times N$	C4 validation, nats/token (GPT-NeoX)	Run; (size, multiplier) cell
OLMo ladder	Excludes input embedding	Logged tokens at final step	C4-en validation, nats/token (dolma2)	(Size, multiplier) cell
Muennighoff et al.	Authors' parameter map (convention unverified)	Tokens; single-epoch runs ($D = U$)	C4 validation, nats/token (GPT-2)	Run
DataDecide	Excludes input embedding	Step $\times$ batch $\times$ 2,048	C4-en validation, ln perplexity	Run; (size, seed) cell
Llama 3	$C/(6D)$, nominal C	Read off the figure	Unstated	IsoFLOP budget (10)
Marin	Configuration; $C_b/(6D)$ for model-free estimates	Reported; budget $C_b = 3 \times$ forward FLOPs	Paloma macro-average	IsoFLOP budget (7–8)
Porian et al.	Four counts: Kaplan, standard, attention, total	Tokens along the loss curve	Smoothed training or validation loss	Bootstrap over loss noise
Step Law	Non-embedding	Tokens, single epoch	Smoothed final training loss	(N, D) cell (17)

Note: “Total” counts all parameters, including embeddings. Porian et al.’s “standard” count adds the output head to the non-embedding count, “attention” also adds attention FLOPs, and “total” adds input and positional embeddings. For Marin, $6ND$ with the configuration count differs from the nominal budget by -7 to $+35$ percent, and the ratio of budget to $6ND$ rises with N within a budget. The Chinchilla clusters apply to the inference of Online Appendix D2. The text of this appendix gives details.

Source: Hoffmann et al. (2022), Besiroglu et al. (2024), Li et al. (2025b), Gadre et al. (2025), Bhagia et al. (2025), Muennighoff et al. (2023), Magnusson et al. (2025), Czech (2026), Marin Community (2026), Porian et al. (2024), Li et al. (2025a).

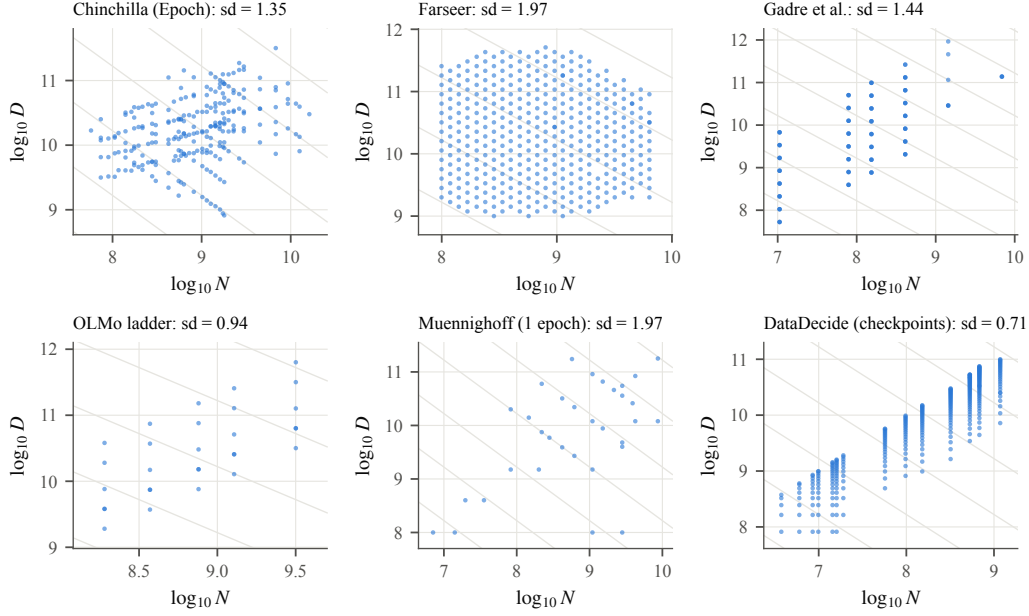


FIGURE B1. DESIGNS OF THE PUBLIC SWEEPS

Note: Each panel plots $\log_{10} D$ against $\log_{10} N$ for one sweep’s estimation sample; gray lines are isocost lines $6ND = \text{constant}$. The panel titles give the off-path spread. The DataDecide panel shows unique (size, step) checkpoints, for which the statistic is 0.71; it is 0.73 over all 21,888 checkpoints (Table B1).

combine nine IsoFLOP profiles with off-profile streaks at fixed model size. Gadre et al. and the OLMo ladder are grids in size and token multiplier. DataDecide’s final checkpoints all lie at $D \approx 85\text{--}110 N$, so only the intermediate checkpoints spread the design.

FARSEER.

The file `1222_full.csv` of Li et al. (2025b) contains 404 runs, the released single-recipe subset of the roughly 1,000 models the authors trained: 25 model sizes, each trained on up to 19 token budgets with one data recipe and with hyperparameters set by the Step Law rules. N excludes both of the model’s untied 65,536-row embedding matrices; we verified that the reported N including embeddings exceeds N by exactly $131,072 h$ for hidden size h . The output is English bits per character on the authors’ validation set (the average of its web, paper and book subsets), which is the output they fit. Three quirks required care. The column labeled `D/N` takes five values unrelated to D/N , so we recompute M from D and N (range 0.31–2,570). The file’s compute column includes attention FLOPs, at 6.6–8.5 FLOP per parameter per token, and we use $6ND$ instead. Four (N, D) pairs were run twice, and we keep both copies.

GADRE ET AL.

The 104 runs of [Gadre et al. \(2025\)](#) are 34 on C4, 35 on RedPajama and 35 on RefinedWeb. N includes embeddings, and $D = 20 \times \text{multiplier} \times N$ exactly. Our main output is the C4 validation loss (174 million tokens), on which every model is evaluated, because tests of neutral data quality need a common output. In-distribution Paloma losses serve as a robustness check. The authors’ token-level confidence intervals imply a sampling standard error of $\ln L$ of only 0.0016. The residual standard deviation of a fitted Chinchilla form is 0.023–0.030, so the residuals reflect misspecification rather than evaluation noise.

OLMO LADDER.

We take the final logged step of 30 runs: five sizes trained to 0.5, 1, 2, 5 and 10 times the Chinchilla token budget, plus five reruns ([Bhagia et al., 2025](#)). N is OLMo’s count, which excludes only the input embedding; we verified it against the ladder code. D is the logged token count at the final step. The output is C4-en validation cross-entropy, and task bits per byte serve as a robustness output. Reruns are clustered with their originals.

MUENNIGHOFF ET AL.

We parse 229 final losses from the authors’ notebook with their own code. The aggregated third-party file mislabels the tokens-per-epoch and epoch columns, so we rebuild N , D and the unique tokens U from the model names. The primary sample is the 33 single-epoch runs ($D = U$). A robustness variant drops two runs with $D/N < 0.4$, following Besiroglu et al.’s rule for Chinchilla. The parameter-count convention in the authors’ map is not documented.

DATADecIDE.

The DataDecide release ([Magnusson et al., 2025](#)) crosses 25 data recipes with 14 model sizes (4M to 1.2B parameters, excluding the input embedding) and three seeds per size (five at 1B), for 1,100 runs. D is $\text{step} \times \text{batch} \times 2,048$, using the batch sizes in the authors’ code, and the output is C4-en validation cross-entropy. We keep 21,888 checkpoints with $D/N \geq 5$ that lie at or before the end of the full schedule. The final checkpoints have $D \approx 85\text{--}110 N$, and even with checkpoints $\text{corr}(\ln N, \ln D) = 0.91$. The checkpoints that break the collinearity are evaluated before the cosine schedule has finished, and schedule progress is itself a function of D/N . The levels of the fitted technology are therefore not interpretable, and we use DataDecide only for comparisons across recipes, which share sizes, steps and schedules. This rests on the assumption that the unfinished schedule affects all recipes alike; for the same reason DataDecide is excluded ex ante from the technologies of Section IV.

SWEEPS WE DID NOT USE.

The Hugging Face release of Kricheli et al. (2026) contains no validation losses, only per-epoch training losses from small multi-epoch runs, so we report no estimates from it. We did not process the open-sci-ref collection of reference runs on eight corpora, which would allow a second multi-recipe neutrality test.

B2. Chinchilla Extraction

The Epoch AI digitization of Figure 4 of Hoffmann et al. (2022) contains 245 runs (Besiroglu et al., 2024), with training compute from 1.4×10^{18} to 1.3×10^{22} FLOP and model sizes from 57 million to 16 billion parameters. Hoffmann et al.’s list of model sizes starts at 44 million parameters; that size is not among the digitized runs, rather than among the dropped ones. N is the total parameter count and C is read off the figure’s horizontal axis, so $D = C/(6N)$ is constructed. Two consequences follow. Digitization error in C passes one for one into $\ln D$ and is correlated with $\ln C$, an errors-in-variables structure that matters mainly for the separation of E from the outer exponent. And if the budgets in the figure count FLOPs differently from $6ND$, for example with embedding and attention terms whose share varies with size, the constructed D inherits an error correlated with N ; Online Appendix D3 bounds its effect on σ^* , and we have not rebuilt D from Hoffmann et al.’s architecture table. The loss is MassiveText validation loss in nats per token, read from a 256-color map and hence quantized to about ± 0.01 . Besiroglu et al. drop the five highest-loss runs, all from the 10^{19} IsoFLOP budget and with 2.0 to 6.8 billion parameters, and we do the same ($n = 240$). Four of the five have $D/N < 0.4$; the fifth has 0.40. Online Appendix D2 shows how the results change when they are kept.

Only part of the extraction consists of IsoFLOP profiles. Digitized $\log_{10} C$ sits systematically 0.018 below the nine nominal budgets, 6×10^{18} to 3×10^{21} FLOP. We estimate this offset as the median deviation of runs within 0.1 of a budget and remove it. A run belongs to an IsoFLOP profile if its corrected $\log_{10} C$ is within 0.045 of a budget. This rule gives 137 profile runs of 245 (132 of 240). The other 108 runs form 35 streaks of fixed model size, the training-horizon runs that Hoffmann et al. pool into their Approach 3. In the 240-run sample, 33 percent of the variance of centered $(\ln N, \ln D)$ lies across the expansion path: 37 percent among the profile runs and 21 percent among the off-profile runs. Assigning every run to its nearest budget, as a nine-cluster bootstrap requires, puts off-profile runs from a 10^{22} -FLOP trunk into the 3×10^{21} IsoFLOP experiment. Our cluster schemes therefore group profile runs by budget and off-profile runs by their model-size trunk, or treat the latter as singletons (Table B2); Online Appendix D2 compares all of them. The 123-point Chinchilla series in the open IsoFLOP compilation of Czech et al. (2026) is a snapped subset of the same digitization, on eight budgets, rather than an independent source; we use it only to cross-check the IsoFLOP path in Online Appendix F5.

B3. Laboratory IsoFLOP Profiles and Hyperparameter Sweeps

LLAMA 3.

The 133 points of Meta’s IsoFLOP figure (Grattafiori et al., 2024) come from the digitization of Czech (2026), distributed in the compilation of Czech et al. (2026); we checked that the two agree row by row. There are ten budgets from 6×10^{18} to 10^{22} FLOP. Two of them, 3×10^{21} and 10^{22} FLOP, have six runs each and do not bracket their minimum with two runs on each side; the model-free estimator drops them, and Online Appendix F5 shows what they do to Meta’s path. D is read off the figure and $N = C/(6D)$ uses the nominal budget, which assumes that Meta’s budgets are $6ND$. The loss units are not stated; the losses lie between 0.69 and 0.93. Parametric levels from these runs are therefore not comparable with other sweeps, but the model-free estimator and the wedge are invariant to monotone transformations of the output.

MARIN.

The Marin project’s scaling ladders (Marin Community, 2026) train a Llama-2-style architecture on three corpora (Comma, DCLM and Nemotron-CC): 85, 85 and 88 runs on seven or eight budgets from 1.8×10^{18} to 3×10^{20} FLOP. The output is the macro-averaged Paloma loss. The nominal budgets are three times the forward FLOPs and differ from $6ND$ with the configuration parameter count by -7 to $+35$ percent. The difference is strongly negatively correlated with N within a budget, so FLOP accounting, as well as the finite IsoFLOP grid, affects IsoFLOP-parabola estimates on these data. The model-free estimates use $N = C_b/(6D)$, which makes each profile an exact isocost; with the configuration count they are 0.04 to 0.07 lower. We also estimate the checkpoint-interpolated FineWeb/C4 profiles of Li, Kudugunta and Zettlemoyer (2025) (176 points) for the duality tests of Online Appendix D2 and the allocative comparisons of Online Appendix E; because their points come before the learning-rate schedule ends, they are excluded from the technologies of Section IV.

PORIAN ET AL.

The released results of Porian et al. (2024) contain 975 runs (772 on Refined-Web, 203 on OpenWebText2): 264 base, 61 tuned, 566 hyperparameter-sweep and 84 seed runs of 16 architectures (width 96–1,504, depth 3–30, vocabulary 50,432, sequence length 2,048). We count parameters and FLOPs without the output head (Kaplan), with it (standard) and with attention FLOPs (attention), as the authors do, and add a fourth count that also includes the input and positional embeddings (total). Our re-implementation of the authors’ IsoFLOP pipeline reads each run’s loss along its loss curve at twelve budgets $1.25 \times 10^{16} \times 2^i$ ($i = 0, \dots, 11$), takes the minimum over runs for each N , locates the minimizing N by spline interpolation, and reproduces the published path estimates to within 0.003. The

first step of their decomposition uses training loss and the remaining steps validation loss. All runs use a constant learning rate without cooldown, so the profiles are losses of unfinished schedules.

STEP LAW.

The dense-model file of [Li et al. \(2025a\)](#) contains 1,911 runs in 17 (N, D) cells, the released subset of the roughly 3,700 models the authors trained. N takes five values between 215M and 1.07B non-embedding parameters, D ranges from 4B to 100B tokens, and M ranges from 19 to 466. Each cell has a grid of up to 12 learning rates and 10 batch sizes; one cell has only 5 learning rates and 47 runs. Batch size is in sequences of 2,048 tokens, and the output is the smoothed final training loss of a single-epoch run. We classify 9.5 percent of runs as diverged: those whose log loss exceeds the cell minimum by more than 0.30.

B4. The Controlled Experiment

CORPORA AND TOKENIZER.

The two corpora are FineWeb-Edu and FineWeb ([Penedo et al., 2024](#)); FineWeb-Edu keeps the documents of the same web crawl that an educational-quality classifier rates highly. We use the first two files of each sample-10BT release for the main grid. The first 20,000 documents of each corpus are held out for validation (24.6 million FineWeb-Edu tokens and 16.7 million FineWeb tokens). The remaining 1.76 billion and 1.70 billion training tokens exceed the largest main-grid budget (800 million), so no run repeats data. The high- M runs append two further files of each release (1.78 billion and 1.72 billion tokens) after the original permutation, so they also see every token once. A byte-level BPE tokenizer ([Sennrich, Haddow and Birch, 2016](#)) with 8,192 tokens, including an end-of-text token that separates documents, is trained on about 300 million characters of each training corpus. Because both corpora share the tokenizer, losses are in the same units. On the validation sets a token covers 3.89 bytes of FineWeb-Edu and 3.74 bytes of FineWeb. We use these ratios to convert losses to bits per byte, $L/(3.89 \ln 2)$ and $L/(3.74 \ln 2)$, which removes the difference in how well the shared tokenizer compresses the two corpora.

ARCHITECTURE AND INPUT COUNTS.

The models are decoder-only transformers with RMSNorm pre-normalization ([Zhang and Sennrich, 2019](#)), rotary position embeddings ([Su et al., 2024](#)), bias-free linear layers, a four-times-wide GELU feed-forward block and an input embedding tied to the output projection. Width d_{model} determines the number of layers and of heads, both $d_{\text{model}}/64$, so the head dimension is always 64. Table B3 lists the shapes. We record both parameter counts. The non-embedding count is $n_{\text{layer}}(12d_{\text{model}}^2 + 2d_{\text{model}}) + d_{\text{model}}$, and the tied embedding adds $8,192d_{\text{model}}$,

TABLE B3—DESIGN OF THE CONTROLLED EXPERIMENT

Width	Layers	N (millions)		Embedding share	Peak LR (10^{-3})	Budgets D_k (M)	$M = D/N$		FLOP per $6N_{\text{ne}}D$
		Non-emb.	Total				Non-emb.	Total	
128	2	0.394	1.44	0.73	5.72	25–800	63–2,031	17–555	3.83
192	3	1.33	2.90	0.54	3.97	25–800	19–602	8.6–276	2.29
256	4	3.15	5.25	0.40	3.07	25–800	7.9–254	4.8–153	1.75
320	5	6.15	8.77	0.30	2.51	25–800	4.1–130	2.9–91	1.49
384	6	10.6	13.8	0.23	2.13	25–800	2.4–75	1.8–58	1.35
448	7	16.9	20.5	0.18	1.85	25–400	1.5–24	1.2–19	1.27
512	8	25.2	29.4	0.14	1.64	25–400	0.99–16	0.85–14	1.21
640	10	49.2	54.4	0.10	1.35	25–200	0.51–4.1	0.46–3.7	1.14
<i>High-M runs with a four-layer floor (extension data)</i>									
128	4	0.788	1.84	0.57	5.72	200–3,200	254–4,063	109–1,743	2.50
256	4	3.15	5.25	0.40	3.07	200–3,200	64–1,017	38–610	1.75

Note: Main grid: each width is trained once per corpus (FineWeb-Edu and FineWeb) with a constant-learning-rate trunk. The token budgets D_k are taken from {25, 50, 100, 200, 400, 800} million; each budget below the largest is a cooldown branch started at $0.8D_k$, and the largest is the trunk’s own cooldown, which gives 44 endpoints per corpus. There are $d_{\text{model}}/64$ attention heads of dimension 64; sequence length 256; batch 64×256 tokens. Non-embedding parameters are $n_{\text{layer}}(12d_{\text{model}}^2 + 2d_{\text{model}}) + d_{\text{model}}$ for width d_{model} and n_{layer} layers; the tied embedding adds $8,192 d_{\text{model}}$. The peak learning rate is $3.07 \times 10^{-3} (d_{\text{model}}/256)^{-0.90}$. The last column is training FLOP per token, including attention and the output projection, relative to $6N_{\text{ne}}$; relative to $6N$ with the total count it is 1.03–1.07 at every shape. High- M runs: budgets {200, 800, 1,600, 3,200} million on a longer data stream (the width-256 shape on FineWeb-Edu only). Further runs, all on both corpora unless stated: learning-rate calibration for FineWeb (widths 128, 320 and 512 at four peak rates, 50 million tokens); peak rates of 0.5 and 2 times the rule at width 128 over the full budget ladder and at width 640 at 25 and 50 million tokens; a second seed at width 128 with 200–800 million tokens; and two further seeds at widths 128, 256 and 384 with 50–200 million tokens.

which is 73 percent of all parameters at $d_{\text{model}} = 128$ and 10 percent at 640. This range makes the counting convention matter far more than in production models, which is the input-measurement problem of [Pearce and Song \(2024\)](#) and [Porian et al. \(2024\)](#). Training FLOP per token, including attention and the output projection, is within 7 percent of $6N_{\text{total}}$ at every shape but 1.14 to 3.83 times $6N_{\text{ne}}$. Initialization is Gaussian with standard deviation $1/\sqrt{\text{fan-in}}$. Residual-branch output projections are scaled down by $1/\sqrt{2n_{\text{layer}}}$, and the embedding has standard deviation 0.02.

OPTIMIZATION AND SCHEDULE.

All runs use AdamW ([Loshchilov and Hutter, 2019](#)) with decay rates 0.9 and 0.99 for the first and second moments, stability constant 10^{-8} , decoupled weight decay 0.1 and gradient clipping at norm 1. The sequence length is 256 and the batch is 64 sequences (16,384 tokens). The learning rate warms up linearly over 250 steps (4.1 million tokens) to its peak and then stays constant on the trunk. For each budget D_k the run takes $S_k = \lceil D_k/16,384 \rceil$ steps. At step $0.8S_k$ the model and optimizer state are copied, and the copy is trained to S_k with learning rate $\text{lr}^*(1 - \sqrt{(s - 0.8S_k)/(0.2S_k)})$. The largest budget is the trunk’s own cooldown. This warmup-stable-decay design with cooldown branches follows [Hägele et al.](#)

(2024), who find that it performs on par with cosine schedules tuned to each budget. One trunk thus delivers a whole column of the (N, D) design, and every endpoint is annealed. Endpoints of one shape share the trunk up to their branch points, so their noise is positively correlated across budgets, and inference clusters by width.

LEARNING RATE.

Before the main grid, we trained widths 128, 320 and 512 at four peak learning rates $(1, 2, 4 \text{ and } 8 \times 10^{-3})$ for 50 million FineWeb-Edu tokens and evaluated them on FineWeb-Edu. For each width, the optimal rate is the minimum of a quadratic in the log learning rate, allowed to extrapolate at most 0.35 log points beyond the tested range. The optima are 5.75×10^{-3} , 2.47×10^{-3} and 1.66×10^{-3} . A log-linear fit gives $\text{lr}^*(d_{\text{model}}) = 3.07 \times 10^{-3} (d_{\text{model}}/256)^{-0.90}$, which we apply to both corpora and every budget. The learning rate is thus a flexible input optimized over width at one budget and on one corpus. By the argument of Online Appendix D6, any remaining inefficiency that varies with D or with the corpus would bias the data exponent or the measured factor bias. Two additions measure it: the same calibration repeated on FineWeb, and runs at half and twice the rule’s learning rate at width 128 over the full budget ladder and at width 640 at 25 and 50 million tokens, on both corpora. The batch size is the same at every width.

HIGH- M RUNS.

In the main grid, ratios above 600 are reached only by the two- and three-layer models. Four-layer models of width 128 (both corpora) and 256 (FineWeb-Edu) are therefore trained to budgets of 200 million, 800 million, 1.6 billion and 3.2 billion tokens, which raises the largest M to 4,063 in non-embedding and 1,743 in total parameters.

COMMON RANDOM NUMBERS, SEEDS AND EVALUATION.

Within a corpus, every shape in the main grid uses initialization seed 0 and the same permutation of training sequences, so differences across (N, D) cells are not driven by different data draws. The shared seed does not create common random numbers in the weights, because the shapes differ; only the data order is common. The two corpora necessarily differ in data. Seed replicates rerun widths 128, 256 and 384 with two further seeds, each changing both the initialization and the data order, to budgets of 50, 100 and 200 million tokens in both corpora, and width 128 with one further seed to 200, 400 and 800 million tokens, the high- M corner. With the main grid they give three replicates per cell at low M and two at the corner, which measure the variance of pure noise in the output. Every endpoint is evaluated on the first 1,048,576 tokens (4,096 sequences of 256) of both validation sets, so each corpus’s models have an in-distribution and a cross-distribution loss

and the corpora can be compared on a common output. Most added runs are also evaluated on the validation and test sets of WikiText-103 (Merity et al., 2016), 655 thousand tokens drawn from neither corpus. For each endpoint we record N under both conventions, the tokens trained, compute under three conventions, all validation losses and the trunk’s training loss. Training curves of the constant-learning-rate trunks are logged every 1,000 steps.

PRE-REGISTRATION.

The pre-analysis plan (`paper/notes/m9_preanalysis_plan.md`) was committed to the project’s version-control repository at 03:12 (UTC+3) on 24 September 2026 (commit `bd5c0ad`); its header gives 03:15, the time of writing. At that moment part of the FineWeb-Edu main grid had finished and its raw endpoint losses had appeared in run logs, but no model had been fitted and no statistic computed; the FineWeb grid, the additions above and the high- M runs had not started. The plan fixes the output (bits per byte; a corpus effect counts as quality only if it has the same sign on the other corpus’s validation set and on WikiText-103), both parameter conventions with the non-embedding count primary for the extrapolation test, the robustness samples (without the 25-million-token budget and without the two- and three-layer shapes), the estimators, the inference (a wild cluster bootstrap by width and a parametric bootstrap with the within-trunk covariance from the seed replicates), six questions with decision rules, a power calculation to be computed before any estimate, and the commitment to report every answer. One deviation was made for time after the plan was committed: the plan called for both four-layer shapes on both corpora and for continuing the two-layer width-128 model to 3.2 billion tokens, but FineWeb trains only the width-128 four-layer shape and the continuation was dropped.

HARDWARE.

All models were trained in MLX (Hannun et al., 2023) on a single Apple M5 Max. Parameters and optimizer moments are kept in 32-bit floating point, and every matrix product is computed in bfloat16 by casting the weights on the fly, so that small updates late in the cooldown are not lost to rounding. [TBD-m9: completed endpoints by corpus and tag (main, high- M , learning-rate checks, seeds), any diverged or failed runs, and total wall-clock hours.]

B5. Released Models

THE EPOCH UNIVERSE.

We start from the 2,070 language-model rows of the Epoch AI database (Epoch AI, 2026) (snapshot of September 23, 2026) and exclude 1,728 of them: derived models such as fine-tunes and post-training updates (531); rows without N (361) or released outside January 2019–September 2026 (234); rows whose token count

cannot be recovered (224); multimodal and non-text models, whose D would include other tokens (133); benchmark-scale research models with unique corpora below 5×10^9 tokens (100); encoders and encoder-decoders (58); post-trained variants identified by name (16); mixture-of-experts models without an active-parameter count (27); rows with implausible epochs (> 20) or $D/N > 30,000$ (15); and 29 manual exclusions with recorded reasons, such as post-training updates listed as base models (Llama 3.3 70B), systems-paper benchmark runs and leaked GPT-4 figures. This leaves 342 models.

D is the dataset size times the number of epochs, with missing epochs set to 1. Epoch records some older datasets in words. Where Epoch’s notes indicate a word count, we convert it to tokens, using operation-counted compute when it is informative and a ratio of 4/3 tokens per word otherwise (7 rows). Where the dataset size is missing but compute is operation-counted and at least “likely”, we set $D = C/(6N)$ (67 rows); such rows inherit any error in Epoch’s compute and enter the conduct tests only if Epoch rates them “Confident”. Six token counts are corrected from primary sources, each documented in Epoch’s own notes. GPT-3 175B was trained on 300B tokens, and GLM-130B on 400B rather than 152B. For Cerebras-GPT-13B (257B) and BloombergGPT (569B), Epoch’s dataset size already counts tokens seen. Polyglot-Ko-12.8B saw 167B tokens rather than 96B words, and for GPT-SW3 3.5B we use 102B tokens. For mixture-of-experts models, active parameters come from the name, Epoch’s notes, or operation-counted compute. Eleven production-scale rows still have D inconsistent with Epoch’s operation-counted or reported compute ($|\ln C/(6ND)| > \ln 1.5$); they are flagged.

THE VERIFIED SAMPLE.

The verified sample starts from the 148 base models in the ObsScaling evaluation file (Ruan, Maddison and Hashimoto, 2024), with nine documented corrections and fills. For example, OPT’s D is 300B tokens, not the 180B-token corpus, and TinyLlama v1.1 was trained on 2T tokens, not 3T. We drop 17 models whose token counts are undisclosed (Mistral 7B and NeMo, Mixtral, Jamba, DeciLM, Qwen1.5-110B, GPT-2) or ambiguous (InternLM2 7B and 20B), upcycled, duplicated or instruction-tuned. We replace the deduplicated Pythia suite with the standard suite (Biderman et al., 2023). We then add 57 models released in 2024–2026, whose token counts come from model cards or technical reports. For 47 of them, a regular expression matches the training-token statement in the saved model card. This gives 188 base models in 73 families. Excluding 15 code-specialized models gives the 173 general-purpose models, in 68 families (67 when Llama 3 and 3.1 are merged into one herd), used in Section IV. The ObsScaling file lists models on a public leaderboard, which selects on popularity; this matters for tests whose outcome is usage, not for the wedge itself.

For 143 of the 188 models, N is the exact count of floating-point tensors in the Hugging Face checkpoint. Integer buffers such as stored causal masks are excluded; they inflate the count of the smallest GPT-NeoX models by up to

36 percent. The 417M-parameter vision encoder of the multimodal Gemma 3 checkpoints is subtracted. For the remaining models the reported size is used. Mixture-of-experts models enter with active parameters, for example 37B of 671B for DeepSeek-V3, and with total parameters in the bounds of Online Appendix F4. Embedding counts, from each model’s configuration file, give the non-embedding N that the FarSeer and other non-embedding technologies require; for MPT-7B and 30B, BTLM-3B and Falcon-180B, whose configuration files require authentication, we use the published architecture. Of the 119 verified models that also appear in Epoch, 5 have token counts that differ from Epoch’s by more than 0.05 in $\log_{10}$. All five differences are documented; Phi-2, for instance, saw 1.4T tokens, whereas Epoch lists the 250B-token dataset. The release date is Epoch’s publication date when matched and otherwise the developer’s announcement or the Hub upload date.

THE CLEAN INFERENCE-DEMAND SAMPLE.

Nine steps, defined before any wedge was computed, reduce the verified sample to 77 models from 36 families and 18 developers, released 2023–2025: one observation per pretraining run; no research suites, replications or data benchmarks whose D was fixed by design; no mixture-of-experts, distilled, pruned or derived-initialization, multimodal, instruction-only or non-transformer models; and token counts with a primary source. Online Appendix F2 lists every step with its sources. The sample’s families are classified by whether all members share a token budget (within 10 percent), and its developers by whether they served their own language models at the model’s release (Online Appendix F7).

PRODUCTION-SCALE UNIVERSE.

Trends and conduct tests use models trained with $6ND \geq 10^{21}$ FLOP. This universe applies the same threshold to both sources. It consists of Epoch rows that are not speculative, not code models and not matched to the verified sample (176), plus the verified general-purpose models above the threshold (159): 335 models released in 2019–2026, 271 of them open weight, from 94 developers after we merge name variants (for example, all Alphabet units into Google). Three subsets enter Section IV. The test of open against closed models and the trends use the 186 models with confident parameter and token counts (Epoch’s “Confident” class or documented by the developer), without mixture-of-experts or non-transformer architectures, and with the clean-sample exclusions applied to verified models; 141 of them are open weight. The bunching test uses the 220 open-weight models released in 2023–2026, other than non-transformers, with their total parameter counts.

USAGE.

From the Hugging Face model tree (the `childrenModelCount` field of the Hub interface) we count, for each clean model, the quantizations, fine-tunes, adapters and merges derived from its base repository, its official post-trained releases and its same-run re-releases (74 models; MosaicML’s and Cerebras’s repositories require authentication). The per-model token volumes that OpenRouter displays are not available through its documented interface, and its terms of service prohibit scraping; we use the developer-level totals published by [Aubakirova et al. \(2026\)](#) instead. We also recorded download counts, OpenRouter listings and LMArena vote counts, which the analysis does not use.

UNITS.

Wedges are computed in each technology’s own conventions: the total count for the Chinchilla, Gadre et al. and laboratory technologies, the non-embedding count for FarSeer, MiniCPM and the non-embedding Chinchilla refit, OLMo’s count for the OLMo ladder, and non-embedding FLOPs per token for DeepSeek’s law. D is in each model’s own tokens, while each technology is in its sweep’s tokens. Under the reference technology $\partial \ln w / \partial \ln D = 0.43$ and $\partial \ln w / \partial \ln N = -0.42$, so a 20 percent difference in tokenizer efficiency moves $\ln w$ by about 0.08.

B6. Benchmark Panel

The benchmark panel of Online Appendix E combines the ObsScaling base-model file with the Sloth data ([Maia Polo et al., 2025](#)). Sloth adds base models absent from ObsScaling and fills missing scores. Five Sloth rows that duplicate ObsScaling’s deduplicated Pythia scores are dropped. Parameter counts are Hub floating-point tensor counts where available (17 models were affected by the causal-mask correction), and reported token counts are corrected much as in the verified sample. The outputs are HellaSwag (10-shot), ARC-Challenge (25-shot) and Winogrande (5-shot) accuracies from the same evaluation harness. We map accuracy to the log odds of above-chance success, $q = \text{logit}((\text{acc} - \text{chance}) / (1 - \text{chance}))$, with chance-adjusted accuracy clipped to $[0.01, 0.99]$; the composite is the logit of the mean chance-adjusted accuracy of the three tasks. The main sample contains 128 dense models with N , D and all three outputs, from 38 families and 22 developers, released March 2021–July 2024. Mixture-of-experts models are excluded. The overlap sample ($N \leq 14\text{B}$, $D \leq 5\text{T}$) has 90 models. Inside the convex hull of the experimental designs lie 20 models, or 57 if the released OLMo-2 7B and 13B models are counted as benchmark runs (Online Appendix E).

The panel’s design is revealing. $C = 6ND$ holds in every row by construction. Within families the standard deviation of $\ln D$ is 0.42, against 1.29 for $\ln N$. Only 18 families (66 models) vary D at all, and $\text{corr}(\ln N, \ln D) = 0.46$. The experimental benchmarks are the OLMo ladder, the 104 Gadre et al. models, the

Pythia checkpoints and the final DataDecide checkpoints; the OLMo-2 releases, whose inputs their developer chose, enter only a robustness design. They were evaluated with their developers’ own harnesses, so the correspondence between output formats is part of the maintained null hypothesis in Online Appendix E.

B7. Algorithmic-Progress Data

We re-implement line by line the sample construction of [Ho et al. \(2024\)](#) from their public notebooks, starting from the 408-row sheet. We keep rows with numeric N and D , not marked as excluded, outliers or uncertain, and with at least one perplexity. We stack the rows by benchmark, drop four named systems (μ P and LoRA fine-tunes), and keep the three best-performing models per paper. This yields 231 observations from 144 papers: 103 on WikiText-103, 80 on Penn Treebank and 48 on WikiText-2. D is the dataset size in unique tokens, not tokens processed. Epochs are known for 155 of the 231 rows; we impute the rest by the median within bins of $\log_{10} D$, and set epochs to one for $D \geq 10^{10}$. The effective-data variant follows [Muennighoff et al. \(2023\)](#). The output is log perplexity, mostly per word; its units vary with each evaluation’s tokenization.

For the allocative-efficiency analysis we build a separate sample of Epoch language models ([Epoch AI, 2026](#)). Of the 792 models with N and C , 544 report a dataset size and 454 are not fine-tunes. Of these, 428 are dense. In 339 of them, $C/(6N)$ lies within a factor of three of dataset size times epochs, and 247 of those were trained for at most four epochs. We set $D = C/(6N)$ so that $6ND = C$ exactly.

B8. Licenses

Epoch AI data are released under a CC-BY license. ObsScaling, the OLMo ladder, Pythia, DataDecide and the Muennighoff et al. data are released under Apache-2.0 or ODC-BY licenses, and Porian et al., Sloth and Gadre et al. under MIT. The FarSeer and Step Law repositories and Epoch’s Chinchilla-extraction repository carry no license. We use them for estimation but redistribute only derived aggregates. The Llama 3 digitization, the Marin ladders and the Ho et al. data sheet, which carries no explicit license (their code is MIT-licensed), are used on the same basis. FineWeb and FineWeb-Edu are released under ODC-BY and WikiText-103 under CC BY-SA; we redistribute neither. Model cards, Hub metadata and model-tree counts were saved only to verify token counts and to count derivatives, and are not redistributed.

MONTE CARLO DESIGN AND RESULTS

This appendix documents the two Monte Carlo designs behind Section II and Online Appendix E. Design A asks what alternative experimental designs identify when they use the same total compute. Design B simulates an industry of labs that choose compute and allocations, and asks which production-function estimators recover the technology from the models they release. Everything is simulated. The truth is the technology of Besiroglu et al. (2024): $E = 1.8172$, $A = 482.01$, $B = 2085.43$, $\alpha = 0.3478$ and $\beta = 0.3658$, so that $a = 0.513$, $\gamma = 0.178$, $\sigma^* = 0.737$, and the compute-optimal ratio at 10^{24} FLOP is $M^* = 18.1$ tokens per parameter ($\ln M^* = 2.898$). “Bias” is the Monte Carlo mean error and “RMSE” the root-mean-squared error around the truth. No estimator is started at the true parameters. The first version of this paper included the truth among the nine starting values of the Chinchilla-form fit, and its industry estimators started at the truth; Table C2 and the Design B discussion report what removing those starts changes.

C1. Design A: What Alternative Designs Identify

Setup.—Every design has 90 runs, ten on each of nine compute budgets that match Chinchilla’s nominal IsoFLOP budgets (6×10^{18} to 3×10^{21} FLOP), and hence the same total training compute, 5.1×10^{22} FLOP. The designs differ only in where the runs sit on their isocost lines (Table C1): near the expansion path, as in a compute-optimal ladder; spread by IsoFLOP profiles or a factorial grid, as in a designed experiment; spread by allocation errors with standard deviation v in $\ln(D/N)$ at given compute, as by optimizing labs that err; or tilted systematically by labs that follow Kaplan’s allocation rule. Log loss carries normal noise with standard deviation 0.0075, the residual standard deviation of the Besiroglu fit on the 240-run Chinchilla extraction. The residuals are heavy-tailed (their MAD-based standard deviation is 0.0049) and bundle seed noise, digitization error and misspecification, so 0.0075 is an upper value for pure seed noise. Noise is realized after inputs are chosen, the case of Zellner, Kmenta and Drèze (1966).

We compare three estimators. The primal estimator is the Huber ($\delta = 10^{-3}$) fit of $\ln L$ used by Hoffmann et al. (2022) and Besiroglu et al. (2024), with $\kappa = 1$, minimized inside a parameter box from nine starting values drawn at random over the region of Hoffmann et al.’s initialization grid. The dual estimator reads the path slope a from the optima (or the IsoFLOP minima) and γ from the frontier $L^*(C)$, and recovers $\alpha = \gamma/a$ and $\beta = \gamma/(1 - a)$; it identifies the exponents only through optimality and $\kappa = 1$. The third is a profile likelihood for σ^* in the generalized family $L = E + (AN^{-a_1} + BD^{-b_1})^\kappa$ with κ free, on a 29-point grid over $[0.05, 0.99]$, with the likelihood ratio measured against the unrestricted optimum. At each grid point the optimizer starts from the members of the observationally equivalent family of Proposition A1 that reproduce the on-path fits of the primal and dual estimators; without these starts it produced spurious rejections at the

edges of the grid.

On-path data do not identify the technology through the primal estimator.—When every run is exactly compute-optimal ($v = 0$), 87 percent of fits end on a corner, with one power-law term switched off, and the RMSE of $\hat{\sigma}^*$ is 0.27, against 0.004 for an IsoFLOP design spanning $\pm 16\times$ the optimal size and 0.005 for the factorial grid at the same compute (Table C1, Panel A). The corners are not optimization failures: no fit at $v = 0$ has a higher Huber objective than the true parameters, so the data are fitted as well by a technology with one input switched off as by the truth, the non-identification of Proposition 3(i)–(ii). Near the path the failure is a ridge: at $v = 0.02$ the median error of $\hat{\sigma}^*$ is -0.12 , and its 5–95 percent range is $[0.34, 0.75]$. The normalized condition number of the information matrix at the truth rises from 1.4×10^3 in the designed experiments to 1.5×10^7 at $v = 0.02$ and is infinite at $v = 0$. As v rises to 0.05, 0.1 and 0.2, the RMSE of $\hat{\sigma}^*$ falls to 0.089, 0.025 and 0.015 (Figure 2, panel A). The compute-optimal ratio is the more demanding object once optimality is not imposed: the RMSE of $\ln \hat{M}^*(10^{24})$ is 28 at $v = 0$, 4.6 at $v = 0.1$, 2.0 at $v = 0.3$ and 0.59 at $v = 1$, against 0.16 under the IsoFLOP design. At $v = 0.3$ the 5–95 percent range of $\ln \hat{M}^*$ is $[-0.40, 6.34]$, a factor of about 850 in M^* . These are the finite-sample counterparts of the information rates in Proposition 3(iii).

Starting values matter only near the path.—With the truth among the starts, as in the first version, 26 percent of on-path fits and 51 percent at $v = 0$ ended on a corner; with nine random starts, 72 and 87 percent do (Table C2). With 64 random starts the corner shares fall to 18 and 56 percent, and the RMSEs of $\hat{\sigma}^*$ on the path, at $v = 0$ and at $v = 0.05$ (0.19, 0.30 and 0.071) are within 10 percent of the first version’s. For $v \geq 0.1$ and for the designed experiments the start set makes no difference. The truth therefore did not flatter the near-path designs much; what it hid is that on a ridge the estimate depends on where the optimizer starts. With nine starts, 7 percent of fits at $v = 0.05$ stop at a higher objective than that of the truth, and 64 starts remove these failures. We keep nine starts, the number used in the first version, as the baseline.

With $\kappa = 1$ imposed, apparent precision at intermediate v is functional form.—For $0.1 \leq v \leq 0.3$ the primal RMSE of $\hat{\sigma}^*$ (0.013 to 0.025) is no better than that of the dual estimator (0.012), which uses only the path. The reason is analytic: with $\kappa = 1$, $\alpha + \beta = \gamma/[a(1 - a)]$, and $a(1 - a)$ is flat near $a = 1/2$ (0.2498 at the truth), so $\sigma^* = 2/(2 + \gamma/[a(1 - a)]) \approx 2/(2 + 4\gamma)$ is pinned down by the on-path frontier elasticity. Freeing κ removes that information (Table C1, Panel B; Figure C1). The 95 percent profile set for σ^* is the whole grid in 91 percent of on-path replications, 95 percent at $v = 0$ and 79 percent at $v = 0.05$. Larger allocation errors first rule out poor substitutes. At $v = 0.1$ the set still contains all of $[0.50, 0.95]$ in 93 percent of replications but excludes some value at or below 0.5 in 72 percent; at $v = 0.3$ its median bounds are $[0.42, 0.99]$. The upper end goes last: the set contains every $\sigma^* \geq 0.9$ in 76 percent of replications at $v = 0.3$ and 45 percent at $v = 0.5$. The asymmetry follows from the Harberger-triangle

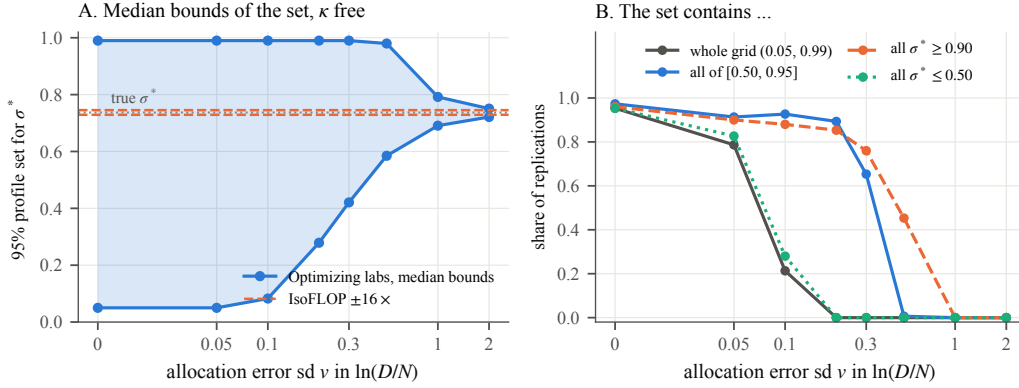
approximation of Section II.A: an error δ in $\ln M$ costs a Farrell loss of about $(1 - \sigma^*)\delta^2/(4\sigma^*)$, which is large when inputs are poor substitutes and vanishes as $\sigma^* \rightarrow 1$, so small losses are consistent with any elasticity near one. The set becomes informative at $v = 1$ (median bounds $[0.69, 0.79]$) and in the designed experiments ($[0.729, 0.745]$ for IsoFLOP $\pm 16\times$ and $[0.727, 0.746]$ for the factorial grid), and it covers the truth in 93 to 100 percent of replications in every design. Under Kaplan beliefs the set is the whole grid in 21 percent of replications. On the range searched in the first version, $[0.50, 0.95]$, the flat shares match the first version except on the path (94 against 88 percent) and under Kaplan beliefs (84 against 77 percent), where the additional starting value from the dual fit removes optimizer failures at the edges of that range.

The dual estimator is precise on the path but inherits any systematic allocation error.—On the path its RMSEs are 0.0005 for a , 0.011 for σ^* and 0.009 for $\ln M^*$. When labs optimize against Kaplan’s beliefs ($N \propto C^{0.73}$), the same estimator is biased by +0.217 in a , -0.088 in σ^* and -3.92 in $\ln M^*$, understating M^* about 50-fold. The primal estimator on the same data is roughly unbiased but weakly identified (RMSE of $\hat{\sigma}^*$ 0.10; the Wald covariance is computable in 77 percent of replications).

On the path, standard inference is either wrong or uninformative.—The Wald covariance is numerically singular, or the fit is at a corner, in 79 percent of on-path replications. The pairs bootstrap as usually implemented, with every draw warm-started at the full-sample estimate, covers σ^* in 44 percent of 150 replications (Monte Carlo standard error 4 percentage points). In a separate check with 60 replications and 49 draws, restarting every draw also from the nine random starting values raises coverage from 33 to 78 percent, still short of nominal, and yields a median interval width of 0.43, against 0.016 under the IsoFLOP design. With designed variation all methods agree: Wald and bootstrap coverage is 95 to 97 percent for the IsoFLOP $\pm 16\times$ and factorial designs and at $v = 1$.

Noise.—What counts as near-optimal depends on v relative to the noise and on the number of runs (Table C3). The RMSE of $\hat{\sigma}^*$ under the IsoFLOP design is 0.0024, 0.0036 and 0.0078 at noise standard deviations of 0.005, 0.0075 and 0.015; at $v = 0.3$ the profile set contains all of $[0.50, 0.95]$ in 38, 65 and 90 percent of replications. The seed noise we measure in our own experiment ([TBD-m9: seed standard deviation of $\ln L$]) places it within this range. Digitizing the losses as in the Epoch extraction barely matters for an IsoFLOP design (RMSE 0.0041 against 0.0036).

Heteroskedastic and clustered noise leaves the ranking of designs intact but breaks inference that assumes independent errors. In this variant the noise standard deviation is proportional to $(ND)^{-1/4}$, larger for small models and short runs, and half of each run’s variance is shared within its compute budget (shared data order) or, in the factorial grid, within its model size (one training trunk). The RMSE of $\hat{\sigma}^*$ is 0.0042 under the IsoFLOP design, 0.0052 under the factorial grid, 0.022 at $v = 0.3$ and 0.25 on the path, and the profile set is still the

FIGURE C1. PROFILE-LIKELIHOOD SETS FOR σ^* WITH THE OUTER EXPONENT FREE

Note: Design A, first 150 replications per design; log-loss noise standard deviation 0.0075; 90 runs per design. The 95 percent set inverts the profile likelihood ratio for σ^* in $L = E + (AN^{-a_1} + BD^{-b_1})^\kappa$ with κ free, on a 29-point grid over $[0.05, 0.99]$, against the unrestricted optimum. Panel A: medians of the lower and upper bounds of the set for optimizing labs, against the standard deviation v of their allocation errors in $\ln(D/N)$; dashed lines, the IsoFLOP $\pm 16\times$ design at the same compute; dotted line, the truth. Panel B: share of replications whose set is the whole grid, contains all of $[0.50, 0.95]$ (the range searched in the first version), contains every $\sigma^* \geq 0.90$, or contains every $\sigma^* \leq 0.50$.

Source: Authors' simulations; replication code `code/analysis/m6_montecarlo`.

whole grid in 61 percent of on-path replications. But Wald intervals that assume independence cover σ^* in only 82, 74 and 61 percent of replications under the IsoFLOP, factorial and $v = 0.3$ designs, and profile sets calibrated by χ_1^2 in 71, 82 and 91 percent. Inference must allow for dependence within budgets or training trunks, as the preregistered analysis of our experiment does with a wild cluster bootstrap over trunks (Section III.E).

C2. Design B: A Simulated AI Industry

Setup.—Forty labs each produce six generations of models. In each generation a lab releases one to four models whose compute budgets are roughly $8\times$ apart. Productivity is Hicks-neutral and equals $\omega_{ft} = 0.15(t-1) + x_{ft}$, where the lab component follows $x_{ft} = 0.8x_{f,t-1} + \xi_{ft}$ with stationary standard deviation 0.25; seed noise has standard deviation 0.05 in $y = -\ln(L-E)$. The dispersion of ω is assumed, chosen to match the 41-fold within-developer 90–10 ratio in effective compute of Mertens, Fischl-Lanzoni and Thompson (2026); TFP growth of 0.15 per generation is also assumed. A lab-generation's log budget is $c_{ft} = \bar{c}_t + h_f - p_{ft} + \nu_{ft}$, where $\bar{c}_t$ grows by $\ln 3$ per generation from 10^{21} FLOP, h_f is lab size (standard deviation 0.7), p_{ft} is an observed compute-price shifter (0.4), and ν_{ft} is noise (0.4). Four compute rules map productivity into budgets: (a) exogenous budgets; (b) funding that responds to current productivity ($+2x_{ft}$); (c) a loss target that the lab reaches by buying compute; and (d) compute provisioned one generation ahead ($+2x_{f,t-1}$), the timing assumption of Akerberg, Caves and

TABLE C1—MONTE CARLO DESIGN A: WHAT ALTERNATIVE DESIGNS IDENTIFY AT EQUAL COMPUTE

Design	Primal estimator ($\kappa = 1$)				Dual estimator	
	a	γ	σ^*	$\ln M^*(10^{24})$	σ^*	$\ln M^*(10^{24})$
<i>Panel A. Bias [root-mean-squared error]</i>						
(i) On path, $v = 0.02$	0.050 [0.296]	0.058 [0.096]	−0.166 [0.222]	−1.08 [23.8]	0.000 [0.011]	0.00 [0.01]
(ii) IsoFLOP, $\pm 16\times$	0.001 [0.007]	0.000 [0.003]	0.000 [0.004]	−0.02 [0.16]	0.000 [0.016]	0.02 [0.14]
(ii′) IsoFLOP, $\pm 4\times$	0.000 [0.015]	0.000 [0.008]	0.000 [0.009]	0.00 [0.31]	−0.001 [0.016]	0.02 [0.36]
(iii) Factorial grid	−0.001 [0.012]	0.000 [0.005]	0.000 [0.005]	0.02 [0.28]	— —	— —
(iv) Optimizing, $v = 0$	0.140 [0.331]	0.093 [0.113]	−0.240 [0.272]	−11.8 [27.7]	0.000 [0.012]	0.00 [0.00]
(iv) Optimizing, $v = 0.1$	0.014 [0.225]	−0.032 [0.051]	−0.006 [0.025]	−0.25 [4.58]	0.000 [0.012]	0.00 [0.05]
(iv) Optimizing, $v = 0.3$	0.000 [0.098]	−0.005 [0.016]	−0.002 [0.013]	0.00 [1.97]	0.000 [0.012]	−0.01 [0.14]
(iv) Optimizing, $v = 1$	−0.002 [0.029]	0.000 [0.011]	−0.001 [0.012]	0.04 [0.59]	−0.001 [0.015]	−0.04 [0.48]
(v) Kaplan beliefs	−0.028 [0.265]	−0.028 [0.072]	−0.023 [0.102]	0.34 [4.97]	−0.088 [0.088]	−3.92 [3.92]
<i>Panel B. Inference for σ^* and the profile likelihood with κ free</i>						
	Corner	Wald	Bootstrap	95% profile set, κ free		
	fits (%)	coverage (%)	coverage (%)	Median bounds	Flat (%)	Flat′ (%)
(i) On path, $v = 0.02$	72	100 [†]	44 [§]	[0.05, 0.99]	91	94
(ii) IsoFLOP, $\pm 16\times$	0	96	97	[0.729, 0.745]	0	0
(ii′) IsoFLOP, $\pm 4\times$	0	94	—	[0.714, 0.760]	0	0
(iii) Factorial grid	0	95	97	[0.727, 0.746]	0	0
(iv) Optimizing, $v = 0$	87	100 [†]	—	[0.05, 0.99]	95	97
(iv) Optimizing, $v = 0.1$	0	99 [†]	—	[0.08, 0.99]	21	93
(iv) Optimizing, $v = 0.3$	0	93	96 [§]	[0.42, 0.99]	0	65
(iv) Optimizing, $v = 1$	0	95	95	[0.69, 0.79]	0	0
(v) Kaplan beliefs	0	92 [†]	—	[0.08, 0.99]	21	84

Note: Designs, truth ($a = 0.513$, $\gamma = 0.178$, $\sigma^* = 0.737$, $\ln M^*(10^{24}) = 2.90$) and noise as described in the text: 90 runs on nine budgets with equal total compute, 500 replications per design; (v) labs follow Kaplan’s rule $N \propto C^{0.73}$ with errors $v = 0.1$. Primal: Huber ($\delta = 10^{-3}$) fit of $\ln L$ from nine starting values drawn at random over the region of Hoffmann et al. (2022)’s initialization grid, none at the truth, inside a parameter box (Table C2 compares start sets). Dual: a from the (noisy) optima or IsoFLOP minima, γ from the frontier (eight starting values on a grid), $\alpha = \gamma/a$, $\beta = \gamma/(1 - a)$; not computed for the factorial grid. Panel B: corner = estimate on the box boundary or one power-law term below 2 percent of reducible loss at every design point; Wald = LAD-sandwich delta-method 95 percent interval ([†]: covariance singular or corner in more than 5 percent of replications, 79 percent on the path; coverage among computable ones); bootstrap = pairs-bootstrap percentile interval (149 draws, first 150 replications), each draw warm-started at the replication’s estimate ([§]: in a separate check with 60 replications and 49 draws, restarting every draw also from the nine starting values moves on-path coverage from 33 to 78 percent and the median width from 0.04 to 0.43; at $v = 0.3$ both give 100 percent); 95 percent profile set = likelihood-ratio confidence set for σ^* in $L = E + (AN^{-a_1} + BD^{-b_1})^\kappa$ on a 29-point grid over [0.05, 0.99], with the likelihood ratio measured against the unrestricted optimum (first 150 replications): medians of the lower and upper bounds, share of replications whose set is the whole grid (flat) and share whose set contains all of [0.50, 0.95], the range searched in the first version (flat′). Monte Carlo standard errors of coverage rates near 95 percent are 1.0, 1.8 and 2.8 percentage points with 500, 150 and 60 replications.

Source: Authors’ simulations; replication code `code/analysis/m6_montecarlo`.

TABLE C2—MONTE CARLO DESIGN A: SENSITIVITY OF THE PRIMAL ESTIMATOR TO ITS STARTING VALUES

Design	Truth + 8 random (first version)			9 random (baseline)			64 random		
	Corner	RMSE		Corner	RMSE		Corner	RMSE	
	(%)	σ^*	$\ln M^*$	(%)	σ^*	$\ln M^*$	(%)	σ^*	$\ln M^*$
On path, $v = 0.02$	26	0.185	16.9	72	0.222	23.8	18	0.189	15.5
Optimizing, $v = 0$	51	0.282	24.0	87	0.272	27.7	56	0.302	24.8
Optimizing, $v = 0.05$	1	0.067	5.87	13	0.089	10.2	1	0.071	5.78
Optimizing, $v = 0.1$	0	0.025	4.49	0	0.025	4.58	0	0.025	4.51
Optimizing, $v = 0.2$	0	0.015	2.99	0	0.015	2.99	0	0.015	3.00
Kaplan beliefs	0	0.101	4.97	0	0.102	4.97	0	0.104	5.00

Note: Design A (Table C1), 500 replications per design, identical simulated data in every column. Primal estimator: Huber ($\delta = 10^{-3}$) fit of $\ln L$ with $\kappa = 1$, minimized by L-BFGS-B from each starting value in the set; the lowest objective is kept. First version: the true parameters plus eight values drawn at random over the region of Hoffmann et al. (2022)’s initialization grid ($\ln A, \ln B \in [0, 25]$, $\ln E \in [-1, 1]$, $\alpha, \beta \in [0.1, 1.2]$). Baseline (all other tables and Figure 2): the same eight plus a ninth random value. 64 random: the baseline nine plus 55 further random values. Corner = estimate on the box boundary or one power-law term below 2 percent of reducible loss. RMSE of $\hat{\sigma}^*$ and of $\ln \hat{M}^*(10^{24})$. The objective at the estimate exceeds the objective at the true parameters in 2.4, 0, 7.0 and 0.2 percent of replications (on path, $v = 0, 0.05$ and 0.1) with nine random starts, and in none with 64. For the IsoFLOP and factorial designs and for $v \geq 0.2$ the first-version and baseline estimates are identical up to optimizer tolerance (Table C1).

Source: Authors’ simulations; replication code `code/analysis/m6_montecarlo`.

Frazer (2015). Labs minimize lifetime compute given inference demand $T = 3D^*(c)\theta$ with $\ln \theta \sim N(0, 0.8^2)$, so that the mean log wedge is 0.46, and make allocation errors with standard deviation 0.2 in $\ln(D/N)$. Variants add release selection (about 18 percent of trained models unreleased), classical measurement error in $\ln D$ (standard deviation 0.15), data-augmenting lab heterogeneity, and a flagship-only sample. The estimators, applied to released models, are listed in Table C4. E is known to all of them except the pooled Huber fit that mimics current practice in machine learning. Every estimator starts from data-based values; relative to the first version, whose least-squares estimators started at the truth, the biases of $\hat{\gamma}$ and of TFP growth change by at most 0.003. Under the loss-target rule the pooled criterion has several local optima, and the bias of the pooled estimate of a moves from -0.03 to -0.06 .

Transmission bias is diluted but not removed by common fixes.—Pooled non-linear least squares with generation effects overstates γ by 0.023 (13 percent) under funding and 0.018 under predetermined budgets, and understates it by 0.053 (30 percent) under a loss target (Table C4, Panel A). With one model per lab-generation under funding the bias is $+0.118$, matching the closed form of Proposition A5 applied to this design (0.118). Lab fixed effects remove about two-thirds of the bias under funding and four-fifths under predetermined budgets, but not under the target rule (-0.022), and almost none of it with flagships only ($+0.113$), because ω is persistent rather than fixed. The bias loads on the frontier elasticity, not on the allocation: the regression of $\ln N$ on $\ln C$ recovers a with $|\text{bias}| \leq 0.003$ in every scenario (Panel B, IV row).

TABLE C3—MONTE CARLO DESIGN A: SENSITIVITY TO THE NOISE PROCESS

Design	Primal estimator ($\kappa = 1$)			95% profile set, κ free			
	RMSE σ^*	RMSE $\ln M^*(10^{24})$	Wald coverage (%)	Covers truth (%)	Flat (%)	Flat' (%)	Median width
<i>Noise: Normal, sd 0.005</i>							
IsoFLOP, $\pm 16\times$	0.0024	0.120	95	94	0	0	0.010
Optimizing, $v = 0.3$	0.0078	1.30	97	97	0	38	0.464
<i>Noise: Normal, sd 0.0075 (baseline)</i>							
On path, $v = 0.02$	0.222	23.8	100 [†]	100	91	94	0.940
IsoFLOP, $\pm 16\times$	0.0036	0.162	96	94	0	0	0.016
Factorial grid	0.0050	0.277	95	95	0	0	0.019
Optimizing, $v = 0.1$	0.025	4.58	99 [†]	97	21	93	0.903
Optimizing, $v = 0.3$	0.013	1.97	93	95	0	65	0.551
Optimizing, $v = 1$	0.012	0.590	95	94	0	0	0.104
<i>Noise: Normal, sd 0.015</i>							
IsoFLOP, $\pm 16\times$	0.0078	0.341	95	95	0	0	0.031
Optimizing, $v = 0.3$	0.027	3.66	96	97	0	90	0.695
<i>Noise: Digitized as in the Epoch extraction</i>							
IsoFLOP, $\pm 16\times$	0.0041	0.198	94	94	0	0	0.017
<i>Noise: Heteroskedastic and clustered</i>							
On path, $v = 0.02$	0.247	25.2	78 [†]	73	61	65	0.940
IsoFLOP, $\pm 16\times$	0.0042	0.090	82	71	0	0	0.013
Factorial grid	0.0052	0.268	74	82	0	0	0.016
Optimizing, $v = 0.1$	0.037	4.16	74 [†]	92	11	69	0.851
Optimizing, $v = 0.3$	0.022	1.47	61	91	0	42	0.488
Optimizing, $v = 1$	0.019	0.429	60	87	0	0	0.089

Note: Design A (Table C1) under alternative noise processes for $\ln L$; 500 replications per cell for the primal estimator and 150 for the profile likelihood. Normal: independent normal noise with the stated standard deviation; 0.0075 is the residual standard deviation of the Besiroglu et al. (2024) fit on the Chinchilla extraction and 0.005 is close to its MAD-based value (0.0049). Digitized: losses rounded to a 0.012-nat grid, 2 percent pixel error in N and C , and D imputed as $C/(6N)$. Heteroskedastic and clustered: the standard deviation for run i is proportional to $(N_i D_i)^{-1/4}$, larger for small models and for short runs, and scaled so that the design-average variance equals 0.0075^2 ; at a given budget the two effects offset, so the standard deviation falls with compute (4.7 times from the smallest to the largest budget) and, in the factorial grid, varies along both inputs. Half of each run's variance is a shock shared by its cluster (correlation 0.5): the nine compute budgets (on-path, optimizing and IsoFLOP designs; shared data order) or the nine model sizes (factorial grid; one training trunk per size). Wald: LAD-sandwich interval assuming independent errors ([†]: singular or corner in more than 5 percent of replications). Covers truth: share of replications whose 95 percent profile set, calibrated by χ_1^2 as if noise were independent, contains the true σ^* . Flat: set is the whole grid $[0.05, 0.99]$; flat': set contains all of $[0.50, 0.95]$.

Source: Authors' simulations; replication code `code/analysis/m6_montecarlo`.

Within-family variation is the observational analog of an IsoFLOP experiment.—Lab $\times$ generation fixed effects, which use only the size menu within a lab-generation, are unbiased for γ , a and TFP growth under every compute rule ($|\text{bias}| \leq 0.002$), because siblings share ω and their tier and inference-demand differences are exogenous. The estimator needs at least two released siblings, is unavailable with flagships only, and is hurt by measurement error in D (bias in $\ln M^*$ of -0.33 at an error standard deviation of 0.15 , similar to lab fixed effects, -0.28). Estimators in the spirit of [Akerberg, Caves and Frazer \(2015\)](#) remove the transmission bias when the timing assumption is right ($|\text{bias}| \leq 0.002$ for γ with lagged instruments only) and fail when it is wrong (-0.027 under the target rule if current compute is treated as predetermined). Both variants, however, use the current within-family deviations of inputs as instruments, without which the lagged moment set is under-identified, so they borrow the same within-family variation; the lagged-instrument version is also about five times less precise for a (RMSE 0.12 to 0.13 against 0.025 to 0.028). The simulation therefore does not show that proxy or timing estimators work on flagship-only panels.

First-order-condition estimators mistake inference demand for technology.—Stacking the within-family loss equation with the training-only first-order condition ($w = 1$) leaves the exponents essentially unbiased but biases $\ln M^*$ by $+1.25$ to $+1.28$ under every compute rule (Panel D), overstating M^* about 3.5-fold. This is close to the closed form $2 \mathbb{E}[\ln w]/(\alpha + \beta) = 2 \times 0.457/0.714 = 1.28$: the first-order-condition residual is the wedge of Proposition [A8](#), and ignoring it makes over-trained models look like evidence that data are cheap. A noisy usage proxy for T (log noise 0.3) removes the bias (-0.02 to -0.03 , RMSE at most 0.05). The instrumental-variable estimator that reads M^* off the observed path inherits the same contamination (median bias $+1.27$ to $+1.32$), although its median bias for γ is small (-0.002 to $+0.005$; median absolute error 0.027 to 0.031) when releases are not selected.

Current practice absorbs more than half of algorithmic progress into the scaling law.—The pooled Huber fit with E free and no time effects, the specification common in the machine-learning literature, understates TFP growth by 0.081 to 0.090 of a true 0.150 per generation under rules (a), (b) and (d) (Panel C), and overstates $\ln M^*$ by 0.53 to 0.64 . Holding E at the truth isolates the scale-attribution channel of [Nagy et al. \(2013\)](#): the same pooled regression still misses 36 to 43 percent of TFP growth and steepens the frontier by 0.050 to 0.059 . Roughly two-thirds of the absorbed progress is attributed to scale and one-third to a downward-biased asymptote $\hat{E}$.

Selection and its correction interact with transmission.—With exogenous budgets and a notability threshold for release, pooled estimates understate γ by 0.008 and overstate TFP growth by 0.011 ; a [Heckman \(1979\)](#) correction, which assumes that the inputs of unreleased models are observed and that a release-policy shifter is excluded, roughly halves both. Under predetermined budgets, selection ($-$) and transmission ($+$) offset (net $+0.01$), and the correction raises the bias to $+0.018$.

by removing only the selection part. Selection also invalidates the compute-price instrument (median bias -0.022 and -0.033). The industry calibration is illustrative, not estimated; only signs and rankings should be read from Design B, and the magnitudes scale with the assumed ratio of productivity-driven to total variance in compute.

C3. Verification of the Transmission and Selection Formulas

Table C5 checks the transmission-bias formula of Proposition A5 against simulations with 10^6 draws per case. The simulated slopes match the closed form to within 10^{-3} in all twelve cases, on the expansion path and with lifetime-optimal allocations. The sign of the bias follows the compute rule: exogenous budgets give γ , funding and predetermined budgets overstate it, and a common loss target drives the slope to zero. The forward and reverse regressions bracket γ under exogenous budgets and dispersed targets, but under funding and predetermined budgets both exceed it. In samples of 240 lab-generations the nominal 95 percent OLS interval covers γ in 95 percent of samples under exogenous budgets, but in 53 percent under mild funding, at most 4 percent under stronger funding, and 16 percent under predetermined budgets with $\lambda = 1$; these coverage rates depend on the assumed exogenous dispersion of compute. The residual dispersion also behaves as predicted: under funding with $\lambda = 2$ the standard deviation of the OLS residual is 0.229, against 0.255 for productivity plus noise, so cross-lab regressions understate TFP dispersion.

Outcome-based release biases the compute slope toward zero (Proposition A6). With exogenous budgets and an absolute release threshold, the slope falls from 0.178 to 0.112, 0.082 and 0.057 as 25, 50 and 75 percent of runs are dropped. Combined with a funding rule ($\lambda = 2$), the net bias moves from $+0.100$ without selection to -0.006 and -0.046 with 50 and 75 percent dropped, so its sign is ambiguous, as the theory implies. The allocation identities with factor-augmenting productivity and the wedge identity of Proposition A8 hold in the simulation to within 10^{-14} .

Reproducibility.—The code is in `code/analysis/m6_montecarlo` (entry point `run_v2.py`; the start-value checks are `designA_startcheck.py` and `designB_startcheck2.py`). Every design cell seeds its own random-number generator, so results do not depend on the number of processes, and the simulated data are identical to those of the first version. On four CPU processes Design A takes 18 minutes, the bootstrap check 3 minutes, Design B 6 minutes and the 64-start check 15 minutes.

THE TECHNOLOGY: ADDITIONAL RESULTS

This appendix supports Sections I through III. Appendix D1 gives the full dictionary between scaling-law practice and production economics. Appendix D2 reports estimators, inference, exclusion rules and duality tests on the Chinchilla

TABLE C4—MONTE CARLO DESIGN B: PRODUCTION-FUNCTION ESTIMATORS IN A SIMULATED AI INDUSTRY

Compute rule:	(a) Exogenous	(b) Funding	(c) Target	(d) Predetermined
<i>Panel A. Frontier elasticity γ (truth 0.178)</i>				
Pooled Huber, E free (ML practice)	0.019 [0.031]	0.027 [0.036]	−0.004 [0.022]	0.025 [0.035]
Pooled NLS + generation FE	−0.001 [0.010]	0.023 [0.024]	−0.053 [0.056]	0.018 [0.019]
Lab FE	0.000 [0.006]	0.008 [0.009]	−0.022 [0.024]	0.004 [0.007]
Lab $\times$ generation FE	0.000 [0.002]	0.000 [0.003]	0.000 [0.002]	0.000 [0.002]
ACF, current c as instrument	0.000 [0.010]	0.007 [0.009]	−0.027 [0.034]	0.000 [0.010]
ACF, lagged instruments only	−0.001 [0.011]	−0.001 [0.011]	−0.002 [0.011]	−0.001 [0.011]
IV (compute price) + path [‡]	−0.002 [0.031]	0.005 [0.031]	−0.002 [0.027]	−0.002 [0.028]
FOC system, T ignored	−0.002 [0.003]	−0.002 [0.003]	−0.003 [0.003]	−0.002 [0.003]
FOC system, T proxied	−0.001 [0.001]	−0.001 [0.001]	−0.001 [0.001]	−0.001 [0.001]
<i>Panel B. Allocation exponent a (truth 0.513)</i>				
Pooled Huber, E free (ML practice)	−0.006 [0.111]	−0.004 [0.115]	−0.004 [0.082]	−0.010 [0.108]
Pooled NLS + generation FE	−0.002 [0.085]	0.004 [0.042]	−0.060 [0.191]	0.000 [0.045]
Lab FE	0.001 [0.066]	0.001 [0.043]	−0.010 [0.112]	0.001 [0.060]
Lab $\times$ generation FE	0.000 [0.027]	0.001 [0.028]	0.000 [0.025]	0.002 [0.027]
ACF, current c as instrument	0.004 [0.117]	0.001 [0.054]	−0.055 [0.241]	0.009 [0.121]
ACF, lagged instruments only	−0.001 [0.126]	−0.001 [0.121]	0.000 [0.126]	0.003 [0.130]
IV (compute price) + path [‡]	0.000 [0.003]	0.000 [0.002]	−0.003 [0.004]	0.000 [0.002]
FOC system, T ignored	−0.002 [0.004]	−0.002 [0.004]	−0.004 [0.005]	−0.002 [0.004]
FOC system, T proxied	−0.001 [0.003]	−0.001 [0.003]	−0.001 [0.003]	−0.001 [0.003]
<i>Panel C. TFP growth per generation (truth 0.150)</i>				
Pooled Huber, E free (ML practice)	−0.081 [0.081]	−0.090 [0.090]	−0.036 [0.039]	−0.088 [0.088]
Pooled NLS + generation FE	0.002 [0.012]	−0.024 [0.027]	0.054 [0.055]	−0.018 [0.021]
Lab FE	0.001 [0.010]	−0.009 [0.013]	0.023 [0.025]	−0.003 [0.010]
Lab $\times$ generation FE	0.001 [0.009]	0.000 [0.009]	0.000 [0.009]	0.001 [0.009]
ACF, current c as instrument	0.002 [0.010]	−0.007 [0.012]	0.024 [0.026]	0.002 [0.010]
ACF, lagged instruments only	0.003 [0.010]	0.002 [0.009]	0.002 [0.009]	0.003 [0.010]
IV (compute price) + path [‡]	0.005 [0.035]	−0.003 [0.032]	0.001 [0.032]	0.003 [0.032]
FOC system, T ignored	0.003 [0.009]	0.002 [0.010]	0.003 [0.009]	0.004 [0.010]
FOC system, T proxied	0.001 [0.009]	0.001 [0.009]	0.001 [0.009]	0.002 [0.009]
<i>Panel D. $\ln M^*(10^{24})$ (truth 2.898)</i>				
Pooled Huber, E free (ML practice)	0.53 [1.61]	0.55 [1.60]	−0.04 [1.41]	0.64 [1.50]
Pooled NLS + generation FE	0.01 [1.17]	0.23 [0.73]	0.10 [2.17]	0.27 [0.80]
Lab FE	0.00 [0.87]	0.11 [0.65]	−0.06 [1.35]	0.06 [0.83]
Lab $\times$ generation FE	0.00 [0.37]	−0.01 [0.38]	−0.01 [0.33]	−0.02 [0.39]
ACF, current c as instrument	0.03 [1.47]	0.10 [0.79]	0.52 [2.83]	−0.03 [1.50]
ACF, lagged instruments only	0.08 [1.58]	0.02 [1.51]	0.03 [1.50]	0.03 [1.60]
IV (compute price) + path [‡]	1.27 [1.27]	1.28 [1.28]	1.32 [1.32]	1.28 [1.28]
FOC system, T ignored	1.25 [1.25]	1.25 [1.25]	1.28 [1.28]	1.25 [1.25]
FOC system, T proxied	−0.02 [0.05]	−0.02 [0.05]	−0.03 [0.04]	−0.03 [0.05]

Note: Bias [root-mean-squared error] over 400 replications per compute rule. Simulated industry and compute rules: (a) exogenous budgets; (b) funding, $\ln C$ rises by $2x_{ft}$; (c) a loss target; (d) predetermined budgets, $\ln C$ rises by $2x_{f,t-1}$; setup in the text. E is known to every estimator except the first. Both ACF rows use current within-lab-generation deviations of n and d as instruments, valid here by construction; without them the lagged moment set is under-identified. [‡]Median bias [median absolute error], because just-identified 2SLS has no finite moments; median first-stage F is 17 to 18 under rules (a), (b), and (d) and 13 under (c). FOC system: the within-family loss equation stacked with the training-only first-order condition ($w = 1$), ignoring or proxying inference demand T (proxy with log noise of standard deviation 0.3). TFP growth: mean change in generation means of $y - \hat{F}(n, d)$. Every estimator is started from data-based values, none at the truth: nine random values for the pooled Huber fit, and nine combinations of $\alpha, \beta \in \{0.2, 0.5, 1\}$ with levels matched to mean output for the nonlinear least-squares estimators, keeping the lowest criterion; ACF starts from pooled NLS and the FOC systems from the lab $\times$ generation fit. *Source:* Authors' simulations; replication code `code/analysis/m6_montecarlo`.

TABLE C5—TRANSMISSION BIAS IN THE RETURNS TO COMPUTE: CLOSED FORM AND SIMULATION

Compute rule		On the path ($T = 0$), $n = 10^6$			$T > 0$	$n = 240$	
		Formula	Simulated	Reverse	Simulated	Mean	Coverage (%)
Exogenous		0.1783	0.1781	0.542	0.1784	0.179	95
Funding, $\lambda =$	0.5	0.2091	0.2095	0.512	0.2099	0.209	53
Funding, $\lambda =$	1	0.2371	0.2370	0.481	0.2370	0.237	4
Funding, $\lambda =$	2	0.2783	0.2782	0.429	0.2784	0.278	0
Funding, $\lambda =$	4	0.3033	0.3034	0.359	0.3032	0.303	0
Predetermined, $\lambda =$	1	0.2253	0.2254	0.487	0.2257	0.225	16
Predetermined, $\lambda =$	2	0.2583	0.2582	0.435	0.2586	0.258	0
Predetermined, $\lambda =$	4	0.2783	0.2783	0.359	0.2789	0.278	0
Target, $\text{sd}(\bar{y}) =$	0	0.0000	0.0000	—	0.0000	0.000	0
Target, $\text{sd}(\bar{y}) =$	0.05	0.0069	0.0068	0.358	0.0068	0.007	0
Target, $\text{sd}(\bar{y}) =$	0.1	0.0246	0.0246	0.223	0.0246	0.025	0
Target, $\text{sd}(\bar{y}) =$	0.25	0.0891	0.0892	0.186	0.0886	0.089	0

Note: Slope of $y = -\ln(L - E)$ on $c = \ln C$ across labs. Formula: $\gamma + \pi_1 \text{Var}(\omega) / (\pi_1^2 \text{Var}(\omega) + \text{Var}(\eta))$ (Proposition A5) with $\gamma = 0.1783$, $\text{sd}(\omega) = 0.25$, $\text{sd}(\eta) = 1$, persistence 0.8. Funding: $c = \lambda\omega + \eta$; predetermined: $c = \lambda\omega_{-1} + \eta$; target: labs buy the compute that reaches a target $\bar{y}$ with the stated dispersion (a common target gives slope 0). Simulated: OLS on 10^6 draws, seed noise 0.05 (simulation standard errors at most 0.0003). Reverse: $\text{Var}(y)/\text{Cov}(c, y)$; “—” where $\text{Cov}(c, y) = 0$ and it is undefined. $T > 0$: lifetime-optimal allocations with log-normal inference demand; under target rules an extra term of -0.002 to -0.001 appears because high- T labs buy more compute. $n = 240$: mean OLS slope and coverage of the nominal 95 percent OLS interval over 2,000 samples of 240 lab-generations; coverage rates are specific to $\text{sd}(\eta) = 1$ and rise with more exogenous dispersion in compute.

Source: Authors’ simulations; replication code `code/analysis/m6_montecarlo`.

data, and the same comparisons on the IsoFLOP designs that labs have published. Appendix D3 documents the finite-grid properties and the specification sensitivity of the model-free estimator of Proposition 4 and its application to Farseeer’s grid. Appendix D4 collects robustness results for the parametric technologies across sweeps, Appendix D5 translates the elasticity of substitution into the compute cost of over-training, and Appendix D6 treats input measurement and flexible inputs.

D1. A Dictionary

Table D1 lists nineteen correspondences between scaling-law concepts and concepts of production theory and IO econometrics. The entries in panel A are identities of duality theory once training compute is measured by the FLOP-accounting approximation $C \approx 6ND$. The others are analogies: the econometric problem is the same, but its source can differ.

Three analogies break down. First, parameters and tokens have no market prices. FLOP accounting gives every lab the same log-cost weights, which is why optimizing labs share one expansion path, but a FLOP is not a dollar, and size-dependent prices of training and serving FLOPs enter the first-order conditions as wedges (Proposition 2). Second, returns to scale measured in loss depend on how output is cardinalized, so “diminishing returns to compute” has no unit-free content; the elasticity of substitution, the optimal mix and the wedge do. Third,

the inputs of a designed sweep are assigned, not chosen. An IsoFLOP sweep is designed variation in the input mix, not a shock to relative prices, and it elicits no behavioral response; the endogeneity machinery of IO applies to cross-lab data (Online Appendix E), not to single-lab experiments.

Two correspondences are narrower than their names suggest. The collinearity of inputs along the expansion path is the problem of optimizing firms that face common prices (Marschak and Andrews, 1944), in the sharp form of Bond and Söderbom (2005): identification requires relative-price variation, adjustment frictions or optimization errors. It is not the functional dependence of Akerberg, Caves and Frazer (2015), which concerns the first stage of proxy estimators and arises here only in cross-lab data (Online Appendix A, Proposition A7). And the wedge $w = \varepsilon_N/\varepsilon_D$ is the ratio statistic of Raval (2023) with FLOP-cost weights in place of revenue shares. Raval reads a ratio different from one as misspecification; Proposition 2 reads it as a value of compactness. Cross-lab factor bias is therefore the first threat to the inversion (Online Appendix F).

D2. The Chinchilla Data

Design.—Epoch AI’s digitization of Figure 4 of Hoffmann et al. (2022) contains 245 runs with training compute from 1.4×10^{18} to 1.3×10^{22} FLOP (Besiroglu et al., 2024). Only 137 of them lie on the nine IsoFLOP profiles, whose budgets run from 6×10^{18} to 3×10^{21} FLOP. The other 108 are off-profile runs that form 35 streaks of fixed model size: the training-horizon runs that Hoffmann et al. pool into their Approach 3. The baseline sample drops the five highest-loss runs ($n = 240$, with 132 profile runs). In it, 33 percent of the variance of centered $(\ln N, \ln D)$ lies across the expansion path: 37 percent among the profile runs and 21 percent among the off-profile runs. The IsoFLOP-minima objects below use the 132 profile runs; the primal uses all 240.

Estimators.—Table D2 fits the Chinchilla form with five estimators. Huber loss with $\delta = 10^{-3}$ on log loss, the estimator of Besiroglu et al. (2024), is least absolute deviations in practice: 84 percent of its residuals lie in the linear region of the loss, and the two agree to the third decimal. Least squares, on log loss or in levels, weights the few large residuals more heavily and gives a larger data exponent; the paired differences in β , 0.039 and 0.060, are two to three bootstrap standard errors. The implied compute-optimal ratios at 5.76×10^{23} FLOP are lower by 3.7 and 7.6 tokens per parameter, less than one and a half standard errors. Estimating the loss equation jointly with the nine IsoFLOP argmins under the cross-equation restrictions of Proposition 1 leaves the estimates unchanged; the restrictions are not rejected ($p = 0.60$). Panel B restates Proposition 3 on these data. The 41 runs within 0.15 log points of the expansion path carry less than 1 percent of their input variance across the path, and the condition number of the Jacobian rises from 62 on the full design to 413 on them, against 69 to 95 for random subsamples of the same size. In raw parameters the condition number of the full design is about 2,000, which is why the standard errors that Besiroglu

TABLE D1—SCALING LAWS IN THE LANGUAGE OF PRODUCTION ECONOMICS: FULL DICTIONARY

Scaling-law concept	Production / IO concept	What the mapping delivers (where used)	Key reference
<i>Panel A. Technology and cost</i>			
Iso-loss contour	Isoquant	σ , optimal mix and w are ordinal (Lemma 1)	Brandfonbrener et al. 2025
Chinchilla law	Additive; CES if $\alpha = \beta$; quasi-homothetic	$\sigma(w)$, $\sigma^* = 2/(2 + \alpha + \beta)$; interior optima imply $\sigma^* < 1$ (Lemma 1)	Hao and Merrill 2026
Outer exponent κ	Two-level aggregator	$\kappa = 1$ ties IsoFLOP curvature to frontier and path slopes (Prop. 4)	Sato 1967
IsoFLOP line, $C \approx 6ND$	Isocost with common log-cost weights	Optimality equates output elasticities; FLOP prices that vary with size are wedges (Prop. 2)	Shephard 1953
Compute-optimal $N^*(C)$, $D^*(C)$	Conditional factor demands	Allocation exponent $a = \beta/(\alpha + \beta)$ (Prop. 1)	Uzawa 1964
IsoFLOP-minima frontier	Inverse cost function	Frontier elasticity γ (Prop. 1)	Nerlove 1963
Hoffmann et al.’s Approaches 1–3	Cost function and factor demands (1, 2); primal (3)	Design-consistent duality tests (Appendix D2)	Hoffmann et al. 2022
<i>Panel B. Identification</i>			
Compute-optimal or fixed-ratio designs	Collinearity under common prices	a and γ identified; σ^* only by functional form, at rate $n^{-1/4}$ (Prop. 3)	Bond and Söderbom 2005
IsoFLOP sweeps; factorial grids	Assigned inputs, not price shocks	σ^* from IsoFLOP curvature and the frontier slope, form-free (Prop. 4)	Box and Lucas 1959
Wedges of released models	Relative-price and constraint variation	Information on $\ln M^*$ second order, on σ^* fourth order in the log wedges (Prop. 3)	Kiefer and Wolfowitz 1959
Lab “secret sauce”	Transmitted Hicks-neutral productivity	Biases γ across labs; given compute, the mix cannot proxy for it (Appendix E)	Ackerberg, Caves and Frazer 2015
Hyperparameters	Flexible inputs	Scale-correlated mis-tuning biases a (Appendix D6)	Gandhi, Navarro and Rivers 2020
<i>Panel C. Measurement</i>			
Omitted embedding or head parameters	Scale-correlated input error	Upward bias in local a that vanishes at scale (Appendix D6)	Collard-Wexler and De Loecker 2016
Loss per token; tokenizer	Units of output and of the data input	Exponents survive proportional rescaling; E , A , B , M^* do not	Tao et al. 2024
Benchmark accuracy; Elo	Bounded monotone measurement map	Ordinal objects survive the link; γ and returns to compute do not	Gadre et al. 2025
<i>Panel D. Productivity and wedges</i>			
Algorithmic progress	Factor-augmenting technical change	Drift of M^* signs the Hicks bias $\beta g_D - \alpha g_N$ (Appendix E)	Ho et al. 2024
Rebalancing after Chinchilla	Farrell allocative efficiency	An allocative, not a technical, gain (Appendix E)	Gundlach et al. 2025
Over-training, $M > M^*(C)$	Input wedge (Raval’s ratio statistic)	Value of compactness $s = (w - 1)/w$; serving share if the developer serves (Prop. 2)	Raval 2023
Cross-lab scaling regressions	Transmission and selection bias	Bias signed by the compute regime; low-power benchmark (Appendix E)	Whitfill 2025

Note: Panel A’s entries are identities of duality theory once compute is measured by the FLOP-accounting approximation $C \approx 6ND$; the others are analogies in which the econometric problem is the same but its source can differ. $M^*(C)$: compute-optimal D/N ; g_N , g_D : rates of parameter- and data-augmenting technical change.

Source: Authors’ compilation.

et al. (2024) report for A and B (125 and 1,293) are large: they measure the levels at $N = D = 1$, far outside the design, and are not evidence that the design fails to identify the technology.

Inference.—Table D3 compares six bootstrap schemes for the Huber refit. Because it behaves as a median-type estimator, the wild bootstraps use the weights of Feng, He and Hu (2011). The restriction $\kappa = 1$ is rejected at 1 percent under every scheme except nine budget clusters that assign each off-profile run to the nearest budget ($p = 0.052$). That assignment is not a model of the dependence in the data, because an off-profile run from a 10^{22} -FLOP trunk is not part of the 3×10^{21} IsoFLOP experiment. Clustering the off-profile runs by their model-size trunk lowers the standard error of β from 0.050 to 0.042 and moves $p(\kappa = 1)$ to 0.004. It also narrows the 95 percent interval for M^* at 5.76×10^{23} FLOP from $[3.7, 179]$ to $[5.2, 106]$. Treating the off-profile runs as singleton clusters gives nearly the same answers. Our preferred scheme is the design-conditional wild bootstrap; the budget-and-trunk clusters are the conservative check.

The CES restriction $\alpha = \beta$ is not rejected by any test that is robust to the heavy tails of the residuals, whose MAD-based standard deviation is 0.0049 against a standard deviation of 0.0076. The Gaussian likelihood ratio of 10.4 has a χ_1^2 p -value of 0.001, but its restricted-bootstrap 95 percent critical value is 20.2, not 3.84, which gives $p = 0.17$. Heteroskedasticity-robust Wald tests give 0.060 and 0.089, so the evidence against CES is borderline at 10 percent and absent at 5. The Chinchilla-70B wedge contains one under every scheme, and it excludes the values of 0.62 to 0.71 that Hoffmann et al.’s published parameters imply only under the pairs and wild schemes. Because DeepMind sized that model with Approaches 1 and 2 on these same runs, $w(70B) \approx 1$ is an internal-consistency check with little power, not a test of revealed preference.

Profiles.—Figure D1 plots the profile likelihood over σ^* with κ free, relative to the unconstrained optimum. On the full design, the Huber (Laplace quasi-likelihood) profile has its minimum at $\hat{\sigma}^* = 0.701$ and a χ^2 -based 95 percent set of $[0.69, 0.71]$. A restricted wild bootstrap of the statistic widens the set to $[0.68, 0.72]$ and rejects the value that $\kappa = 1$ implies, 0.74 (likelihood ratio 29.0, $p = 0.003$, the smallest value the bootstrap can deliver). The Gaussian profile has its minimum at 0.680 and a set of $[0.67, 0.69]$, which belongs with the Gaussian point estimate rather than the Huber one. On the 41 runs within 0.15 log points of the expansion path, the profile is flat: the quasi-likelihood ratio stays below 1.21 between 0.25 and 0.94, and the 95 percent set reaches the upper edge of the grid. On-path runs carry no information about σ^* , as Proposition 3 implies.

Exclusion rules.—Table D4 shows how the estimates move as the k highest-loss runs are dropped. The data exponent β falls almost monotonically, from 0.453 with all 245 runs to 0.347 at $k = 15$, while α stays between 0.345 and 0.349. The five runs that Besiroglu et al. (2024) drop are the most data-starved runs of the 10^{19} budget (D/N between 0.04 and 0.40), with log residuals 9 to 45 times the residual standard deviation. Their influence is leverage in the $D/N \rightarrow 0$ corner of

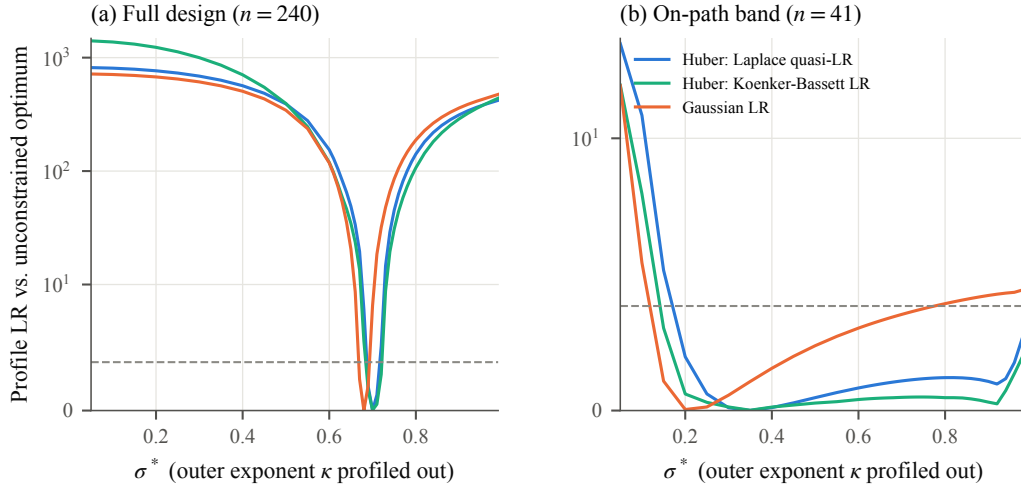


FIGURE D1. PROFILE LIKELIHOOD OVER THE ELASTICITY OF SUBSTITUTION: FULL DESIGN AND ON-PATH RUNS

Note: Profile of the κ -family objective over $\sigma^* = 2/(2 + a_1 + b_1)$ on the Chinchilla data, each point re-estimating E , A , B and κ , measured relative to the unconstrained optimum (log scale above one). Panel (a): the 240-run design. Panel (b): the 41 runs within 0.15 log points of the expansion path. Laplace quasi-LR and Koenker-Bassett statistics come from the Huber objective; the Gaussian LR from least squares on $\ln L$. Dashed line: χ^2_1 95 percent critical value; restricted-bootstrap critical values are larger (7.7–11.4 on the full design).

Source: Besiroglu et al. (2024); authors' calculations.

the design, not truncation bias: a truncated-Gaussian likelihood that conditions on the exclusion rule (Hausman and Wise, 1977) returns the estimate that ignores truncation ($\beta = 0.406$ in both), and dropping runs by their data-to-parameter ratio rather than their loss gives the same answer as dropping them by loss. The Chinchilla-70B wedge is 1.54 with the outliers. Across twelve rules that remove them it lies between 0.95 and 1.06 for ten; dropping the whole 10^{19} budget gives 0.78, and the IsoFLOP-profile runs alone give 1.16.

Duality.—Table D5 reports the objects that each estimation approach delivers and tests the cross-equation restrictions of Proposition 1. We call the curve fitted through the nine IsoFLOP minima the IsoFLOP-minima frontier. It is not Hoffmann et al.’s Approach 1, whose training-curve envelopes the digitization does not contain and which reports allocation exponents as well as a frontier. Tests must be design-consistent. Comparing the IsoFLOP argmins with the analytic primal path rejects even our own refit under the stratified bootstrap, because parabolas fitted on a finite grid are biased (Czech et al., 2026). Against design-consistent nulls, our refit and Besiroglu et al.’s parameters are consistent with the argmins ($p \geq 0.16$ in every bootstrap-calibrated and classical variant). Hoffmann et al.’s published Approach 3 is marginally inconsistent (classical $p = 0.040$; calibrated $p = 0.060$), mainly because its compute-optimal model size at the center of the design is about 12 percent below the argmins. The frontier restrictions are rejected under the stratified bootstrap but not under the wild bootstrap. The IsoFLOP-minima frontier lies 0.005 to 0.008 nats below every primal frontier, a gap of the order of the quantization of the digitized losses, which are read from a color map, and we do not interpret it.

Lab-own designs.—Table D6 applies the same comparison to the IsoFLOP experiments that labs have published. Our argmin fit reproduces Meta’s Llama 3 law exactly. The primal fitted to the same runs implies a steeper path ($a = 0.527$ against 0.463), in which M^* falls rather than rises with compute; the difference is significant at 5 percent once the primal’s own sampling error is included ($t = -2.11$) and not under the wild bootstrap. Two of Meta’s ten budgets do not bracket their minimum. Fitted on the other eight, the path slope is 0.50 and $M^*(10^{24})$ is 15 instead of 31 (Online Appendix Table F5). For the Marin ladders the design-consistent tests do not reject (p between 0.33 and 0.54). About three-quarters of the naive discrepancy is finite-grid parabola bias and the rest is FLOP accounting: $6ND$ differs from Marin’s nominal budgets by -7 to $+35$ percent. The Marin primal exponents (α between 0.64 and 0.69) are far from Chinchilla’s, a first indication that exponents are specific to the recipe.

D3. The Model-Free Estimator

Finite grids.—A least-squares quadratic fitted to an IsoFLOP profile on a finite grid overstates the curvature at the minimum, because the profile’s quartic term loads on the quadratic coefficient. To leading order the bias is $(f^{(4)}/24)\kappa_4(h)$, where κ_4 depends only on the moments of the grid (Online Appendix A, remark

TABLE D2—CHINCHILLA DATA: ESTIMATORS AND ON-PATH FUNCTIONAL DEPENDENCE

<i>Panel A. The Chinchilla form under five estimators ($n = 240$)</i>					
	Huber (1)	LAD (2)	Gaussian (3)	NLS, levels (4)	System (5)
α	0.347 (0.016)	0.348 (0.016)	0.360 (0.013)	0.358 (0.013)	0.348 (0.014)
β	0.367 (0.021)	0.366 (0.021)	0.406 (0.025)	0.428 (0.026)	0.365 (0.016)
a	0.514 (0.020)	0.513 (0.020)	0.530 (0.017)	0.545 (0.017)	0.512 (0.015)
γ	0.178 (0.006)	0.178 (0.006)	0.191 (0.007)	0.195 (0.007)	0.178 (0.005)
σ^*	0.737 (0.006)	0.737 (0.006)	0.723 (0.008)	0.718 (0.008)	0.737 (0.005)
$M^*(5.76 \times 10^{23})$	17.9 (6.9)	18.3 (7.3)	14.3 (4.9)	10.4 (3.8)	18.5 (6.0)
$w(70B)$	1.04 (0.16)	1.03 (0.17)	1.14 (0.15)	1.30 (0.18)	1.03 (0.12)
Difference from (1): β		−0.001 (0.004)	0.039 (0.018)	0.060 (0.020)	
$M^*(5.76 \times 10^{23})$		0.4 (1.6)	−3.7 (4.9)	−7.6 (5.3)	
<i>Panel B. Functional dependence on the expansion path (Huber, Chinchilla form)</i>					
	Full design ($n = 240$)	On-path band ($n = 41$)	Random subsamples (four, $n = 41$ each)		
Transverse variance share	0.33	0.008	0.29–0.42		
Condition number, normalized	62	413	69–95		
Condition number, raw	2,003	14,736			

Note: $L = E + AN^{-\alpha} + BD^{-\beta}$ on the 240-run Chinchilla sample; $a = \beta/(\alpha + \beta)$, $\gamma = \alpha\beta/(\alpha + \beta)$, $\sigma^* = 2/(2 + \alpha + \beta)$; $M^*(C)$ is the compute-optimal D/N at C FLOP; $w(70B)$ is the wedge $\varepsilon_N/\varepsilon_D$ at Chinchilla-70B ($N = 70B$, $D = 1.4T$). Estimators: (1) Huber loss ($\delta = 10^{-3}$) on $\ln L$, the estimator of Besiroglu et al. (2024), 84 percent of whose residuals lie in the linear region of the loss; (2) least absolute deviations on $\ln L$; (3) Gaussian nonlinear least squares on $\ln L$; (4) nonlinear least squares on L ; (5) the loss equation (Laplace errors) estimated jointly with the nine IsoFLOP argmins, predicted design-consistently under the cross-equation restrictions (nested likelihood ratio of the restrictions 1.01, $p = 0.60$). Standard errors in parentheses: pairs bootstrap over runs (400 draws; for M^* , interquartile range/1.349); column 5, within-budget stratified bootstrap (300 draws); design-conditional alternatives for column 1 are in Online Appendix Table D3. Differences from (1) are paired, on the shared bootstrap draws. Panel B: on-path band, runs with $|\ln N - \ln N^*(C)| \leq 0.15$; random subsamples, four draws of 41 runs from the full design; transverse share, share of the variance of centered $(\ln N, \ln D)$ orthogonal to the expansion path; condition numbers of the Jacobian of the Huber fit in column 1 for each sample, with parameters normalized at the geometric means of the inputs (Klump, McAdam and Willman, 2007) or in raw units (A and B at $N = D = 1$).

Source: Hoffmann et al. (2022); Besiroglu et al. (2024); authors' estimates.

TABLE D3—INFERENCE ON THE CHINCHILLA DATA: CLUSTERS, WILD BOOTSTRAP AND PROFILES

<i>Panel A. Estimates, standard errors and 95 percent intervals ($n = 240$)</i>							
Scheme (clusters)	β	σ^*	$M^*(5.76 \times 10^{23})$	$w(70B)$	κ	$p(\kappa = 1)$	$p(\alpha = \beta)$
Estimate	0.367	0.737	17.9	1.04	0.774		
Pairs (runs)	(0.021)	(0.006)	[7.4, 35.4]	[0.82, 1.43]	(0.057)	<0.001	0.49
Nine budgets (9)	(0.050)	(0.011)	[3.7, 179]	[0.48, 1.89]	(0.116)	0.052	0.77
Budgets, trunks (44)	(0.042)	(0.009)	[5.2, 106]	[0.58, 1.65]	(0.079)	0.004	0.72
Budgets, singletons (117)	(0.041)	(0.009)	[5.9, 109]	[0.58, 1.56]	(0.076)	0.003	0.71
Wild, Feng–He–Hu (runs)	(0.016)	(0.005)	[10.5, 37.4]	[0.80, 1.26]	(0.054)	<0.001	0.40
Wild cluster, Webb (44)	(0.029)	(0.011)	[9.2, 37.3]	[0.81, 1.32]	(0.062)	<0.001	0.50
<i>Panel B. Likelihood-ratio tests</i>							
	Laplace quasi-LR		Gaussian LR			Koenker–Bassett	
Null hypothesis	Statistic	p_{boot}	Statistic	p_{χ^2}	p_{boot}	Statistic	p_{χ^2}
$\kappa = 1$	31.2	0.001	46.2	<0.001	0.001	20.7	<0.001
$\alpha = \beta$	1.8	0.40	10.4	0.001	0.17	1.2	0.27
<i>Panel C. Profile over σ^* with κ free, relative to the unconstrained optimum</i>							
Sample	Objective	$\hat{\sigma}^*$	95% set (χ_1^2)	Grid edge	Max. LR		
Full design ($n = 240$)	Laplace quasi-LR	0.701	[0.69, 0.71]	none	817		
Full design ($n = 240$)	Gaussian LR	0.680	[0.67, 0.69]	none	718		
On-path band ($n = 41$)	Laplace quasi-LR	0.342	[0.20, 0.99]	upper	24		
On-path band ($n = 41$)	Gaussian LR	0.215	[0.15, 0.77]	none	16		

Note: Epoch AI’s digitization of the Chinchilla runs: 245 runs, 137 on nine IsoFLOP profiles and 108 off-profile runs in 35 model-size trunks, training FLOP 1.4×10^{18} – 1.3×10^{22} ; the five highest-loss runs are dropped ($n = 240$: 132 profile runs). Estimator: Huber loss ($\delta = 10^{-3}$) on $\ln L$. Panel A: 999 draws per scheme; standard errors in parentheses, percentile intervals in brackets. Nine budgets: every run assigned to the nearest IsoFLOP budget; budgets, trunks: profile runs clustered by budget and off-profile runs by model size; wild: design held fixed, $y^* = \hat{y} + v|\tilde{r}|$ around the κ -free fit with the weights and leverage-adjusted residuals of [Feng, He and Hu \(2011\)](#); wild cluster: one weight of [Webb \(2023\)](#) per cluster. p -values: Wald with the scheme’s standard error. The standard error of α is 0.014–0.025 across schemes. Panel B: p_{boot} from a restricted wild (Feng–He–Hu) bootstrap that generates the data under the null, 999 draws (smallest attainable $p = 0.001$); heteroskedasticity-robust Wald tests of $\alpha = \beta$ for the Gaussian fit give $p = 0.060$ (HC1) and 0.089 (HC3). Panel C: $\sigma^* = 2/(2 + a_1 + b_1)$ in the κ family, grid [0.05, 0.99]; on-path band: runs with $|\ln N - \ln N^*(C)| \leq 0.15$. Restricted-bootstrap 95 percent critical values of the Laplace quasi-LR are 7.7–11.4 on the full design (299 draws each), which widens its set to [0.68, 0.72].

Source: [Hoffmann et al. \(2022\)](#); [Besiroglu et al. \(2024\)](#); authors’ calculations.

TABLE D4—CHINCHILLA DATA: DROPPING HIGH-LOSS RUNS AND OTHER EXCLUSION RULES

Sample	n	α	β	a	$M^*(10^{21})$	w (70B)
<i>Panel A. Drop the k highest-loss runs (Huber)</i>						
$k = 0$	245	0.349 (0.018)	0.453 (0.054)	0.565 (0.032)	15.5 (3.7)	1.54 (0.50)
$k = 1$	244	0.345 (0.016)	0.412 (0.041)	0.544 (0.027)	17.6 (3.6)	1.30 (0.31)
$k = 2$	243	0.346 (0.016)	0.403 (0.035)	0.538 (0.025)	18.4 (3.5)	1.24 (0.24)
$k = 3$	242	0.347 (0.015)	0.382 (0.028)	0.524 (0.022)	20.0 (3.3)	1.12 (0.21)
$k = 4$	241	0.347 (0.016)	0.371 (0.024)	0.516 (0.022)	21.0 (3.5)	1.06 (0.19)
$k = 5$ (Besiroglu)	240	0.347 (0.015)	0.367 (0.021)	0.514 (0.020)	21.4 (3.5)	1.04 (0.17)
$k = 6$	239	0.347 (0.013)	0.363 (0.018)	0.511 (0.018)	21.9 (3.2)	1.02 (0.13)
$k = 8$	237	0.346 (0.013)	0.358 (0.019)	0.509 (0.019)	22.2 (3.5)	1.00 (0.13)
$k = 10$	235	0.346 (0.015)	0.358 (0.018)	0.509 (0.018)	22.2 (3.5)	1.00 (0.14)
$k = 15$	230	0.347 (0.016)	0.347 (0.020)	0.500 (0.021)	23.3 (4.0)	0.95 (0.14)
<i>Panel B. Alternative exclusion rules (Huber)</i>						
Drop 6 lowest-loss runs	239	0.359 (0.014)	0.479 (0.057)	0.572 (0.032)	14.6 (3.6)	1.67 (0.54)
Drop 5 largest residuals	240	0.350 (0.015)	0.373 (0.024)	0.516 (0.020)	21.1 (3.4)	1.05 (0.18)
Drop $D/N < 0.4$	241	0.347 (0.016)	0.371 (0.024)	0.516 (0.021)	21.0 (3.5)	1.06 (0.18)
Drop $D/N < 1$	234	0.343 (0.015)	0.342 (0.019)	0.500 (0.021)	22.7 (3.9)	0.96 (0.14)
Drop $D/N < 2$	224	0.331 (0.013)	0.347 (0.023)	0.512 (0.022)	20.2 (3.4)	1.05 (0.15)
Drop the 10^{19} budget	211	0.357 (0.014)	0.313 (0.022)	0.467 (0.024)	27.4 (3.9)	0.78 (0.12)
IsoFLOP-profile runs only	137	0.383 (0.030)	0.490 (0.055)	0.561 (0.030)	15.8 (3.3)	1.56 (0.58)
IsoFLOP-profile runs, $k = 5$	132	0.363 (0.026)	0.402 (0.028)	0.526 (0.022)	18.8 (3.3)	1.16 (0.19)
<i>Panel C. Gaussian likelihood and truncated regression</i>						
Gaussian, all runs	245	0.372	0.612	0.622	9.6	3.08
Gaussian, $k = 5$	240	0.360	0.406	0.530	20.8	1.14
Truncated Gaussian, $k = 5$	240	0.361	0.406	0.530	20.8	1.14

Note: Huber ($\delta = 10^{-3}$) estimates on the 245 digitized Chinchilla runs, with pairs-bootstrap standard errors (200 draws) in parentheses; Online Appendix Table D3 reports design-conditional alternatives for the $k = 5$ sample. w is the wedge $\varepsilon_N/\varepsilon_D$ at Chinchilla-70B ($N = 70\text{B}$, $D = 1.4\text{T}$), which equals 1 if that model is training-compute-optimal under the estimated technology. Panel A: $k = 5$ is the sample of Besiroglu et al. (2024); the five dropped runs are the most data-starved runs of the 10^{19} IsoFLOP budget ($D/N = 0.04\text{--}0.40$), with log residuals 9 to 45 times the residual standard deviation. Panel B: selection on the regressor (D/N or budget) instead of the outcome; the IsoFLOP-profile runs are the 137 runs within 0.045 of a nominal budget in $\log_{10} C$. Across the rules that remove the five outliers, $w(70\text{B})$ lies between 0.78 and 1.16. Panel C: the truncated-Gaussian likelihood (Hausman and Wise, 1977) conditions on L below the fifth-highest loss (3.447). A paired bootstrap of the $k = 0 \rightarrow 5$ change (400 draws, rule applied within each draw) gives $\Delta\beta = 0.086$ (s.e. 0.052) and $\Delta a = 0.051$ (s.e. 0.029).

Source: Epoch AI digitization of Hoffmann et al. (2022), Figure 4; authors' estimates.

TABLE D5—CHINCHILLA DATA: COST FUNCTION, CONDITIONAL FACTOR DEMAND, AND PRIMAL ESTIMATES

<i>Panel A. Dual objects implied by each approach</i>					
	IsoFLOP	Minima,	Approach 3 (primal)		
	minima	E fixed	Analytic	Design	
Path slope a	0.498 (0.020)	0.498	0.514 (0.018)	0.515 (0.020)	
Path level $\ln N^*(\bar{C})$	20.589 (0.027)	20.589	20.528 (0.036)	20.560 (0.045)	
Frontier elasticity γ	0.168 (0.008)	0.181	0.179 (0.005)	0.177 (0.006)	
Frontier level $L^*(\bar{C})$	2.557 (0.002)	2.554	2.562 (0.001)	2.563 (0.002)	
<i>Panel B. Tests of the cross-equation restrictions</i>					
	Path ($a, \ln N^*$)		Frontier (γ, L^*)		Classical
	$\chi^2_2(p)$	Calibrated p	$\chi^2_2(p)$	Calibrated p	$F_{2,7}: p$
Null technology					
Our refit, analytic	6.8 (0.033)	0.053 / 0.357	16.7 (<0.001)	0.007 / 0.489	—
Our refit, design	2.9 (0.238)	0.233 / 0.698	20.7 (<0.001)	0.007 / 0.414	0.653
Hoffmann, analytic	53.5 (<0.001)	0.010 / 0.055	7.7 (0.022)	0.037 / 0.848	—
Hoffmann, design	15.6 (<0.001)	0.060 / 0.209	14.4 (<0.001)	0.017 / 0.696	0.040
Besiroglu, analytic	13.5 (0.001)	0.027 / 0.406	32.0 (<0.001)	0.003 / 0.195	—
Besiroglu, design	4.3 (0.118)	0.176 / 0.746	42.5 (<0.001)	0.003 / 0.150	0.664

Note: The 240-run Chinchilla sample. IsoFLOP minima: a parabola in $\ln N$ fitted to the 132 IsoFLOP-profile runs of each of the nine budgets, with the argmins regressed on $\ln(C/6)$ (conditional factor demand, Hoffmann et al.’s Approach 2) and $L_b^* = E + K(C_b/6)^{-\gamma}$ fitted to the nine minima (the IsoFLOP-minima frontier, an inverse cost function), with E free or fixed at the primal value. This frontier is not Hoffmann et al.’s Approach 1, which uses envelopes of training curves that the digitization does not contain and which also reports allocation exponents. Approach 3 is the Huber primal on all 240 runs. Hoffmann and Besiroglu denote the published parameters of Hoffmann et al. (2022) and Besiroglu et al. (2024). “Analytic” uses the closed forms $a = \beta/(\alpha + \beta)$, $\gamma = \alpha\beta/(\alpha + \beta)$, $N^* = G(C/6)^a$ and $L^* = E + K(C/6)^{-\gamma}$; “design” (design-consistent) applies the IsoFLOP-minima procedure to the primal’s fitted values at the observed design, which removes the finite-grid bias of parabola fits (Czech et al., 2026). $\bar{C}$ is the geometric mean of the budgets. Standard errors in parentheses: within-budget stratified pairs bootstrap (300 draws). Panel B: Wald statistics with an outlier-robust bootstrap covariance and χ^2_2 p -values; calibrated p -values take critical values from the bootstrap distribution of the same statistic (stratified pairs / fixed-design wild bootstrap); the classical test regresses the nine argmin gaps on $[1, \ln C_b]$ and treats the null technology as known. For published parameters only the IsoFLOP-minima side is random. With all 245 runs, the design-consistent path test of our refit gives a calibrated p of 0.054 (stratified) and a classical p of 0.041.

Source: Epoch AI digitization of Hoffmann et al. (2022); authors’ estimates.

TABLE D6—LAB-OWN TECHNOLOGIES FROM PUBLISHED ISOFLOP EXPERIMENTS

	Llama 3	Marin (2026), by corpus			(Mis)Fitting
	(Meta)	Comma	DCLM	Nemotron-CC	FineWeb/C4
<i>Panel A. Primal (Approach 3, Huber) on the IsoFLOP runs</i>					
Runs (budgets)	133 (10)	85 (7)	85 (7)	88 (8)	176 (26)
α	0.284 (0.050)	0.666 (0.085)	0.686 (0.061)	0.639 (0.079)	0.610 (0.051)
β	0.316 (0.024)	0.350 (0.038)	0.405 (0.034)	0.400 (0.040)	0.705 (0.040)
a	0.527 (0.032)	0.345 (0.026)	0.371 (0.018)	0.385 (0.020)	0.536 (0.019)
σ^*	0.769 (0.018)	0.663 (0.022)	0.647 (0.017)	0.658 (0.022)	0.603 (0.014)
$M^*(10^{21})$	13.9 (2.7)	26.6 (5.3)	17.9 (2.7)	16.7 (2.7)	13.8 (3.1)
$M^*(3.8 \times 10^{25})$	7.9	707	270	189	6.4
<i>Panel B. IsoFLOP argmins and IsoFLOP-minima frontier (α, β via $\kappa = 1$)</i>					
a	0.463 (0.006)	0.405 (0.032)	0.424 (0.030)	0.422 (0.027)	0.517 (0.041)
γ	0.136 (0.003)	0.051 (0.011)	0.076 (0.023)	0.075 (0.020)	0.165 (0.078)
$M^*(3.8 \times 10^{25})$	40.9	116	56.7	61.8	10.2
<i>Panel C. Tests of the IsoFLOP argmins against the primal: statistic (p-value)</i>					
Path, design-consistent χ^2_2	16.9 (<0.001)	2.2 (0.332)	1.5 (0.466)	1.3 (0.535)	4.5 (0.108)
Path, calibrated p	0.015	0.517	0.507	0.602	0.149
Slope, t with primal s.e.	-2.11 (0.035)	0.17 (0.867)	0.13 (0.895)	-0.13 (0.896)	0.88 (0.378)
System LR, small-sample p	0.050	0.246	0.391	0.351	0.135

Note: IsoFLOP runs compiled by [Czech et al. \(2026\)](#): Meta’s Llama 3 IsoFLOPs digitized from [Grattafiori et al. \(2024\)](#) ($N = C/(6D)$; loss units not stated), the Marin Llama-2-architecture ladders on three corpora ([Marin Community, 2026](#)) (6ND differs from the nominal budgets by -7 to $+35$ percent), and the checkpoint-interpolated IsoFLOPs of [Li, Kudugunta and Zettlemoyer \(2025\)](#). Loss levels are not comparable across columns; exponents are. Panel A: cluster-bootstrap standard errors over budgets (200 draws). Panel B: robust within-budget stratified-bootstrap standard errors; the classical standard error of the Llama 3 path slope is 0.018. Our argmin fit reproduces Meta’s published law, $D^* = 0.2994 C^{0.5368}$ against $0.299 C^{0.537}$, although two of Meta’s ten budgets (3×10^{21} and 10^{22} FLOP) do not bracket the minimum with two runs on each side. M^* at 3.8×10^{25} FLOP extrapolates 3.5 (Llama 3) to 5 (Marin) orders of magnitude beyond the largest budget and is shown only to illustrate how far the primal and dual extrapolations diverge. Panel C: the calibrated p -value takes critical values from the bootstrap distribution of the path statistic; the slope t adds the primal’s bootstrap standard error to the classical standard error of the argmin slope; the small-sample p -value calibrates the nested system likelihood ratio (level and slope shift of the argmin equation) with an F form.

Source: [Czech et al. \(2026\)](#) and [Czech \(2026\)](#); authors’ estimates.

after Proposition A10). Table D7 evaluates it for nine runs centered at the minimum of the Chinchilla profile at 10^{21} FLOP. A grid spanning a factor of 9 in N overstates the curvature by 1.3 percent and gives $\hat{\sigma}^* = 0.7345$ against a true 0.7370. A grid spanning a factor of 100 overstates it by 5.9 percent ($\hat{\sigma}^* = 0.7257$). A quartic removes the bias in both cases, at the cost of variance. Within a span of about 3 in N , the bias of a quadratic is below 0.3 percent of the curvature. The argmin is biased too; this is the finite-grid bias of IsoFLOP argmins documented by Czech et al. (2026), and it is why the duality tests of Appendix D2 use design-consistent nulls.

Specification.—The primary estimator of Section III fits a quadratic in $\ln N$ to the runs within one log point of a pooled Approach-2 path, on budgets whose minimum is bracketed by at least two runs on each side, and pools the budgets by a random-effects mean. Table D8 reports its sensitivity. Pooled F tests of a cubic term never reject at 5 percent. Across window widths, polynomial orders, frontier smoothers and window centering, the random-effects σ^* ranges over 0.665–0.714 for Chinchilla (0.603 with the narrowest window, which keeps seven budgets), 0.642–0.677 for Llama 3 and 0.64–0.72 for Marin. An independent reimplementation that shares no code gives 0.669–0.694 on the same five designs. In a Monte Carlo at each design’s own fitted κ -family technology, the random-effects interval covers the truth in 92 to 98 percent of replications, with a bias of at most 0.005. The pooled design-conditional interval under-covers (82 to 91 percent), which is why the random-effects summary is primary. Without noise, the pooled bias is at most 0.006 in absolute value on every design except the strongly curved κ -family fit to Porian et al.’s profiles (+0.047 to +0.050).

The estimator needs exact isocosts, so each design is analyzed in the parameter convention in which its budgets equal $6ND$. The convention matters where FLOP accounting varies with size. On Marin, whose accounting is observed, the configuration count gives σ^* of 0.64 to 0.66, 0.04 to 0.07 below the FLOP-implied count. If Chinchilla’s FLOPs per token varied with N with elasticity η_C , the measured curvature would rescale by $(1 + \eta_C)^{-2}$ and σ^* would move by about $0.44\eta_C$, at most 0.04 for $|\eta_C| \leq 0.1$. Rebuilding D from Hoffmann et al.’s architecture table would remove this uncertainty; we have not done so.

Farseer’s local expansion path.—Farseer’s grid has no IsoFLOP profiles, so we estimate the loss surface locally. A Gaussian product-kernel local quadratic of $\ln L$ in $(\ln N, \ln D)$, with bandwidths of 0.24 and 0.74 in logs from leave-one-out cross-validation, gives the two output elasticities at every point and hence the local wedge $w = (L_c - L_x)/(L_c + L_x)$ of Proposition 4(i), which is invariant to monotone transformations of the loss. On ten isocosts from 10^{18} to 10^{21} FLOP we locate the point inside the design where $w = 1$; it exists on eight, from 5×10^{18} FLOP up, with compute-optimal ratios of 25 to 45. Two estimators of σ^* follow. The first evaluates the elasticity of substitution from the local gradient and Hessian at these points: a random-effects mean of 0.703 (standard error 0.013) that falls by 0.015 per decade of compute (0.008). Second derivatives need more smoothing than the

TABLE D7—FINITE-GRID BIAS OF THE MODEL-FREE ESTIMATOR OF σ^*

Half-width in $\ln N$	Span of N (ratio)	Curvature bias (%)	Actual/leading- order bias	$\hat{\sigma}^*$, quadratic	$\hat{\sigma}^*$, quartic	Argmin bias in $\ln N$
0.05	1.1	0.00	1.000	0.7370	0.7370	−0.00001
0.20	1.5	0.04	1.000	0.7369	0.7370	−0.00009
0.50	2.7	0.27	1.001	0.7365	0.7370	−0.00055
1.10	9.0	1.32	1.005	0.7345	0.7370	−0.00267
2.30	100	5.92	1.022	0.7257	0.7372	−0.01174
3.40	900	13.25	1.050	0.7122	0.7377	−0.02562

Note: Nine equally spaced runs in $\ln N$, centered at the minimum of the Besiroglu et al. (2024) IsoFLOP profile at 10^{21} FLOP, without noise; true $\sigma^* = 0.7370$. The estimator is $1/\hat{\sigma}^* - 1 = \hat{L}_{nn}/(2|dL^*/dc|)$ with $\hat{L}_{nn}$ twice the quadratic (or quartic) coefficient and the exact frontier slope. Leading-order bias of the quadratic coefficient: $(f^{(4)}/24)\kappa_4$ with $\kappa_4 = (m_6 - m_2m_4)/(m_4 - m_2^2)$ and m_k the grid’s k -th moment; argmin bias $-(f'''/6)(m_4/m_2)/f''$ (Online Appendix A, remark after Proposition A10). The table reports bias only: with seed and evaluation noise, quartic fits and narrow grids have higher variance. *Source:* Authors’ calculations.

surface itself, and this estimate moves between 0.66 and 0.73 with the bandwidth; at the narrower cross-validated optimum on an extended grid it is 0.664. The second estimator uses only first derivatives. By Proposition 4(iii), the slope of $\ln w$ in $u = \ln(M/M^*(C))$ at $u = 0$ equals $1/\sigma^* - 1$ for any smooth technology, so a regression of the local $\ln w$ on u and u^2 gives $\sigma^* = 1/(1 + b_1)$ from the linear coefficient b_1 : 0.708 (wild cluster standard error by model size 0.001), between 0.701 and 0.711 under every bandwidth, and 0.670 with embedding parameters counted. Section III reports this estimate; its standard error treats the grid points as the design and excludes smoothing bias. The quadratic coefficient, 0.020 (0.001), shows that $\ln w$ is convex in u , which no member of the κ family allows, since each makes $\ln w$ linear in u ; Online Appendix F6 uses the convexity to sign the error of parametric wedges at high M .

Drift with compute.—Within Chinchilla and Llama 3 the per-budget estimates fall with compute, by 0.072 (s.e. 0.026) and 0.053 (0.010) per decade. Marin shows no drift over its two decades, and Porian et al.’s unannealed profiles drift the other way. Neither the Chinchilla form nor the κ family lets σ^* vary along the path, and under those constant- σ^* technologies fitted to Chinchilla and Llama 3 the estimator’s noise-free per-budget bias lies within $[-0.005, +0.002]$, so the drift is not a finite-grid artifact. It is, however, window-sensitive in Chinchilla (−0.034 to −0.096 per decade), and no design spans more than 2.7 decades of compute.

Porian et al.’s profiles.—These give σ^* near 0.51, well below every other design, and are window-sensitive: 0.45 to 0.55 across windows and 0.33 with a global quadratic. They are read off constant-learning-rate runs without cooldown. The gap between such a loss and an annealed loss plausibly varies with D/N at fixed compute, which bends the profile. We therefore treat them as IsoFLOP-like rather than IsoFLOP designs and keep them out of the range reported in Section III; with them, the random-effects mean across designs falls from 0.695 to 0.646 and heterogeneity becomes large ($Q = 189$ on 7 degrees of freedom).

TABLE D8—THE MODEL-FREE ELASTICITY: SPECIFICATION SENSITIVITY AND MONTE CARLO

<i>Panel A. Random-effects σ^* under alternative specifications</i>									
	Primary	Half-width h			Global	Cubic	Frontier	Iterated	Indep.
Design	(s.e.)	0.8	1.25	1.5	quadratic	($h = 1$)	smoothers	window	code
Chinchilla	0.673 (0.027)	0.669	0.714	0.691	0.671	0.665	0.675–0.679	0.701	0.690
Llama 3	0.660 (0.023)	0.677	0.657	0.654	0.642	0.655	0.660–0.663	0.666	0.669
Marin, Comma	0.700 (0.027)	0.685	0.721	0.723	0.699	0.668	0.702–0.706	0.707	0.690
Marin, DCLM	0.713 (0.027)	0.703	0.691	0.696	0.679	0.688	0.710–0.713	0.682	0.694
Marin, Nemotron	0.705 (0.023)	0.691	0.696	0.703	0.689	0.662	0.703–0.705	0.692	0.691
Porian, RW	0.518 (0.017)	0.554	0.513	0.474	0.332	0.511	0.518–0.519	0.492	0.473
Porian, OWT2	0.505 (0.014)	0.512	0.524	0.451	0.341	0.488	0.498–0.503	0.506	0.484
<i>Panel B. Monte Carlo at each design's fitted κ-family technology</i>									
	True σ^*	Bias		Coverage, 95%		Noise-free bias (per budget)			
		RE	Pooled	RE	Pooled				
Chinchilla	0.667	−0.005	−0.004	0.92	0.85	−0.004 to +0.002			
Llama 3	0.693	−0.001	0.000	0.98	0.91	0.000 to +0.002			
Marin, DCLM	0.663	+0.005	+0.001	0.93	0.82	−0.007 to +0.010			

Note: Model-free σ^* from $1/\sigma^* - 1 = L_{nn}|_C/(2|dL^*/dc|)$ (Proposition 4), estimated per IsoFLOP budget from a local polynomial in $\ln N$ on runs within h of a pooled Approach-2 path (primary: quadratic, $h = 1$; log-cubic frontier in $\ln C$, log-quadratic for Marin Comma) and pooled across budgets by a DerSimonian–Laird random-effects (RE) mean (DerSimonian and Laird, 1986). Budgets are valid if the minimum is interior with at least two runs on each side: 9/9 (Chinchilla), 8/10 (Llama 3), 5/7, 7/7, 7/8 (Marin) and 12/12 (Porian). Standard errors: design-conditional wild bootstrap with budget-level shocks, 999 draws (199 for Panel A’s variants). Global quadratic: all runs of a budget. Frontier smoother: log-quadratic, cubic in levels and power-law frontiers. Iterated window: centered on each budget’s own argmin. Independent code: an equal-weighted re-implementation that shares no code with the primary. RW: RefinedWeb; OWT2: OpenWebText2. Pooled F tests of a cubic term never reject at 5 percent (p from 0.055 to 0.85). Marin with configuration N instead of $C_b/(6D)$: 0.658, 0.642 and 0.655. Porian et al.’s profiles are read off constant-learning-rate runs without cooldown. Panel B: 100 replications with 149 bootstrap draws each; the truth is each design’s fitted κ -family technology plus Gaussian noise at the design’s robust residual standard deviation; pooled: ratio of weighted sums with a basic bootstrap interval. Noise-free bias: range over budgets under that technology without noise.

Source: Hoffmann et al. (2022); Czech (2026); Marin Community (2026); Porian et al. (2024); authors’ calculations.

D4. The Technology across Sweeps

Specification tests.—Table D9 reports the tests for the seven sweep–corpus technologies from five studies. The restriction $\kappa = 1$ is rejected in every sweep. CES is not rejected by the Huber-based Wald test in any sweep (p between 0.22 and 0.90). The Gaussian likelihood ratio rejects it on Chinchilla and is borderline on Farseeer ($p = 0.052$) and the OLMo ladder (0.067), but on Chinchilla the rejection does not survive calibration for heavy tails (Appendix D2). Rank-one curvature, which the κ family imposes on a translog in $\ln(L - E)$, is rejected in two sweeps when E comes from the Chinchilla form and in one (Farseeer) when it comes from the κ family.

The outer exponent free.—Table D10 reports σ_κ^* with design-conditional wild standard errors and under changes of sample, convention and irreducible loss. On Farseeer it is stable across size exclusions and samples (0.694 to 0.716) except for the parameter-count convention: counting both embedding matrices lowers it to 0.668. On Chinchilla it falls to 0.690 and 0.648 when the smallest tenth and quarter of the runs are dropped, and to 0.664 when the five outliers are kept. The small sweeps are fragile. On Gadre et al.’s corpora σ_κ^* moves between 0.48 and 0.70 when one or two of the six sizes are dropped, leaving 18 to 27 runs. The trade-off between E and κ is immaterial inside the confidence sets for E on Chinchilla, Farseeer and the OLMo ladder (at most 0.006) but material on Gadre et al.’s RedPajama (0.606 to 0.685) and on Muennighoff et al. (0.503 to 0.555). Only one variant moves a small sweep near 0.70: RedPajama without its two smallest sizes, from 19 runs with a standard error of 0.097. The heterogeneity of σ_κ^* across sweeps in Section III may therefore partly reflect how the κ family fits small designs without IsoFLOP profiles, rather than technology.

The Chinchilla form.—Table D11 reports the $\kappa = 1$ estimates under alternative conventions, outputs and samples. Conventions matter most for the allocation exponent. Counting Farseeer’s two untied embedding matrices moves a from 0.526 to 0.411 and σ^* from 0.772 to 0.724. Dropping the two single-epoch runs in which Muennighoff et al. (2023) trained 1.1 and 2.8 billion parameters on 100 million tokens moves a from 0.535 to 0.360 and $M^*(10^{21})$ from 60 to 139. With task bits per byte instead of C4 cross-entropy, the OLMo ladder’s σ^* falls from 0.825 to 0.763 (and σ_κ^* from 0.544 to 0.512). Ordinality guarantees invariance only to transformations of the same output; two outputs can have different isoquants, so such differences can reflect technology as well as sampling noise and misspecification.

Neutrality of data quality.—Table D12 nests hypotheses about how better data enter. What matters for allocation and for the wedge is the tilt $\ln(A_r/B_r)$, which is ordinal; asymptote shifts and neutral scalings of reducible loss leave the isoquant map unchanged. Across the three web corpora of Gadre et al. (2025), a Hicks-neutral scaling of reducible loss with corpus-specific asymptotes is not rejected ($p = 0.90$) and explains 96.8 percent of the cross-corpus variation. The test has little power: the data- and parameter-augmenting alternatives are not

TABLE D9—SPECIFICATION TESTS ACROSS SWEEPS: OUTER EXPONENT, CES AND RANK-ONE CURVATURE

Sweep, corpus	n	Outer exponent		CES ($\alpha = \beta$)			Rank one, p	
		$\hat{\kappa}$ (s.e.)	$p(\kappa = 1)$	$\hat{\alpha} - \hat{\beta}$ (s.e.)	Wald p	LR p	$\hat{E}_{\kappa=1}$	$\hat{E}_{\kappa}$
Chinchilla	240	0.774 (0.060)	<0.001	−0.020 (0.029)	0.49	0.001 ^a	0.007	0.69
Farseer	404	0.472 (0.014)	<0.001	−0.031 (0.025)	0.22	0.052	≤0.003	≤0.003
Gadre, C4	34	0.353 (0.080)	<0.001	−0.010 (0.080)	0.90	0.65	0.68	0.21
Gadre, RedPajama	35	0.395 (0.139)	<0.001	−0.030 (0.055)	0.59	0.35	0.97	0.82
Gadre, RefinedWeb	35	0.355 (0.120)	<0.001	−0.067 (0.057)	0.24	0.48	0.73	0.80
OLMo ladder	30	0.223 (0.022)	<0.001	0.110 (0.146)	0.45	0.067	0.95	0.053
Muennighoff et al.	33	0.422 (0.104)	<0.001	−0.051 (0.178)	0.77	0.58	0.65	0.75

Note: Seven sweep–corpus technologies from five studies. Outer exponent: $L = E + (AN^{-a_1} + BD^{-b_1})^\kappa$ fitted by Huber loss on $\ln L$, pairs-bootstrap standard errors (m2 estimates; design-conditional wild standard errors are in Online Appendix Table D10); $p(\kappa = 1)$: Wald. CES: Wald test of $\alpha = \beta$ in the Chinchilla form with Huber loss, and the Gaussian likelihood-ratio (LR) test against χ^2_1 . The corresponding σ^* are in Online Appendix Table D10. ^aHeavy tails make the Gaussian LR miscalibrated on Chinchilla: its restricted-bootstrap p -value is 0.17 and heteroskedasticity-robust Wald p -values are 0.060 (HC1) and 0.089 (HC3) (Online Appendix Table D3). Rank one: bootstrap test that the second-order matrix of a translog in $\ln(L - \hat{E})$ has rank one, as the κ family implies, with $\hat{E}$ from the Chinchilla form or from the κ family; 300 null draws, so the smallest attainable p is 0.003. Rank-one curvature is rejected at 5 percent in two of seven sweeps with $\hat{E}_{\kappa=1}$ and in one (Farseer) with $\hat{E}_{\kappa}$.

Source: Hoffmann et al. (2022); Li et al. (2025b); Gadre et al. (2025); Bhagia et al. (2025); Muennighoff et al. (2023); authors’ calculations.

rejected either ($p = 0.13$ and 0.15). Across the 25 DataDecide recipes (Magnusson et al., 2025) every restriction is rejected, at the resolution of the bootstrap (0.01 with 99 draws, 0.05 with 19 draws). Asymptote shifts alone explain 98.1 percent of the variation, and a neutral scale brings this to 99.7 percent. The remaining tilt is small in fit but not in allocation: it spans 0.22 to 0.26 across recipes, implies compute-optimal ratios that differ by a factor of 2.9 to 3.4, and moves revealed wedges by a factor of 1.25 to 1.29 at the same inputs. Because DataDecide’s checkpoints come before the learning-rate schedule ends, the comparison assumes that the unfinished schedule affects all recipes alike. DeepSeek-AI et al. (2024) report the related finding that higher-quality data shift the compute-optimal allocation toward model size.

Functional form on Farseer.—Table D13 compares the Chinchilla form with richer forms on the Farseer grid (Li et al., 2025b). The Chinchilla form extrapolates poorly in compute: its out-of-sample error on the top decile of compute is 8 to 10 times that of Farseer’s nine-parameter form. On the local expansion path, the model-free estimate of σ^* is 0.708, against 0.710 with κ free and 0.772 in the Chinchilla form.

D5. The Compute Cost of Over-Training

Practitioners use the elasticity of substitution through one quantity: the extra training compute needed to reach the loss of the compute-optimal model while training at k times the compute-optimal ratio M^* . Table D14 reports it. Over-training by a factor of 10 costs 39 percent ($\sigma^* = 0.8$) to 124 percent ($\sigma^* = 0.6$)

TABLE D10—ROBUSTNESS OF THE ELASTICITY OF SUBSTITUTION WITH THE OUTER EXPONENT FREE

Technology	Variant	n	σ_κ^* (s.e.)	$\hat{\kappa}$	σ^* ,		σ_κ^* with E	
					$\kappa = 1$	at $\hat{E}_{\kappa=1}$	over E set	
Chinchilla	Baseline	240	0.701 (0.012)	0.77	0.737	0.703	0.697–0.703	
	Drop smallest 10% of N	216	0.690 (0.014)	0.75	0.729	0.690		
	Drop smallest 25% of N	180	0.648 (0.017)	0.69	0.702	0.654		
	All 245 runs	245	0.664 (0.016)	0.70	0.714	0.671		
Farseer	Baseline	404	0.710 (0.004)	0.47	0.772	0.696	0.709–0.711	
	Drop smallest size	389	0.710 (0.004)	0.47	0.773	0.694		
	Drop 2 smallest sizes	374	0.709 (0.004)	0.48	0.770	0.692		
	Drop 5 smallest sizes	324	0.710 (0.005)	0.46	0.774	0.687		
	N with embeddings	404	0.668 (0.004)	0.52	0.724	0.648		
	Drop $M > 1,000$	389	0.716 (0.004)	0.48	0.785	0.705		
	Drop $M < 2$	355	0.694 (0.002)	0.43	0.797	0.690		
Gadre, C4	Baseline	34	0.607 (0.045)	0.35	0.795	0.602	0.600–0.614	
	Drop smallest size	26	0.528 (0.050)	0.41	0.720	0.535		
	Drop 2 smallest sizes	18	0.493 (0.093)	0.30	0.729	0.542		
Gadre, RedPajama	Baseline	35	0.623 (0.052)	0.39	0.795	0.679	0.606–0.685	
	Drop smallest size	27	0.574 (0.071)	0.38	0.771	0.568		
	Drop 2 smallest sizes	19	0.696 (0.097)	0.57	0.783	0.658		
Gadre, RefinedWeb	Baseline	35	0.594 (0.051)	0.36	0.796	0.595	0.580–0.605	
	Drop smallest size	27	0.545 (0.082)	0.38	0.758	0.545		
	Drop 2 smallest sizes	19	0.484 (0.080)	0.26	0.764	0.480		
OLMo ladder	Baseline	30	0.544 (0.019)	0.22	0.825	0.541	0.541–0.545	
	Drop smallest size	24	0.548 (0.027)	0.24	0.810	0.526		
	Drop 2 smallest sizes	18	0.518 (0.018)	0.32	0.709	0.506		
	Task bits per byte	30	0.512 (0.059)	0.33	0.763	0.487		
Muennighoff	Baseline	33	0.511 (0.047)	0.42	0.735	0.558	0.503–0.555	
	Drop smallest size	32	0.527 (0.051)	0.50	0.701	0.527		
	Drop 2 smallest sizes	31	0.536 (0.050)	0.52	0.693	0.514		
	Drop $D/N < 0.4$	31	0.566 (0.058)	0.51	0.726	0.612		

Note: $\sigma_\kappa^* = 2/(2 + a_1 + b_1)$ from $L = E + (AN^{-a_1} + BD^{-b_1})^\kappa$ fitted by Huber loss on $\ln L$. Standard errors: design-conditional wild bootstrap (Feng–He–Hu weights on leverage-adjusted residuals; weights of Webb (2023) by model size for Farseer and by (size, multiplier) cell for the OLMo ladder), 499 draws for baselines and 199 for variants. σ^* , $\kappa = 1$: the Chinchilla form on the same sample. At $\hat{E}_{\kappa=1}$: E fixed at the Chinchilla-form estimate and the rest refitted. Over E set: range of σ_κ^* over the values of E whose profile Laplace quasi-LR is below the χ_1^2 95 percent critical value (descriptive; a 36-point grid in E refined with 25 points around the set). Farseer’s N with embeddings counts both untied embedding matrices.

Source: Hoffmann et al. (2022); Li et al. (2025b); Gadre et al. (2025); Bhagia et al. (2025); Muennighoff et al. (2023); authors’ calculations.

TABLE D11—ROBUSTNESS OF THE TECHNOLOGY ESTIMATES ACROSS SWEEPS

Sweep and specification	Huber			Gaussian NLS
	σ^*	a	$M^*(10^{21})$	σ^*
Farseer: baseline	0.772 (0.010)	0.526 (0.022)	24.8	0.769 (0.006)
N including embeddings	0.724 (0.011)	0.411 (0.017)	19.8	0.721 (0.007)
Training loss (nats per token)	0.785 (0.012)	0.535 (0.026)	18.1	0.786 (0.007)
Clusters = model size	0.772 (0.009)	0.526 (0.014)	24.8	0.769 (0.005)
Gadre et al., C4: baseline	0.795 (0.020)	0.509 (0.072)	6.9	0.803 (0.020)
In-distribution evaluation	0.799 (0.019)	0.487 (0.078)	9.8	0.806 (0.017)
Non-embedding N	0.806 (0.016)	0.534 (0.069)	11.0	0.816 (0.018)
Gadre et al., RedPajama: baseline	0.795 (0.013)	0.529 (0.045)	5.8	0.809 (0.014)
In-distribution evaluation	0.753 (0.018)	0.559 (0.083)	8.0	0.772 (0.018)
Non-embedding N	0.808 (0.015)	0.586 (0.042)	4.8	0.821 (0.014)
Gadre et al., RefinedWeb: baseline	0.796 (0.015)	0.566 (0.055)	3.4	0.806 (0.016)
In-distribution evaluation	0.805 (0.018)	0.564 (0.055)	2.8	0.817 (0.015)
Non-embedding N	0.806 (0.014)	0.606 (0.059)	3.8	0.818 (0.014)
OLMo ladder: baseline	0.825 (0.034)	0.370 (0.177)	48.3	0.833 (0.024)
Task bits per byte as output	0.763 (0.030)	0.353 (0.116)	85.4	0.770 (0.033)
Muennighoff et al.: single epoch	0.735 (0.031)	0.535 (0.126)	59.6	0.723 (0.031)
Single epoch, $D/N \geq 0.4$	0.726 (0.016)	0.360 (0.086)	139	0.711 (0.018)
Up to 4 epochs	0.732 (0.014)	0.522 (0.067)	71.6	0.719 (0.014)

Note: Chinchilla functional form ($\kappa = 1$); bootstrap standard errors in parentheses (400 draws for baselines, 200 for variants). Farseer (Li et al., 2025b): N including embeddings adds the two untied 65,536-row embedding matrices to the non-embedding count; training loss replaces English bits per character. Gadre et al. (2025): in-distribution evaluation uses the training corpus’s Paloma subset instead of C4. OLMo ladder (Bhagia et al., 2025): task bits per byte instead of C4 cross-entropy. Muennighoff et al. (2023): $D/N \geq 0.4$ drops the two single-epoch runs trained on 100 million tokens with 1.1 and 2.8 billion parameters, the exclusion rule that Besiroglu et al. (2024) apply to the Chinchilla data; up to 4 epochs treats repeated passes as fresh tokens.

Source: Run-level data released with each sweep; authors’ estimates.

TABLE D12—IS BETTER DATA HICKS-NEUTRAL? NESTED TESTS ACROSS CORPORA AND RECIPES

Model for corpus or recipe r	Parameters	Share explained (%)	LR	df	p (wild)
<i>Panel A. Gadre et al. (2025): three web corpora (104 runs, 37 clusters)</i>					
(0) One technology	5	0.0	30.7	6	≤ 0.002
(iv) Asymptote shift E_r only	7	89.9	2.6	4	0.004
(i) Hicks-neutral, common E	7	72.7	8.6	4	≤ 0.002
(ii) Data-augmenting, common E	7	64.1	11.4	4	≤ 0.002
(iii) Parameter-augmenting, common E	7	75.0	7.8	4	≤ 0.002
(i') Hicks-neutral + E_r	9	96.8	0.0	2	0.896
(ii') Data-augmenting + E_r	9	95.4	0.5	2	0.130
(iii') Parameter-augmenting + E_r	9	95.9	0.4	2	0.150
Common exponents (A_r, B_r, E_r free)	11	96.9	1.2	4	0.090
(v) Exponents vary	15	100.0			
<i>Panel B. DataDecide: 25 recipes (21,888 checkpoints, 44 clusters)</i>					
(0) One technology	5	0.0	58,185	72	≤ 0.050
(iv) Asymptote shift E_r only	29	98.1	4,465	48	≤ 0.010
(i) Hicks-neutral, common E	29	90.4	17,732	48	≤ 0.050
(ii) Data-augmenting, common E	29	92.6	14,600	48	≤ 0.050
(iii) Parameter-augmenting, common E	29	89.4	18,964	48	≤ 0.050
(i') Hicks-neutral + E_r	53	99.7	145	24	≤ 0.010
(ii') Data-augmenting + E_r	53	99.7	69	24	≤ 0.010
(iii') Parameter-augmenting + E_r	53	99.5	738	24	≤ 0.010
Common exponents (A_r, B_r, E_r free)	77	99.8	705	48	≤ 0.010
(v) Exponents vary	125	100.0			

Note: All models are evaluated on the same C4 validation set. Models nest in $L_r = E_r + A_r N^{-\alpha_r} + B_r D^{-\beta_r}$: (i) Hicks-neutral in reducible loss, $A_r = e^{-\omega_r} A$ and $B_r = e^{-\omega_r} B$; (ii) only B_r varies; (iii) only A_r varies; (iv) only E_r varies; primes add recipe-specific E_r . Share explained = $(\text{SSR}_0 - \text{SSR}_m) / (\text{SSR}_0 - \text{SSR}_v)$, where model (0) is one common technology and (v) separate fits. LR = $n \ln(\text{SSR}_m / \text{SSR}_{CE})$ against the common-exponent model (for that row, against (v)), from Gaussian NLS on log loss. p -values from a wild cluster restricted bootstrap with Rademacher weights (clusters: size $\times$ token-multiplier cells for Gadre et al., 499 draws; size $\times$ seed cells for DataDecide, 99 draws for the key tests and 19 for the common- E models; $\leq$ marks the smallest attainable value). Wald tests with a cell-level pairs-bootstrap covariance confirm every DataDecide rejection and, for Gadre et al., the non-rejection of (i') and the rejection of (iv); they disagree across estimators on (ii'). DataDecide uses intermediate checkpoints with $D/N \geq 5$, evaluated before the learning-rate schedule has finished; the comparisons across recipes assume that this schedule effect is common to all recipes.

Source: Gadre et al. (2025); Magnusson et al. (2025); authors' estimates.

TABLE D13—FARSEER: THE CHINCHILLA FORM AGAINST RICHER FUNCTIONAL FORMS

Model	k	In sample		Out of sample: RMSE $\times 10^3$	
		RMSE $\times 10^3$	BIC	Largest N	Top C decile
Chinchilla, Huber	5	8.74	−2,647	12.65	24.75
Chinchilla, NLS	5	8.50	−2,670	13.79	21.95
κ free, NLS	6	2.47	−3,663	3.57	5.23
Translog, E free, NLS	7	2.05	−3,806	3.32	4.69
Farseer Eq. 3, NLS	9	1.76	−3,916	2.63	2.58

Note: The Farseer grid of [Li et al. \(2025b\)](#): 404 runs, output English bits per character; RMSE of log loss; k is the number of parameters. The translog is quadratic in $(\ln N, \ln D)$ for $\ln(L - E)$. Farseer’s Eq. 3 is $L = e^{a_3 N^{\gamma_3} + b_3} + e^{a_2 N^{\gamma_2} + b_2} D - e^{a_1 N^{\gamma_1} + b_1}$, fitted here by NLS on log loss rather than the authors’ differential procedure. Out-of-sample fits start only from training-sample values; the splits hold out the four largest model sizes ($N \geq 3.8 \times 10^9$) or the top decile of $6ND$. The like-for-like model-free comparison is on the local expansion path ($w = 1$): its first-derivative estimate of σ^* is 0.708 (wild cluster standard error 0.001; 0.701–0.711 across bandwidths), against 0.710 for the κ -free form and 0.772 for the Chinchilla form; a Hessian-based estimate on the same path is 0.703 (0.013) at the primary bandwidth but moves between 0.66 and 0.73 with the bandwidth. Eq. 3 implies $\sigma^*(C)$ falling from 0.745 at 10^{18} to 0.672 at 10^{22} FLOP. Eq. 3 implies a compute-optimal ratio M^* of 32, 25, 28, 43, and 82 at 10^{19} to 10^{23} FLOP; the last two are extrapolations.

Source: [Li et al. \(2025b\)](#); authors’ estimates.

more compute; by a factor of 100 it costs 3.5 to 14 times the minimum, and by 1,000 it costs 14 to 129 times. The curvature matters more as k grows. At $k = 100$, $\sigma^* = 0.70$ implies 7.2 times the minimum compute, and $\sigma^* = 0.74$, the value that the Chinchilla form imposes on the Chinchilla data, implies 5.4 times. The entries assume a homothetic split of the exponents; the splits of [Besiroglu et al. \(2024\)](#) and [Hoffmann et al. \(2022\)](#) change them by at most 4.5 and 19 percent for $k \leq 100$.

Figure D2 compares these curves with the compute-overhead curve of [de Vries \(2023\)](#), which lies between our curves for $\sigma^* = 0.74$ and 0.80; with his parameters, a model at 30 percent of the compute-optimal size needs 99 percent more compute, as our formula implies. The implied wedge $w = k^{1/\sigma^* - 1}$ links the table to Section IV. A developer who trains 100 times beyond M^* reveals an expenditure share of the value of compactness, $s = (w - 1)/w$, of 0.68 to 0.95, depending on σ^* .

D6. Measurement and Flexible Inputs

The largest disagreement in the scaling literature concerns the allocation exponent: $N^* \propto C^{0.73}$ in [Kaplan et al. \(2020\)](#) against about 0.5 in [Hoffmann et al. \(2022\)](#). [Porian et al. \(2024\)](#) trace it to a sequence of changes in training practice, and [Pearce and Song \(2024\)](#) to the treatment of embedding parameters. Both are classic problems in production-function estimation. Counting the parameters that cost FLOPs, such as the output head, is input measurement, whose bias depends on how the measurement error correlates with scale ([Collard-Wexler and De Loecker, 2016](#)). Warmup, learning-rate schedule, learning rate and batch size

TABLE D14—THE COMPUTE COST OF OVER-TRAINING

σ^*	Training compute $C/C_{\min}$ at $M = k M^*$				Implied wedge $w = k^{1/\sigma^*} - 1$				Share s
	$k = 2$	$k = 10$	$k = 100$	$k = 1,000$	$k = 2$	$k = 10$	$k = 100$	$k = 1,000$	
0.60	1.08	2.24	14.3	129	1.59	4.64	21.5	100	0.95
0.70	1.05	1.73	7.23	49.8	1.35	2.68	7.20	19.3	0.86
0.74	1.04	1.57	5.42	31.3	1.28	2.25	5.04	11.3	0.80
0.80	1.03	1.39	3.52	14.5	1.19	1.78	3.16	5.62	0.68

Note: Compute needed to reach the loss of the compute-optimal model when training at k times the compute-optimal tokens-per-parameter ratio, relative to the minimum, in the Chinchilla family: $C/C_{\min} = ((\alpha + \beta w)/(\alpha + \beta))^{1/\gamma} w^{-1/\alpha}$ with $\ln w = (1/\sigma^* - 1) \ln k$. Entries are for the homothetic case $\alpha = \beta = \rho = 1/\sigma^* - 1$, where $C/C_{\min} = \cosh(\rho \ln k/2)^{2/\rho}$; every entry was checked by direct minimization. At the same σ^* , the α/β split of Besiroglu et al. (2024) changes the entries by at most 4.5 percent and that of Hoffmann et al. (2022) by at most 19 percent for $k \leq 100$ (k relative to M^* at the model's own compute). $s = (w - 1)/w$ is the expenditure share of the value of compactness that rationalizes over-training by k (Proposition 2); for a developer that serves its model it is the planned serving share of lifetime cost. With the parameters of de Vries (2023) ($\alpha = 0.32$, $\beta = 0.28$, $\sigma^* = 0.77$), a model at 30 percent of the compute-optimal size needs 6.6 times the tokens and 99 percent more compute ($k = 21.1$); the formula gives 99 percent.

Source: Authors' calculations; de Vries (2023).

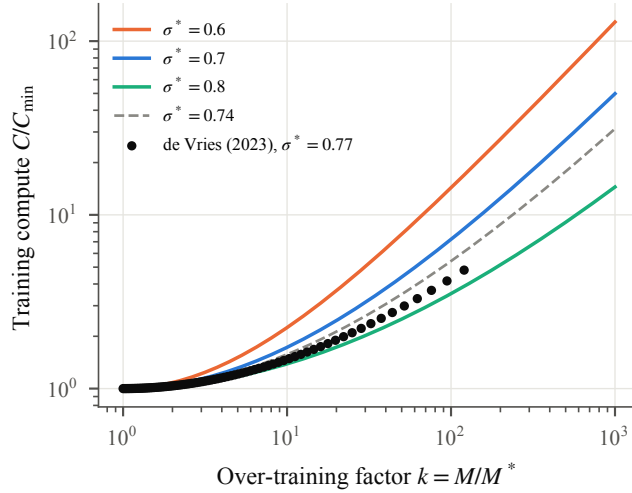


FIGURE D2. THE COMPUTE COST OF OVER-TRAINING

Note: Training compute relative to the minimum needed to reach the compute-optimal loss, against the over-training factor $k = M/M^*$, for four values of σ^* (homothetic case, as in Table D14). Dots: the curve of de Vries (2023), computed from his parameters ($\sigma^* = 0.77$) and expressed in the same units.

Source: Authors' calculations; de Vries (2023).

are flexible inputs; when they are not re-optimized at each scale, they are not concentrated out of the production function (Akerberg, Caves and Frazer, 2015; Gandhi, Navarro and Rivers, 2020).

Decomposition.—Re-estimating Porian et al.’s path from their 975 released runs reproduces all ten published exponents to within 0.003 (Table D15, panel A). The exponent falls from 0.837 to 0.496 on RefinedWeb and from 0.862 to 0.517 on OpenWebText2. Counting the head accounts for 39 and 47 percent of the drop, and flexible inputs for the remaining 53 to 61 percent. Because a sequential decomposition depends on the order of the steps, panel B holds the runs fixed and changes only the count. Counting the head lowers a by 0.12 to 0.17 whether or not the runs are tuned, and tuning lowers it by 0.19 to 0.23 under every count, so the two components are nearly additive. The switch from training to validation loss between steps 1 and 2 accounts for at most 0.005 of the measurement step. These models have 5M to 901M parameters, and the shares need not hold at larger scale.

Input measurement.—Let the true technology be $L = E + AN^{-\alpha} + BD^{-\beta}$, and let the researcher count N_m parameters when the true count is $N = N_m + X(N_m)$. The omitted component has share $q \equiv X/N$ and a constant scale elasticity $\theta_X \equiv d \ln X / d \ln N_m < 1$. The researcher measures compute as $C_m = 6N_m D$ and takes the IsoFLOP argmin in measured units: $n_m \equiv \ln N_m$ minimizes L subject to $\ln D = c_m - \ln 6 - n_m$, with $c_m \equiv \ln C_m$. Because $d \ln N / d n_m = 1 - q(1 - \theta_X) \equiv \vartheta$, the first-order condition is $\alpha u \vartheta = \beta v$, with $u = AN^{-\alpha}$ and $v = BD^{-\beta}$. Differentiating its log along the researcher’s path, with $d \ln u / d n_m = -\alpha \vartheta$, $d \ln v = \beta (d n_m - d c_m)$ and $\zeta \equiv d \ln \vartheta / d n_m$, gives $(\zeta - \alpha \vartheta) d n_m / d c_m = \beta d n_m / d c_m - \beta$, so the local measured allocation exponent is

$$(D1) \quad a_m \equiv \frac{d n_m}{d c_m} = \frac{\beta}{\beta + \alpha \vartheta - \zeta}, \quad \vartheta \equiv 1 - q(1 - \theta_X), \quad \zeta = \frac{(1 - \theta_X) q (\vartheta - \theta_X)}{\vartheta} \geq 0.$$

Since $\vartheta \leq 1$ and $\zeta \geq 0$, $a_m \geq a = \beta / (\alpha + \beta)$, with equality when $q = 0$ or $\theta_X = 1$. A proportional error ($\theta_X = 1$) moves only the level of the path. The bias comes from the part of the error that is correlated with scale, as in Collard-Wexler and De Loecker (2016), and it vanishes at scale as the omitted share falls: a local elasticity estimated over a restricted size range is not the structural one, which is the functional-form lesson of Nerlove (1963). As $q \rightarrow 1$, $a_m \rightarrow \beta / (\beta + \alpha \theta_X)$, the small-scale limit of Pearce and Song (2024). The closed form matches brute-force argmins to within 2×10^{-7} in $\ln N$.

With Pearce and Song’s embedding map ($\theta_X = 1/3$), equation (D1) predicts a measured exponent over Kaplan et al.’s size range of 0.792 under Besiroglu et al.’s parameters and 0.757 under Hoffmann et al.’s, against the 0.78 and 0.74 that Pearce and Song obtain by simulation. The small-scale limit is sensitive to how the omitted component scales: under Besiroglu et al.’s parameters it is 0.76 at $\theta_X = 1/3$ and 0.68 at $\theta_X = 1/2$. On Porian et al.’s ladder, where the omitted component is the head (its share falls from 0.94 at 5M parameters to 0.08 at

901M, with a fitted θ_X of 0.36), the formula gets the sign and order of magnitude of the measurement bias right but over-predicts it, by 20 to 50 percent in a semi-synthetic check on the actual design. The Chinchilla data are not immune: with non-embedding counts the allocation exponent rises from 0.514 to 0.556 (paired difference 0.042, s.e. 0.006) and σ^* from 0.737 to 0.753.

Flexible inputs.—Let observed loss be $L_{\text{obs}} = L^* e^{\iota(N,D)}$, where L^* is the technology with the hyperparameters concentrated out and $\iota \geq 0$ is the inefficiency of the researcher’s hyperparameter policy. Along the IsoFLOP curve at compute C , write $f(n)$ for L^* at $d = c - \ln 6 - n$, with minimizer $n^*(c)$. Then $f' = -\alpha u + \beta v$ and $f'' = \alpha^2 u + \beta^2 v$, and at the minimizer $\alpha u = \beta v = \gamma R^*$, with R^* the reducible loss on the frontier, so $f'' = (\alpha + \beta)\gamma R^*$. The observed argmin solves $f'(n) + \Delta(n) = 0$, where $\Delta \equiv \partial(L_{\text{obs}} - L^*)/\partial n$ at fixed compute. To first order in Δ , the log argmin shifts by $-\Delta(C)/f''(C)$, and differentiating in $\ln C$ gives

$$(D2) \quad a_{\text{obs}} - a = -\frac{\partial [\Delta/f'']}{\partial \ln C}.$$

At the minimizer $\Delta = L^* e^{\iota}(\iota_n - \iota_d)$, with $\iota_n \equiv \partial \iota / \partial \ln N$ and $\iota_d \equiv \partial \iota / \partial \ln D$, so inefficiency that depends on the inputs only through compute ($\iota_n = \iota_d$) leaves a unbiased. With small, constant log-gradients, $L^* = E + R^*$ and $R^* \propto C^{-\gamma}$ give $a_{\text{obs}} - a = -(\iota_n - \iota_d)E/[(\alpha + \beta)R^*(C)]$. The bias in a is first order in the gradient, although by the envelope theorem the effect on the loss level is second order in the hyperparameter error. Under the Besiroglu et al. (2024) technology, equation (D2) matches simulated IsoFLOP argmins to within 0.003 in a .

On Porian et al.’s runs the formula explains about half to all (47 to 114 percent, depending on how f'' is estimated) of the three flexible-input steps. The mechanism is Kaplan et al.’s fixed warmup of 1.57 billion tokens and large batch, which leave short runs with too few useful steps. At 2.5×10^{16} FLOP the untuned runs’ excess loss at the tuned optimum is 1.57 nats (the coincidence with the warmup length is numerical), and it vanishes by about 3×10^{18} FLOP. A penalty of this kind pushes small-budget argmins toward smaller models, by 0.7 to 1.2 log points, and steepens the fitted path (Figure D3, panel b).

Hyperparameters on the Step Law grid.—The Step Law grid of Li et al. (2025a), 1,911 runs in 17 (N, D) cells with up to 120 learning-rate and batch-size configurations per cell, shows the flexible inputs directly. Table D16 shows that this design does not identify the allocation exponent: the sum of squared residuals varies by less than 17 percent for E between 0.6 and 1.7, over which a ranges from 0.76 to 0.45. Holding E fixed, published hyperparameter rules shift a by -0.041 to $+0.076$, all within 0.7 residual-bootstrap standard errors. Choosing one random configuration per cell costs mainly precision for a (median bias -0.3 standard errors) but biases σ^* by -0.051 , or -3.7 standard errors. These data therefore give no reason to regard σ^* as the more robust object: relative to its own sampling error it moves more than a along the profile in E (7.0 against 2.8 standard errors).

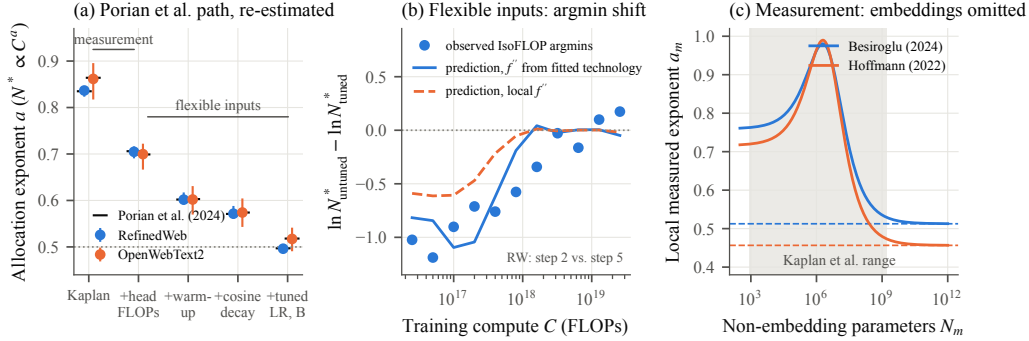


FIGURE D3. MEASUREMENT AND FLEXIBLE INPUTS IN THE ALLOCATION EXPONENT

Note: Panel (a): the allocation exponent re-estimated from Porian et al.’s released runs at each step of their decomposition, with 95 percent intervals, against their published values. Panel (b): the observed shift of the IsoFLOP argmin between untuned (step 2) and tuned (step 5) RefinedWeb runs, and the first-order prediction $-\Delta/f''$ of equation (D2) with f'' from the fitted technology or from local quadratics. Panel (c): the local measured exponent a_m of equation (D1) when embeddings are omitted from N , under two published technologies, with Kaplan et al.’s size range shaded and the true exponents dashed. *Source:* Porian et al. (2024); Pearce and Song (2024); authors’ calculations.

Inefficiency falls with training tokens but not detectably with parameters (Table D17, panel A): the scale of inefficiency has an elasticity of -0.23 to -0.26 with respect to D and an insignificant one with respect to N , so longer horizons flatten the loss surface in the hyperparameters. Because the grid is designed, its minimum measures the frontier directly, and the stochastic frontier adds only this variance function. By equation (D2), the implied bias for a researcher who uses random configurations is small (-0.012 in a). Panel B of the same table estimates the demand for the flexible inputs themselves. The unconditional learning-rate demand is close to Step Law’s published rule. Holding the batch fixed, the positive token elasticity of the optimal learning rate shrinks to between 0.02 and 0.05, the distinction between conditional and unconditional demand of Milgrom and Roberts (1996). This removes the positive sign of Step Law’s elasticity but not the gap to the negative elasticity of Bjorck et al. (2025): conditioning closes 26 to 43 percent of the gap to their joint estimate of -0.32 , and less against their steeper per-size estimates for small models.

The allocation exponent is thus the object most exposed to measurement conventions and to flexible inputs that are not re-optimized at each scale. Section VI draws the implications for how scaling studies should report it.

TABLE D15—MEASUREMENT AND FLEXIBLE INPUTS IN THE ALLOCATION EXPONENT

<i>Panel A. Porian et al.’s path, re-estimated from the 975 released runs</i>						
Step	RefinedWeb			OpenWebText2		
	$\hat{a}$	Published	$\Delta\hat{a}$	$\hat{a}$	Published	$\Delta\hat{a}$
1. Kaplan reproduction	0.837	0.835		0.862	0.864	
	[0.823, 0.845]			[0.817, 0.896]		
2. + head in N and C	0.705	0.706	−0.132	0.700	0.699	−0.162
	[0.691, 0.716]			[0.666, 0.722]		
3. + warmup $\propto N$	0.602	0.602	−0.103	0.602	0.603	−0.098
	[0.591, 0.617]			[0.570, 0.631]		
4. + cosine decay	0.572	0.571	−0.030	0.574	0.574	−0.028
	[0.561, 0.588]			[0.543, 0.604]		
5. + tuned LR, batch, Adam decay	0.496	0.497	−0.075	0.517	0.518	−0.057
	[0.490, 0.504]			[0.491, 0.541]		
Share of step 2 in total			0.39			0.47
<i>Panel B. Same runs, only the parameter count changes (training loss)</i>						
	Kaplan setup	Tuned	Diff.	Kaplan setup	Tuned	Diff.
Non-embedding (Kaplan)	0.837	0.620	0.217	0.862	0.631	0.231
	(0.006)	(0.005)		(0.020)	(0.017)	
+ head (standard)	0.700	0.502	0.198	0.696	0.501	0.194
	(0.006)	(0.004)		(0.017)	(0.012)	
+ head, attention FLOPs	0.717	0.505	0.212	0.713	0.507	0.206
	(0.006)	(0.004)		(0.018)	(0.013)	
+ head, input embeddings	0.661	0.460	0.202	0.670	0.471	0.199
	(0.007)	(0.003)		(0.017)	(0.012)	

Note: a is the exponent of $N^* \propto C^a$ from the IsoFLOP procedure of Porian et al. (2024), re-implemented on their released runs (16 architectures, 5M–901M parameters; budgets $1.25 \times 10^{16} \cdot 2^i$, $i = 0, \dots, 11$). Panel A: step 2 is input measurement, steps 3–5 are flexible inputs; brackets, 95 percent intervals from the authors’ noise-and-interpolate bootstrap (1,000 draws); “Published” = their Table 1. Step 1 uses Kaplan’s count and training loss, steps 2–5 the standard count and validation loss; with training loss throughout, counting the head alone moves a by 0.137 (RefinedWeb) and 0.166 (OpenWebText2) on the step-1 runs (Panel B), so the switch of loss accounts for at most 0.005 of step 2. Step 5 tunes the learning rate, the batch and Adam’s second-moment decay rate at each size, with a constant learning rate. Panel B: “Kaplan setup” = untuned runs with Kaplan’s 1.57-billion-token warmup; “Tuned” = step-5 runs; training loss; bootstrap standard errors in parentheses. “Head, input embeddings” also counts the input and positional embeddings (the “total” count of Pearce and Song (2024)).

Source: Porian et al. (2024); authors’ calculations.

TABLE D16—HYPERPARAMETERS AS FLEXIBLE INPUTS: INEFFICIENCY AND THE ESTIMATED TECHNOLOGY

Policy	Inefficiency ι (percent)			E held at 1.4		E estimated	
	Mean	$\partial\iota/\partial\ln N$	$\partial\iota/\partial\ln D$	a	σ^*	a	σ^*
Frontier (best run)	0.00	0.00	0.00	0.573 (0.108)	0.790 (0.014)	0.563 (0.148)	0.786 (0.040)
Step Law rule	0.10	−0.05 (0.04)	0.01 (0.02)	0.532	0.794	0.517	0.789
Best of 16 random	0.21	−0.04 [0.08]	−0.06 [0.06]	0.558	0.791	0.551	0.788
Best of 4 random	0.70	−0.16 [0.25]	−0.20 [0.20]	0.555	0.786	0.589	0.793
One random	2.58	−0.44 [1.19]	−0.70 [0.71]	0.539	0.739	0.565	0.760
One fixed setting	0.29	−0.08 (0.08)	−0.03 (0.05)	0.619	0.780	0.515	0.748
Porian et al., base	0.41	0.32 (0.16)	−0.23 (0.11)	0.621	0.785	0.564	0.769
Porian et al., N -rule	0.77	0.72 (0.22)	−0.57 (0.14)	0.649	0.785	0.453	0.722
DeepSeek C -rule	0.75	−0.16 (0.06)	−0.02 (0.03)	0.543	0.793	0.488	0.774
Björck et al. rule	0.91	−0.57 (0.07)	0.04 (0.04)	0.576	0.782	0.548	0.773

Note: The Step Law grid of Li et al. (2025a): 1,911 runs in 17 (N, D) cells with N from 215 million to 1.07 billion non-embedding parameters and D from 4 to 100 billion tokens, up to 12 learning rates and 10 batch sizes per cell; the output is the smoothed final training loss. Each policy selects one run per cell (the grid point nearest its target); random rows are medians over 200 Monte Carlo draws that keep the best of k non-diverged configurations per cell (Monte Carlo standard deviations in brackets). Inefficiency gradients: OLS across the 17 cells, HC1 standard errors in parentheses. Technology: the Chinchilla form fitted by NLS on log loss, with residual-bootstrap standard errors for the frontier (200 draws). The profile likelihood in E is flat (sum of squared residuals within 17 percent for E between 0.6 and 1.7, over which a ranges from 0.76 to 0.45), so only differences across rows at fixed E are informative. N follows Step Law’s non-embedding convention, which biases the level of a upward; counting the output head moves the frontier a to 0.465 (E estimated) and 0.550 ($E = 1.4$). The Huber estimator on the frontier gives $E = 0.91$, $a = 0.689$, and $\sigma^* = 0.824$. $\iota = \ln L - \ln L^*$. “One fixed setting” uses the best single (learning rate, batch) pair for all cells. Rules: Step Law, the learning-rate and batch rules of Li et al. (2025a); DeepSeek, the compute-based rules of DeepSeek-AI et al. (2024); Björck et al., the learning-rate rule in (N, D) of Björck et al. (2025) at a fixed 0.5-million-token batch; Porian et al., the base setting and the N -based rule of Porian et al. (2024).

Source: Li et al. (2025a); authors’ estimates.

TABLE D17—THE STEP LAW GRID: INEFFICIENCY AND FLEXIBLE-INPUT DEMAND

<i>Panel A. Scale of inefficiency: elasticities of σ_u (1,730 runs, 17 cells)</i>				
	γ_N	γ_D		
Half-normal inefficiency	−0.038 (0.037)	−0.231 (0.020)		
Exponential inefficiency	−0.063 (0.053)	−0.260 (0.028)		
Cell-level OLS of $\ln \bar{\epsilon}_c$	−0.134 (0.108)	−0.266 (0.028)		
<i>Panel B. Learning-rate demand, $\ln \eta^*$</i>				
	$\ln N$	$\ln D$	$\ln B$	Observations
Unconditional, grid argmin	−0.823 (0.127)	0.288 (0.071)		17
Unconditional, smoothed argmin	−0.693 (0.050)	0.210 (0.025)		17
Conditional on batch, smoothed	−0.534 (0.027)	0.046 (0.015)	0.320 (0.034)	165
Conditional on batch, grid argmin	−0.612 (0.048)	0.016 (0.029)	0.295 (0.038)	157
Step Law’s rule	−0.713 (0.030)	0.307 (0.019)		
Björck et al.’s rule, fixed batch	−0.23	−0.32		
<i>Panel C. Decomposition of $d \ln \eta^* / d \ln D$</i>				
	Smoothed	Grid		
Share due to batch co-scaling	0.91 (0.15)	0.56 (0.24)		
Share of the published gap closed	0.26 (0.02)	0.43 (0.13)		

Note: The Step Law grid of Li et al. (2025a); N non-embedding parameters, D tokens, B batch size in tokens. Panel A: cost-type stochastic frontier $\ln L_{ij} = \mu_c + \epsilon_{ij} + \iota_{ij}$ with 17 unrestricted cell effects μ_c (the concentrated technology), noise $\epsilon \sim N(0, \sigma_\epsilon^2)$ and inefficiency $\iota \geq 0$ with $\ln \sigma_u = \gamma_0 + \gamma_N(\ln N - \bar{\ln N}) + \gamma_D(\ln D - \bar{\ln D})$ (Caudill, Ford and Gropper, 1995), on the non-diverged runs (loss at most 35 percent above the cell minimum); numerical-Hessian standard errors in parentheses; cell-clustered sandwich standard errors of (γ_N, γ_D) are (0.076, 0.032) for half-normal and (0.070, 0.023) for exponential inefficiency. Fixing σ_v at the median within-cell residual standard deviation (0.0017) gives γ_D of -0.229 and -0.262 . Because the grid is designed, its minimum measures the frontier directly: the estimated frontier coincides with the best run in each cell (mean absolute gap at most 0.11 percent), and the stochastic frontier adds only the variance function. Last row: OLS across cells of the log mean inefficiency (HC1 standard errors). Panel B: optimal learning rate in each cell regressed on $\ln N$, $\ln D$ and, holding the batch fixed, $\ln B$. Grid argmin: best run in the cell; smoothed: minimum of a quadratic in $(\ln \eta, \ln B)$ fitted to runs within 2 percent of the cell minimum. Standard errors: two-level bootstrap (cells, then each cell’s optimum; 1,000 draws) for unconditional demands, cell-clustered for conditional ones; Step Law’s own bootstrap for its rule; the rule of Björck et al. (2025) holds the batch fixed. With the batch fixed at 0.52 million tokens the elasticities are -0.495 (0.077) and 0.040 (0.047). The batch-size demand has a D -elasticity of 0.598 (0.021), smoothed, against Step Law’s 0.571 . Panel C: the unconditional elasticity equals the conditional one plus the learning-rate–batch complementarity times the batch elasticity (Milgrom and Roberts, 1996); the published gap is between Step Law’s $+0.307$ and Björck et al.’s -0.32 for models of at least 760 million parameters. Against their steeper per-size exponents for small models (down to -0.70) the shares closed would be about 0.16 and 0.27. Cell-cluster bootstrap standard errors (1,000 draws).

Source: Li et al. (2025a); authors’ estimates.

OBSERVATIONAL PRODUCTION FUNCTIONS AND ALGORITHMIC PROGRESS

The main text reads developers’ choices through a technology measured in designed experiments. This appendix asks the reverse question: what do released models, used as production data, reveal about the technology and about its change over time? Much of what is known about returns to compute at scale, and about the pace of algorithmic progress, comes from such data: regressions of benchmark scores on the inputs of models released by many developers (Ruan, Maddison and Hashimoto, 2024; Maia Polo et al., 2025; Owen, 2024; Mertens, Fischl-Lanzoni and Thompson, 2026), and of perplexity on the inputs and publication dates of published models (Ho et al., 2024). In these data the firms chose the inputs, and Whitfill (2025) and König et al. (2026) warn that the estimates may be confounded. Section E1 collects the results of Online Appendix A that apply to such data, and the remaining sections take them to the data.

There are four findings. First, once the design is matched, cross-lab returns to compute on HellaSwag do not differ detectably from the experimental technology. This is a failure to reject in a low-power joint test, not evidence of unbiasedness, and the split between parameters and data is not reliably estimated. Second, compute is measured well enough that errors in variables do not matter, but the standard remedies for transmission bias (instruments, dynamic panels, proxy inversion) cannot be applied to these data. Third, in the data of Ho et al. (2024) the rate of algorithmic progress is weakly identified: estimates lie on a ridge along which the bias of technical change trades off against its rate, and the published point is not the minimizer of the authors’ objective. Fourth, counting only the elimination of under-training, the move from Kaplan-rule to Chinchilla-rule allocations raised the training-cost efficiency of large models by a factor of 2.0 under the reference technology of Section IV (1.1 to 4.9 across technologies). The ten-fold gain of Gundlach et al. (2025) is the counterfactual cost of keeping the earlier rule at 2025 frontier compute, not a gain realized between eras.

E1. What Cross-Lab Data Can Identify

The results below, proved in Online Appendix A, concern a common technology with Hicks-neutral productivity ω that the developer knows when it chooses inputs, and log output $y = \omega + F(n, d) - \epsilon$, where F is the technology in output units and ϵ is noise.

Transmission.—If compute responds to productivity, $c = \pi_0 + \pi_1\omega + \eta$, the least-squares slope of output on compute is biased with the sign of π_1 (Proposition A5). The bias is upward when more productive developers raise more compute (funding, $\pi_1 > 0$) and downward when every developer trains to a common capability target ($\pi_1 = -1/\gamma$); if the target is hit exactly, cross-lab data carry no information about the frontier elasticity γ . The forward and reverse regressions bracket γ under target and exogenous budgets but not under funding, so the errors-in-variables bracket of Klepper and Leamer (1984) does not survive transmission

from productivity to compute. Hicks-neutral transmission leaves the allocation exponent a unbiased, because productivity does not move the input mix at given compute, and it compresses measured productivity dispersion.

Proposition A5 places runs on the expansion path. Released models are far off it (Section IV), and their log wedges enter output through the Farrell loss of Lemma A5. Two consequences follow by linear projection. When the regression is run on both inputs, the Farrell loss is part of the technology F , and transmission loads on the mix only if developers' wedges covary with their productivity at given compute. If they do not, the projection of ω on (n, d) runs along $c = \ln 6 + n + d$, and transmission shifts the two input elasticities by the same amount. If more productive developers also over-train more (or less), the two biases differ and can have opposite signs. When the regression is run on compute alone, it imposes equal weights on the two inputs, and the productivity residual inherits the Hall-type term of Corollary A5 (Section E2).

Selection.—If a model is released only when its output clears a threshold, least squares on released models is attenuated. This is the classical result for truncation on the dependent variable (Goldberger, 1981; Hausman and Wise, 1977). What Proposition A6 adds is the interaction with transmission. Under funding, mild selection leaves the upward transmission bias dominant and severe selection reverses it. Under target or exogenous budgets the net bias is negative, and the forward–reverse bracket survives in released samples.

Proxies.—Proxy estimators (Olley and Pakes, 1996; Levinsohn and Petrin, 2003; Akerberg, Caves and Frazer, 2015) recover productivity by inverting a policy function, so they need a state variable, a flexible input and a timing assumption. Here the natural state variable is the developer's compute capacity, the provisioned cluster, which is fixed before a run. The flexible input is the run's split of its compute budget between N and D , chosen at the start of the run given capacity and the recipe's quality. Release follows the observed outcome. The flexible input cannot proxy Hicks-neutral productivity, because at given compute the policy function is flat in ω (Proposition A7(ii)). Compute itself fails as a Levinsohn–Petrin-type proxy for a different reason. Under a strictly monotone budget rule the inputs on the path are functions of c alone, so a control function in c absorbs the technology: this is the functional-dependence problem of Akerberg, Caves and Frazer (2015) (Proposition A7(iii)). Identification would then require timing, for example compute committed before a recipe innovation, or an observed capacity investment, which public data do not record. Under a capability target the mix chosen for a given loss does co-move with ω when $\alpha \neq \beta$, but only through compute (Lemma A3(v)). Conditional on c , the mix remains uninformative about ω .

Technical change.—Diamond, McFadden and Rodriguez (1978) show that time series of outputs, inputs and marginal products cannot separate the elasticity of substitution from the bias of technical change. Time series of compute-optimal models still identify the sign of the bias, though not its size. Because the FLOP

accounting fixes the relative cost of a parameter and a token, the compute-optimal ratio $M^*(C, t)$ drifts at given compute as $\partial \ln M^* / \partial t = -\mathcal{B}_t \sigma_t^* / (1 - \sigma_t^*)$. Here $\mathcal{B}_t$, the Hicks bias, is the change over time in the marginal rate of technical substitution at fixed inputs (Proposition A3(i)). Since $0 < \sigma^* < 1$ at interior optima (Lemma 1), M^* falls if and only if technical change is parameter-using ($\mathcal{B}_t > 0$). In the generalized family, with inner exponents a_1 and b_1 and factor augmentation at rates g_N and g_D , the bias is $\mathcal{B} = b_1 g_D - a_1 g_N$, which is $\beta g_D - \alpha g_N$ when $\kappa = 1$. The drift therefore signs $\beta g_D - \alpha g_N$, not $g_D - g_N$. For example, with $(\alpha, \beta) = (0.30, 0.45)$ and $(g_N, g_D) = (0.5, 0.4)$, $\mathcal{B} = 0.03 > 0$ and the compute-optimal D/N falls, although parameters are augmented faster. Within the family g_N and g_D are identified separately, but only through its functional form; the curvature, and with it the size of the bias, is not. The result is methodological. We do not take it to published compute-optimal laws of different vintages, which differ in tokenizers, parameter-counting conventions and outputs. The sign result also requires that the effective relative cost of tokens not drift through data costs, data constraints or serving demand (Proposition 2). A designed IsoFLOP sweep at each vintage on a common output would identify the sign, and through Proposition 4 the size.

Allocative versus technical change.—A model’s log excess compute over best practice splits into an allocative term $\Phi(w) = \ln(C/C_{\min})$, which depends only on its wedge (Lemma A5), and a technical term, the productivity gap in compute-equivalent units (Proposition A4). A reallocation at fixed productivity, such as the move from Kaplan-rule to Chinchilla-rule allocations, is purely allocative. Three corollaries concern growth accounting. First, the compute-equivalent gain of Gundlach et al. (2025) between two technologies is the same at every compute level if and only if they share the irreducible loss E and the frontier elasticity γ (Corollary A2). Second, regressions of frontier loss on compute that omit time effects attribute part of algorithmic progress to scale, overstating the compute elasticity by the factor $1/(1 - s_A)$, where s_A is the algorithmic share of effective-compute growth (Corollary A4; Nagy et al., 2013). Third, a productivity residual computed with equal weights on $\ln N$ and $\ln D$, as $C = 6ND$ suggests, misattributes $\frac{1}{2}(\varepsilon_N - \varepsilon_D)(\Delta n - \Delta d)$ to productivity. When over-trained models ($w > 1$) shift further toward tokens, the residual understates progress (Corollary A5; Hall, 1988).

E2. Returns to Compute: A Design-Matched Validation

Data and specification.—The observational sample has 128 dense base models from 38 families and 22 developers, released between March 2021 and July 2024 (Online Appendix B). The models share no validation loss, so output is the log-odds of above-chance accuracy, $q = \ln[\tilde{a}/(1 - \tilde{a})]$, where $\tilde{a}$ is accuracy net of chance, rescaled to the unit interval and clipped to $[0.01, 0.99]$. The main benchmark is HellaSwag (Zellers et al., 2019), evaluated 10-shot on the Open LLM Leaderboard. We estimate $q_i = \theta_0 + \theta_N n_i + \theta_D d_i + \omega_i + e_i$, which is Cobb–Douglas in the odds

with Hicks-neutral productivity ω_i , and the regression of q_i on c_i alone. Its slope θ_C is what observational scaling studies report.

Why the design must be matched.—A comparison with global experimental elasticities misleads, because the technology is curved and released models sit in a different part of it. In the designed sweeps, HellaSwag log-odds are strongly concave in (n, d) ; the F -statistic for the quadratic terms is 70 in the sweep of Gadre et al. (2025). On the experimental surface described below, at $N = 1\text{B}$ the local data elasticity falls from 0.57 at 20 tokens per parameter to 0.07 at 500. Released models are heavily over-trained (median D/N of 213, ninetieth percentile 1,715), so they sit where an extra token is worth little. Global log-linear fits give observational data elasticities of 0.19 to 0.26, against 0.44 in the OLMo ladder and 0.51 in Gadre et al. (2025), but much of that gap is the technology, not bias.

We therefore match the design. We fit a quadratic surface $\hat{f}(n, d)$ with design intercepts to the designed runs of the OLMo ladder (Bhagia et al., 2025) and of Gadre et al. (2025) with $N \geq 0.1\text{B}$, 86 runs in all ($R^2 = 0.994$). We keep the released models inside the convex hull of the experimental inputs, the common-support trimming of Dehejia and Wahba (1999). We then apply each estimator twice, with the same fixed effects and controls: to the observed outputs, and to the outputs $q_i^* = \hat{f}(n_i, d_i)$ that the experimental technology produces with the same inputs. The difference is the estimator’s bias. The benchmark comes from other developers’ recipes, not from the treated population. The exercise is therefore a validation across populations, in the sense of Todd and Wolpin (2006) and Hotz, Imbens and Mortimer (2005), rather than a comparison with an experiment on the same units as in LaLonde (1986). The null hypothesis is joint: a common technology up to Hicks-neutral shifts, outputs that map one to one across evaluation harnesses, and no transmission or selection bias. Section III documents factor-biased differences across recipes, which would violate the first part, so a null bias can also reflect offsetting failures.

Support.—An earlier version of this paper included the released OLMo-2 7B and 13B models (Team OLMo et al., 2024) among the benchmark runs. Their developer chose their inputs, so they are no longer part of the experimental benchmark. The cost is support. Without them the largest designed D is 921B tokens (one run of Gadre et al., 2025 at 1.4B parameters), and above 3.8B parameters it is 634B. The strict support contains 20 released models from 9 families and 6 developers, all released between March 2021 and September 2023 with $N \leq 6.7\text{B}$ and $D \leq 627\text{B}$; five of the nine families have two or more models, 16 models in all. The benchmark that includes OLMo-2 (88 runs) covers 57 models from 30 families and 19 developers, with $N \leq 8.8\text{B}$. Fifteen of its families have two or more models (42 models) and fifteen are singletons. We report both designs.

Inference.—Inference must allow for few clusters and for the benchmark’s own uncertainty. Standard errors are CV3 jackknife standard errors by developer (MacKinnon, Nielsen and Webb, 2023). The benchmark’s uncertainty comes

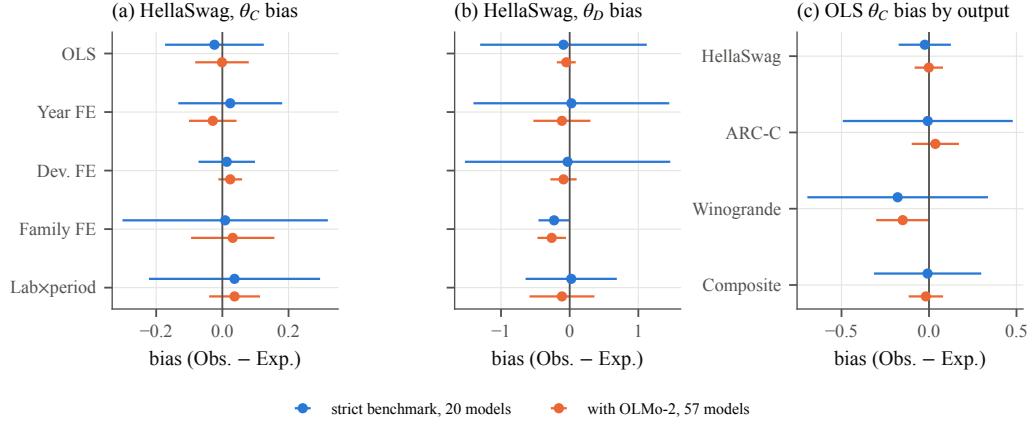


FIGURE E1. OBSERVATIONAL ESTIMATES AGAINST THE EXPERIMENTAL TECHNOLOGY ON THE SAME DESIGN

Note: Bias of each observational estimator relative to the same estimator applied to the outputs of the experimental technology on the same models (Obs. - Exp.), with 95 percent intervals (± 1.96 CV3 standard errors of the bias, including the benchmark's uncertainty). Blue: strict benchmark (OLMo ladder and [Gadre et al., 2025](#); 20 models). Orange: benchmark including the released OLMo-2 7B and 13B models (57 models). Panels (a) and (b): HellaSwag, compute-only elasticity θ_C and data elasticity θ_D . Panel (c): pooled OLS bias in θ_C by output; ARC-Challenge and Winogrande use the censored (Tobit) benchmark. Estimates and conservative few-cluster p -values are in Table E1.

from a design-conditional wild bootstrap of the surface, clustered by design and model size. For each bias we compute three few-cluster p -values that include the benchmark's uncertainty: CV3 with $t(G - 1)$ critical values, and restricted wild-cluster bootstrap- t tests studentized by CR1 or by CV3 ([Cameron, Gelbach and Miller, 2008](#); [Webb, 2023](#)). Table E1 reports the largest of the three, so a rejection requires all three to reject. The effective number of clusters ([Carter, Schnepel and Steigerwald, 2017](#)) is 2.5 to 6.7 of 19 on the 57-model design. On the strict design it is 1.2 to 3.8 of 6; there the three procedures disagree by up to a factor of ten, and none is reliable.

Returns to compute.—Once the design is matched, cross-lab returns to compute on HellaSwag do not differ detectably from the experimental technology (Table E1, Figure E1). On the 57-model design, pooled OLS gives $\theta_C = 0.328$ against 0.329 from the experimental technology on the same models, a bias of -0.001 (0.041) with $p = 0.98$. Across six estimators, from pooled OLS to lab-by-period fixed effects and a notability-propensity control, the biases lie between -0.029 and $+0.037$, and every p -value is at least 0.36. The strict surface evaluated on the same 57 models, an extrapolation for 37 of them, gives $+0.014$ (0.045), so the result does not rest on anchoring the surface with OLMo-2. On the strict support itself the point estimates do not move (from -0.024 to $+0.037$ across five estimators), but with six clusters they carry no inferential weight.

This is a failure to reject in a low-power joint test, not evidence of unbiasedness. On 57 models the 95 percent interval for the OLS bias is $[-0.088, 0.086]$ with $t(18)$

TABLE E1—DESIGN-MATCHED COMPARISON OF OBSERVATIONAL AND EXPERIMENTAL ELASTICITIES

	Compute only, θ_C			θ_N	θ_D
	Obs. (1)	Exp. (2)	Bias (3)	Bias (4)	Bias (5)
<i>Panel A. HellaSwag, strict benchmark (20 models, 9 families, 6 developers)</i>					
OLS	0.406 (0.043)	0.430 (0.007)	−0.024 (0.076) [0.77]	0.105 (0.107) [0.37]	−0.090 (0.619) [0.89]
Year FE	0.460 (0.035)	0.437 (0.007)	0.024 (0.080) [0.78]	0.022 (0.147) [0.89]	0.025 (0.727) [0.97]
Developer FE	0.449 (0.014)	0.436 (0.008)	0.013 (0.044) [0.77]	0.066 (0.148) [0.68]	−0.030 (0.763) [0.97]
Family FE	0.473 (0.243)	0.464 (0.010)	0.008 (0.158) [0.96]	0.185 (0.046) [0.14]	−0.227 (0.116) [0.15]
Lab × period FE	0.476 (0.239)	0.440 (0.008)	0.037 (0.132) [0.79]	0.053 (0.149) [0.74]	0.023 (0.339) [0.95]
<i>Panel B. HellaSwag, benchmark with OLMo-2 (57 models, 30 families, 19 developers)</i>					
OLS	0.328 (0.046)	0.329 (0.007)	−0.001 (0.041) [0.98]	0.079 (0.091) [0.43]	−0.050 (0.070) [0.58]
Year FE	0.319 (0.076)	0.348 (0.007)	−0.029 (0.037) [0.47]	0.101 (0.107) [0.36]	−0.113 (0.212) [0.60]
Developer FE	0.393 (0.054)	0.370 (0.008)	0.024 (0.018) [0.66]	0.141 (0.070) [0.14]	−0.090 (0.098) [0.57]
Family FE	0.441 (0.048)	0.409 (0.013)	0.031 (0.064) [0.63]	0.203 (0.059) [0.048]	−0.262 (0.106) [0.027]
Lab × period FE	0.442 (0.032)	0.405 (0.010)	0.037 (0.039) [0.36]	0.165 (0.077) [0.13]	−0.113 (0.241) [0.65]
Notability-propensity control	0.335 (0.071)	0.350 (0.006)	−0.015 (0.043) [0.80]	0.076 (0.125) [0.55]	−0.057 (0.072) [0.53]
<i>Panel C. OLS, other outputs and supports</i>					
HellaSwag, strict surface, 57	0.328 (0.046)	0.314 (0.014)	0.014 (0.045) [0.76]	0.090 (0.091) [0.37]	−0.032 (0.073) [0.73]
ARC-Challenge, 20	0.466 (0.117)	0.473 (0.018)	−0.007 (0.248) [0.98]	0.218 (0.188) [0.50]	−0.123 (1.429) [0.93]
ARC-Challenge, 57, with OLMo-2	0.449 (0.055)	0.413 (0.027)	0.036 (0.069) [0.61]	0.171 (0.182) [0.49]	−0.045 (0.129) [0.74]
Winogrande, 20	0.371 (0.055)	0.550 (0.038)	−0.179 (0.263) [0.53]	−0.040 (0.165) [0.90]	−0.251 (1.341) [0.86]
Winogrande, 57, with OLMo-2	0.351 (0.047)	0.501 (0.044)	−0.150 (0.077) [0.084]	−0.027 (0.194) [0.89]	−0.225 (0.202) [0.28]

Note: Output: log-odds of above-chance accuracy. Obs.: estimator on the observed outputs of released base models. Exp.: the same estimator, with the same fixed effects and controls, on the outputs that the experimental technology produces for the same models: a quadratic in $(\ln N, \ln D)$ fitted to the OLMo ladder and [Gadre et al. \(2025\)](#) ($N \geq 0.1B$; 86 runs); Panel B adds the released OLMo-2 7B and 13B models (88 runs), as an earlier version of this paper did. Each panel uses the models inside the convex hull of its own runs; ‘strict surface, 57’ evaluates the Panel A surface on the Panel B models, 37 of which lie outside its support. Bias = Obs. − Exp. Parentheses: CV3 jackknife s.e. by developer (Obs.; bias: of Obs. − Exp., with the benchmark’s s.e. added in quadrature) and the design-conditional wild-cluster bootstrap s.e. of the surface (9,999 draws; Exp.). Brackets: the largest of three few-cluster p -values for zero bias that include the benchmark’s uncertainty (CV3 with $t(G - 1)$; restricted wild-cluster bootstrap- t , Webb weights, 9,999 draws, studentized by CR1 or by CV3). Effective numbers of clusters: 1.2–3.8 of 6 in Panel A (no p -value is reliable there); 2.5–6.7 of 19 in Panel B. ARC-Challenge and Winogrande: Tobit surface censored at the floor (52 and 16 of 86 runs are at it), level matched to the observed outputs, counterfactual clipped at the same floor. Notability-propensity control (selection on observables): quadratic in the estimated propensity of Epoch notability; Panel A has no notable model. *Source:* Authors’ calculations. Observational outputs: Open LLM Leaderboard v1 via [Ruan, Maddison and Hashimoto \(2024\)](#) and [Maia Polo et al. \(2025\)](#); parameter counts from Hugging Face; release dates from [Epoch AI \(2026\)](#). Experimental runs: [Bhagia et al. \(2025\)](#), [Team OLMo et al. \(2024\)](#) (Panel B only) and [Gadre et al. \(2025\)](#).

critical values, about ± 26 percent of the benchmark. The bias detectable with 80 percent power is 0.12, or 37 percent of the benchmark; on the strict support it is 0.27, or 62 percent. The test covers only models with at most 8.8B parameters, and its benchmark is the technology of the AI2 and OpenLM recipes.

Other benchmarks.—ARC-Challenge and Winogrande are less informative, because many small designed runs score at or below chance: 52 of the 86 runs are at the floor of chance-adjusted accuracy on ARC-Challenge and 16 on Winogrande, against 2 on HellaSwag. A least-squares surface fitted to the clipped log-odds makes the observational compute elasticity on ARC-Challenge look attenuated, with a bias of -0.098 on 57 models (as reported in an earlier version of this paper). The attenuation does not survive modeling the censoring. We fit a Tobit surface censored at the floor, match its level to the observed outputs, and clip the counterfactual at the same floor; the bias is then $+0.036$ (0.069), with $p = 0.61$ (Table E1, Panel C). The Tobit benchmark extrapolates a Gaussian latent surface below the floor for 60 percent of the runs, so we do not read it as showing that the attenuation was an artifact. Winogrande keeps a borderline attenuation: -0.150 (0.077) on 57 models, with p -values between 0.068 and 0.084 across the three procedures. On the strict support the Winogrande estimate is similar but uninformative (-0.179 , $p = 0.53$). The three-benchmark composite gives -0.018 (0.050).

Two parts of the joint null are not tested. Observational outputs come from the Open LLM Leaderboard (HellaSwag 10-shot, ARC-Challenge 25-shot, Winogrande 5-shot), whereas the designed runs were evaluated in the OLMES and LLM-foundry harnesses with other prompts and shot counts; we harmonize them only through the log-odds of chance-adjusted accuracy. And benchmark contamination, or classifier-filtered corpora that favor benchmark-like text, raise output at given inputs; family effects absorb such shifts only when they are common to a family.

The split between parameters and data.—The split is not reliably estimated. With family fixed effects on the 57-model design, the observational parameter elasticity exceeds the benchmark by 0.203 (0.059) and the data elasticity falls short of it by 0.262 (0.106). The conservative p -values are 0.048 and 0.027, and dropping any one developer leaves both signs unchanged (Table E2). But the identifying variation is thin. In the full panel the within-family standard deviation of $\ln D$ is 0.42, against 1.29 for $\ln N$. The effective number of clusters is about 4 of 19, and two families supply 65 percent of the leverage for θ_D : Cerebras-GPT, trained at exactly 20 tokens per parameter, and Phi, trained on synthetic data. Under the strict benchmark the biases have similar size ($+0.185$ and -0.227) but conservative p -values of 0.14 and 0.15. Biases of opposite sign are also hard to reconcile with transmission. If wedges are unrelated to productivity at given compute, transmission shifts both elasticities by the same amount (Section E1). Opposite signs point instead to factor-biased recipe changes within families or to errors in D , which the benchmark cannot distinguish from bias. We read them

TABLE E2—FEW-CLUSTER INFERENCE FOR THE FAMILY FIXED-EFFECTS BIASES

	θ_N bias = 0.203			θ_D bias = -0.262		
	SE Obs.	SE bias	p	SE Obs.	SE bias	p
CR1, all k (earlier version), normal	0.082	0.086	0.019	0.161	0.163	0.11
CR1, singletons dropped, normal	0.071	0.077	0.008	0.140	0.142	0.065
CR1, nested k , $t(18)$	0.055	0.063	0.005	0.110	0.112	0.031
CV3 jackknife, $t(18)$	0.052	0.059	0.003	0.104	0.106	0.024
WCR, benchmark fixed (earlier version)			0.046			0.024
WCR, benchmark fixed, nested k			0.044			0.025
WCR, with benchmark draws			0.048			0.028
WCU, with benchmark draws			<0.0001			0.010
WCR, family weights			0.031			0.024
WCR, model weights			0.010			0.030
WCR, CV3-studentized, with benchmark draws			0.048			0.011
Effective clusters G^* (of 19)		4.3			4.1	
Bias, one developer left out		[0.157, 0.224]			[-0.339, -0.193]	

Note: Family fixed-effects rows of Table E1, Panel B (HellaSwag, 57 models, benchmark including OLMo-2); only the inference differs across rows. ‘Earlier version’: the procedures of an earlier version of this paper. SE (bias) adds the design-conditional bootstrap standard error of the benchmark (0.029 for θ_N , 0.020 for θ_D). The CR1 small-sample factor $\frac{G}{G-1} \frac{n-1}{n-k}$ is 2.24 on the variance when $k = 32$ counts the family dummies (15 of the 30 families are singletons), 1.64 with singletons dropped, and 1.04 when fixed effects nested within developer clusters are not counted (Cameron and Miller, 2015). A bootstrap- t is invariant to this constant; a normal reference is not, which is why the earlier version paired a p -value of 0.02 with a t -ratio of 1.6 for θ_D . WCR/WCU: restricted/unrestricted wild-cluster bootstrap- t , developer clusters, Webb weights, 9,999 draws (Cameron, Gelbach and Miller, 2008; Webb, 2023); subcluster: weights drawn by family or by model with developer-clustered standard errors (MacKinnon and Webb, 2018); CV3-studentized: the bootstrap statistic uses the CV3 standard error of Obs. – Exp. in the sample and in every draw. CV3: jackknife by developer (MacKinnon, Nielsen and Webb, 2023). G^* : Carter, Schnepel and Steigerwald (2017) with $\rho = 0$. Identifying variation for θ_D (partial leverage) by developer: Cerebras 37%, Microsoft 28%, Meta 12%, BigScience 7%; 7 of 19 developers contribute nothing (they have only singleton families) and 2 more under 0.1%; by family: Cerebras-GPT 37%, Phi 28%, OPT 9%. *Source:* Authors’ calculations; see Table E1.

as a within-family disagreement driven by two families, not as a finding.

Table E2 also explains why an earlier version of this paper paired a t -ratio of 1.6 with a wild-bootstrap p -value of 0.02 for the same θ_D bias. The benchmark’s uncertainty is not the cause; its standard error (0.020) is small against the observational one. The cause is the small-sample factor of the CR1 variance. That version counted the 29 family dummies in k although families are nested within developer clusters, which raises $(n-1)/(n-k)$ from 1.04 to 2.24 (Cameron and Miller, 2015). The standard error was therefore 1.47 times too large; a bootstrap- t is invariant to that constant, and a normal reference is not. With the nested convention, the CV3 jackknife or any restricted wild-cluster variant, the p -values are 0.01 to 0.03 for θ_D and 0.003 to 0.05 for θ_N .

A Hall-type bias.—Both worlds agree that $\theta_N > \theta_D$ on this design: 0.42 against 0.27 under the experimental technology on the 57 models, and $\theta_N - \theta_D = 0.21$ (0.08) in the observational data on all 128. Over-trained models have wedges above one, so their output elasticity with respect to parameters exceeds that with respect to data. A regression on $\ln C = \ln 6ND$ alone therefore imposes equal weights that the data reject, and the residual of such an equal-weight index

misattributes $\frac{1}{2}(\theta_N - \theta_D)(\Delta n - \Delta d)$ to productivity. This is the Hall-type bias of Corollary A5: as models shift further toward tokens, measured productivity understates progress.

How large could transmission bias be?—We treat the 25 pretraining recipes of Magnusson et al. (2025) as developers with real outcomes and simulate how they choose model size. Pooled OLS is then unbiased under exogenous budgets. It overstates θ_C by 0.044 when funding rises with productivity rank, and understates it by 0.18 (40 percent) when each developer trains to a common capability target. Cross-family productivity dispersion in the observational panel is about five times that of the recipes, which scales the funding bias to about +0.2 and the target bias beyond −0.18. The 95 percent interval of the matched OLS bias excludes both. The data are therefore consistent with budgets roughly orthogonal to productivity, or with offsetting funding and release selection; the simulated rules are stylized. In the simulated industry of Online Appendix C, pooled estimation overstates the frontier elasticity by 13 percent under funding, understates it by 30 percent under a capability target, and overstates it by 66 percent with one flagship per developer and generation. Estimation within developer and generation, the observational analogue of an IsoFLOP sweep, is unbiased under every rule, but it requires at least two released siblings.

E3. Measurement of Compute and the Standard Remedies

Measurement error.—Measurement error in compute is small, but showing it requires an independent measure. Compute in the panel is $C = 6ND$ from reported N and D , and Epoch’s compute figures for these models are mostly $6ND$ itself (operation counting) or developer-reported counts of the same kind (Epoch AI, 2026). Only hardware-time records measure compute independently of the reported inputs. We rebuild compute from them as chip-hours times peak dense 16-bit throughput times a common utilization of 0.3. Epoch’s utilization field is not used, because it is back-calculated from Epoch’s own compute estimate in every record we checked. Across the 27 models with such records, from 12 families, $\ln(C_{\text{hw}}/6ND)$ has standard deviation 0.29, with 95 percent interval [0.20, 0.41] (Table E3). This bounds the error in $\ln 6ND$ from above, because it also contains differences in true utilization and errors in chip-hours. The implied reliability of $\ln C$ on the 57-model design is at least 0.98 pooled and 0.955 within families (0.96 and 0.91 at the upper end of the interval). Correcting for errors in variables raises the pooled compute elasticity by 0.007 and the within-family one by 0.021, or by 0.042 at the upper end (Table E4).

An earlier version of this paper reported a within-family errors-in-variables estimate of 0.858 against a benchmark of 0.409. It combined two errors. It used Epoch’s ‘Confident’ uncertainty class ($\sigma_u = \ln 3/1.645 = 0.67$), which describes the uncertainty of Epoch’s own estimate rather than the error in $6ND$. And it computed the within-family variance of $\ln C$ with an $n - 1$ denominator, ignoring the 30 family means. With the correct variance and the hardware-time bound,

TABLE E3—HOW WELL IS TRAINING COMPUTE MEASURED?

<i>Panel A. Discrepancy between two compute measures, $\ln(C_{alt}/6ND)$</i>					
	Models (fam.)	Mean	S.d.	95% interval	MAD s.d.
Hardware time (independent)	27 (12)	0.017	0.292	[0.197, 0.409]	0.296
Epoch ‘Hardware’/‘Reported’	14 (8)	0.074	0.195	[0.033, 0.244]	0.051
Epoch operation counting	51 (22)	0.000	0.129	[0.029, 0.205]	0.015
Developer-reported FLOP	5 (3)	−0.015	0.033	[0.000, 0.038]	0.025
<i>Panel B. Implied reliability of $\ln C$, $\lambda = 1 - \sigma_u^2 / \text{Var}(\ln C)$</i>					
	Models	σ_u	Pooled	Within family	Within, $n - 1$
Hardware-time s.d.	57	0.292	0.980	0.955	0.907
upper 95% bound	57	0.409	0.961	0.912	0.818
Epoch ‘Hardware’/‘Reported’	57	0.195	0.991	0.980	0.959
Epoch ‘Confident’ class	57	0.668	0.895	0.766	0.514
Hardware-time s.d.	20	0.292	0.973	0.969	0.947
upper 95% bound	20	0.409	0.948	0.939	0.895
Epoch ‘Hardware’/‘Reported’	20	0.195	0.988	0.986	0.976
Epoch ‘Confident’ class	20	0.668	0.861	0.838	0.721
Hardware-time s.d.	128	0.292	0.988	0.973	0.961
upper 95% bound	128	0.409	0.977	0.946	0.924
Epoch ‘Hardware’/‘Reported’	128	0.195	0.995	0.988	0.983
Epoch ‘Confident’ class	128	0.668	0.939	0.857	0.798

Note: Panel A compares $C = 6ND$ (reported N and D ; exact Hugging Face parameter counts where available) with other compute measures for main-sample models matched to Epoch (Epoch AI, 2026). Only hardware time is independent of reported N and D : chip-hours (or chips $\times$ training hours) $\times$ peak dense 16-bit FLOP/s $\times$ a common utilization of 0.3. Epoch’s utilization field is back-calculated from its own compute estimate for most rows and is not used. Epoch’s reported compute is itself $6ND$ (operation counting), a developer-reported count (usually $6ND$), or a geometric mean of $6ND$ and hardware time; the ‘Hardware’/‘Reported’ set, used as an independent comparison in an earlier version of this paper, mixes all three. The hardware-time s.d. also contains the dispersion of true utilization and any error in chip-hours, so it bounds the error s.d. of $\ln 6ND$ from above; its within-family value is 0.208 (6 families with two or more records). Intervals: family-cluster bootstrap, 9,999 draws; MAD s.d.: $1.4826 \times$ median absolute deviation. Panel B: within-family variance $\sum \hat{e}_i^2 / (n - F)$, net of the F family means; the last column divides by $n - 1$, as an earlier version of this paper did. Within-family reliabilities assume errors independent across a family’s models; an error in a family-wide token count is absorbed by the family effect.

Source: Authors’ calculations.

the within-family bias is +0.052 (0.049), with $p = 0.30$.

Reverse regression.—The reverse regression of $\ln C$ on output gives an inverse slope of 0.560. By construction this equals the forward slope, 0.328, divided by the forward R^2 of 0.585 (with family effects, $0.441/0.760 = 0.580$). Its distance from the same statistic computed on the experimental outputs, 0.352, therefore measures output variation not explained by compute, that is, productivity dispersion and noise, not a bias. Under funding the forward–reverse pair need not bracket the truth at all (Proposition A5(iii)).

Instruments, panels and proxies.—The standard remedies for transmission bias cannot be used, for reasons that concern the data rather than the estimators. The price–performance of the best available data-center accelerator is a step function of calendar time. On the 57-model design its first-stage F is 0.90 (0.01 with developer fixed effects), and the Anderson–Rubin sets are unbounded. Applied to the experimental outputs, the same instrument recovers 0.012 against a design-matched elasticity of 0.33: the instrument fails, not the data. The price–performance of each model’s own training hardware is strong only on 10 models from 7 developers ($F = 14.3$), and it takes three values, so it acts as a vintage indicator; its Anderson–Rubin set is $[-1.50, 0.34]$. An export-control instrument (Chinese developers after October 2023, with developer and year effects) is strong among the 90 models with at most 14B parameters and 5T tokens ($F = 10.6$). It gives $\theta_C = 0.90$ (0.25), with Anderson–Rubin set $[0.49, 2.23]$, which is 1.7 to 2.3 times the least-squares and fixed-effects estimates on that sample. Its exclusion restriction fails because recipes and data changed at the same time. Dynamic-panel estimators (Arellano and Bond, 1991; Blundell and Bond, 1998) and the moments of Proposition A5(v) need three consecutive generations, which only six product-line cells have. Proxy inversion is unavailable for the reasons given in Section E1. The notability-propensity control of Table E1 (a quadratic in the estimated propensity of Epoch notability) is selection on observables, not an Olley–Pakes correction; it barely moves the matched bias (-0.015 , $p = 0.80$).

E4. Productivity Dispersion

How dispersed is productivity across model families? The answer depends on the units in which it is measured (Table E5). Let Δ be the gap between the 90th and 10th percentiles of family productivity in HellaSwag log-odds, net of inputs. In output units, the 90–10 ratio of the odds, e^Δ , is 1.9 to 3.0 across samples and netting methods. In reducible-loss units, $e^{\gamma\Delta/\theta_C}$ with the frontier elasticity $\gamma = 0.178$ of Besiroglu et al. (2024), it is 1.3 to 1.8. Both ranges bracket the 90–10 ratio of 1.92 within US manufacturing industries (Syverson, 2004, see also Syverson, 2011). In compute-equivalent units, e^{Δ/θ_C} , the compute that the 10th-percentile family needs to match the 90th, the ratio is 7 to 24 with the 57-model benchmark $\theta_C = 0.33$, or 4 to 13 with the strict benchmark’s 0.43. Within developers, in the design of Mertens, Fischl-Lanzoni and Thompson (2026), it is 11.1 in compute units (90 percent interval $[4.1, 18.1]$), against their 41 on another

TABLE E4—COMPUTE-ONLY ESTIMATORS: MEASUREMENT ERROR, REVERSE REGRESSION AND INSTRUMENTS

	Obs. (1)	Exp. (2)	Bias (3)	λ , R^2 , F (4)	AR set; $\hat{\theta}/R^2$ (5)
<i>Panel A. HellaSwag, 57 models (benchmark including OLMo-2)</i>					
OLS, EIV, $\sigma_u = 0.29$	0.335 (0.038)	0.329 (0.007)	0.006 (0.039) [0.90]	0.980	
Family FE, EIV, $\sigma_u = 0.29$	0.461 (0.047)	0.409 (0.013)	0.052 (0.049) [0.30]	0.955	
Family FE, EIV, $\sigma_u = 0.67$	0.576 (0.059)	0.409 (0.013)	0.166 (0.060) [0.12]	0.766	
Family FE, EIV, $\sigma_u = 0.67$, $n - 1$	0.858 (0.088)	0.409 (0.013)	0.448 (0.089) [0.016]	0.514	
Reverse regression	0.560 (0.126)	0.352 (0.006)	0.209	0.585	0.328/0.585
Reverse, family FE	0.580 (0.122)	0.432 (0.012)	0.147	0.760	0.441/0.760
IV, frontier FLOP/\$ ($n = 57$)	0.724	0.012 (0.047)	0.712	0.90	$(-\infty, \infty)$
IV, frontier, developer FE ($n = 57$)	8.535	-2.287 (0.190)	10.822	0.01	$(-\infty, \infty)$
IV, own hardware ($n = 10$)	-0.090	0.195 (0.014)	-0.285	14.3	$[-1.50, 0.34]$ $[-0.94, 0.27]$
<i>Panel B. HellaSwag, 20 models (strictly experimental benchmark)</i>					
OLS, EIV, $\sigma_u = 0.29$	0.417 (0.019)	0.430 (0.007)	-0.013 (0.020) [0.44]	0.973	
Family FE, EIV, $\sigma_u = 0.29$	0.488 (0.055)	0.464 (0.010)	0.024 (0.056) [0.65]	0.969	
Reverse regression	0.551 (0.129)	0.442 (0.006)	0.109	0.737	0.406/0.737
Reverse, family FE	0.505 (0.081)	0.476 (0.009)	0.029	0.937	0.473/0.937
IV, frontier FLOP/\$ ($n = 20$)	0.008	0.384 (0.014)	-0.376	5.15	$(-\infty, \infty)$ $[-1.85, 0.27]$
IV, frontier, developer FE ($n = 20$)	-0.237	0.332 (0.017)	-0.569	3.09	$(-\infty, \infty)$ $(-\infty, \infty)$
IV, own hardware	fewer than 8 models with the instrument				

Note: Compute-only estimators of θ_C on the design-matched samples of Table E1. EIV: slope divided by the reliability $\lambda = 1 - \sigma_u^2 / \text{Var}(\ln C)$ (column 4), with $\text{Var}(\ln C)$ pooled or within family; the within-family variance is $\sum \hat{c}_i^2 / (n - F)$, net of the F family means. $\sigma_u = 0.29$ is the standard deviation of $\ln(C_{\text{hw}}/6ND)$ across the 27 main-sample models with hardware-time records (Table E3), an upper bound on the error in $\ln 6ND$; $\sigma_u = \ln 3/1.645$ is Epoch’s ‘Confident’ class, which describes the uncertainty of Epoch’s own estimate, not of $6ND$. ‘ $n - 1$ ’: the within-family variance computed with an $n - 1$ denominator, as in an earlier version of this paper (reliability 0.514 instead of 0.766 for the same σ_u). Exp. applies the same estimator to the experimental outputs, which are built from the same measured inputs and need no correction. Brackets: restricted wild-cluster bootstrap- t p -values for zero bias (CR1-studentized, developer clusters, Webb weights, 9,999 draws, each with an independent draw of the benchmark). Reverse regression: $1/b$ from regressing $\ln C$ on the output. By construction $1/b = \hat{\theta}_C/R^2$ of the forward regression (column 5), so its gap to Exp. measures the output variance not explained by compute (productivity dispersion and noise), not a bias. IV: $\ln C$ instrumented by the log FLOP/s per dollar of the best data-center accelerator released six months before the model, or of the model’s own training hardware (Epoch); F : cluster-robust first-stage statistic; Anderson–Rubin sets by inversion over $\theta_C \in [-3, 4]$ with $t(G - 1)$ critical values (second line: normal critical values); $(-\infty, \infty)$: the set covers the whole grid.

Source: Authors’ calculations; see Table E1; hardware records from Epoch AI (2026).

benchmark.

Manufacturing’s ratio is both an output-unit and an input-unit ratio, because returns to scale are near one. For language models the two differ by the factor $1/\theta_C$ in logs, between 2.3 and 3, because returns to compute are small. The comparison with manufacturing is therefore a choice of cardinalization, not a finding.

Three caveats apply. First, family effects absorb benchmark contamination and the similarity of training data to the benchmark, as well as productivity. Second, training compute omits the compute of teachers and synthetic-data generators. With inputs netted within families, the synthetic-data families Phi and SmolLM rank first and third of 38, so measured productivity credits the student with an intermediate input, the gross-output problem of [Gandhi, Navarro and Rivers \(2020\)](#). Third, the conversion to compute rests on a single cardinal elasticity from one benchmark; the intervals in Table E5 draw θ_C from its bootstrap. Transmission, finally, would make all of these ratios understate dispersion (Proposition A5(ii)).

E5. Algorithmic Progress

[Ho et al. \(2024\)](#) estimate that the compute needed to reach a given perplexity halves every 8.4 months, from 231 model–benchmark observations in 144 papers. Their preferred model is the technology of Section I with $E = 0$ and factor-augmenting productivity growing linearly in time, $\psi_N = g_N t$ and $\psi_D = g_D t$:

$$\ln \text{ppl} = \exp(\alpha_{0b} - \alpha g_N t - \alpha \ln N) + \exp(\beta_{0b} - \beta g_D t - \beta \ln D),$$

where b indexes three benchmarks and t is time in years. Effective compute grows at $g_C = g_N + g_D$ per year and doubles every $\tau_C = 12 \ln 2 / g_C$ months; Hicks neutrality is $\alpha g_N = \beta g_D$.

The ridge.—The rate is weakly identified because the sample is not on the expansion path, a problem of the kind studied by [Diamond, McFadden and Rodriguez \(1978\)](#). To first order the data pin down how fast loss falls at fixed inputs, an average of αg_N and βg_D weighted by the mean shares of the two terms in loss (0.55 and 0.45). The doubling time instead weights the two rates by $1/\alpha$ and $1/\beta$. The two combinations coincide only on the expansion path, along which parameters would grow with compute at elasticity $a = 0.37$ under their exponents. Off the path, rate pairs that fit equally well imply different doubling times.

Figure E2 shows the resulting ridge. The bootstrap correlation between αg_N and βg_D is -0.73 with the authors’ code and -0.82 at the converged estimates. The profile-likelihood interval for τ_C is $[4.1, 40.5]$ months. The parameter-augmenting share g_N/g_C can lie anywhere in $[-0.44, 3.37]$ (95 percent profile interval), and over that range τ_C runs from 5.3 to 39.6 months (Table E6). These likelihood ratios assume independent Gaussian errors, although observations are clustered by paper. The profile over g_N/g_C also has a second local minimum

TABLE E5—PRODUCTIVITY DISPERSION ACROSS MODEL FAMILIES IN OUTPUT AND INPUT UNITS

	Output units odds	Compute $\theta_C = 0.33^a$	Loss	Compute $\theta_C = 0.43$	Loss	θ used
	(1)	(2)	(3)	(4)	(5)	(6)
<i>All 128 models (38 families)</i>						
Family effects, θ_C netting	2.68 [1.94, 5.27]	19.9 [7.5, 160.7]	1.70 [1.43, 2.47]	13.1 [5.1, 69.8]	1.58 [1.34, 2.13]	0.33
Family effects, within netting	2.83 [2.02, 5.59]	23.7 [8.5, 193.1]	1.76 [1.46, 2.56]	11.2 [5.1, 53.7]	1.54 [1.34, 2.03]	0.33
Within-developer residuals	2.78 [1.77, 3.19]	11.1 [4.1, 18.1]	1.54 [1.29, 1.68]			0.43
<i>Without distilled and synthetic-data models</i>						
Family effects, θ_C netting	2.57 [1.90, 5.13]	17.6 [7.0, 143.9]	1.67 [1.42, 2.43]	13.1 [5.1, 58.9]	1.58 [1.34, 2.07]	0.33
Family effects, within netting	2.60 [1.87, 5.90]	18.3 [6.8, 216.6]	1.68 [1.41, 2.61]	9.2 [4.3, 58.9]	1.49 [1.30, 2.07]	0.33
Within-developer residuals	2.84 [1.78, 3.21]	11.3 [4.0, 18.0]	1.54 [1.28, 1.67]			0.43
<i>Also without code families (32 families)</i>						
Family effects, θ_C netting	2.07 [1.70, 2.29]	9.1 [4.9, 12.7]	1.48 [1.33, 1.57]	6.9 [3.5, 10.4]	1.41 [1.25, 1.52]	0.33
Family effects, within netting	1.90 [1.61, 5.10]	7.0 [4.2, 143.4]	1.41 [1.29, 2.42]	4.4 [3.3, 37.3]	1.30 [1.23, 1.91]	0.33
Within-developer residuals	2.51 [1.71, 3.04]	8.4 [3.8, 14.2]	1.46 [1.27, 1.61]			0.43
Manufacturing (Syverson, 2004)	1.92	1.92	1.92	1.92	1.92	≈ 1

Note: 90–10 ratios of Hicks-neutral productivity on HellaSwag log-odds q . A gap Δ between the 90th and 10th percentiles is reported as an odds ratio e^Δ (column 1, output units), as compute-equivalents e^{Δ/θ_C} (input units: the compute the 10th-percentile family needs to match the 90th), and in reducible-loss units $e^{\gamma\Delta/\theta_C}$ with $\gamma = 0.178$ (Besiroglu et al., 2024); $\gamma = 0.155$ (Hoffmann et al., 2022) is in the replication files. $\theta_C = 0.33$: the design-matched benchmark on 57 models (benchmark including OLMo-2); $\theta_C = 0.43$: the strictly experimental benchmark on 20 models (Table E1). Family effects net inputs either with the benchmark θ_C or with the within-family (θ_N, θ_D); within-developer residuals follow Mertens, Fischl-Lanzoni and Thompson (2026) (developer and year effects, single-model developers dropped) and use the regression’s own θ_C (column 6) in columns 2–3 (^a). Brackets: 90 percent intervals from a developer-cluster bootstrap (999 draws) that also draws θ_C from its benchmark bootstrap. Manufacturing’s 1.92 is an output-unit ratio and, with returns to scale near one, also an input-equivalent ratio; for LLMs the two differ by the factor $1/\theta_C$ in logs, because returns to compute are small. Family effects also absorb benchmark contamination, similarity of training data to the benchmark and omitted teacher compute (distillation, synthetic data).

Source: Authors’ calculations; outputs from Ruan, Maddison and Hashimoto (2024) and Maia Polo et al. (2025).

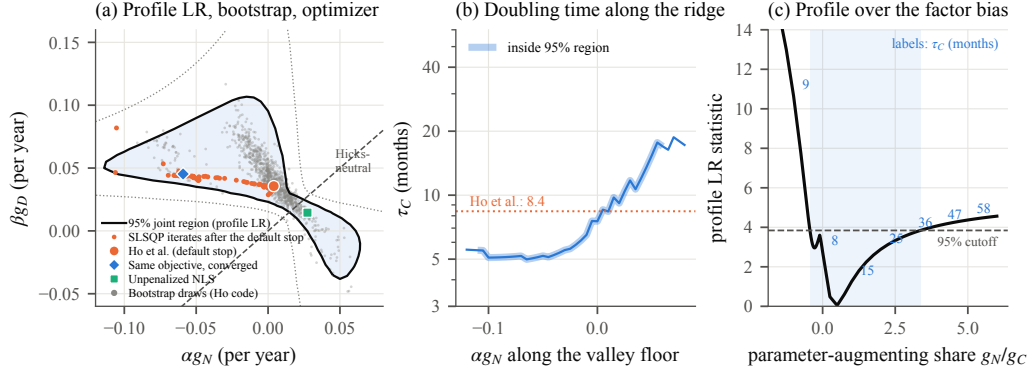


FIGURE E2. THE RIDGE IN ESTIMATES OF ALGORITHMIC PROGRESS

Note: Data of Ho et al. (2024), 231 model-benchmark observations. Panel (a): 95 percent joint profile-likelihood region for the year terms $(\alpha_{g_N}, \beta_{g_D})$, bootstrap draws from the authors' code, the optimizer's iterates from its default stopping point to the converged minimum of the same objective, the unpenalized least-squares minimum, and the Hicks-neutral line. Panel (b): doubling time of effective compute τ_C along the floor of the profile; the thick segment lies inside the 95 percent region. Panel (c): profile likelihood-ratio statistic over the parameter-augmenting share g_N/g_C , with the implied τ_C in months at selected points; the dashed line is the 95 percent cutoff and the shaded band the 95 percent interval, $[-0.44, 3.37]$. Likelihood ratios assume independent Gaussian errors.

near -0.28 ; a missed global minimum could only widen the interval.

Neutrality.—Precision about the rate is bought with an assumption about the bias. Hicks neutrality is not rejected (bootstrap $p = 0.29$ and 0.94 for the converged penalized and the unpenalized estimators), which reflects low power rather than evidence of neutrality. Imposing it gives 9.2 months with unpenalized least squares, with interval $[6.3, 14.3]$, or 11.6 months with the converged penalized estimator: the tightest intervals among the specifications that the data do not reject. As Diamond, McFadden and Rodriguez (1978) predict, the rate is identified here by the restriction on the bias. Even that restriction is fragile. If the true technology has $E > 0$, neutrality imposed within the $E = 0$ form is misspecified. In simulations on Ho et al.'s design with a true doubling time of 12 months, it yields 4.2 to 4.4 months, so the neutral estimates may overstate progress.

The published estimate.—The published point also depends on the optimizer. We reproduce Ho et al.'s estimates exactly: the same objective value (0.0518), every parameter within 4×10^{-7} , and a bootstrap median of 8.44 months under their protocol. The published 8.4 months is that median; the point estimate is 8.7. It is not the minimizer of the authors' objective, mean squared error plus an L_1 penalty. SciPy's default stopping rule halts the optimizer after 18 iterations, part of the way along the ridge. Run to convergence, the objective falls to 0.0507 , all 300 random starts end below the published value, and $\tau_C = 6.1$ months with a paper-cluster interval of $[3.0, 22.7]$, about twice as wide as published. Mean squared error changes by only 0.27 percent: almost all of the improvement in the objective comes from the penalty on constants whose values depend on an arbi-

TABLE E6—ALGORITHMIC PROGRESS IN THE DATA OF HO ET AL. (2024): REPLICATION AND RIDGE

	τ_C (months)	95% interval	MSE	Objective
<i>Panel A. Replication of Ho et al.'s preferred model (231 observations, 144 papers)</i>				
Published (bootstrap median)	8.4	[4.5, 14.3]	—	0.0518129
Their code and protocol, median	8.44	[4.4, 14.1]	0.0464	0.0518129
point estimate, cluster interval	8.68	[4.1, 16.9]		
Same objective, run to convergence	6.08	[3.0, 22.7]	0.0463	0.0507227
share of 300 starts below published				1.00
Unpenalized least squares (NLS)	10.2	[4.1, 27.6]	0.0453	
profile-likelihood interval		[4.1, 40.5]		
Benchmark-specific E_b estimated	10.5	[2.9, 26.3]	0.0451	
<i>Panel B. The ridge: what the data identify</i>				
Corr($\alpha g_N, \beta g_D$), their code / converged	−0.73 / −0.82			
Share g_N/g_C , 95% profile interval	[−0.44, 3.37]			
range of τ_C over that interval	5.3–39.6			
Hicks neutrality p : converged / NLS	0.29 / 0.94			
Hicks neutrality imposed, converged	11.6	[6.6, 17.6]	0.0465	
Hicks neutrality imposed, NLS	9.2	[6.3, 14.3]	0.0454	
Effective data (converged)	25.7	[3.9, ∞]	0.0484	
Exponents of Besiroglu et al. (2024) imposed	$g_C < 0$	[72.5, ∞]	0.1126	
Year terms $\div$ exponents (Whitfill)	76.5	[46.2, ∞]	—	

Note: Ho et al.'s model: $\ln \text{ppl} = \exp(\alpha_{0b} - \alpha g_N t - \alpha \ln N) + \exp(\beta_{0b} - \beta g_D t - \beta \ln D)$, irreducible loss zero, t in years; $\tau_C = 12 \ln 2 / (g_N + g_D)$ is the doubling time of effective compute in months. Their estimator minimizes $\text{MSE} + 0.0025 \sum |\theta|$ by SLSQP from zero with scipy's default tolerance, which stops after 18 iterations; 'converged' runs the same objective and algorithm to convergence. The objective gap (0.0518 against 0.0507) is almost all penalty: MSE changes by 0.27 percent, and the penalty falls on constants whose values depend on an arbitrary normalization. Intervals: paper-cluster bootstrap (400 draws), percentiles of $g_N + g_D$ mapped to τ_C (∞ when non-positive growth lies inside); the first two rows use Ho et al.'s iid bootstrap. Profile intervals invert an iid Gaussian likelihood ratio and are indicative, because papers are clustered. The interval for g_N/g_C is located on a grid of step 0.025 near its boundaries (an earlier version's coarser grid gave [−0.39, 3.37], and its range of 5 to 38 months was computed over the grid points [−0.25, 3.25]); the profile has a second local minimum near −0.28, and a missed global minimum could only widen the interval. Hicks neutrality: $\alpha g_N = \beta g_D$. Effective data: D measured as effective data under repetition (Muennighoff et al., 2023).

Source: Authors' calculations from the data and code of Ho et al. (2024).

trary normalization. Unpenalized least squares gives 10.2 months, with interval [4.1, 27.6]. Estimators of the same model thus give 6.1 to 10.2 months, and the data cannot arbitrate between them. The finding concerns numerical optimization, not the order of magnitude of progress. It parallels the finding of Besiroglu et al. (2024) for the third approach of Hoffmann et al. (2022), and convergence problems in demand estimation (Knittel and Metaxoglou, 2014; Dubé, Fox and Su, 2012).

Attenuated exponents.—Ho et al.’s data exponent is 0.04, against about 0.37 in experiments, and Whitfill (2025) reads the gap as evidence that progress is overstated about ninefold: dividing their year coefficients by experimental exponents gives 76.5 months. Imposing experimental exponents in estimation is rejected (mean squared error rises by a factor of 1.3 to 4.8 across 14 experimental technologies) and implies negative or implausibly slow progress. Two specification choices explain much of the exponent gap while moving τ_C much less (Table E7). First, fitting the $E = 0$ form to the Chinchilla sweep itself lowers the frontier elasticity from 0.178 to 0.053, close to the sweep’s mean elasticity of total rather than reducible loss (0.054). The like-for-like gap is therefore about 2 to 2.5-fold, not 7 to 9-fold. Dropping E rescales exponents and year coefficients by similar factors, so rates barely move: imposing a common E from 0 to 1.5 nats per word doubles $\hat{\gamma}$ but moves τ_C only from 10.2 to 12.4 months. Ho et al.’s data themselves prefer $E = 0$ within this form, so this argument rests on the sweep and on simulations; in the simulations the authors’ procedure can also stop at a local optimum that overstates progress. Second, their D is dataset size, while small models were trained for a median of 62 to 140 epochs. Measuring effective data under repetition (Muennighoff et al., 2023) raises τ_C by a factor of 1.5 to 4, with an unbounded upper confidence limit.

Scale dependence and frontier regressions.—Two further questions need designed data. Whether the compute-equivalent gain of transformers over earlier architectures depends on scale turns on whether their frontier elasticities differ (Corollary A2). In Ho et al.’s data the difference is 0.008, with interval $[-0.012, 0.034]$, and bootstrap intervals for the implied gains span six orders of magnitude (Table E8). And regressions of loss on compute that omit time effects overstate the compute elasticity by a factor of 1.15 on all 231 rows and 1.44 among 36 record-setting releases (Corollary A4). Among record setters, whose compute moves closely with time, independently measured growth rates imply an algorithmic share of 0.37 and a predicted factor of 1.58, consistent with the estimate. On all rows the prediction fails, because compute and time are only moderately correlated (0.62) and the ordinary omitted-variable formula applies instead.

The missing data.—Proposition A3 names the data that would settle these questions: IsoFLOP sweeps at several vintages on a common output, whose drift in the compute-optimal M^* signs the bias of technical change and whose curvature fixes its size. Our controlled experiment (Section III) supplies the cross-sectional

TABLE E7—WHY IS THE CROSS-LAB DATA EXPONENT SMALL? CHANNELS OF ATTENUATION

	n	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\gamma}$	τ_C [95% CI]	Default stop	
						$\hat{\beta}$	τ_C
<i>Panel A. Cross-lab model on alternative data</i>							
Baseline (model 7)	231	0.138	0.025	0.021	6.1 [3.0, 26.5]	0.040	8.7
Benchmark-specific perplexity	231	0.139	0.023	0.020	5.7 [2.8, 21.2]	0.030	9.0
(a) Tokens seen, epochs = 1	231	0.065	0.035	0.023	8.0 [1.1, ∞]	0.018	2.6
(a) Tokens seen, epochs imputed	231	0.121	0.028	0.023	4.7 [2.4, ∞]	0.027	4.7
(a) Effective data	231	0.057	0.067	0.031	25.7 [3.9, ∞]	0.052	12.7
(a) Word-level vocabulary	134	0.040	0.000	0.000	n.d.	0.008	0.9
(a) Vocabulary control	185	0.135	0.021	0.018	3.2 [1.4, 18.7]	0.036	4.6
(b) Published 2018 or later	180	0.052	0.077	0.031	7.8 [2.0, ∞]	0.040	4.3
(b) Transformers only	104	0.061	0.039	0.024	5.2 [1.6, ∞]	0.032	11.3
(b) No year terms	231	0.063	0.084	0.036	—	0.037	—
(c) Benchmark-specific E	231	0.072	0.042	0.026	10.5 [2.7, 44.3]	0.042	10.5
(d) All models per paper	261	0.055	0.064	0.030	15.3 [2.7, 79.3]	0.038	7.6
(d) Top model per paper	144	0.143	0.025	0.022	6.7 [2.5, 56.9]	0.042	9.6
(d) Flagged outliers kept	238	0.150	0.023	0.020	5.2 [2.5, 17.9]	0.030	7.5
(d) No zero-shot evaluations	207	0.049	0.091	0.032	16.6 [3.1, 23.5]	0.059	11.4
<i>Panel B. Chinchilla sweep</i>							
E estimated, Huber	240	0.347	0.367	0.178	[0.169, 0.188]		
$E = 0$, Huber	240	0.375	0.061	0.053	[0.051, 0.055]		
$E = 0$, least squares	240	0.061	0.377	0.053			
Total-loss elasticities	240	0.053	0.056	0.054			
<i>Panel C. Monte Carlo on the cross-lab design</i>							
	Global optimum			MC		Authors' code	
	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\gamma}$	τ_C [5, 95]		$\hat{\beta}$	τ_C
Correct specification	0.348	0.366	0.178	9.9 [7.4, 12.8]		0.366	12.0
(c) E dropped	0.046	0.388	0.041	11.2 [8.8, 15.5]		0.346	11.1
(c) and neutrality imposed	0.046	0.489	0.042	4.4 [3.7, 5.4]		0.446	4.2
(c)+(a) $D = \text{dataset size}$	0.032	0.389	0.029	10.4 [7.9, 16.5]		0.048	4.6
(c)+(a)+(d) best 3	0.036	0.415	0.033	10.9 [7.6, 18.1]		0.280	11.9
(c)+(a) random 3	0.033	0.398	0.030	10.7 [7.5, 17.9]		0.303	10.5
(c)+(a), $\rho(\psi_D, \ln D) = -0.5$	0.031	0.370	0.028	9.8 [5.3, 16.5]		0.045	4.6
(c)+(a), $\rho(\psi_D, \ln D) = +0.5$	0.025	0.384	0.023	10.5 [5.8, 21.4]		0.288	10.1

Note: Panel A: the model and penalized objective of [Ho et al. \(2024\)](#) (mean squared error plus $0.0025 \sum |\theta|$, $E = 0$), minimized to convergence on the indicated data; the last two columns (default stop) stop the optimizer at the default tolerance used in the authors' code. τ_C is the doubling time of effective compute, with a 95 percent paper-cluster bootstrap interval (400 draws, each started at zero; " ∞ " when nonpositive growth is inside the interval). Rows: (a) measurement of data, with D as tokens seen (missing epoch counts set to 1 or imputed from dataset size), as effective data under repetition ($R^* = 15.4$; [Muennighoff et al., 2023](#)), or with word-level vocabularies only or a vocabulary control; 33 percent of epoch counts are imputed; (b) time and collinearity; (c) benchmark-specific irreducible loss estimated by NLS; (d) selection, using all models of each paper, only the best, keeping flagged outliers, or dropping zero-shot evaluations. Panel B: the [Besiroglu et al. \(2024\)](#) estimator on the Chinchilla sweep with E estimated or set to zero, with 95 percent bootstrap intervals for $\gamma = \alpha\beta/(\alpha + \beta)$; the last row gives the sample means of the total-loss elasticities $\alpha u/L$, $\beta v/L$, and $\gamma(L - E)/L$ implied by the full model. Panel C: pseudo-true values (at the global optimum and with the authors' code) from noise-free data when the truth is the Besiroglu technology with Hicks-neutral progress ($\tau_C = 12$ months) evaluated at the actual inputs of the sample; MC median and 5th–95th percentiles of τ_C from 204 replications with noise of standard deviation 0.215; rows switch on (c) dropping E , (a) measuring D as dataset size, (d) keeping the best three models per paper, and a data-augmenting productivity shock ψ_D correlated with $\ln D$. n.d.: not defined because $\hat{\beta} = 0$.

Source: Data of [Ho et al. \(2024\)](#); Epoch AI digitization of [Hoffmann et al. \(2022\)](#); authors' estimates.

TABLE E8—COMPUTE-EQUIVALENT GAINS OF TRANSFORMERS AND THE BIAS OF FRONTIER REGRESSIONS

<i>Panel A. Gain of transformers over other architectures</i>						
	Frontier elasticity		Difference	Gain $f(C)$ at $C =$		
	Other	Transformer	[95% CI]	10^{17}	10^{19}	10^{23}
Penalized	0.0133	0.0217	0.0085 [−0.012, 0.034]	0.02	0.45	163
Unpenalized	0.0197	0.0210	0.0013 [−0.011, 0.028]	10.6	14.5	27.1
<i>Panel B. Log loss on log compute, with and without a time trend</i>						
	Compute slope			Inflation	Compute	Implied
	Rows	No trend	Trend	[95% CI]	growth	s_A
All rows	231	0.0282 (0.0020)	0.0246 (0.0018)	1.15 [1.10, 1.21]	1.12	0.13
Record setters	36	0.0329 (0.0024)	0.0228 (0.0027)	1.44 [1.24, 1.78]	2.64	0.31

Note: Panel A: model 13 of [Ho et al. \(2024\)](#) (architecture-specific exponents, common factor-augmenting year terms, $E = 0$), estimated with the penalized objective run to convergence or by unpenalized NLS, on their 231-row sample, WikiText-103 in 2023. $f(C)$ is the compute a non-transformer lab needs to match the transformer frontier loss at compute C (FLOP), relative to C ; with $E = 0$, $\ln f$ is linear in $\ln C$ and constant only if the two frontier elasticities coincide, consistent with the condition in Corollary A2 (equal E and equal γ). Non-transformer models span $C \leq 3 \times 10^{18}$ FLOP, so f above that is extrapolated; bootstrap intervals of f span more than six orders of magnitude. Panel B: OLS of the log of log perplexity on $\ln(6ND)$ and benchmark indicators, with and without a linear publication-year trend; paper-clustered standard errors in parentheses; inflation is the ratio of the two slopes, with a paper-cluster bootstrap interval; compute growth is the within-sample growth of log compute per year; the implied algorithmic share s_A , one minus the ratio of the slope with trend to the slope without, is backed out of the inflation factor and is not an independent test. Record setters are releases with the best perplexity on the benchmark to date.

Source: Data of [Ho et al. \(2024\)](#); authors' estimates.

analogue for a change in data quality at one date. [TBD-m9: one sentence on the shift in the compute-optimal M^* at fixed compute between the two corpora of our experiment, and hence the sign of the Hicks bias of the data-quality difference (Proposition A3(i) applied across corpora rather than dates); delete if m9 cannot answer it.]

E6. Allocative versus Technical Change

Measured progress, the fall in the compute needed to reach a given loss, mixes shifts of the frontier with movements toward it. By Proposition A4, the move from the allocations of Kaplan et al. (2020) to those of Hoffmann et al. (2022) changed the mix, not the technology, so its gain is allocative; under the Besiroglu et al. (2024) parameters, Gopher’s allocation used about twice the compute its loss required. Gundlach et al. (2025) define the compute-equivalent gain from such rebalancing and count it as algorithmic progress.

Measurement.—Measuring the gain actually realized between eras requires a choice. Training-cost efficiency, the Farrell efficiency against the training-only frontier, $C_{\min}(L(N, D))/6ND$, counts deliberate over-training as waste, although over-training reflects the value that developers place on compact models (Section IV). Serving demand cannot produce a wedge below one (Proposition 2). We therefore count only the elimination of under-training: training-cost efficiency evaluated at $\min(w, 1)$, which within the Chinchilla family depends on the wedge alone. The truncated measure attributes every wedge below one to allocative error. A wedge below one can also reveal a binding data constraint or data-augmenting productivity specific to the developer (Proposition A8(ii) and (iv)). For 2020–21 models trained on a few hundred billion tokens, a belief in the Kaplan et al. (2020) law is the more plausible reading, but it remains an assumption.

Realized gains.—We compare models in the Epoch database with at least 10^{23} FLOP: 7 released in 2020–21 (era 1) and 92 in 2022–24 (era 2), all dense, trained for at most four epochs, with $D = C/6N$ (Table E9, Figure E3). Every era-1 model has $w < 1$ under every technology, and the median ratio of tokens to parameters rose from 1.7 to 92 between the eras. Counting only the elimination of under-training, geometric-mean training-cost efficiency rose by a factor of 2.0 under the reference technology (Chinchilla with κ free) and 1.7 under Besiroglu et al. (2024), with 95 percent interval [1.21, 2.44]. Across all 14 estimated technologies the factor ranges from 1.1 to 4.9. Era-2 releases closed 80 to 94 percent of the era-1 under-training gap, in logs. The positive sign is close to mechanical, because every era-1 model is under-trained and truncation can only raise era-2 efficiency; the content is in the size. Without truncation the gain ranges from 0.55 to 4.3 across technologies: counting over-training as waste can reverse its sign.

The intervals resample models within era; only the Besiroglu interval also draws the technology’s parameters, and with seven era-1 models all intervals are rough. Technologies estimated on other sweeps use their own conventions for N , D and

TABLE E9—ALLOCATIVE GAINS FROM REBALANCING, WITH AND WITHOUT TRUNCATING THE WEDGE AT ONE

	$w < 1$ (%)	Untruncated		Truncated		Gap	Kaplan
	era 1/2 (1)	Gain (2)	95% CI (3)	Gain (4)	95% CI (5)	closed (6)	CEG (7)
<i>Chinchilla sweep</i>							
Besiroglu et al. (2024)	100 / 18	1.05	[0.57, 1.75]	1.68	[1.21, 2.44]	0.92	11.7
Hoffmann et al. (2022)	100 / 45	1.91	[1.53, 2.50]	2.34	[1.87, 3.03]	0.88	32.4
Hoffmann et al. (2022), rounded	100 / 50	2.41	[1.86, 3.24]	2.77	[2.16, 3.69]	0.86	46.7
Chinchilla, κ free (reference)	100 / 26	1.19	[0.92, 1.56]	1.97	[1.58, 2.53]	0.92	20.3
<i>Other public sweeps, Chinchilla form (Section III)</i>							
Farseeer, total N	100 / 42	1.58	[1.27, 2.04]	2.06	[1.66, 2.60]	0.89	38.7
Farseeer, non-embedding N	100 / 18	1.03	[0.86, 1.25]	1.54	[1.32, 1.84]	0.92	7.3
Gadre et al. (2025), C4	100 / 12	0.60	[0.51, 0.70]	1.14	[1.06, 1.26]	0.94	3.5
OLMo ladder	100 / 66	2.52	[1.99, 3.20]	2.63	[2.07, 3.33]	0.81	57.0
Muennighoff et al. (2023)	100 / 38	1.74	[1.38, 2.27]	2.33	[1.87, 2.97]	0.90	18.1
<i>Laboratory IsoFLOP designs, Chinchilla form (Online Appendix D)</i>							
Llama 3 IsoFLOPs (Meta)	100 / 12	0.74	[0.62, 0.90]	1.30	[1.16, 1.50]	0.93	5.1
Marin, Comma	100 / 62	4.31	[2.95, 6.60]	4.95	[3.42, 7.50]	0.80	388.5
Marin, DCLM	100 / 50	3.04	[2.09, 4.45]	4.00	[2.83, 5.77]	0.85	230.7
Marin, Nemotron-CC	100 / 49	2.57	[1.87, 3.68]	3.52	[2.55, 5.11]	0.86	166.3
(Mis)Fitting, FineWeb/C4	100 / 12	0.55	[0.40, 0.75]	1.64	[1.31, 2.12]	0.93	15.0
<i>Cross-lab regression</i>							
Ho et al. (2024), model 7	0 / 0	0.09	[0.07, 0.12]	1.00	[1.00, 1.00]	—	—
<i>Other samples, Besiroglu et al. (2024) technology</i>							
$C \geq 10^{23}$, cleaned (7, 79)	100 / 19	1.09	[0.60, 1.78]	1.70	[1.23, 2.43]	0.94	
Top five per year (10, 15)	70 / 40	1.27	[0.91, 1.72]	1.31	[1.05, 1.76]	0.73	
Epoch frontier models (7, 6)	86 / 33	1.56	[1.07, 2.19]	1.54	[1.12, 2.19]	0.82	

Note: Models in the Epoch database with $C \geq 10^{23}$ FLOP: 7 released in 2020–21 (era 1) and 92 in 2022–24 (era 2); dense, at most four epochs, $D = C/6N$. Training-cost efficiency, the Farrell efficiency against the training-only frontier, $CE = C_{\min}(L(N, D))/6ND$, depends only on the wedge $w = \varepsilon_N/\varepsilon_D$ within the Chinchilla family. Untruncated gain: ratio of era geometric means of CE . Truncated gain: the same with $CE(\min(w, 1))$, which counts only the elimination of under-training: over-training ($w > 1$) can reflect the value of compactness, including serving demand, and is not treated as inefficiency, whereas serving demand cannot produce $w < 1$ (Proposition 2). Column 1: share of models with $w < 1$. Column 6: $\ln(\text{truncated gain})$ divided by the era-1 log compute lost to under-training. Column 7: compute-equivalent gain of abandoning the Kaplan rule ($N \propto C^{0.73}$ from GPT-3) at 5×10^{26} FLOP, the counterfactual of Gundlach et al. (2025); the Kaplan allocations have $w < 1$ under every technology, so truncation does not change it. Intervals: percentile bootstrap over models within era (999 draws); only the Besiroglu et al. (2024) row also draws the technology’s parameters, so the other intervals hold the technology fixed, and with seven era-1 models all intervals are rough. A wedge below one is counted as allocative error; by Proposition A8(ii) and (iv) it can also reflect a binding data constraint or data-augmenting productivity specific to the developer. Technologies from other sweeps use each sweep’s own N and D conventions and loss corpus, so their levels are indicative. Group labels give the data on which each technology is estimated: the Chinchilla sweep (the reference technology of Section IV is Chinchilla with κ free), Chinchilla-form fits to the other public sweeps and to laboratory IsoFLOP designs, and Ho et al.’s cross-lab model, under which every model has $w > 1$.

Source: Authors’ calculations from the Epoch AI models database (Epoch AI, 2026).

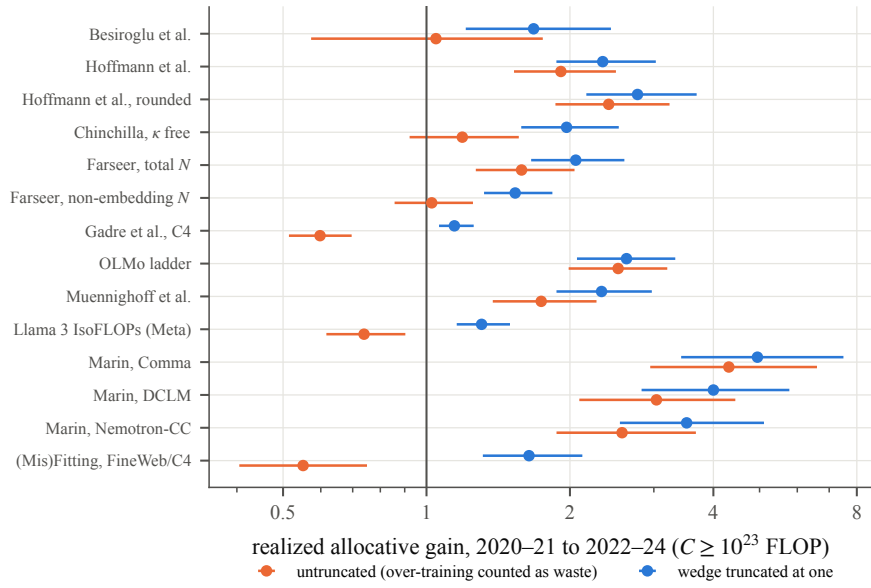


FIGURE E3. REALIZED ALLOCATIVE GAINS FROM REBALANCING, BY TECHNOLOGY

Note: Ratio of geometric-mean training-cost efficiency, 2022–24 against 2020–21, for Epoch models with at least 10^{23} FLOP (7 and 92 models), under each technology, with 95 percent percentile-bootstrap intervals over models within era (log scale). Orange: efficiency against the training-only frontier, which counts over-training as waste. Blue: the wedge truncated at one, which counts only the elimination of under-training. Only the Besiroglu et al. (2024) interval also reflects uncertainty in the technology's parameters. Estimates in Table E9.

the loss corpus, so their levels are indicative. The Marin fits, whose parameter exponents are about twice Chinchilla’s, give the largest gains (3.5 to 4.9). Samples of the five largest models per year and of Epoch’s frontier models give smaller and noisier truncated gains (1.3 and 1.5 under Besiroglu).

Comparison with progress.—Against the effective-compute gain implied by Ho et al.’s published rate over the same interval, 8.9-fold over 2.28 years, the realized allocative gain amounts to 6 to 73 percent in logs across technologies (31 percent under the reference technology). The denominator itself ranges from 6.4 to 22.5 across the estimators of Section E5. Gundlach et al. (2025) report that the Kaplan-to-Chinchilla rebalancing is worth about a factor of 10 at the 2025 frontier. Our counterfactual agrees: under the Besiroglu technology, continuing Kaplan’s rule from GPT-3 would cost a factor of 1.6 at GPT-3’s compute, 5.2 at 3.8×10^{25} FLOP and 11.7 at 5×10^{26} FLOP. Because Kaplan-rule allocations have $w < 1$ under every technology, truncation does not affect this counterfactual. What we add is the contrast with the gain realized between eras, 1.1 to 4.9, which is smaller than both. Under Ho et al.’s own exponents, finally, every model has $w > 1$. An attenuated data exponent makes tokens nearly worthless, so a regression with those exponents cannot credit rebalancing to inputs and loads it onto the time trend.

Summary.—Cross-lab data thus deliver returns to compute once the design is matched, though only as a low-power failure to reject, and measures of productivity dispersion whose size depends on the units. They do not deliver the split between inputs, the bias of technical change, or its rate without a restriction on the bias. In each case the missing variation is the transverse variation that designed IsoFLOP sweeps supply.

THE REVEALED VALUE OF COMPACTNESS: ROBUSTNESS

This appendix supports Section IV. Appendix F1 states what the wedge identifies under alternative objectives of the developer. Appendices F2 through F5 document the clean inference-demand sample, the 32 technologies, parameter-count conventions and the developers’ own technologies. Appendix F6 treats extrapolation beyond the designs, and Appendices F7 through F10 report the conduct tests, family token budgets, the cost side, validation and trends. Unless stated otherwise, wedges are evaluated under the reference technology, the Chinchilla form with the outer exponent free ($\sigma_\kappa^* = 0.701$), on the 77 models of the clean inference-demand sample, and intervals for sample statistics come from the joint distribution across bootstrap draws of the technology: each draw recomputes the median and the shares for the whole sample.

F1. What the Wedge Identifies

Table F1 lists what the wedge $w = \varepsilon_N/\varepsilon_D$ and the share $s = (w - 1)/w$ measure under the objectives and cost structures of Proposition A8. The rows

fall into three groups. In the first, a developer bears some or all of the cost of serving its model, and $w - 1 = \lambda\phi X_S/X_T$ is internalized serving expenditure relative to training expenditure. This ratio needs no conversion of FLOPs into dollars; a count of serving tokens needs the price ratio p as well. In the second group, a binding memory or tier cap or a latency objective multiplies the wedge by one plus the constraint’s normalized multiplier. The wedge then bounds the value of compactness from above without identifying it, and any wedge can be rationalized with no serving demand at all. In the third group, data costs, data caps and logit distillation add to the cost of tokens, so the wedge understates the value of compactness; a binding data cap can push it below one. Serving demand alone cannot produce $w < 1$ as long as serving cost rises with size at given quality.

The last row is the factor-bias critique of markup estimation. If the first-order conditions of the developer involve a technology or an output different from the one the econometrician uses, the estimated wedge is the true one times the ratio of the two technologies’ wedges at the same inputs. The developer’s own technology (Appendix F5) and the spread across 32 technologies (Appendix F3) are the two ways the paper addresses it. The cases also make different predictions about conduct (Remark A8), which Appendix F7 tests.

F2. The Clean Inference-Demand Sample

The clean sample was defined before any wedge was computed. It starts from the 173 general-purpose open-weight base models of the verified sample, whose parameter and token counts are documented by their developers (Online Appendix B). Nine steps, applied in order, drop models whose (N, D) is not the choice that Proposition 2 describes; a model is dropped at the first step that applies. The replication package lists every flag with its reason and source.

- (a) *One observation per pretraining run* (8 models; the first release is kept): Llama 3.1 8B and 70B, which continue the pretraining of the Llama 3 runs; Qwen1.5-72B, with the same N and D as Qwen-72B; OpenLLaMA v2 3B and 7B; Yi-1.5 6B and 34B, which the model card describes as continued pretraining of Yi on 500 billion tokens; and H2O-Danube2-1.8B, initialized from H2O-Danube-1.8B.
- (b1) *Research suites whose D is fixed by design* (45): Pythia, OPT, BLOOM, RWKV-4-Pile, XGLM, Cerebras-GPT and GPT-Neo, GPT-J and GPT-NeoX; the natural-language stage of CodeGen; and Falcon-RW-1B, a data ablation.
- (b2) *Replications and data benchmarks* (8): OpenLLaMA, RedPajama-INCITE, LLM360 Amber and K2, and DCLM-7B.
- (c) *Mixture-of-experts models* (12), reported with bounds in Appendix F4.
- (d) *Distilled, pruned or derived initialization* (10): Gemma 2 2B and 9B and every Gemma 3 size, which their technical reports describe as trained with

TABLE F1—WHAT THE WEDGE IDENTIFIES UNDER ALTERNATIVE DEVELOPER OBJECTIVES

Developer objective or cost structure	Wedge $w = \varepsilon_N/\varepsilon_D$	$s = (w - 1)/w$ measures	$\hat{T}$ vs. λpT
Lifetime compute, developer serves ($\lambda = p = \phi = 1$)	$1 + T/(3D)$	Serving share of lifetime compute	Equal
(a) Serving cost borne at rate λ , price ratio p , cost elasticity ϕ	$1 + \lambda\phi pT/(3D)$	$\lambda\phi X_S/(X_T + \lambda\phi X_S)$	$\times \phi$
(a') Open release, adoption value $v(A)$, users bear serving	$1 + v'A \varepsilon_{A,P} /X_T$	Adoption value of compactness	Not T
(b) Binding memory or tier cap, multiplier ν	$(1 + \nu)(1 + m_N)$	Upper bound on serving share	Overstates
(c) Binding latency SLO, $\tau(N) \leq \bar{\tau}$	$(1 + \nu_\tau)(1 + m_N)$	Upper bound on serving share	Overstates
(d) Data cost c_D per token	$(1 + m_N)/(1 + m_D)$	$(X_S - X_D)/(X_T + X_S)$	Understates
(d') Binding data cap, multiplier μ	$(1 + m_N)/(1 + m_D + \mu)$	Below serving share; $w < 1$ possible	Understates
(e) Logit distillation, teacher N_T	$\frac{1 + T/(3D)}{1 + N_T/(3N)}$	$(X_S - X_{\text{teach}})/(X_T + X_S)$	Understates
(f) Dollar cost $\propto N^{1+\delta}D$; serving $\propto N^{1+\delta_S}$	$1 + \delta + \lambda(1 + \delta_S)pT/(3D)$	Mixes δ with serving	$+3D\delta + \lambda\delta_S pT$
(g) Family token budget, sizes chosen	$\sum_i \pi_i(1 + m_{N,i}) = (1 + m_D)\bar{w}_H$	$1 - 1/\bar{w}_H = \sum_i \omega_i s_i$	Only $\sum_i \pi_i \hat{T}_i$ identified
Lab's technology differs (factor bias, beliefs, output)	$\hat{w} = w w_L/w_{L'}$	Contaminated	Over iff $w_L > w_{L'}$

Note: The developer chooses (N, D) to maximize $V(L, N)$ minus training expenditure $X_T = c_T 6ND$ (and data cost $c_D D$). $m_N \equiv -(\partial V/\partial \ln N)_L/X_T$ is the value of compactness at fixed quality per unit of training expenditure; $m_D \equiv c_D D/X_T$; X_S is serving expenditure, λ the share of it the developer internalizes, p the price of a serving FLOP relative to a training FLOP, ϕ the size elasticity of serving cost; $\varepsilon_{A,P}$ is the elasticity of adoption with respect to users' serving cost; $\nu, \nu_\tau, \mu \geq 0$ are normalized multipliers; δ (δ_S) is the size elasticity of the price of a training (serving) FLOP; $\pi_i \propto \omega_i/w_i$ with ω_i the member's compute share, $\bar{w}_H$ the compute-weighted harmonic mean of member wedges; w_L and $w_{L'}$ are the wedges $\varepsilon_N/\varepsilon_D$ at the same inputs under the econometrician's and the lab's technology or output. The last column compares $\hat{T} \equiv 3D(\hat{w} - 1)$ with λpT , the internalized serving demand in training-FLOP units (rows other than (a) and (f) take $\phi = 1$); "Over iff $w_L > w_{L'}$ " means that $\hat{T}$ overstates, which under factor bias is $\chi < 0$. Every row was verified by brute-force maximization of the objective (maximum relative error 2×10^{-7} ; Online Appendix A, Propositions A8 and A9).

Source: Authors' derivations.

knowledge distillation; Llama 3.2 1B and 3B, pruned from Llama 3.1 8B and distilled from larger models’ logits; and Yi-1.5-9B, continued from a depth-upscaled Yi-6B. A keyword search of all 179 saved model cards and of these reports found no other case.

- (e) *Multimodal pretraining*: Gemma 3 4B–27B and Llama 4, all already removed at (c) or (d).
- (f) *No base checkpoint* (7): Phi-3 mini, small and medium, Phi-3.5-mini, Phi-4-mini, phi-4, and Qwen3-32B.
- (g) *Non-transformer or hybrid architectures*: RWKV, already removed at (b1).
- (h) *Token counts without a primary source* (6): Qwen1.5 0.5B to 32B, whose counts appear only in the ObsScaling file (Ruan, Maddison and Hashimoto, 2024).

The result is 77 models from 36 families and 18 developers, released 2023–2025. Table F2, panel A, follows the headline through the steps. The median wedge stays between 3.77 and 4.46: dropping the research suites and replications, whose wedges are lower, raises it, and dropping mixture-of-experts, distilled and instruction-only models lowers it again. The sign is less robust than the point estimates suggest: every one of the 32 technologies puts the point estimate above one for 64 percent of clean models, but the union of their 95 percent intervals lies above one for only 18 percent.

Panel B splits the clean sample. Families that train every size on one token budget, whose member wedges are not revealed preferences (Appendix F8), have a median wedge of 3.90, against 4.01 for families with size-specific budgets; without them the median share is 0.75. Dropping the nine models trained mainly on teacher-generated synthetic data (phi-1.5, phi-2 and SmolLM 1–3) gives a median share of 0.74. Models of developers that served their own language models at release have a median share of 0.80 and a median wedge of 4.95, against 0.74 and 3.80 for the others; the 12 models with a stated on-device or local target, from three developers, have 0.86. The last rows compare models inside and outside the ranges of M that designs observe. Inside Chinchilla’s range ($M \leq 341$) the median share is 0.56 and no model has all technologies’ intervals above one; only 2 of the 77 models lie inside Chinchilla’s design in both M and compute.

F3. Technologies

The ex-ante rule.—The set of technologies was fixed before any wedge was computed. It contains (i) every technology estimated in Section III and Online Appendix D from final-checkpoint losses of a designed sweep with the primary estimator (Huber loss on log loss), with each corpus and parameter-count convention as its own row, under $\kappa = 1$ and, where estimated, with κ free; (ii) for each IsoFLOP design, the Approach-2 path with its $\kappa = 1$ curvature, the primal

TABLE F2—THE CLEAN INFERENCE-DEMAND SAMPLE: CONSTRUCTION AND SUBSAMPLES

<i>Panel A. Sequential cleaning steps (reference technology: Chinchilla, κ free)</i>								
Step (models dropped)	n	Groups	Median w	Share $w > 1$	All points > 1	Band > 1	Median s	Median $w, \kappa = 1$
Verified sample	173	67	3.77	0.93	0.61	0.17	0.73	3.19
(a) One per pretraining run (−8)	165	65	3.79	0.93	0.62	0.18	0.74	3.19
(b1) Research suites (−45)	120	56	4.20	0.97	0.70	0.16	0.76	3.51
(b2) Replications, benchmarks (−8)	112	51	4.46	0.98	0.72	0.17	0.78	3.70
(c) Mixture of experts (−12)	100	43	4.30	0.98	0.69	0.18	0.77	3.61
(d) Distilled, pruned, derived (−10)	90	40	4.03	0.98	0.66	0.17	0.75	3.42
(e) Multimodal (−0)	90	40	4.03	0.98	0.66	0.17	0.75	3.42
(f) No base checkpoint (−7)	83	37	3.99	0.98	0.64	0.18	0.75	3.34
(g) Non-transformer (−0)	83	37	3.99	0.98	0.64	0.18	0.75	3.34
(h) D undocumented (−6): clean	77	36	3.99	0.97	0.64	0.18	0.75	3.34
<i>Panel B. Subsamples of the clean sample</i>								
Common- D families	32	11	3.90	1.00		0.13	0.74	
Size-specific- D families	30	10	4.01	0.93		0.30	0.75	
Singletons	15	15	3.93	1.00		0.07	0.75	
All but common- D families	45	25	3.99	0.96		0.22	0.75	
No synthetic-data models	68	31	3.80	0.97		0.15	0.74	
Developer serves its models	33	7	4.95	1.00			0.80	
Developer does not serve	44	12	3.80	0.95			0.74	
On-device or local target	12	3	7.59	1.00			0.86	
$M \leq 341$ (Chinchilla’s range)	30		2.28	0.93		0.00	0.56	
$M > 341$	47		6.57	1.00		0.30	0.85	
In Farseer’s range of M^a	57		3.10	1.00		0.04	0.68	
Beyond Farseer’s range ^a	20		10.8	1.00		0.60	0.91	

Note: Panel A: sequential steps from the 173 verified open-weight base models; each row describes the sample that remains after the step; a model is dropped at the first step that applies. Reference technology: the Chinchilla form with the outer exponent free (inner exponents 0.424 and 0.431, $\sigma_\kappa^* = 0.701$). $s = (w - 1)/w$. All points > 1 : every one of the 32 ex-ante technologies of Table F3 puts the point estimate above one. Band > 1 : the union of the 95 percent intervals over those technologies lies above one. Last column: the median under the Chinchilla form with $\kappa = 1$. Groups: families, except developers in the three serving and target rows of panel B. Common- D families train every member on token budgets within 10 percent of each other. Synthetic data: phi-1.5, phi-2 and SmoLLM 1–3 dropped. Serving: the developer operated a commercial API or consumer product serving its own language models at the model’s release (Meta counts from 2023-09-27). ^aFarseer technology with κ free, non-embedding M (design maximum 2,570); its median over all 77 models is 3.91.

Source: Model cards, technical reports and Hugging Face configuration files; authors’ calculations.

fit, and the path combined with the design’s model-free curvature; and (iii) the literature’s estimates on the Chinchilla data and the allocation laws published by Meta, DeepSeek (DeepSeek-AI et al., 2024) and MiniCPM (Hu et al., 2024). Excluded ex ante are DataDecide and the checkpoint-interpolated IsoFLOPs of Li, Kudugunta and Zettlemoyer (2025), whose losses come before the learning-rate schedule ends, an artifact that varies with D/N , and estimator or sample variants of a design already in the set, which are reported below as sensitivity checks. Table F3 lists the 32 technologies.

When a path and a curvature come from different sources, the inner exponents are $\alpha = S(1 - a)$ and $\beta = Sa$ with $S = 2(1/\sigma^* - 1)$; the wedge depends on the technology only through S , the path slope a and the path level (Corollary A3). DeepSeek’s law is stated in non-embedding FLOPs per token, which we compute for other developers’ models from their configuration files; MiniCPM’s in non-embedding parameters. MiniCPM’s compute-optimal ratio of 192 tokens per parameter at 10^{21} FLOP lies outside its own design, whose ratios run from 10 to 60, so it is itself an extrapolation. The reference technology and the model-free rows are bootstrapped design-conditionally (wild bootstrap with the weights of Feng, He and Hu 2011); the other rows reuse the pairs or stratified draws of their source estimates, and published laws enter as points.

What drives the dispersion.—The median share ranges from 0.17 (MiniCPM) to 0.85 (Muennighoff et al.’s runs with κ free). Technologies that let the data choose the curvature give 0.69 to 0.79. The low medians come from three sources: $\kappa = 1$ curvatures far above the model-free estimates, as in Marin’s Approach-2 laws (σ^* of 0.87 to 0.90, from designs spanning two decades of compute on which the frontier’s E and γ are not separately identified); the OLMo ladder with $\kappa = 1$, whose $M^*(10^{24})$ of 291 puts a quarter of the sample below one; and MiniCPM’s zero point. The high medians come from the κ -free fits of the small sweeps, whose curvatures (0.51 to 0.62) are heterogeneous.

Curvature governs levels, not signs. Holding the reference zero point fixed, lowering σ^* from 0.80 to 0.60 raises the median wedge from 2.25 to 8.64 and the median share from 0.55 to 0.88, and Llama 3 8B’s wedge from 3.0 to 19.5. The share of models with $w > 1$, 0.974, does not move. At $\sigma^* = 0.70$ the median wedge is 4.00, almost exactly the reference value.

Estimator variants.—On the Chinchilla data, least squares in levels gives a median wedge of 4.58, keeping the five outliers gives 5.50, and the Chinchilla form with $\kappa = 1$ gives 3.34 (Table F3). The bootstrap matters for intervals, not for points. For the reference, pairs resampling, which changes the design, gives a 95 percent interval for the median wedge of [3.25, 6.04], against [3.21, 4.86] from the design-conditional wild bootstrap; for the $\kappa = 1$ refit, a cluster bootstrap over the nine budgets gives [1.83, 6.48]. A separate implementation of the κ -free fit differs from ours in the fourth decimal; over all 173 verified models it gives a median wedge of 3.78, against 3.77.

TABLE F3—THE REVEALED WEDGE UNDER EVERY EX-ANTE TECHNOLOGY

Technology	σ^*	$M^*(10^{24})$	Median w	Share $w > 1$	Median s
<i>Chinchilla data</i>					
κ free (reference)	0.701	21.6	3.99 [3.21, 4.86]	0.97	0.75 [0.69, 0.79]
$\kappa = 1$	0.737	17.6	3.34 [2.64, 4.81]	1.00	0.70 [0.62, 0.79]
Besiroglu et al. parameters	0.737	18.1	3.31	1.00	0.70
Hoffmann et al., Approach 3	0.762	62.0	2.12	0.88	0.53
Non-embedding N , $\kappa = 1$	0.753	9.7	3.65 [2.83, 4.71]	1.00	0.73 [0.65, 0.79]
<i>Farseer</i>					
Non-embedding N , $\kappa = 1$	0.772	17.3	2.74 [2.38, 3.20]	1.00	0.63 [0.58, 0.69]
Total N , $\kappa = 1$	0.724	67.7	2.64 [2.23, 3.10]	0.88	0.62 [0.55, 0.68]
Non-embedding N , κ free	0.710	18.3	3.91 [3.77, 4.06]	1.00	0.74 [0.73, 0.75]
<i>Gadre et al.</i>					
RefinedWeb, $\kappa = 1$	0.796	1.4	4.41 [2.19, 7.67]	1.00	0.77 [0.54, 0.87]
C4, $\kappa = 1$	0.795	6.0	3.18 [1.12, 7.66]	1.00	0.69 [0.11, 0.87]
RedPajama, $\kappa = 1$	0.795	3.9	3.50 [2.52, 6.94]	1.00	0.71 [0.60, 0.86]
RefinedWeb, κ free	0.594	61.0	4.76 [3.49, 15.39]	0.88	0.79 [0.71, 0.94]
C4, κ free	0.607	78.5	3.95 [2.06, 10.60]	0.88	0.75 [0.51, 0.91]
RedPajama, κ free	0.623	32.3	5.64 [3.48, 12.88]	0.94	0.82 [0.71, 0.92]
<i>OLMo ladder</i>					
$\kappa = 1$	0.825	291	1.32 [0.45, 2.66]	0.77	0.24 [−1.22, 0.62]
κ free	0.544	142	4.68 [3.80, 6.37]	0.84	0.79 [0.74, 0.84]
<i>Muennighoff et al.</i>					
$\kappa = 1$	0.735	36.7	2.55 [0.54, 10.02]	0.92	0.61 [−0.85, 0.90]
κ free	0.511	100.0	6.87 [1.15, 37.30]	0.86	0.85 [0.13, 0.97]
<i>Meta (Llama 3 IsoFLOPs)</i>					
Meta’s law (A2, $\kappa = 1$)	0.785	31.3	2.30 [1.95, 2.65]	0.92	0.57 [0.49, 0.62]
Primal, $\kappa = 1$	0.769	9.6	3.26 [2.36, 4.04]	1.00	0.69 [0.58, 0.75]
A2 path, model-free σ^*	0.660	31.3	4.79 [3.60, 6.58]	0.92	0.79 [0.72, 0.85]
<i>Marin ladders</i>					
Nemotron-CC, primal	0.658	81.8	3.52 [2.45, 6.84]	0.87	0.72 [0.59, 0.85]
DCLM, primal	0.647	106	3.31 [2.41, 6.40]	0.86	0.70 [0.59, 0.84]
Comma, primal	0.663	228	2.13 [1.48, 3.97]	0.78	0.53 [0.32, 0.75]
Nemotron-CC, A2, $\kappa = 1$	0.867	35.1	1.63 [1.36, 2.17]	0.92	0.39 [0.27, 0.54]
DCLM, A2, $\kappa = 1$	0.866	32.5	1.66 [1.26, 2.40]	0.92	0.40 [0.21, 0.58]
Comma, A2, $\kappa = 1$	0.904	58.0	1.33 [1.13, 1.57]	0.88	0.25 [0.11, 0.36]
Nemotron-CC, model-free σ^*	0.705	35.1	3.78 [2.48, 5.79]	0.92	0.74 [0.60, 0.83]
DCLM, model-free σ^*	0.713	32.5	3.71 [2.37, 6.16]	0.92	0.73 [0.58, 0.84]
Comma, model-free σ^*	0.700	58.0	3.21 [2.40, 5.38]	0.88	0.69 [0.58, 0.81]
<i>Published laws</i>					
DeepSeek LLM, reference σ^*	0.701	13.9	4.57 [3.96, 5.31]	1.00	0.78 [0.75, 0.81]
MiniCPM	0.794	426	1.20	0.64	0.17
Range, 32 technologies		1.4–426	1.20–6.87	0.64–1.00	0.17–0.85

Note: Clean inference-demand sample (77 models). Brackets (median w and s): 2.5 and 97.5 percentiles over the technology’s bootstrap draws, recomputing the sample statistic in every draw (one technology draw shared by all models); no bracket: published point estimate; the share with $w > 1$ under the reference has the interval [0.92, 1.00]. A2: IsoFLOP-argmin (Approach 2) path. $M^*(10^{24})$: compute-optimal tokens per parameter at 10^{24} FLOP in the technology’s own convention, an extrapolation for every row. The ex-ante inclusion rule and the bootstrap draws are described in the text. Model-free rows combine the design’s own IsoFLOP path with its model-free σ^* (Online Appendix Table D8). Non-embedding and OLMo-convention technologies are evaluated at each model’s own count.

Source: Hoffmann et al. (2022); Besiroglu et al. (2024); Li et al. (2025b); Gadre et al. (2025); Bhagia et al. (2025); Muennighoff et al. (2023); Grattafiori et al. (2024); Marin Community (2026); DeepSeek-AI et al. (2024); Hu et al. (2024); authors’ calculations.

F4. Parameter Conventions and Mixture-of-Experts Models

Each technology is evaluated at the model’s own parameter count in the technology’s convention: the total count for the Chinchilla, Gadre et al. and laboratory technologies, the non-embedding count, from each model’s configuration file, for Farseeer, MiniCPM and the non-embedding Chinchilla refit, and OLMo’s count, which excludes only the input embedding, for the OLMo ladder. A non-embedding technology ignores the output head, which costs $2N_{\text{head}}$ FLOPs per token in training and serving alike (N_{head} is vocabulary size times width). Counting it in FLOPs turns the ratio of lifetime to training cost into $w(1 + N_{\text{head}}/N_{\text{ne}})$.

Table F4, panel A, shows that the sample median moves little across conventions, because embeddings are at most 28 percent of parameters in the clean sample. For the eight clean models below one billion parameters, the convention matters under one sweep but not the other. Comparing like with like, the non-embedding count lowers their median wedge by 8 percent under the Chinchilla form (9.13 against 9.96) and by 36 percent under the Farseeer form (6.85 against 10.67). The Farseeer gap exceeds the effect of tokenizer differences on D : under the reference, $\partial \ln w / \partial \ln D = 0.43$, so a 20 percent difference in tokens per byte moves the wedge by about 8 percent. We do not normalize D by bytes.

Mixture-of-experts models are excluded from the clean sample. Panel B bounds their wedges under the reference. A dense technology evaluated at active parameters treats a mixture of experts as parameter-augmenting and overstates its wedge; evaluated at total parameters, it ignores the loss advantage of sparsity (Clark et al., 2022; Krajewski et al., 2024; Abnar et al., 2025). The median of the 12 models is 5.14 at active and 2.07 at total parameters. The bounds are widest for the largest models: DeepSeek-V3 has a wedge of 3.50 or 1.01, and Kimi K2 3.79 or 0.87. Neither bound is an estimate from a mixture-of-experts scaling law.

F5. The Developers’ Own Technologies

Where a developer has published scaling evidence, its own technology is primary for its models. For Meta this is its IsoFLOP path, fitted to all ten budgets of Meta’s figure, which reproduces the published law $D^* = 0.299 C^{0.537}$ (Grattafiori et al., 2024), combined with the model-free curvature of Meta’s own IsoFLOP profiles, 0.660 (standard error 0.023). For AI2 it is the OLMo ladder with κ free (Bhagia et al., 2025); for DeepSeek, its published law with the reference curvature, since DeepSeek published no curvature; and for Marin, the Nemotron-CC ladder path with its model-free curvature (Marin 8B’s model card lists Nemotron-CC and DCLM among its data). These technologies cover 22 models: Meta 10, AI2 9, DeepSeek 2 and Marin 1. For 11 of them the developer’s law was published before the model was planned. For the other 11 (Llama 1 and 2, OLMo 1 and 1.7, Marin 8B) it came later, so it describes the developer’s technology but not the law it planned with.

Table F5, panel A, reports two model-free estimators of the designs’ curvature.

TABLE F4—PARAMETER-COUNT CONVENTIONS AND MIXTURE-OF-EXPERTS MODELS

Panel A. Clean sample (77 models) under technologies in different parameter conventions							
Technology	Convention of N	Median w	Median w , head FLOPs	Share $w > 1$	Median s	Median w , $N < 1\text{B}$ ($n = 8$)	
Chinchilla, κ free (reference)	Total	3.99		0.97	0.75	15.10	
Chinchilla, $\kappa = 1$	Total	3.34		1.00	0.70	9.96	
Farseer, $\kappa = 1$	Total	2.64		0.88	0.62	10.67	
Chinchilla, $\kappa = 1$ (refit)	Non-embedding	3.65	3.74	1.00	0.73	9.13	
Farseer, $\kappa = 1$	Non-embedding	2.74	2.86	1.00	0.63	6.85	
Farseer, κ free	Non-embedding	3.91	4.15	1.00	0.74	14.26	
MiniCPM (published law)	Non-embedding	1.20	1.27	0.64	0.17	3.04	
Panel B. Mixture-of-experts models (excluded from the clean sample), reference technology							
Model (year)	N (B), D		M ,	w		s	
	active / total	(T)	active / total	Active N	Total N	Active N	Total N
OLMoE 1B-7B (2024)	1.3 / 7	5.1	3,946 / 741	9.18	4.51	0.89	0.78
DeepSeek-V2-Lite (2024)	2.4 / 16	5.7	2,375 / 363	7.40	3.34	0.86	0.70
DeepSeekMoE 16B (2024)	2.8 / 16	2.0	714 / 122	4.42	2.09	0.77	0.52
Phi-3.5-MoE (2024)	6.6 / 42	4.9	742 / 117	4.52	2.06	0.78	0.51
DeepSeek-V2 (2024)	21.0 / 236	8.1	386 / 34	3.43	1.23	0.71	0.19
DeepSeek-V3 (2024)	37.0 / 685	14.8	400 / 22	3.50	1.01	0.71	0.01
Qwen3 30B-A3B (2025)	3.3 / 31	36.0	10,909 / 1,179	14.30	5.56	0.93	0.82
Llama 4 Scout (2025)	17.0 / 109	40.0	2,353 / 368	7.46	3.40	0.87	0.71
Qwen3 235B-A22B (2025)	22.0 / 235	36.0	1,636 / 153	6.39	2.34	0.84	0.57
GLM-4.5 (2025)	32.0 / 358	23.0	719 / 64	4.50	1.61	0.78	0.38
Llama 4 Maverick (2025)	17.0 / 402	22.0	1,294 / 55	5.77	1.51	0.83	0.34
Kimi K2 (2025)	32.0 / 1,026	15.5	484 / 15	3.79	0.87	0.74	−0.15
Median (12 models)				5.14	2.07	0.80	0.52

Note: Panel A: medians over the clean inference-demand sample, each technology evaluated at the model’s own count in the technology’s convention (non-embedding counts from each model’s configuration file). With the output head counted in FLOPs, the lifetime-to-training ratio for a non-embedding technology is $w(1 + N_{\text{head}}/N_{\text{ne}})$. Last column: the eight clean models below one billion parameters; comparing like with like, the non-embedding count lowers their median wedge by 8 percent under the Chinchilla form (9.13 against 9.96) and by 36 percent under the Farseer form (6.85 against 10.67). Panel B: w and $s = (w - 1)/w$ under the reference technology (Chinchilla, κ free), with N equal to active or to total parameters. A dense technology evaluated at active parameters treats a mixture-of-experts model as parameter-augmenting and overstates its wedge; at total parameters it ignores the loss advantage of sparsity. The two columns bound the wedge under a dense technology; neither is an estimate from a mixture-of-experts scaling law.

Source: Model cards, technical reports and Hugging Face configuration files; authors’ calculations.

The primary one (Section III) fits local quadratics near the path on budgets whose minimum is bracketed by two runs on each side; the second fits one quadratic to all runs of every budget, including unbracketed ones, and pools a ratio of sums. They agree on Llama 3 (0.660 and 0.653) and differ by 0.05 to 0.08 on Marin; after the second estimator’s simulation-based bias correction (to 0.66–0.68 on every design), the Marin gap is 0.03 to 0.05, within the range of the primary estimator’s own specifications (0.64 to 0.72). We use the primary estimator, whose Monte Carlo bias is negligible (Table D8).

On the 22 models, the median share is 0.61 under the developers’ technologies and 0.65 under the reference (0.61 and 0.67 on the 11 contemporaneous ones). Panel B shows selected models; four points qualify them.

Meta. Two of Meta’s ten budgets (3×10^{21} and 10^{22} FLOP, six runs each) do not bracket their minimum. On the eight that do, the path slope is 0.501 rather than 0.463 and $M^*(10^{24})$ is 15 rather than 31; Llama 3 8B’s wedge becomes 12.2 and the 405B’s 1.64. We keep the ten-budget path because the developer’s technology is meant to be the law Meta planned with, which sized the 405B model; its wedge of 0.97 is on Meta’s path by construction. With the second curvature estimator after its simulation-based correction ($\sigma^* = 0.678$), Llama 3 8B’s wedge is 7.07.

AI2. The κ -free ladder has a steep curvature ($1/\sigma^* - 1 = 0.84$), which makes the wedges of small models extreme: 20.2 for OLMo 2 1B, against 2.0 under the ladder with $\kappa = 1$. OLMo 2 32B’s interval, $[0.96, 1.72]$, contains one.

DeepSeek. The published law gives 3.42 for the 7B and 1.34 for the 67B model; the intervals reflect only the reference curvature.

Marin. The curvature estimator matters: Marin 8B’s wedge is 5.08 with the primary estimate and 6.66 with the corrected second one, and 1.43 to 6.66 across Marin’s own alternatives.

F6. Extrapolation and Partial Identification

Extrapolation in M .—Farseeer’s near-factorial grid reaches $M = 2,570$ in non-embedding parameters, which allows a direct check of parametric wedges at high ratios. Table F6 compares the local wedge of Proposition 4(i), from a kernel-weighted local quadratic of the loss surface, with the wedges of parametric forms fitted to the same runs. Every form understates the wedge at high M , and the gap grows with M . At $M \geq 1,024$, where the local wedge is 7.5, the κ -free form’s wedge is 0.70 times the local one and the Chinchilla form’s 0.44 times; the Chinchilla form fitted only to runs with $M \leq 100$ gives 0.26 times. The sign of the gap survives every bandwidth, but its size does not: for the κ -free form, $\ln(w_{\text{param}}/w_{\text{local}})$ runs from -0.25 at the narrower cross-validated bandwidth to -0.67 at twice the primary one (panel B). The source of the gap is curvature of

TABLE F5—LAB-OWN TECHNOLOGIES: ISOFLOP CURVATURE, PATHS AND WEDGES

<i>Panel A. Model-free curvature and compute-optimal paths of the IsoFLOP designs</i>						
Design	Budgets (valid)	Largest budget	σ^* used (s.e.)	σ^* , global quadratics [95%]	Path slope a	M^* at largest budget [95%]
Llama 3 (Meta)	10 (8)	10^{22}	0.660 (0.023)	0.653 [0.639, 0.666]	0.463	22.3 [20.8, 23.7]
Marin, Comma	7 (5)	3×10^{20}	0.700 (0.027)	0.650 [0.611, 0.669]	0.405	12.4 [10.7, 13.9]
Marin, DCLM	7 (7)	3×10^{20}	0.713 (0.027)	0.629 [0.586, 0.653]	0.424	9.4 [8.0, 10.6]
Marin, Nemotron-CC	8 (7)	3×10^{20}	0.705 (0.023)	0.644 [0.613, 0.666]	0.422	10.0 [8.7, 11.1]
Chinchilla profiles	9 (9)	3×10^{21}	0.673 (0.027)	0.610 [0.588, 0.644]	0.492	22.7 [18.2, 29.4]
<i>Panel B. Wedges of selected models under their developer's own technology</i>						
Model	Lab law vintage	Lab-own w [95%]	Lab-own s	Range over lab's alternatives	Reference w	Reference s
Llama 3 8B	contemp.	8.36 [5.67, 12.76]	0.88	3.09–12.16	6.73	0.85
Llama 3 70B	contemp.	2.51 [2.09, 3.02]	0.60	1.63–3.96	2.67	0.63
Llama 3.1 405B	contemp.	0.97 [0.88, 1.08]	−0.03	0.97–1.64	1.29	0.23
Llama 2 70B	ex post	0.97 [0.91, 1.04]	−0.03	0.97–1.42	1.13	0.12
OLMo 2 1B	contemp.	20.2 [14.5, 35.2]	0.95	2.01–20.2	7.83	0.87
OLMo 2 7B	contemp.	4.05 [3.22, 5.32]	0.75	1.28–4.05	3.99	0.75
OLMo 2 32B	contemp.	1.28 [0.96, 1.72]	0.22	0.91–1.28	2.54	0.61
DeepSeek LLM 7B	contemp.	3.42 [3.05, 3.87]	0.71		2.99	0.67
DeepSeek LLM 67B	contemp.	1.34 [1.31, 1.38]	0.26		1.15	0.13
Marin 8B	ex post	5.08 [3.10, 8.71]	0.80	1.43–6.66	6.26	0.84
Median, 22 models			0.61			0.65
Median, 11 contemp.			0.61			0.67

Note: Panel A: σ^* used is the model-free random-effects estimate of Section III over budgets whose minimum is bracketed by two runs on each side (Online Appendix Table D8). Global quadratics: a second estimator that fits one quadratic to all runs of each budget, including unbracketed ones, and pools a ratio of sums; design-conditional wild bootstrap, 999 draws. Its simulation-based bias correction gives 0.662–0.678, and it is used only in sensitivity rows; for Chinchilla it uses the 123-point snapped subset of the IsoFLOP compilation of Czech et al. (2026) (eight budgets) as a cross-check. Path slope and M^* : Approach-2 argmins regressed on $\ln C$ over all budgets; brackets: wild bootstrap. Meta's path reproduces the published law $D^* = 0.299 C^{0.537}$; fitted on its eight bracketed budgets it has slope 0.501 and $M^*(10^{24}) = 15$ instead of 31. Panel B: lab-own technologies are Meta's IsoFLOP path with the model-free curvature; AI2's OLMo ladder with κ free; DeepSeek LLM's published allocation law in its own units with the reference curvature (no curvature published); and Marin's Nemotron-CC path with the model-free curvature. Vintage: whether the lab's law was published before the model was planned (contemporaneous) or afterward (ex post). Llama 3.1 405B lies on Meta's path by construction, because Meta sized it with this law. Alternatives: Meta's $\kappa = 1$ law, primal and path on eight budgets; AI2's $\kappa = 1$ ladder; Marin's other corpora and $\kappa = 1$ rows. Reference: Chinchilla with κ free. Medians are over the 22 models with a lab-own technology (Meta 10, AI2 9, DeepSeek 2, Marin 1).

Source: Grattafiori et al. (2024); Czech (2026); Marin Community (2026); Bhagia et al. (2025); DeepSeek-AI et al. (2024); authors' calculations.

TABLE F6—IS THE WEDGE EXTRAPOLATED CORRECTLY? LOCAL AND PARAMETRIC WEDGES ON FARSEER

$M = D/N$	Local	$\ln(w_{\text{param}}/w_{\text{local}})$				Farseer Eq. 3
	$\ln w$	Chinchilla form		κ free		
	(w)	All runs	$M \leq 100$	All runs	$M \leq 100$	
<i>Panel A. By bin of M; non-embedding parameters, primary bandwidth</i>						
$M < 16$	−0.69 (0.50)	0.16	0.08	−0.02	0.02	−0.03
16–64	0.06 (1.06)	0.01	−0.23	0.05	−0.04	0.01
64–256	0.68 (1.96)	−0.19	−0.55	0.00	−0.18	−0.01
256–1,024	1.37 (3.93)	−0.49	−0.95	−0.14	−0.42	−0.13
$\geq 1,024$	2.01 (7.48)	−0.82	−1.36	−0.36	−0.70	−0.31
<i>Panel B. The $M \geq 1,024$ bin under alternative bandwidths and conventions</i>						
Extended-grid optimal h_N	1.84	−0.70	−1.22	−0.25	−0.58	−0.20
$h_N \times 0.75$	1.96	−0.78	−1.32	−0.32	−0.66	−0.27
Bandwidths $\times 1.5$	2.16	−0.96	−1.50	−0.49	−0.84	−0.44
Bandwidths $\times 2$	2.35	−1.14	−1.68	−0.67	−1.02	−0.63
With embeddings in N	2.74	−1.17	−1.67	−0.70	−1.00	−0.62

Note: Farseer’s 404 released runs (Li et al., 2025b). Local $\ln w$: Gaussian product-kernel local quadratic of $\ln L$ in $(\ln N, \ln D)$, with $w = \partial_{\ln N} \ln L / \partial_{\ln D} \ln L$ (Proposition 4(i); invariant to monotone transformations of the loss), averaged over grid points inside each model size’s observed token range with a kernel effective sample size of at least 15; primary bandwidths $h_N = 0.24$ and $h_D = 0.74$ in logs, from leave-one-out cross-validation over multiples of the standard deviation (h_N at the smallest multiple on that grid; the optimum on an extended grid is $h_N = 0.148$). Other columns: log ratio of the wedge implied by a parametric form fitted to the same runs to the local wedge at the same points; negative values mean that the form understates the wedge. Chinchilla form and κ family by Huber loss, Farseer’s Eq. 3 by nonlinear least squares; “ $M \leq 100$ ”: fitted only to runs with $M \leq 100$ and extrapolated. In the $M \geq 1,024$ row, the 95 percent intervals are $[-0.90, -0.75]$, $[-1.44, -1.28]$, $[-0.43, -0.28]$, $[-0.79, -0.64]$ and $[-0.38, -0.23]$ in column order, and those of the other rows are narrower: basic intervals from a wild cluster bootstrap by model size (Webb weights, 999 draws, every model refitted); they exclude the smoothing bias of the local estimator, which Panel B shows instead. Only the models of 0.10 to 0.34 billion parameters reach $M \geq 1,024$.

Source: Li et al. (2025b); authors’ calculations.

the wedge itself. In a regression of the local $\ln w$ on $u = \ln(M/M^*(C))$ and u^2 , the quadratic term is 0.020 (standard error 0.001) and positive at every bandwidth, so $\ln w$ is convex in u , whereas every member of the κ family makes it linear.

Applying Farseer’s gaps by bin of M to the 57 clean models inside Farseer’s range of M raises their median wedge under the κ -free Farseer technology from 3.10 to 3.56, and their median share from 0.68 to 0.72; at the narrower bandwidth, to 3.49 and 0.71. Under the Chinchilla form fitted to Farseer the median wedge rises from 2.29 to 3.73. The correction is negative for the 56 percent of these models with $M \geq 256$ and slightly positive, about +0.05 in logs, at M of 16 to 64. This transfers one recipe’s gaps to other developers’ models and to larger N and C , so it signs the error rather than correcting levels. For the 20 clean models beyond Farseer’s range of M there is no model-free evidence, and revealed preference recovers the technology a developer believed, which may be the parametric one (Remark A12).

Extrapolation in compute.—No design observes $M^*(C)$ above 10^{22} FLOP; the largest IsoFLOP budgets are 3×10^{21} (Chinchilla), 10^{22} (Llama 3) and 3×10^{20} FLOP (Marin; DeepSeek LLM). The 402 billion parameters and 16.55 trillion

tokens that Meta reports at 3.8×10^{25} FLOP extrapolate its law and are not used as an observation. We anchor M^* at each design’s largest budget, with a 95 percent wild-bootstrap interval: 22.7 [18.2, 29.4] at 3×10^{21} FLOP for Chinchilla’s profiles, 22.3 [20.8, 23.7] at 10^{22} for Llama 3, 12.4, 9.4 and 10.0 at 3×10^{20} for Marin’s three corpora, and DeepSeek’s published law. Proposition 5 then gives an identified set for $M^*(C)$ once the elasticity of M^* with respect to compute, $e = 1 - 2a$, is bounded. Table F7 reports four sets of assumptions. PI-1 lets e lie anywhere in the range spanned by the IsoFLOP paths and the published laws, including DeepSeek’s three data sets and MiniCPM, $[-0.16, 0.19]$. PI-2 adds that M^* does not fall with compute. PI-3 uses a developer’s own anchor where one exists. PI-4 admits every ex-ante technology’s in-support M^* as an anchor, with the widest range of slopes, $[-0.16, 0.31]$.

By Proposition 5(iii), the sign of $w - 1$ is the sign of $\ln M - \ln M^*(C)$ for any technology with single-peaked IsoFLOP profiles, so over-training is identified whenever M exceeds the upper bound of the set. That is the case for 86 percent of clean models under PI-1 and PI-2, 77 percent under PI-3 and 23 percent under PI-4. No model’s wedge is identified as below one under any set. The 11 models whose sign is not identified under PI-1 all have $M \leq 96$: Llama 2 70B, LLaMA 30B and 65B, Llama 3.1 405B, Qwen-72B, Qwen2-72B, Yi-34B, Falcon 40B and 180B, MPT-30B and DeepSeek LLM 67B. Under PI-3, AI2’s models join them, because the OLMo ladder’s own anchor is imprecise. Magnitudes need a curvature; with $1/\sigma^* - 1$ between 0.40 and 0.52, the range of the κ -free and model-free estimates, the medians of the models’ lower and upper bounds on s are 0.59 and 0.93 under PI-1.

Panel C repeats the logic on the 173 verified models in their own units, with anchors fitted by least squares to the IsoFLOP minima and wild bootstrap- t intervals (Online Appendix A, Remark A11). It illustrates the bounds rather than estimating them, because units are not harmonized. With a cross-sweep range of slopes, over-training is identified for 77 to 81 percent of the models beyond each anchor’s largest budget; with normal inputs alone ($|e| \leq 1$), for about a third. The two anchors’ point extrapolations disagree for Llama 3.1 405B, which is over-trained against the Chinchilla anchor and under-trained against Meta’s, and its sign is not identified under any set-valued assumption. The two Chinchilla anchors differ, 22.7 in Panel A and 20.5 [11.4, 36.8] in Panel C, because Panel A uses the eight-budget compilation of Czech et al. (2026) and a percentile interval, and Panel C all nine budgets of the digitization and a bootstrap- t interval with HC2-rescaled residuals.

F7. Conduct

Table F8 reports the three conduct tests of Section IV.E. All cluster by parent developer and use a restricted wild cluster bootstrap- t with the six-point weights of Webb (2023) (Cameron, Gelbach and Miller, 2008). Because $\ln \hat{w}$ is a function of (N, C) under one technology, every test concerns the choice of size at given

TABLE F7—PARTIAL IDENTIFICATION OF THE COMPUTE-OPTIMAL RATIO AT FRONTIER SCALE

Panel A. Bounds on $M^*(C)$ at frontier compute (harmonized units; total- N anchors)					
Assumption set	e range	$M^*(10^{23})$	$M^*(10^{24})$	$M^*(10^{25})$	Anchors
PI-1: IsoFLOP anchors	$[-0.16, 0.19]$	$[3.2, 57.3]$	$[2.3, 88.8]$	$[1.6, 138]$	5
PI-2: PI-1, nondecreasing	$[0.00, 0.19]$	$[8.0, 57.3]$	$[8.0, 88.8]$	$[8.0, 138]$	5
PI-3: Meta’s own anchor	$[-0.16, 0.19]$	$[14.5, 36.8]$	$[10.1, 57.0]$	$[7.1, 88.3]$	1
PI-4: all technologies	$[-0.16, 0.31]$	$[0.4, 1,609]$	$[0.3, 3,293]$	$[0.2, 6,739]$	30
Panel B. Sign of $w - 1$ on the clean sample ($n = 77$), each anchor in its own convention					
Assumption set	e range	$w > 1$	Not	Median bound on s	
		identified	identified	Lower	Upper
PI-1	$[-0.16, 0.19]$	0.86	0.14	0.59	0.93
PI-2	$[0.00, 0.19]$	0.86	0.14	0.59	0.89
PI-3	$[-0.16, 0.19]$	0.77	0.23	0.55	0.94
PI-4	$[-0.16, 0.31]$	0.23	0.77	-1.43	0.98
Panel C. Illustration in unharmonized units: verified models beyond the anchor’s largest budget					
Assumption	e range	$M^*(10^{24})$	Share of models		
			$w > 1$	$w < 1$	Not identified
Chinchilla anchor ($n = 149$)					
Normal inputs	$[-1.00, 1.00]$	$[0.03, 12,651]$	0.31	0.00	0.69
Cross-sweep slopes	$[-0.14, 0.26]$	$[5.0, 168]$	0.77	0.01	0.21
Anchor’s own slope	$[-0.16, 0.17]$	$[4.3, 100]$	0.83	0.01	0.16
Point extrapolation	0.005	21.1	0.97	0.03	0.00
Llama 3 anchor ($n = 137$)					
Normal inputs	$[-1.00, 1.00]$	$[0.12, 4,020]$	0.34	0.00	0.66
Cross-sweep slopes	$[-0.14, 0.26]$	$[6.4, 133]$	0.81	0.01	0.18
Anchor’s own slope	$[-0.09, 0.24]$	$[7.9, 119]$	0.84	0.02	0.14
Point extrapolation	0.074	31.3	0.90	0.10	0.00

Note: $M^*(C)$ is never observed beyond 10^{22} FLOP: the largest IsoFLOP budgets are 3×10^{21} (Chinchilla), 10^{22} (Llama 3) and 3×10^{20} (Marin; DeepSeek LLM). Anchors: each design’s Approach-2 path at its largest budget with its 95 percent wild-bootstrap interval (DeepSeek: published law). Extrapolation (Proposition 5): $\ln M^*(C) = \ln M^*(C_0) + e \ln(C/C_0)$ with the path elasticity $e = 1 - 2a$ anywhere in the stated range. PI-1: range of e over the IsoFLOP paths and published laws (including DeepSeek LLM’s three data sets and MiniCPM); PI-2 adds M^* nondecreasing in C ; PI-3 uses the model’s own lab’s anchor where it exists (Meta, Marin, DeepSeek, AI2) and PI-1 otherwise; PI-4 uses every ex-ante technology’s in-support M^* and all path slopes. Llama 3’s 402B parameters and $16.55T$ tokens at 3.8×10^{25} FLOP extrapolate Meta’s law and are not used as an anchor. By Proposition 5(iii), $w > 1$ is identified when $\ln M$ exceeds the upper bound for every anchor, for any technology with single-peaked IsoFLOP profiles; no model’s $w < 1$ is identified in Panel B. Bounds on s use $1/\sigma^* - 1 \in [0.40, 0.52]$, the range of the κ -free and model-free estimates, and are medians of the models’ bounds. Panel C repeats the logic on the verified sample with each model’s own parameter and token conventions and anchors fitted to the IsoFLOP minima (wild bootstrap- t intervals with HC2-rescaled residuals, $M^* = 20.5$ $[11.4, 36.8]$ at 2.9×10^{21} for Chinchilla and 22.3 $[12.2, 40.2]$ at 10^{22} for Llama 3); cross-sweep slopes: allocation exponents 0.37 – 0.57 ; anchor’s own slope: its 95 percent interval. It illustrates the bounds, not the paper’s estimate. *Source:* Hoffmann et al. (2022); Grattafori et al. (2024); Marin Community (2026); DeepSeek-AI et al. (2024); Hu et al. (2024); authors’ calculations.

compute.

Open and closed models.—The test uses the production-scale universe of Online Appendix B ($6ND \geq 10^{21}$ FLOP, 2019–2026), restricted to models with confident parameter and token counts and without mixture-of-experts or non-transformer architectures, with the clean-sample exclusions applied to verified models: 186 models from 67 developers, 23 of which released at least one closed model. At given compute and year, open-weight models have log wedges 0.52 higher ($p = 0.011$), and 0.71 higher among releases since 2023 ($p = 0.001$). On the share the premium is 0.29 ($p = 0.097$). Closed developers have disclosed few allocations since 2023: six in 2024 and two in 2025.

Serving footprint and deployment target.—A developer counts as serving if, at the model’s release, it operated a commercial interface or consumer product that served its own language models. Meta is coded as serving from 27 September 2023, the launch of Meta AI, so Llama 1 and 2 are not; Alibaba, Google, Microsoft, IBM, MosaicML, DeepSeek (from 2024), 01.AI (from 2024), Moonshot, Zhipu and Together are coded as serving; TII, Hugging Face, H2O.ai, AI2, Stability AI, Swiss AI, Marin, M-A-P, TinyLlama and EleutherAI are not. Two dates are judgment calls: phi-1.5, released under a research license in September 2023, is coded as served, and 01.AI’s interface is dated to 2024. A model has an on-device target if its card names on-device, mobile or phone use, and a local target if it names laptops or desktops. In the clean sample, 33 models come from 7 serving developers; 9 models have an on-device and 3 a local target, but only two and three developers carry these attributes. With three or fewer treated clusters the wild cluster bootstrap is unreliable (MacKinnon and Webb, 2018), so only the serving test is informative, and with 18 clusters it has little power: 0.45 (standard error 0.26; $p = 0.39$). Adding the research suites and replications raises the sample to 136 models and 26 developers and gives 0.47 ($p = 0.070$), and 0.14 on the share ($p = 0.046$). That sample adds suites whose developers do not serve, however, so the contrast is partly between suites and products.

Tiers.—Windows around quantized memory boundaries (2.4–3.3, 6.5–9.5, 11.5–14.9, 26–32.9 and 65–72.9 billion parameters) hold 51 percent of the 220 open-weight production-scale models released in 2023–2026, against 13 percent under a smooth polynomial counterfactual (Chetty et al., 2011; Kleven, 2016); the excess mass is 83 models (standard error 9.8), 45 of them at 6.5–9.5 billion. Position relative to the tiers does not predict the wedge beyond what the mechanical relation between N and $\ln w$ implies. Models inside a window have lower wedges given compute (−0.39, standard error 0.28), and 99.7 percent of 999 randomly shifted tier menus give a larger coefficient, so the real windows are, if anything, less over-trained than placebo windows. The log distance to the next cap has a coefficient of 0.46, against a placebo mean of 0.45. Tier menus shape the choice of N ; the tier model’s prediction for the wedge is not visible at given compute.

TABLE F8—CONDUCT TESTS AND DERIVATIVE COUNTS

Test	Sample	n	G	Coef.	(s.e.)	p (WCR)	Note
<i>Panel A. Conduct tests; outcome $\ln w$ (reference), controls year effects and $\ln C$</i>							
(a) Open weights	universe	186	67	0.52	(0.18)	0.011	$G_0=23$ closed
2023 and later	universe	150	60	0.71	(0.17)	0.001	$G_0=15$ closed
(b) Developer serves	clean	77	18	0.45	(0.26)	0.390	$G_1=7$
On-device target	clean	77	18	0.22	(0.23)	0.410	$G_1=2$
Local target	clean	77	18	0.04	(0.25)	0.878	$G_1=3$
Developer serves	clean + suites	136	26	0.47	(0.17)	0.070	$G_1=8$
On-device target	clean + suites	136	26	0.57	(0.24)	0.209	$G_1=2$
(c) In a tier window	clean	75	18	-0.39	(0.28)	0.340	placebo p 1.00
	open, 2023+	133	52	-0.16	(0.21)	0.523	placebo p 0.59
In distance to cap	clean	75	18	0.46	(0.20)	0.059	placebo p 0.61
	open, 2023+	133	52	0.62	(0.16)	0.005	placebo p 0.37
(c) Excess mass	open 2023–26	220		82.6	(9.8)		share 0.51 vs 0.13
(c) Excess mass	clean	77		37.2	(5.5)		share 0.58 vs 0.10
<i>Panel B. Hugging Face derivative repositories</i>							
HF quantizations on $\ln M$	clean	74	16	0.61	(0.11)	0.018	
HF fine-tunes on $\ln M$	clean	74	16	0.87	(0.18)	0.029	
HF adapters on $\ln M$	clean	74	16	0.90	(0.25)	0.075	
HF all derivatives on $\ln M$	clean	74	16	0.80	(0.17)	0.031	
HF all on $\ln(T/D)$	clean	74	16	0.60	(0.22)	0.086	

Note: G : developer clusters. s.e.: cluster-robust (CRV1) standard error. p (WCR): restricted wild cluster bootstrap- t with Webb six-point weights, 9,999 draws (floor 10^{-4}), clustered by developer. (a) Production-scale universe ($6ND \geq 10^{21}$ FLOP, 2019–2026; open-weight and closed models), confident N and D only (Epoch ‘Confident’ or documented), no mixture-of-experts or non-transformer models, clean-sample exclusions for verified models; “open, 2023+”: its open-weight models released 2023–2025. (b) Serving: developer operated a commercial API or consumer product serving its own language models at the model’s release (coding and sources in the replication package); on-device/local: stated target in the model card. Both entered jointly; clean + suites adds the research suites and replications. G_1 (G_0): developer clusters with (without) the attribute, e.g. closed models in (a); with $G_1 \leq 3$ the wild cluster bootstrap is unreliable (MacKinnon and Webb, 2018) and the on-device and local rows are not informative. (c) Tier windows at quantized memory boundaries: 2.4–3.3B, 6.5–9.5B, 11.5–14.9B, 26–32.9B, 65–72.9B parameters; placebo = share of 999 shifted tier menus (edges multiplied by e^u , $u \sim U(-0.69, 0.69)$) giving a coefficient at least as favorable to the tier story; because $\ln w$ is a deterministic function of (N, C) under one technology, tier regressors correlate with $\ln w$ mechanically and only the placebo comparison is informative. Excess mass: open-weight production-scale models released 2023–2026 (total parameters; non-transformers excluded) and the clean sample, against a polynomial (degree 5) counterfactual of the $\log_{10} N$ histogram (0.1 bins) excluding the windows; bootstrap s.e. (999). Panel B: HF = number of derivative repositories (quantizations, fine-tunes, adapters, merges) of the base and official post-trained releases, $\ln(1 + \cdot)$.

Source: Epoch AI (Epoch AI, 2026); model cards; Hugging Face model tree; authors’ calculations.

F8. Family Token Budgets

A family has a common token budget if its members' budgets lie within 10 percent of each other; Llama 3 and 3.1 count as one herd. Eleven families of the clean sample, with 32 models, qualify; among all 173 verified models, 23 families with 83 models do.

If sizes are chosen at the margin and D is shared, the family's first-order condition in D identifies one number, a weighted average of the members' values of compactness (Proposition A9). Without data costs, $\sum_i \pi_i m_{N,i} = \bar{w}_H - 1$, where $\bar{w}_H = [\sum_i \omega_i / w_i]^{-1}$ is the harmonic mean of the member wedges weighted by training compute and $\pi_i \propto \omega_i / w_i$. Member wedges then differ only through the technology, $\ln w_i - \ln w_j = -\alpha(\ln N_i - \ln N_j)$, and are not revealed preferences. If sizes come from a menu, the condition in D says nothing about the members' values of compactness. In three family problems solved by brute force, the condition holds to within 2.2×10^{-8} .

Table F9 reports the family shares $1 - 1/\bar{w}_H$. They run from 0.18 for DeepSeek LLM to 0.89 for Qwen3. Because the flagship carries most of a family's compute, the family share lies far below its members' typical shares: the Llama 3 herd's members have wedges from 1.30 to 6.73, but the herd's weighted condition gives 1.42, a share of 0.29. The family number equals the compute-weighted mean of the members' lifetime-to-training ratios when those ratios are uncorrelated with $1/w_i$, and it is a lower bound when smaller members, which are more over-trained, are also planned for more serving.

F9. The Cost Side

FLOPs are not costs. Table F10 lets the price of a training FLOP vary with model size (elasticity δ , for example because utilization changes with size) and serving cost per token scale as N^ϕ , with a price p of a serving FLOP relative to a training FLOP (Proposition A8(iii)(f)). The first-order condition is $w = 1 + \delta + \phi X_S / X_T$, so the share $s = (w - 1 - \delta) / (w - 1 - \delta + \phi)$ does not depend on p . It moves little: the median is 0.72 to 0.77 for δ between -0.1 and 0.1 with $\phi = 1 + \delta$, and 0.70 to 0.86 for ϕ between 0.5 and 1.25 with $\delta = 0$. Token counts do depend on p : at the median, planned serving tokens are 9.0 times training tokens at $p = 1$ and 0.9 times at $p = 10$, and for Llama 3 8B 17.2 and 1.7 times.

F10. Validation and Trends

Token volumes.—OpenRouter's documented interface reports no usage for any model, and its terms of service (updated 31 August 2026) prohibit scripts that scrape its activity pages, so per-model token volumes cannot be collected. The developer-level totals of Aubakirova et al. (2026) for November 2024 to November 2025 give an order of magnitude. One year of OpenRouter traffic equals 4.6 percent of the lifetime tokens that DeepSeek's verified releases imply at $p = 1$, 0.7 percent for Moonshot, 0.5 percent for Zhipu, 0.2 percent for Meta, 0.08 percent

TABLE F9—FAMILIES WITH A COMMON TOKEN BUDGET: WHAT THE FAMILY REVEALS

Family	Members	Max./min. D	Member wedges w_i (range)	Compute-weighted mean of w_i	Family $\bar{w}_H$	Family share $1 - 1/\bar{w}_H$
Qwen3	5	1.00	7.57–29.56	10.05	9.13	0.89
Granite 3.0	2	1.00	6.07–9.97	6.99	6.68	0.85
Qwen2.5	7	1.00	2.86–23.75	4.11	3.53	0.72
StableLM-Alpha	2	1.00	2.65–3.79	2.99	2.91	0.66
Apertus	2	1.00	2.67–6.72	3.09	2.85	0.65
Olmo 3	2	1.08	2.44–4.73	2.89	2.70	0.63
Yi	2	1.00	1.81–3.79	2.11	1.97	0.49
Llama 3 herd	3	1.04	1.29–6.73	1.58	1.42	0.29
MPT	2	1.00	1.20–2.22	1.39	1.31	0.24
Llama 2	3	1.00	1.13–3.04	1.45	1.29	0.23
DeepSeek LLM	2	1.00	1.15–2.99	1.32	1.22	0.18

Note: The eleven families of the clean inference-demand sample whose members were trained on token budgets within 10 percent of each other (32 models). Wedges under the reference technology. With a common D and sizes chosen at the margin, the family’s first-order condition in D identifies one number, a weighted average of the members’ values of compactness: $\sum_i \pi_i m_{N,i} = \bar{w}_H - 1$ without data costs, where $\pi_i \propto \omega_i/w_i$, ω_i is member i ’s share of family training compute and $\bar{w}_H = [\sum_i \omega_i/w_i]^{-1}$ is the compute-weighted harmonic mean of the member wedges, computed from the technology (Proposition 2(iii); Online Appendix Proposition A9); $1 - 1/\bar{w}_H = \sum_i \omega_i s_i$ is the family share. Member wedges then differ only through the technology, $\ln w_i - \ln w_j = -\alpha(\ln N_i - \ln N_j)$, and are not revealed-preference objects. If sizes come from a menu, the condition in D says nothing about the members’ values of compactness. In three family problems solved by brute force, the condition holds to within 2.2×10^{-8} .

Source: Model cards and technical reports; authors’ calculations.

TABLE F10—SENSITIVITY OF THE REVEALED SERVING SHARE TO THE COST SIDE

Case	δ	ϕ	Share s		Median T/D		Llama 3 8B T/D	
			median	Llama 3 8B	$p = 1$	$p = 10$	$p = 1$	$p = 10$
A	−0.1	0.90	0.77	0.87	10.3	1.0	19.4	1.9
A	0.0	1.00	0.75	0.85	9.0	0.9	17.2	1.7
A	0.1	1.10	0.72	0.84	7.9	0.8	15.3	1.5
B	0.0	0.50	0.86	0.92	17.9	1.8	34.4	3.4
B	0.0	0.75	0.80	0.88	11.9	1.2	22.9	2.3
B	0.0	1.00	0.75	0.85	9.0	0.9	17.2	1.7
B	0.0	1.25	0.70	0.82	7.2	0.7	13.7	1.4

Note: Lifetime cost $6N^{1+\delta}D + 2pN^\phi T$ in training-FLOP units at the model’s own size: δ is the elasticity of the training cost of a FLOP with respect to N (model FLOPs utilization falling or rising with size), ϕ the elasticity of serving cost per token with respect to N , and p the price of a serving FLOP relative to a training FLOP. The first-order condition is $w = 1 + \delta + \phi X_S/X_T$, with X_S and X_T serving and training expenditure, so $X_S/X_T = (w - 1 - \delta)/\phi$, $s = (w - 1 - \delta)/(w - 1 - \delta + \phi)$ and $T/D = 3(w - 1 - \delta)/(p\phi)$. Case A: $\phi = 1 + \delta$, so that $w = (1 + \delta)(1 + pT/3D)$. Case B: $\delta = 0$. The share s does not depend on p . Clean inference-demand sample, reference technology.

Source: Authors’ calculations.

for Google and 0.04 percent for Alibaba. The platform carries a small part of a model’s lifetime use, and a test of levels is not possible with public data.

Derivative repositories.—From the Hugging Face model tree we count the quantizations, fine-tunes, adapters and merges derived from each model’s base repository, its official post-trained releases and same-run re-releases (74 clean models from 16 developers; MosaicML’s and Cerebras’s repositories require authentication). At given compute and release year, models with higher M have more derivatives: elasticities of 0.61 for quantizations, 0.87 for fine-tunes, 0.90 for adapters and 0.80 for all derivatives (Table F8, panel B). The elasticity with respect to the planned T/D is 0.60 ($p = 0.086$), below one. Smaller models of equal compute are adapted more, as planned demand predicts but also as accessibility predicts, so the counts do not validate levels.

Trends.—Figure F1 plots the median share by release year in the universe of the open–closed test. Among open-weight models it rises from 0.51 in 2023 to 0.81 in 2024 and 0.85 in 2025. Closed models with disclosed allocations had negative median shares in 2021–2023 and 0.64 in 2024, from six models; there are too few after 2024 to plot. The compute-weighted aggregates are in Table 3, panel C. On the clean sample the aggregate share rises from 2023 to 2025 under every one of the 32 technologies. Under the developers’ own technologies where available it is 0.27, 0.56 and 0.82 in 2023, 2024 and 2025, against 0.28, 0.58 and 0.83 under the reference. Llama 3.1 405B accounts for 55 percent of the clean sample’s 2024 compute; without it the 2024 share is 0.73.

What is not done.—Four checks remain open. A test of the production approach on a second margin, in the spirit of Raval (2023), such as the sparsity of mixture-of-experts models, the number of key-value heads or the vocabulary, which trade quality against serving cost at different training costs, has not been run. Token counts are not normalized by bytes. Durability, the obsolescence of models that shortens the horizon over which serving demand is planned, is part of T but is not modeled. And token counts reported as corpus sizes, such as the 18 trillion tokens stated for every Qwen2.5 size, have not been audited against the tokens each model processed.

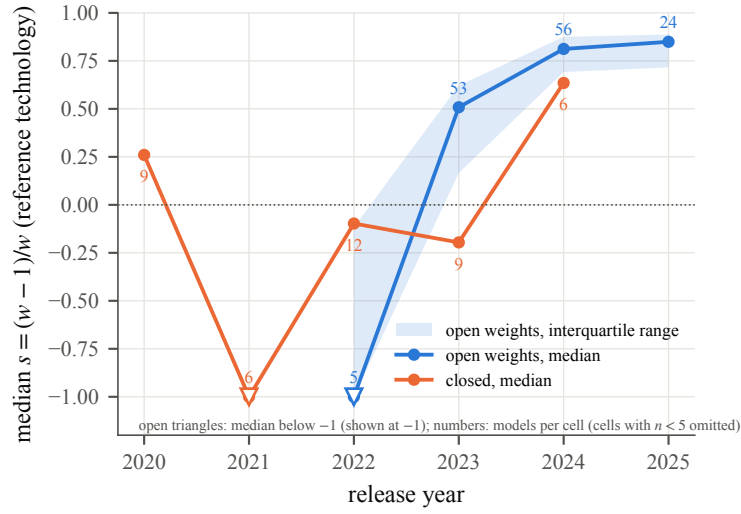


FIGURE F1. REVEALED EXPENDITURE SHARES BY RELEASE YEAR: OPEN AND CLOSED MODELS

Note: Median of $s = (w - 1)/w$ under the reference technology by release year, for open-weight and closed models of the production-scale universe ($6ND \geq 10^{21}$ FLOP) with confident parameter and token counts, excluding mixture-of-experts and non-transformer models and applying the clean-sample exclusions to verified models (186 models). Shaded: interquartile range of open-weight models. Numbers: models per cell; cells with fewer than five models are omitted; medians below -1 are drawn at -1 (open triangles). For developers that serve their models, s is the planned serving share of lifetime cost; for others, the internalized value of compactness.

Source: Epoch AI ([Epoch AI, 2026](#)); model cards; authors' calculations.